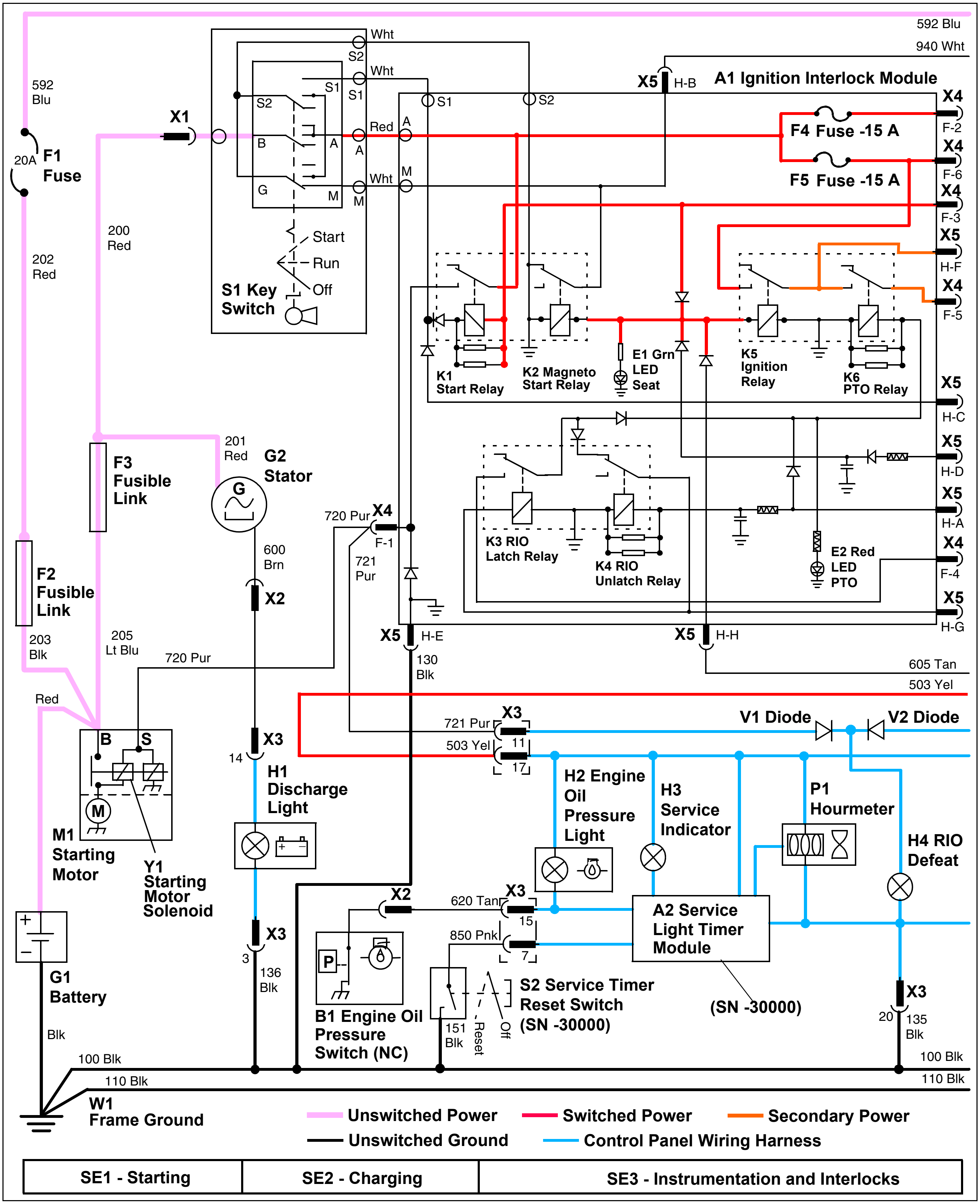
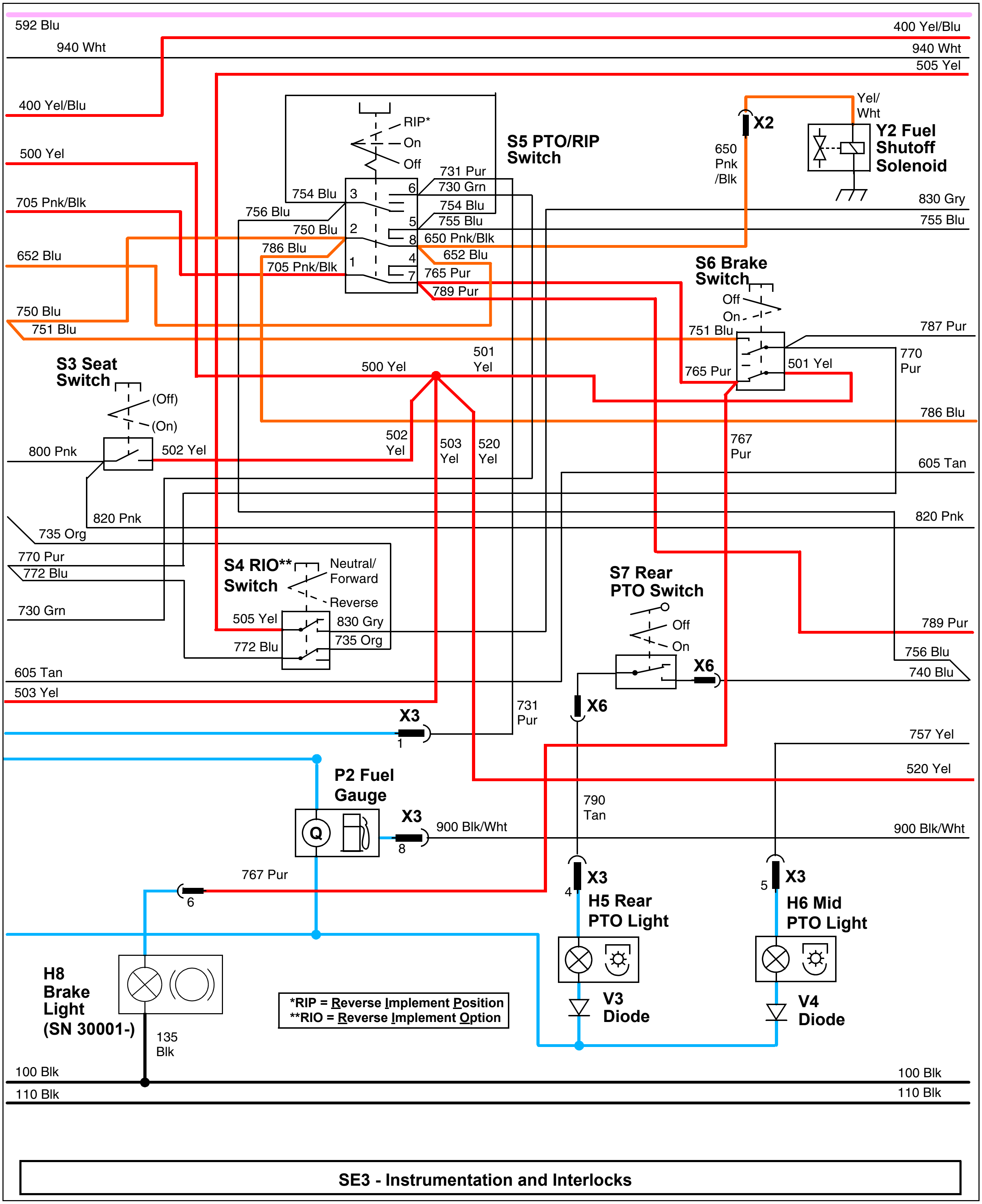
Electrical Schematic - X465 (1 of 3)



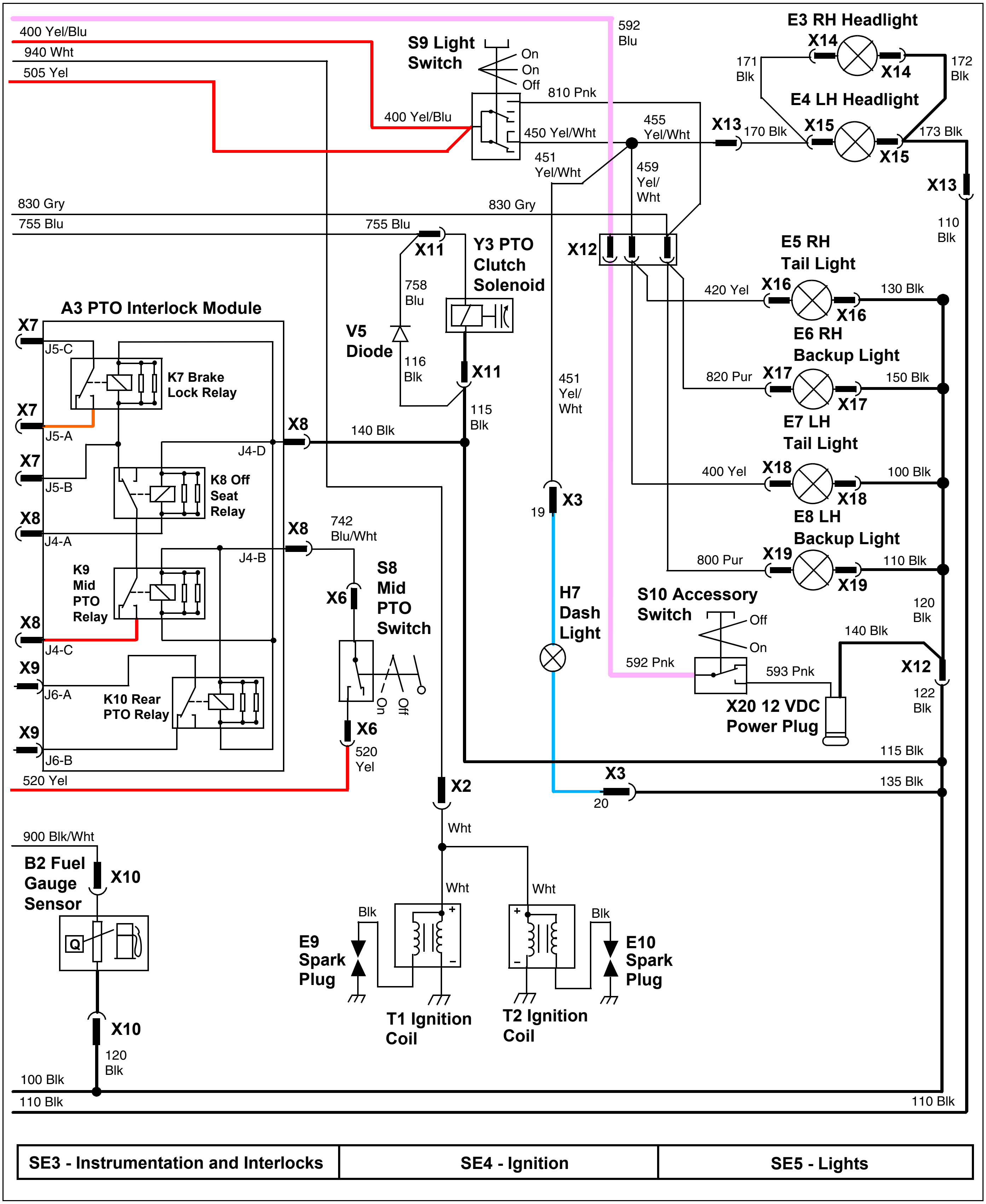
MIF

Electrical Schematic - X465 (2 of 3)



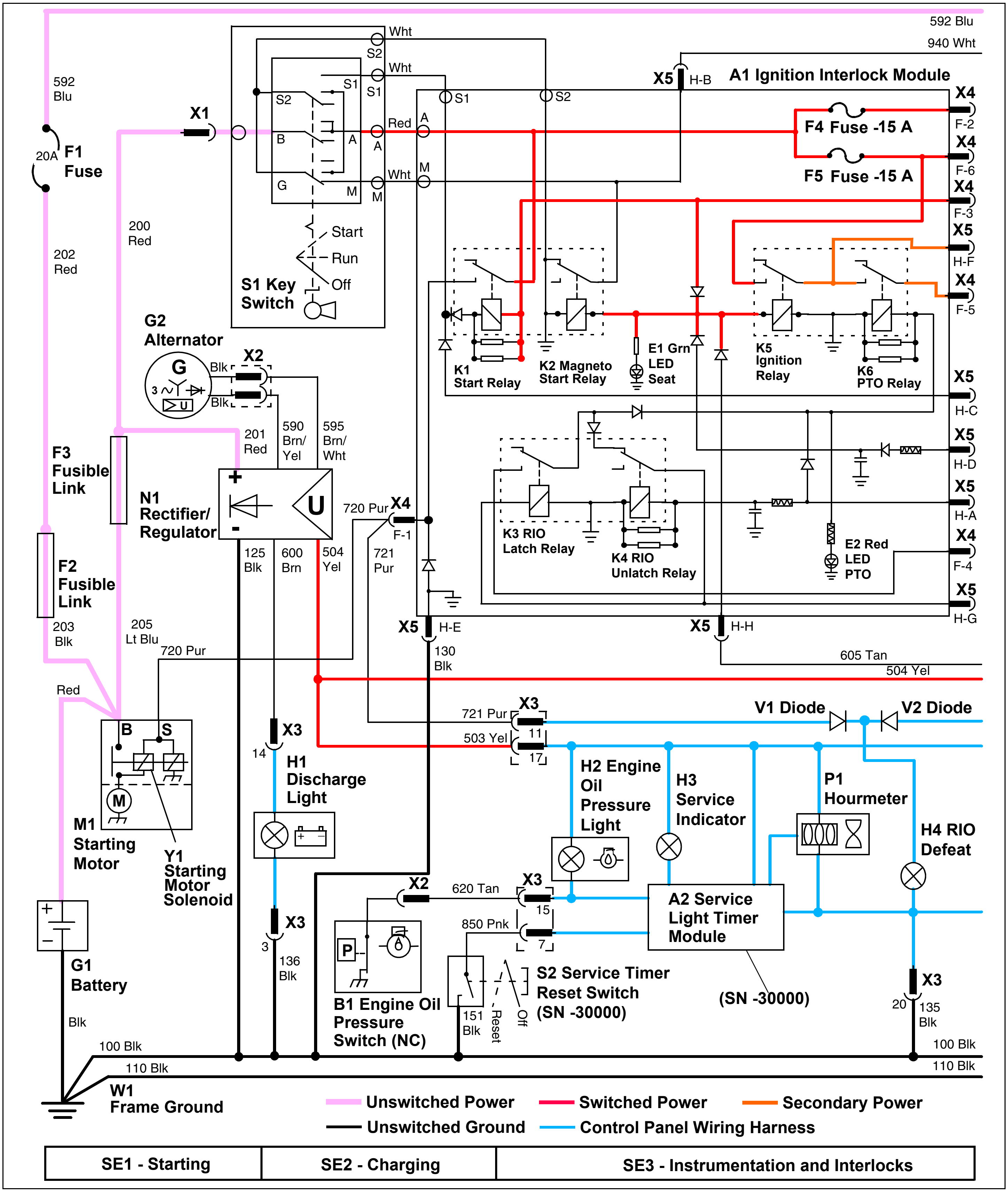
MIF

Electrical Schematic - X465 (3 of 3)



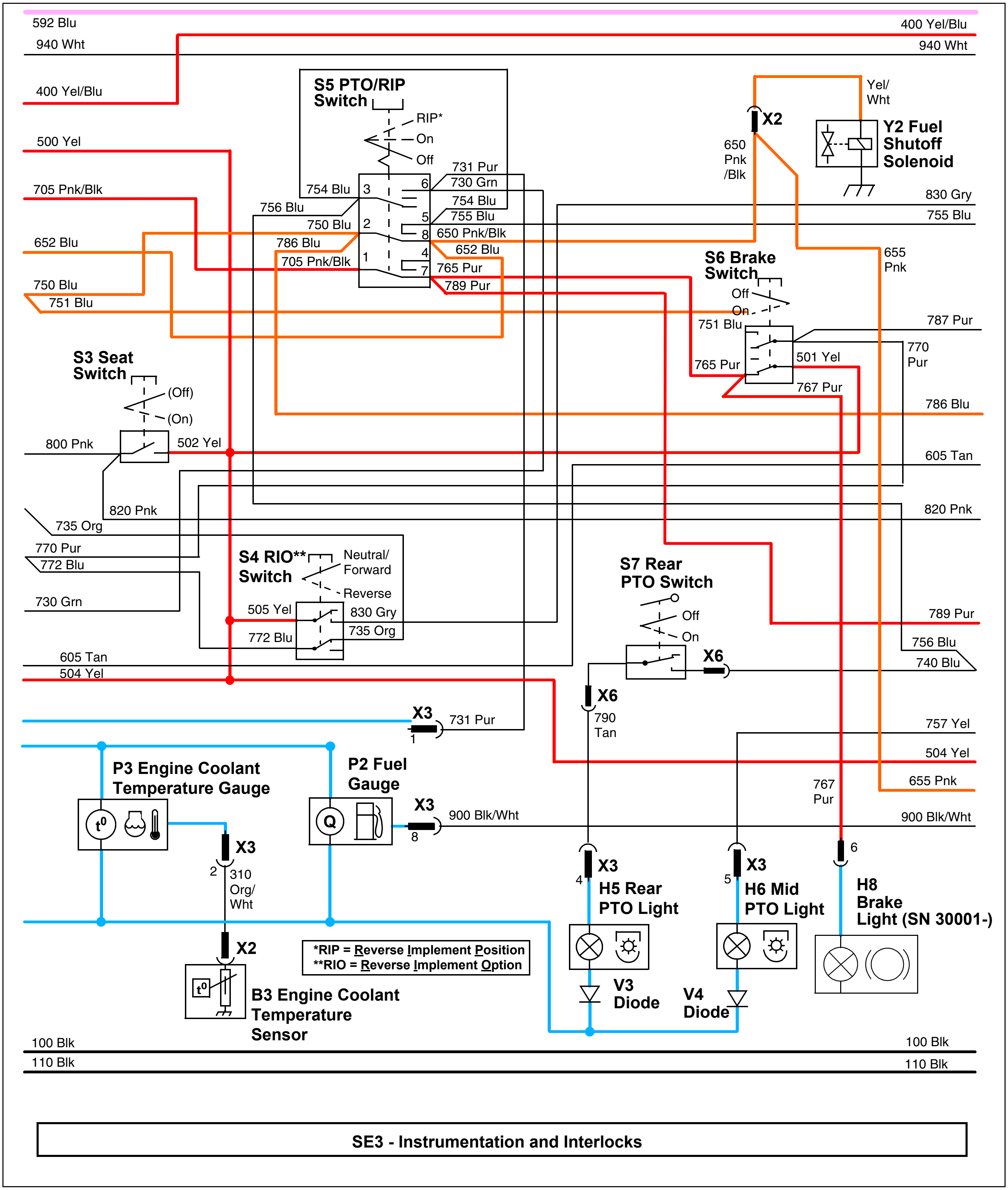
MIF

Electrical Schematic - X475 and X575 (1 of 3)



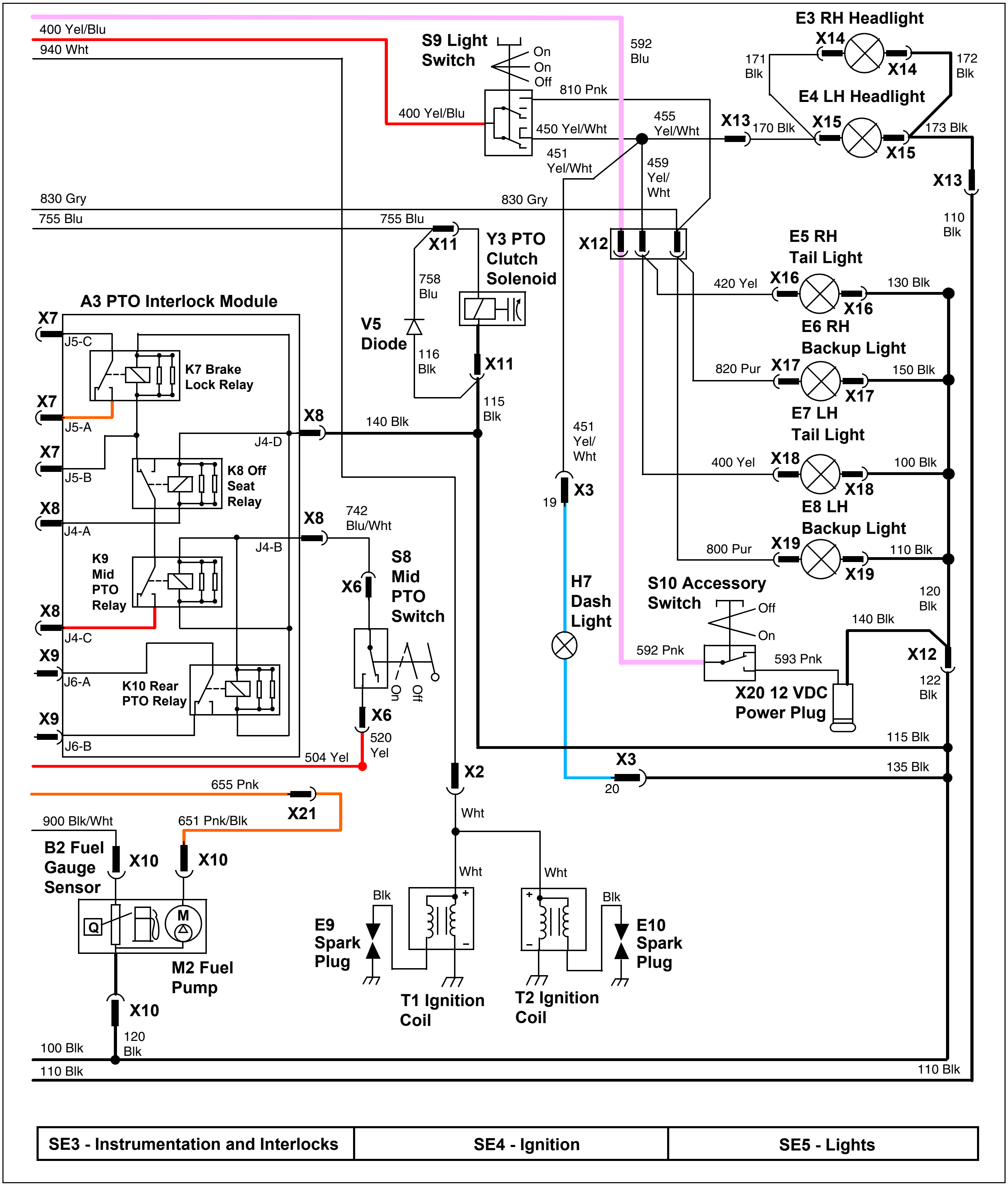
MIF

Electrical Schematic - X475 and X575 (2 of 3)



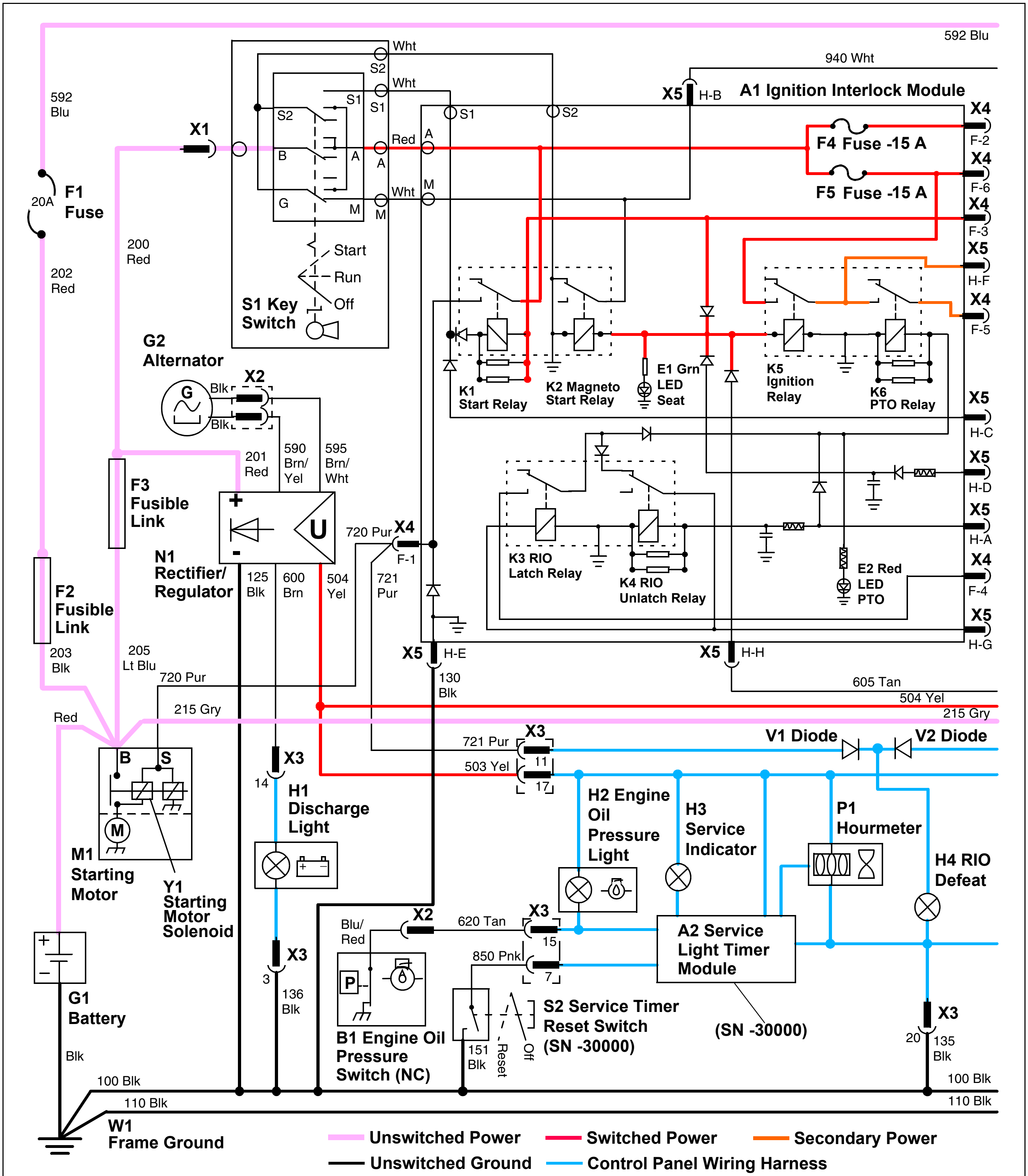
MIF

Electrical Schematic - X475 and X575 (3 of 3)



MIF

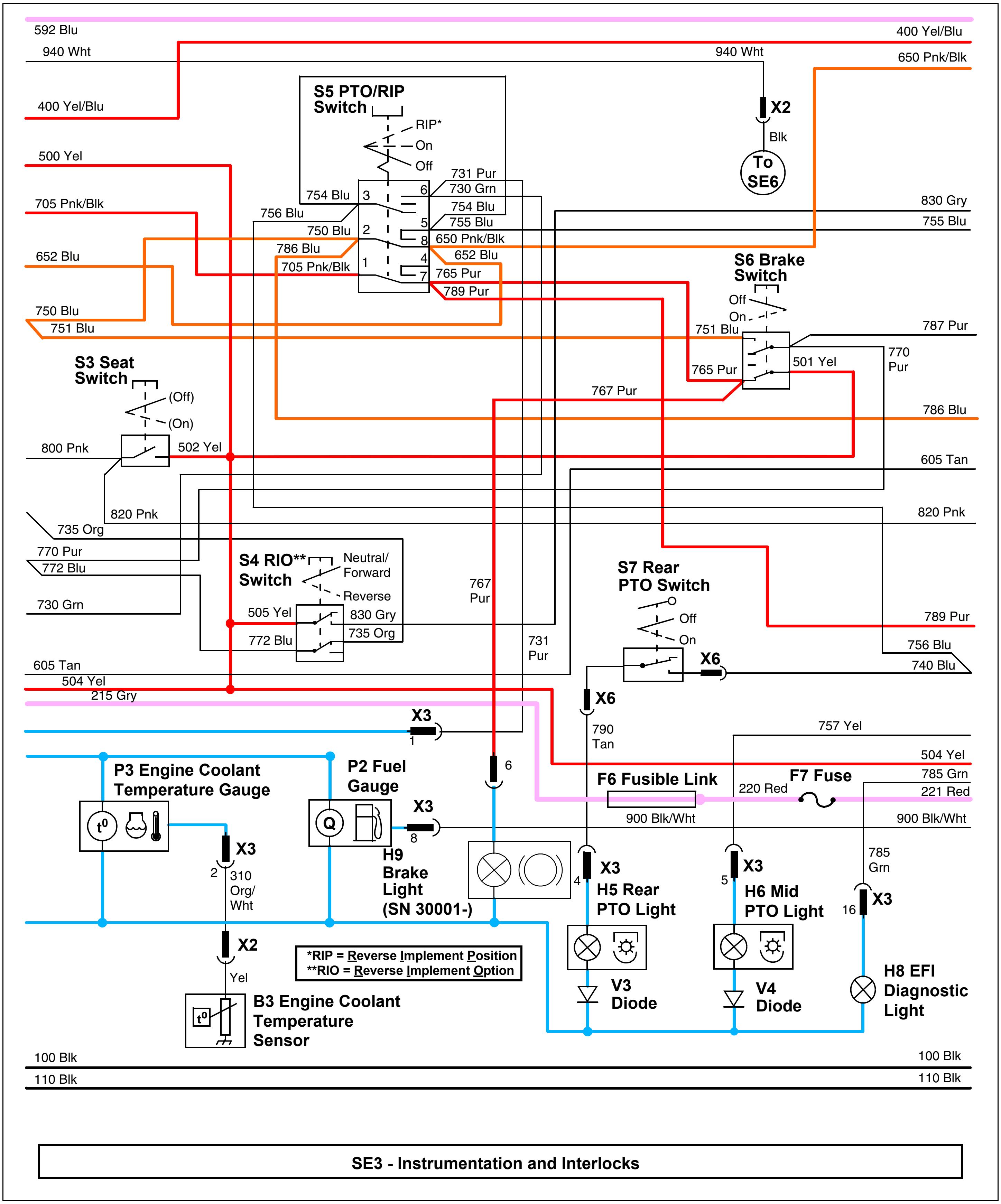
Electrical Schematic - X485 and X585 (1 of 4)



SE1 - Starting	SE2 - Charging	SE3 - Instrumentation and Interlocks

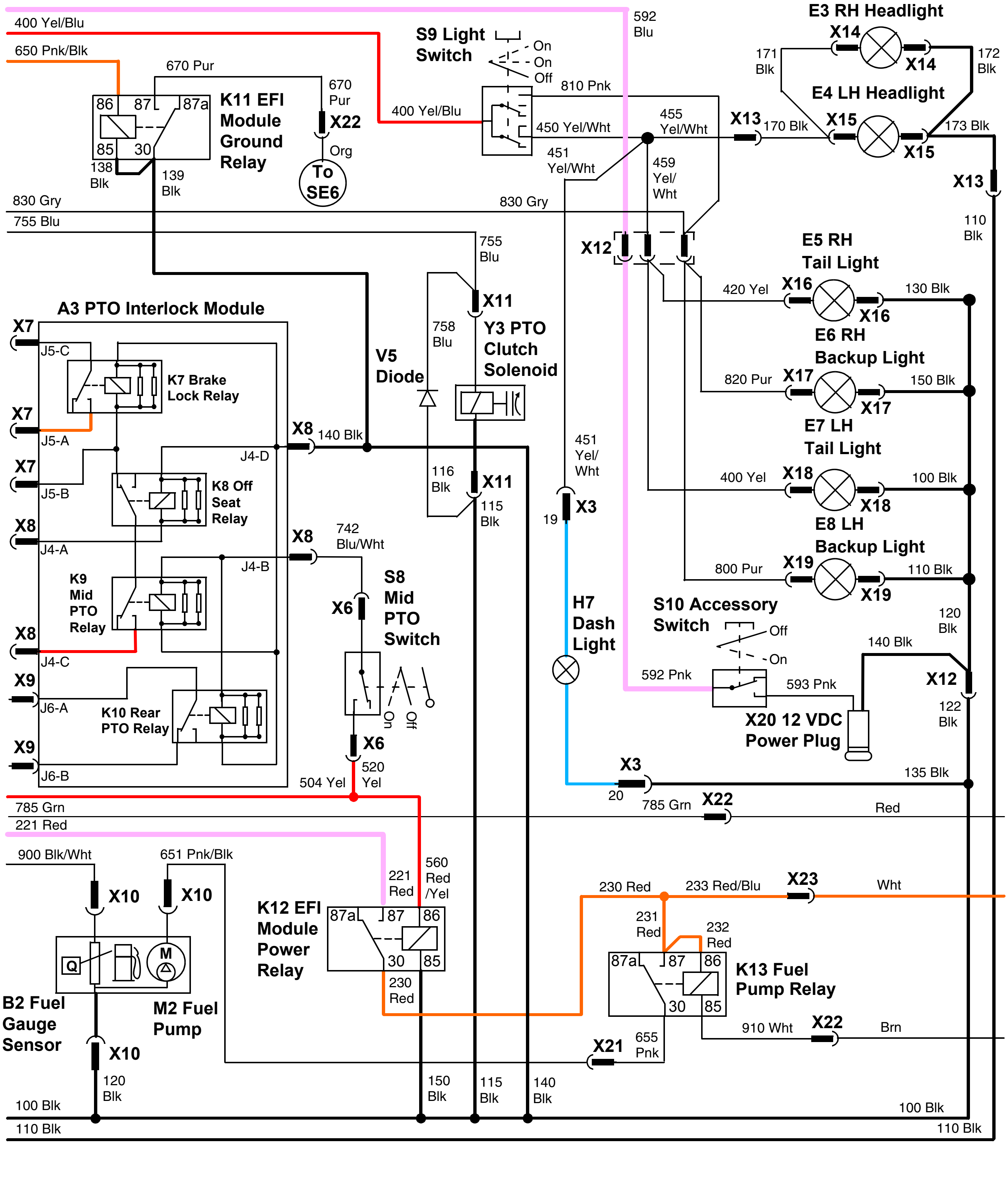
MIF

Electrical Schematic - X485 and X585 (2 of 4)



MIF

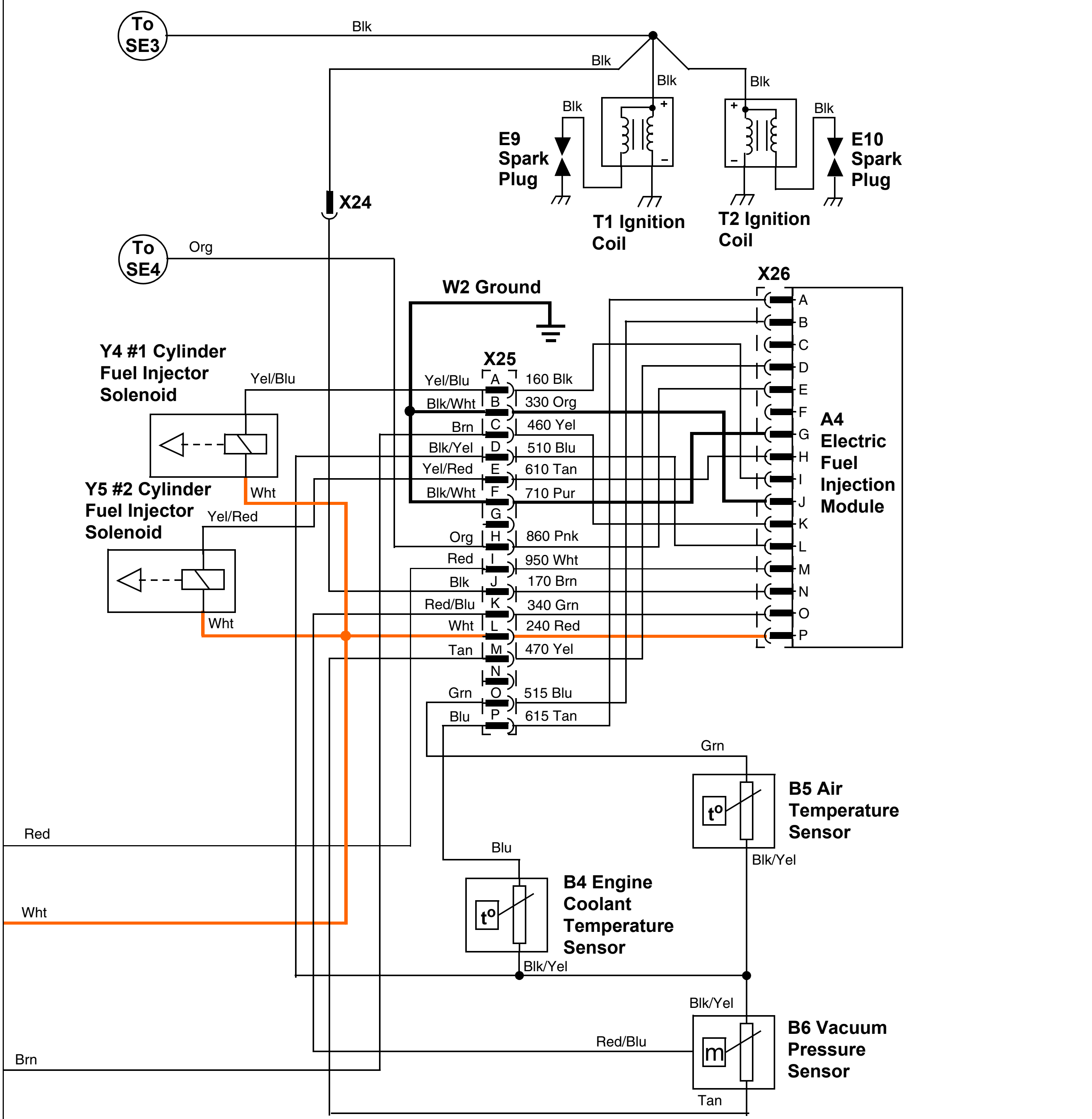
Electrical Schematic - X485 and X585 (3 of 4)



SE3 - Instrumentation and Interlocks	SE4 - Ignition	SE5 - Lights

MIF

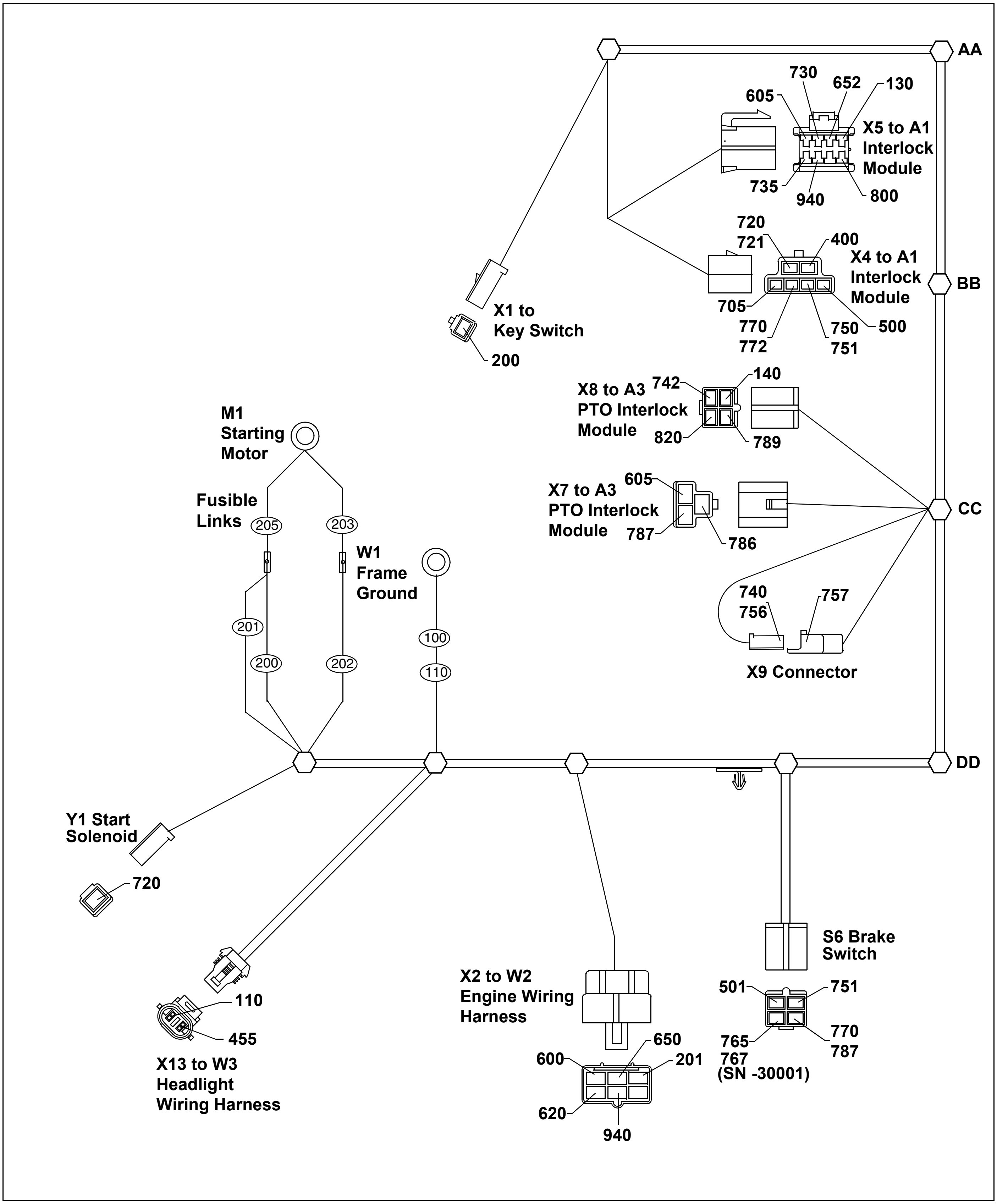
Electrical Schematic - X485 and X585 (4 of 4)



SE6 - Ignition and Carburetion - X485 and X585

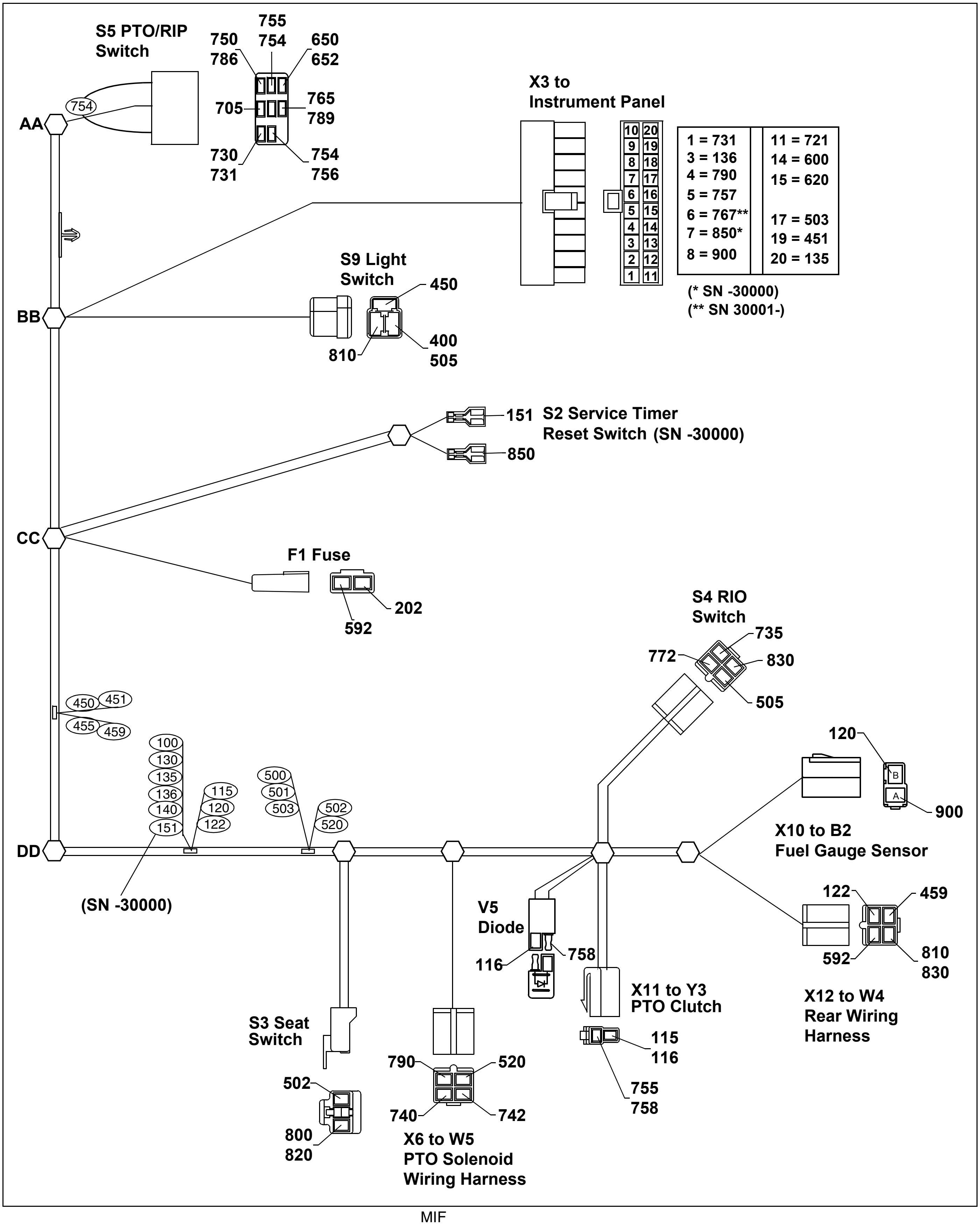
MIF

W1 Main Wiring Harness - X465 (1 of 2)

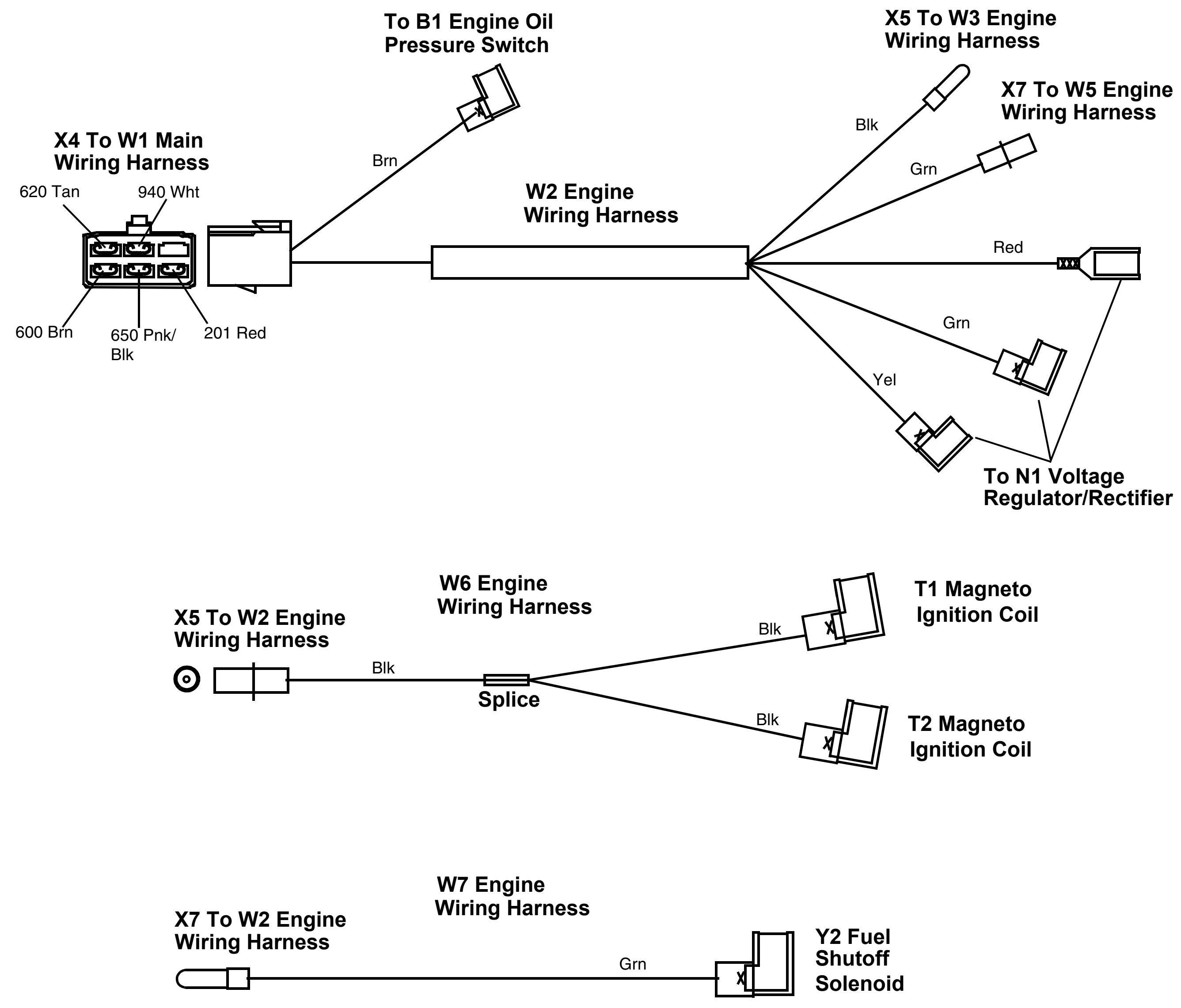


MIF

W1 Main Wiring Harness - X465 (2 of 2)

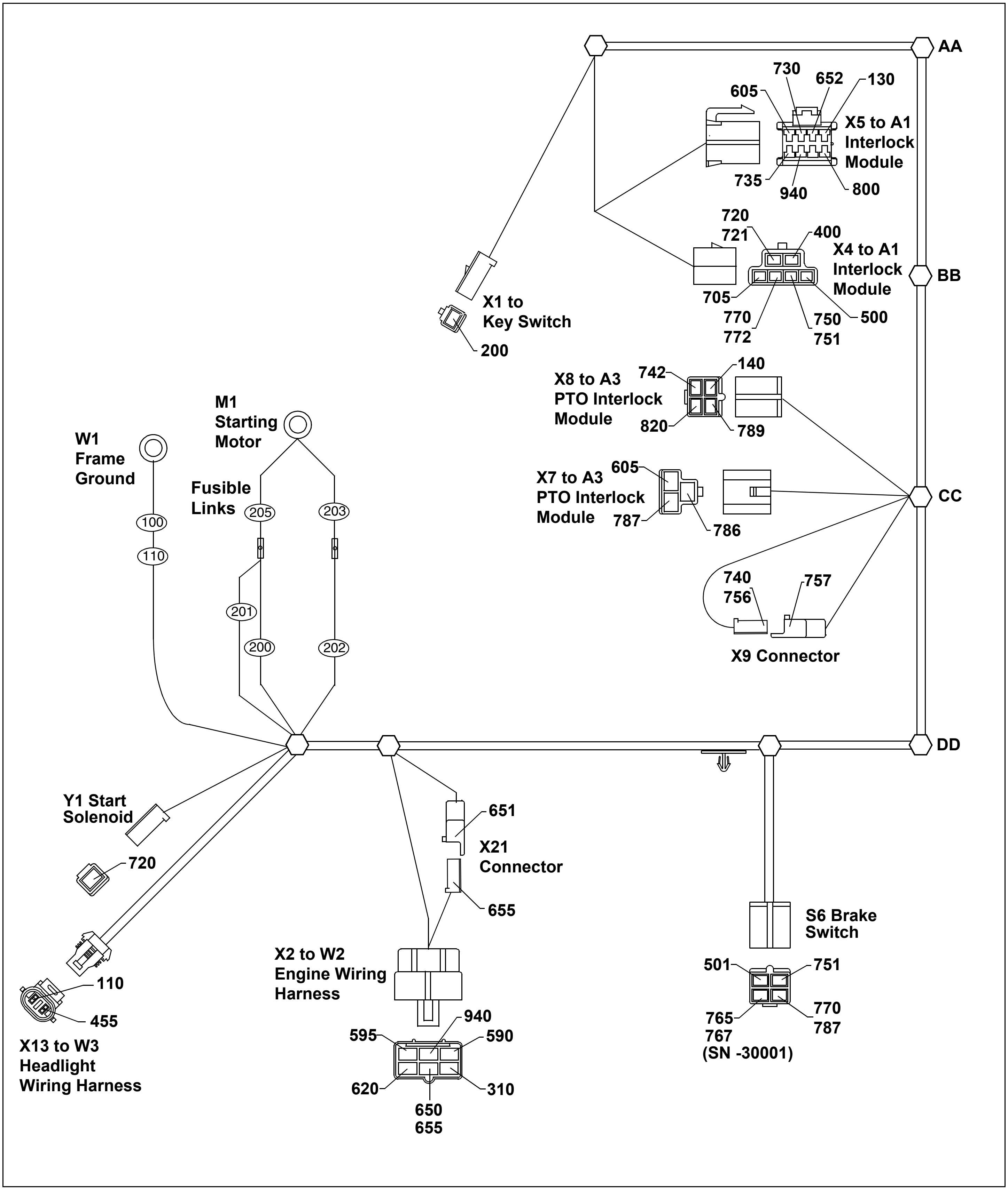


W2, W6 and W7 Engine Wiring Harnesses - X465



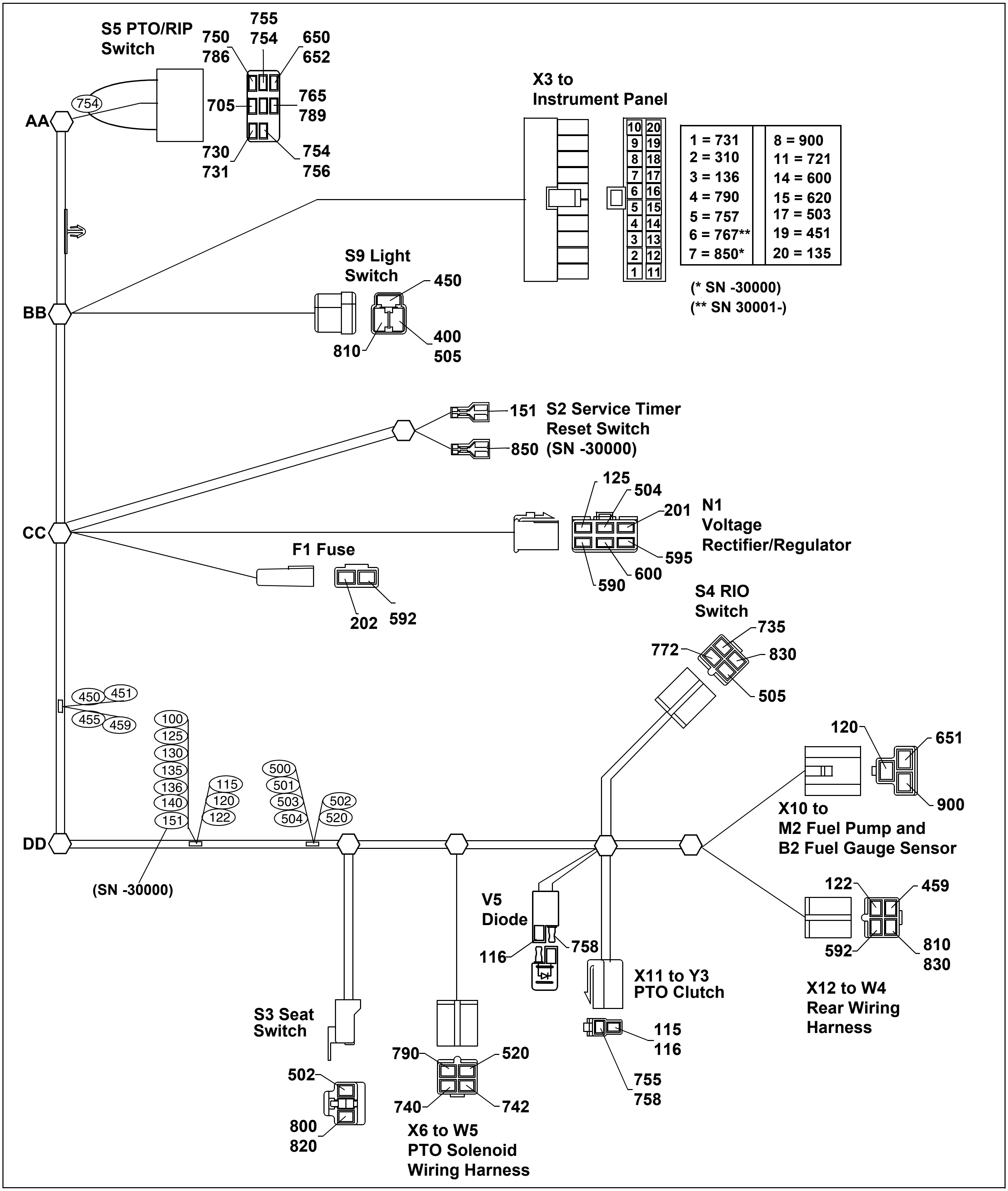
MIF

W1 Main Wiring Harness - X475 and X575 (1 of 2)



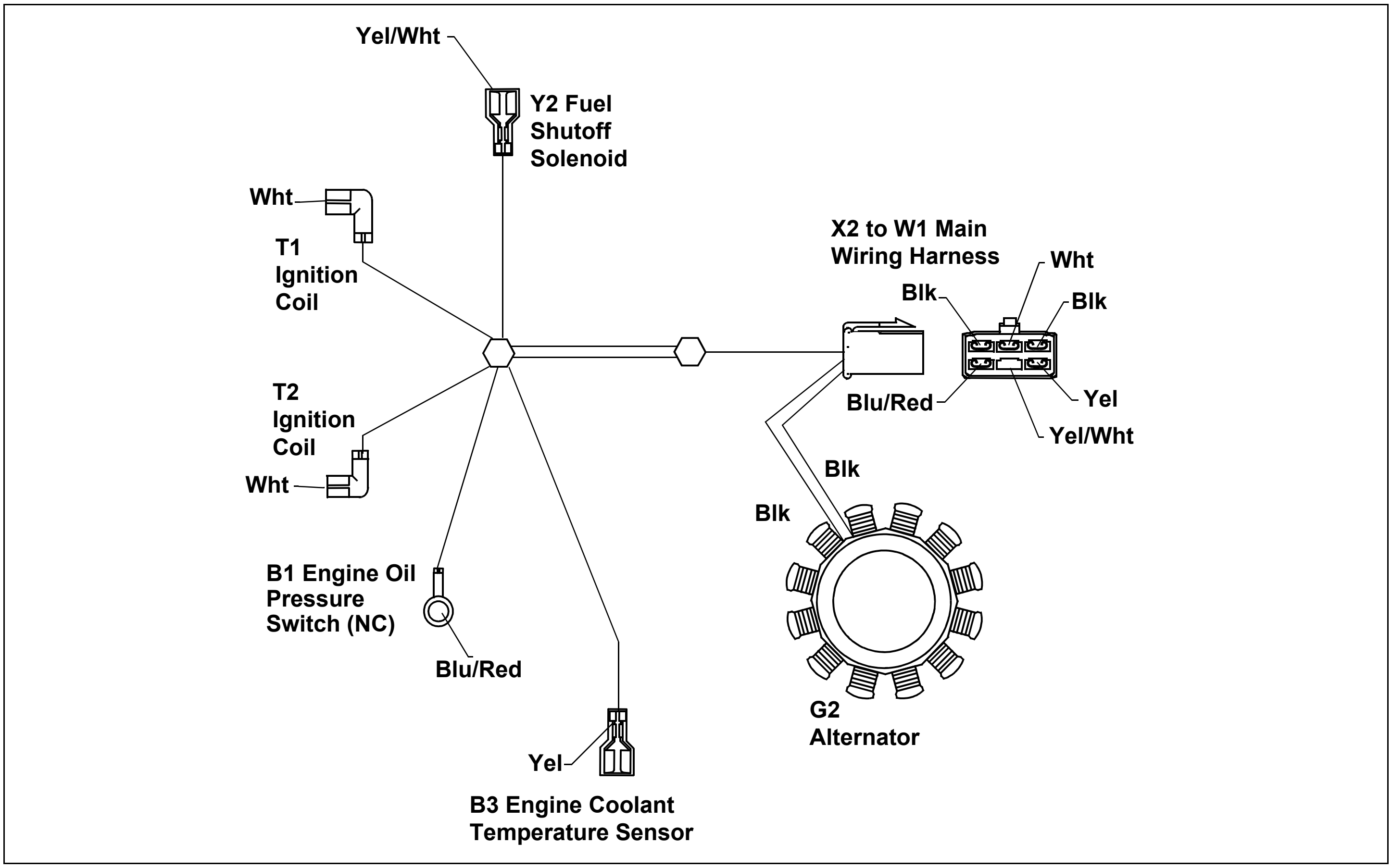
MIF

W1 Main Wiring Harness - X475 and X575 (2 of 2)



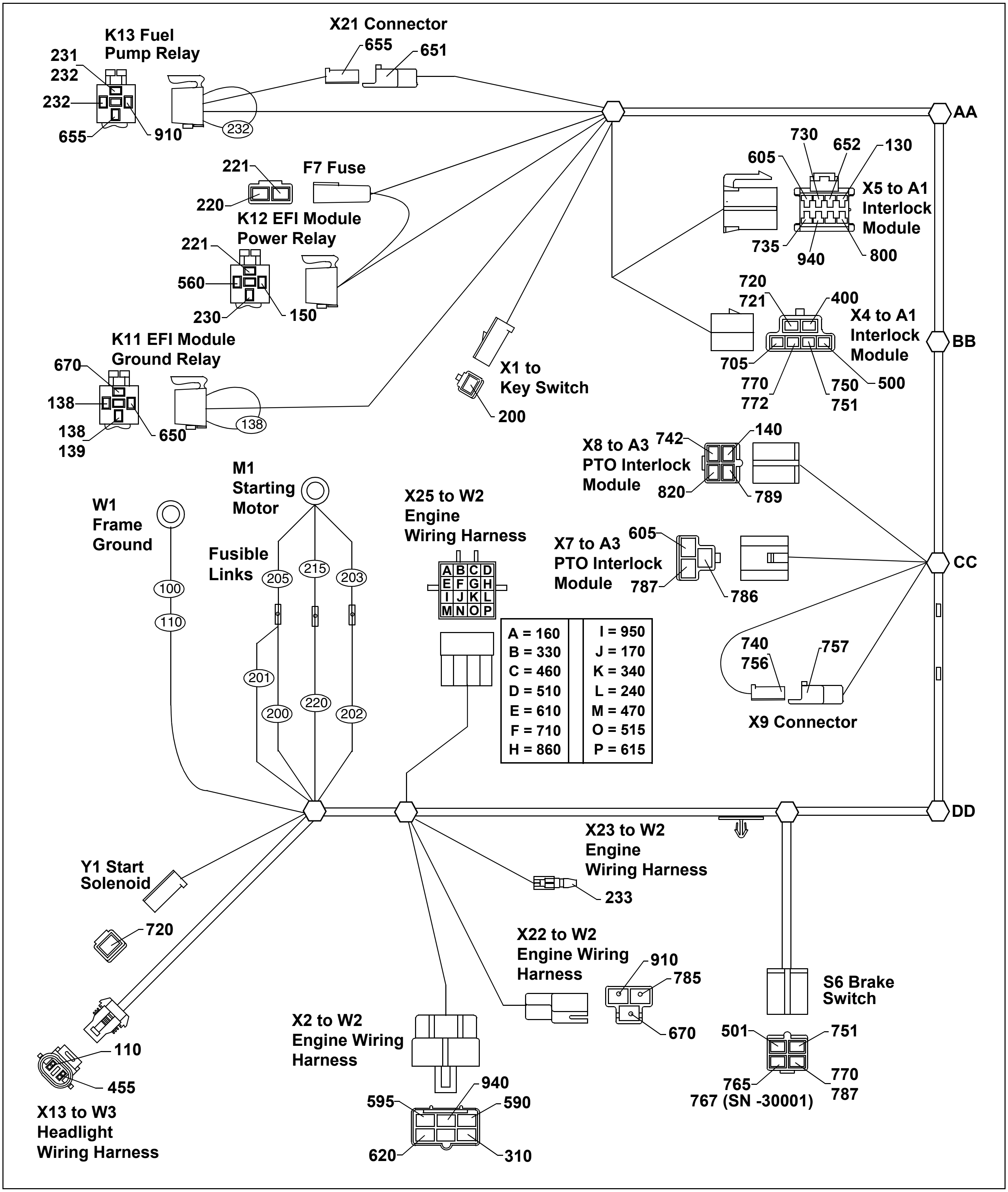
MIF

W2 Engine Wiring Harness - X475 and X575



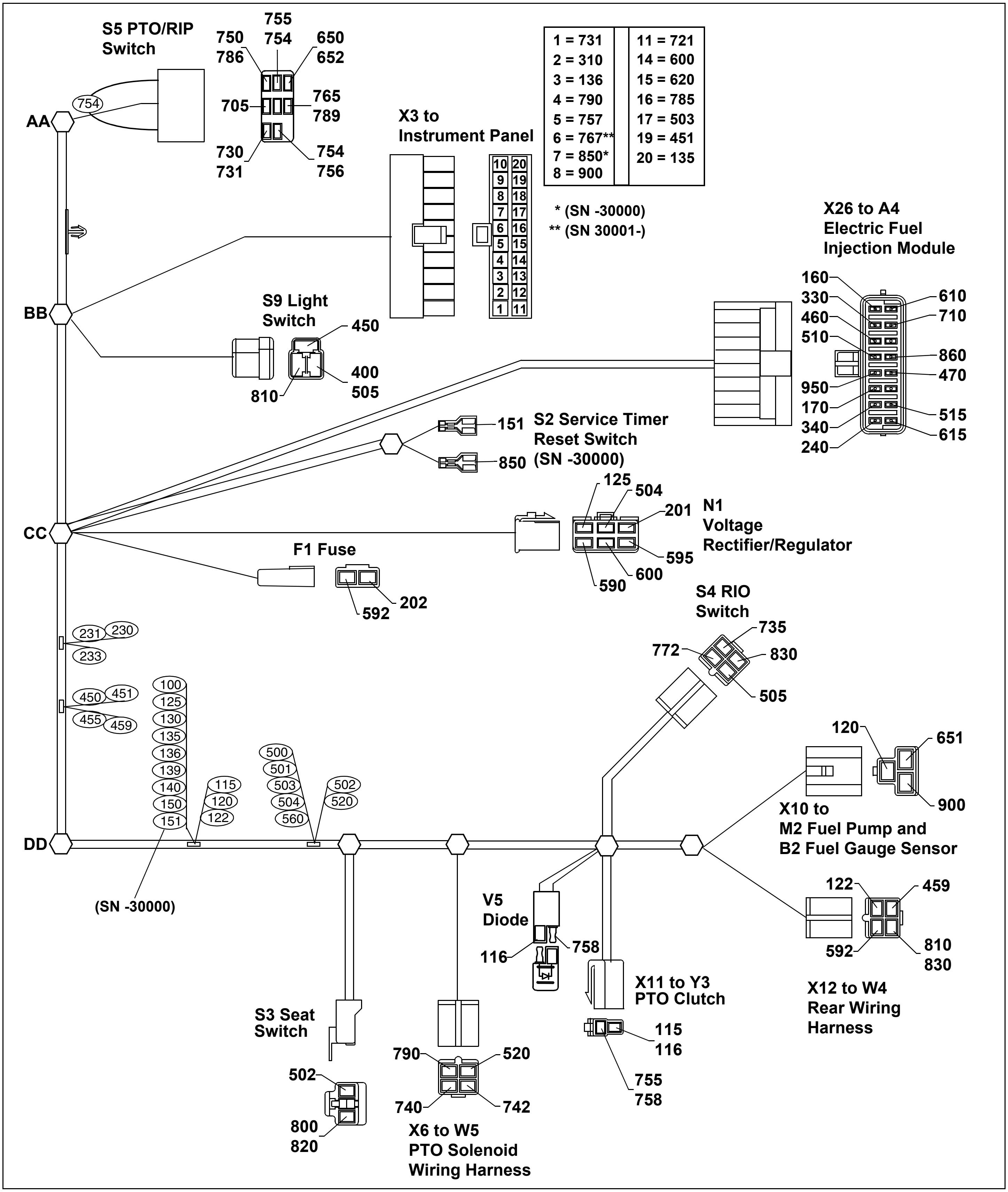
MIF

W1 Main Wiring Harness - X485 and X585 (1 of 2)



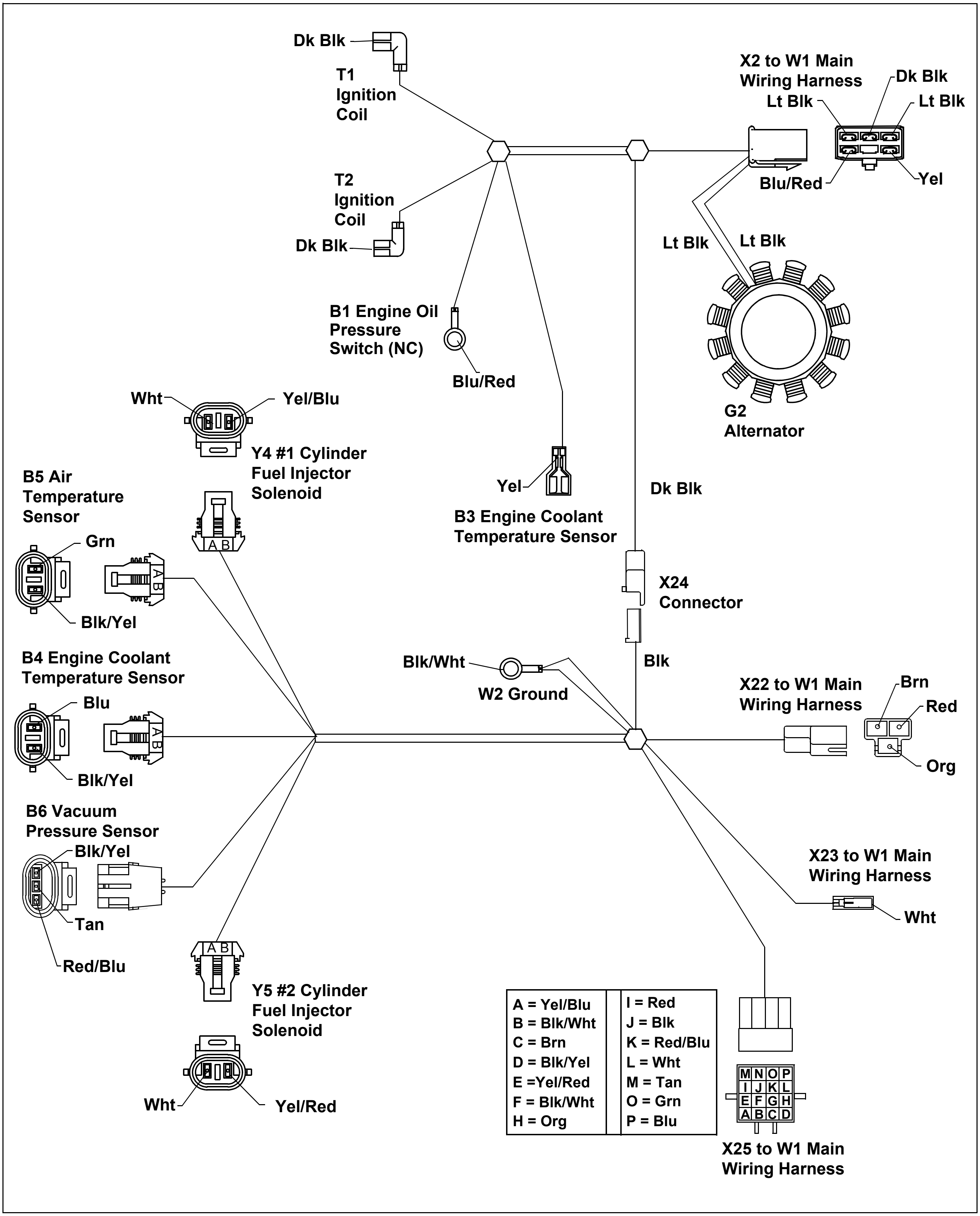
MIF

W1 Main Wiring Harness - X485 and X585 (2 of 2)

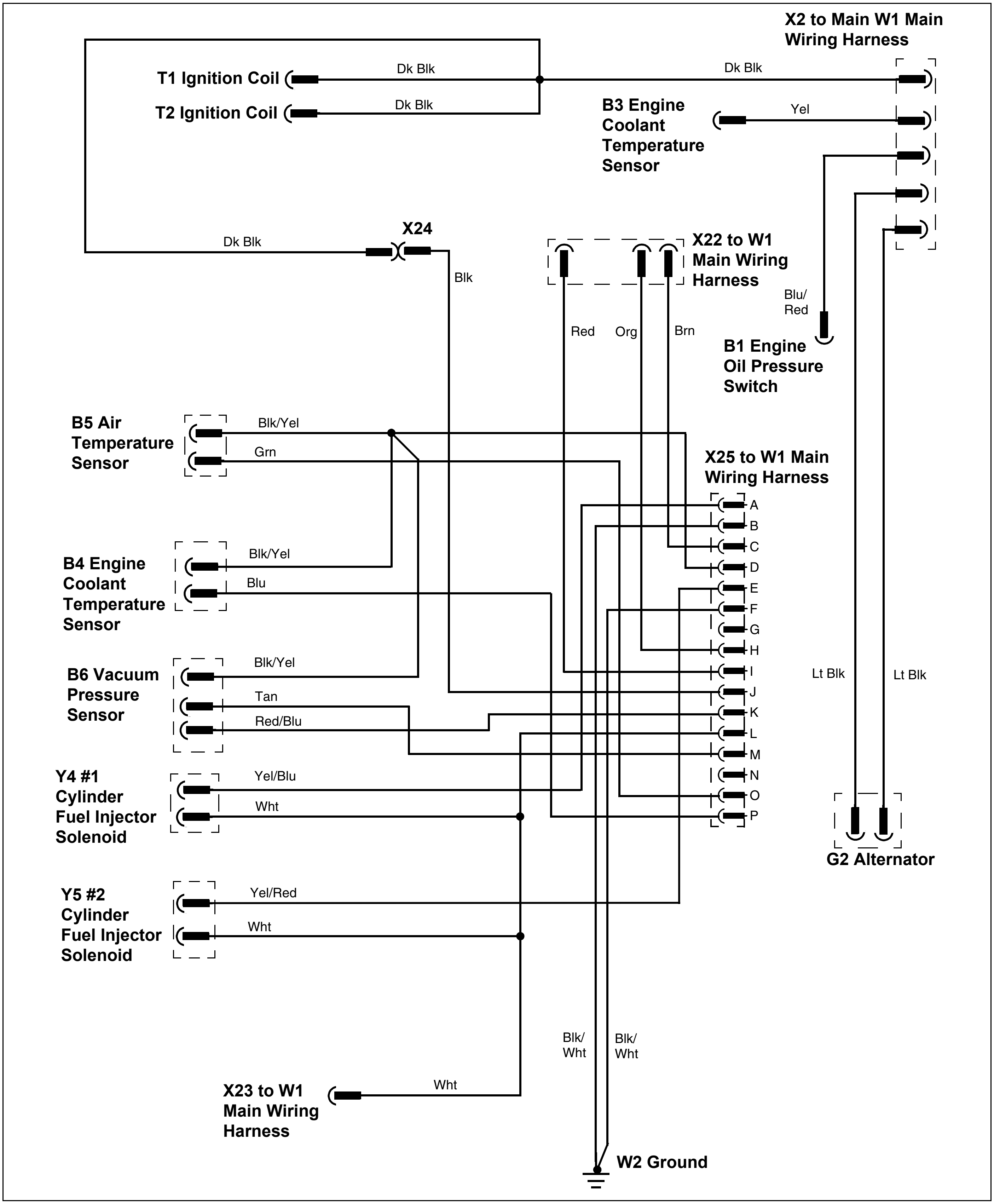


MIF

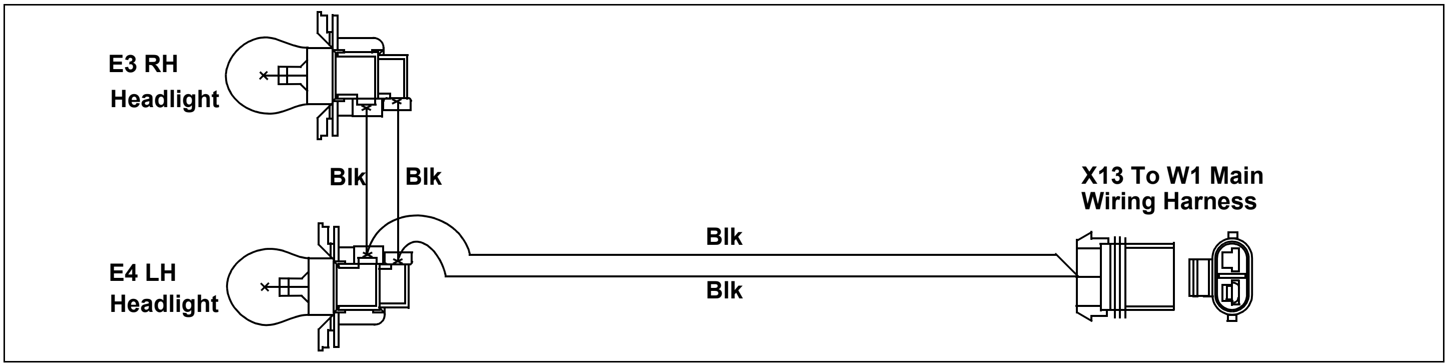
W2 Engine Wiring Harness - X485 and X585



W2 Engine Wiring Harness Circuit Schematic - X485 and X585

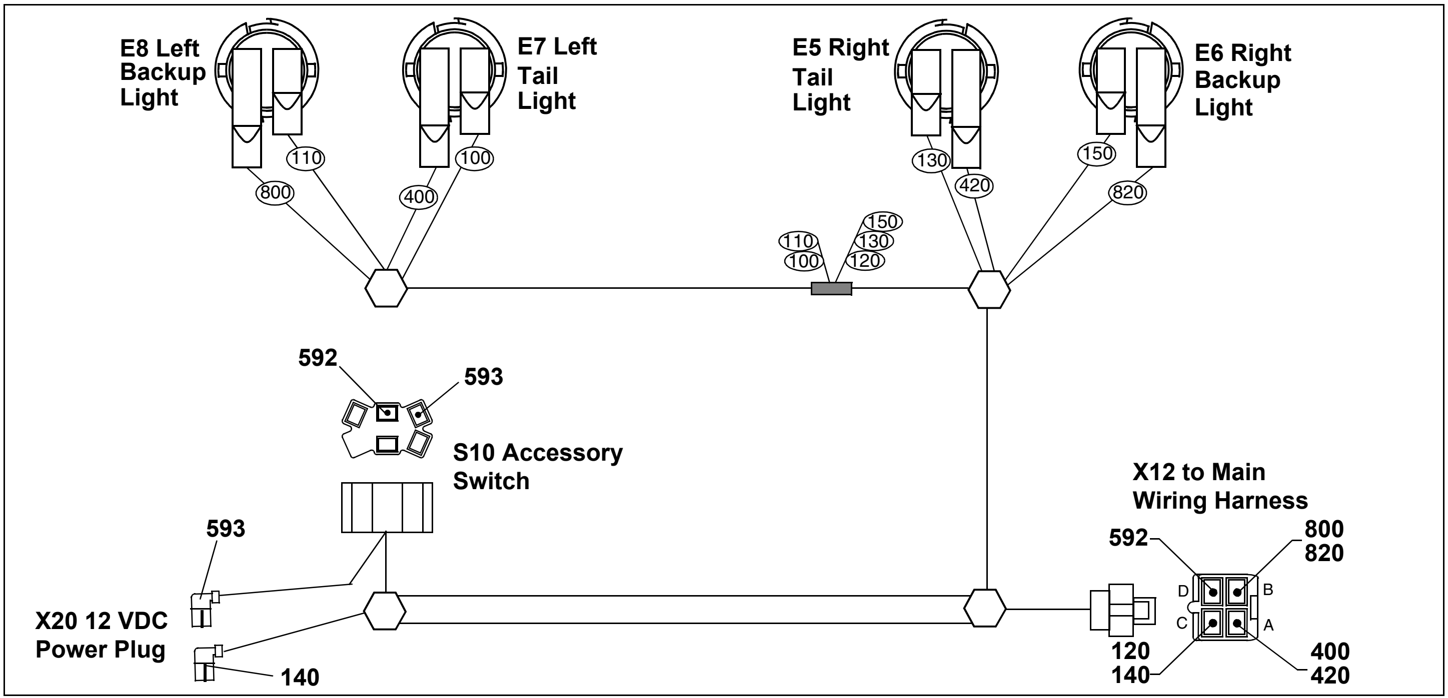


W3 Headlight Wiring Harness



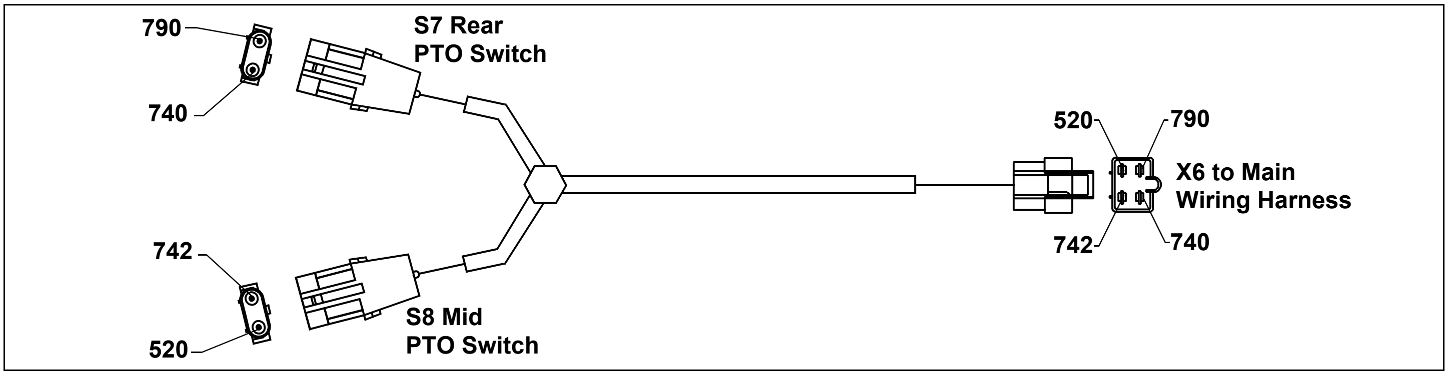
MIF

W4 Rear Wiring Harness



MIF

W5 PTO Option Wiring Harness



MIF

W1 Main Wiring Harness Color Codes - X465

Size/No./Color	Wire Connection Points
2.0 100 Blk	W1, Splice
1.0 110 Blk	W1, X13
0.8 115 Blk	Splice, X11
0.8 116 Blk	V5, X11
0.8 120 Blk	Splice, X10
3.0 122 Blk	Splice, X12
0.5 130 Blk	Splice, X5
0.5 135 Blk	Splice, X3
0.5 136 Blk	Splice, X3
0.5 140 Blk	Splice, X8
0.8 151 Blk	Splice, S2 (SN -30000)
3.0 200 Red	F3, X1
2.0 201 Red	F3, N1
3.0 202 Red	F2, F1
1.0 203 Blk	Fusible Link F2, M1
0.8 205 Lt Blu	Fusible Link F3, M1
1.0 400 Yel/Blu	X4, S9
1.0 450 Yel/Wht	S9, Splice
0.5 451 Yel/Wht	Splice, X3
1.0 455 Yel/Wht	Splice, X13
0.8 459 Yel/Wht	Splice, X12
1.0 500 Yel	X4, Splice
0.5 501 Yel	Splice, S6
0.5 502 Yel	Splice, S3
0.5 503 Yel	Splice, X3
0.8 505 Yel	S9, S4
0.5 520 Yel	Splice, X6
3.0 592 Blu	X12, F1
0.5 600 Brn	N1, X3
0.5 605 Tan	X5, X7
0.5 620 Tan	X2, X3

Size/No./Color	Wire Connection Points
0.8 650 Pnk/Blk	S5, X2
0.8 652 Blu	X5, S5
0.5 705 Pnk/Blk	X4, S5
1.0 720 Pur	Y1, X4
0.8 721 Pur	X4, X3
0.8 730 Grn	S5, X5
0.8 731 Pur	S5, X3
0.8 735 Org	X5, S4
0.5 740 Blu	X6, X9
0.5 742 Blu/Wht	X6, X8
0.8 750 Blu	S5, X4
0.8 751 Blu	X4, S6
0.8 754 Blu	S5, S5
0.8 755 Blu	S5, X11
0.8 756 Blu	X9, S5
0.5 757 Yel	X9, X3
0.8 758 Blu	X11, V5
0.8 765 Pur	S5, S6
0.8 767 Pur	S6, H8
0.8 770 Pur	S6, X4
0.8 772 Blu	X4, S4
0.5 786 Blu	S5, X7
0.5 787 Pur	X7, S6
0.5 789 Pur	S5, X8
0.5 790 Tan	X6, X3
0.8 800 Pnk	X5, S3
0.8 810 Pnk	X12, S9
0.5 820 Pnk	S3, X8
0.8 830 Gry	X12, S4
0.8 850 Pnk	S2, X3 (SN -30000)
0.5 900 Blk/Wht	X10, X3
0.8 940 Wht	X2, X5

W1 Main Wiring Harness Color Codes - X475 and X575

Size/No./Color	Wire Connection Points
3.0 100 Blk	W1, Splice
1.0 110 Blk	W1, X13
0.8 115 Blk	Splice, X11
0.8 116 Blk	V5, X11
0.8 120 Blk	Splice, X10
3.0 122 Blk	Splice, X12
2.0 125 Blk	Splice, N1
0.5 130 Blk	Splice, X5
0.5 135 Blk	Splice, X3
0.5 136 Blk	Splice, X3
0.5 140 Blk	Splice, X8
0.8 151 Blk	Splice, S2 (SN -30000)
2.0 200 Red	F3, X1
2.0 201 Red	F3, N1
3.0 202 Red	F2, F1
1.0 203 Blk	Fusible Link F2, M1
0.8 205 Lt Blu	Fusible Link F3, M1
0.8 310 Org	X2, X3
1.0 400 Yel/Blu	X4, S9
1.0 450 Yel/Wht	S9, Splice
0.5 451 Yel/Wht	Splice, X3
1.0 455 Yel/Wht	Splice, X13
0.8 459 Yel/Wht	Splice, X12
1.0 500 Yel	X4, Splice
0.5 501 Yel	Splice, S6
0.5 502 Yel	Splice, S3
0.5 503 Yel	Splice, X3
0.5 504 Yel	Splice, N1
0.8 505 Yel	S9, S4
0.5 520 Yel	Splice, X6

Size/No./Color	Wire Connection Points
2.0 590 Brn/Yel	N1, X2
3.0 592 Blu	X12, F1
2.0 595 Brn/Wht	N1, X2
0.5 600 Brn	N1, X3
0.5 605 Tan	X5, X7
0.5 620 Tan	X2, X3
0.8 650 Pnk/Blk	S5, X2
0.8 651 Pnk/Blk	X10, X21
0.8 652 Blu	X5, S5
0.8 655 Pnk	X2, X21
0.5 705 Pnk/Blk	X4, S5
1.0 720 Pur	Y1, X4
0.8 721 Pur	X4, X3
0.8 730 Grn	S5, X5
0.8 731 Pur	S5, X3
0.8 735 Org	X5, S4
0.5 740 Blu	X6, X9
0.5 742 Blu/Wht	X6, X8
0.8 750 Blu	S5, X4
0.8 751 Blu	X4, S6
0.8 754 Blu	S5, S5
0.8 755 Blu	S5, X11
0.8 756 Blu	X9, S5
0.5 757 Yel	X9, X3
0.8 758 Blu	X11, V5
0.8 765 Pur	S5, S6
0.8 767 Pur	S6, X3 (Pin 6 - instrument panel) (SN 30001-)
0.8 770 Pur	S6, X4
0.8 772 Blu	X4, S4
0.5 786 Blu	S5, X7
0.5 787 Pur	X7, S6
0.5 789 Pur	S5, X8

Size/No./Color	Wire Connection Points
0.5 790 Tan	X6, X3
0.8 800 Pnk	X5, S3
0.8 810 Pnk	X12, S9
0.5 820 Pnk	S3, X8
0.8 830 Gry	X12, S4
0.8 850 Pnk	S2, X3 (SN -30000)
0.5 900 Blk/Wht	X10, X3
0.8 940 Wht	X2, X5

W1 Main Wiring Harness Color Codes - X485 and X585

Size/No./Color	Wire Connection Points
3.0 100 Blk	W1, Splice
1.0 110 Blk	W1, X13
0.8 115 Blk	Splice, X11
0.8 116 Blk	V5, X11
0.8 120 Blk	Splice, X10
3.0 122 Blk	Splice, X12
2.0 125 Blk	Splice, N1
0.5 130 Blk	Splice, X5
0.5 135 Blk	Splice, X3
0.5 136 Blk	Splice, X3
1.0 138 Blk	K11, K11
1.0 139 Blk	Splice, K11
0.5 140 Blk	Splice, X8
0.8 150 Blk	Splice, K12
0.8 151 Blk	Splice, S2 (SN -30000)
0.8 160 Blk	X25, X26
0.8 170 Brn	X25, X26
2.0 200 Red	F3, X1
2.0 201 Red	F3, N1
3.0 202 Red	F2, F1
1.0 203 Blk	Fusible Link F2, M1

Size/No./Color	Wire Connection Points
0.8 205 Lt Blu	Fusible Link F3, M1
0.5 215 Gry	Fusible Link F4, M1
1.0 220 Red	F6, F7
1.0 221 Red	F7, K12
0.8 230 Red	K12, Splice
0.8 231 Red	Splice, K13
0.8 232 Red	K13, K13
0.5 233 Red/Blu	Splice, X23
0.8 240 Red	X25, X26
0.8 310 Org	X2, X3
0.8 330 Org	X25, X26
0.8 340 Grn	X25, X26
1.0 400 Yel/Blu	X4, S9
1.0 450 Yel/Wht	S9, Splice
0.5 451 Yel/Wht	Splice, X3
1.0 455 Yel/Wht	Splice, X13
0.8 459 Yel/Wht	Splice, X12
0.8 460 Yel	X25, X26
0.8 470 Yel	X25, X26
1.0 500 Yel	X4, Splice
0.5 501 Yel	Splice, S6
0.5 502 Yel	Splice, S3
0.5 503 Yel	Splice, X3
0.5 504 Yel	Splice, N1
0.8 505 Yel	Splice, S4
0.8 510 Blu	X25, X26
0.8 515 Blu	X25, X26
0.5 520 Yel	Splice, X6
0.8 560 Red/Yel	Splice, K12
2.0 590 Brn/Yel	N1, X2
3.0 592 Blu	X12, F1
2.0 595 Brn/Wht	N1, X2

Size/No./Color	Wire Connection Points
0.5 600 Brn	N1, X3
0.5 605 Tan	X5, X7
0.8 610 Tan	X25, X26
0.8 615 Tan	X25, X26
0.5 620 Tan	X2, X3
0.8 650 Pnk/Blk	S5, K11
0.8 651 Pnk/Blk	X10, X21
0.8 652 Blu	X5, S5
0.8 655 Pnk	K13, X21
1.0 670 Pur	K11, X22
0.5 705 Pnk/Blk	X4, S5
0.8 710 Pur	X25, X26
1.0 720 Pur	Y1, X4
0.8 721 Pur	X4, X3
0.8 730 Grn	S5, X5
0.8 731 Pur	S5, X3
0.8 735 Org	X5, S4
0.5 740 Blu	X6, X9
0.5 742 Blu/Wht	X6, X8
0.8 750 Blu	S5, X4
0.8 751 Blu	X4, S6
0.8 754 Blu	S5, S5
0.8 755 Blu	S5, X11
0.8 756 Blu	X9, S5
0.5 757 Yel	X9, X3
0.8 758 Blu	X11, V5
0.8 765 Pur	S5, S6
0.8 767 Pur	S6, X3 (SN 30001-)
0.8 770 Pur	S6, X4
0.8 772 Blu	X4, S4
0.5 785 Grn	X22, X3
0.5 786 Blu	S5, X7

Size/No./Color	Wire Connection Points
0.5 787 Pur	X7, S6
0.5 789 Pur	S5, X8
0.5 790 Tan	X6, X3
0.8 800 Pnk	X5, S3
0.8 810 Pnk	X12, S9
0.5 820 Pnk	S3, X8
0.8 830 Gry	X12, S4
0.8 850 Pnk	S2, X3 (SN -30000)
0.8 860 Pnk	X25, X26
0.5 900 Blk/Wht	X10, X3
0.5 910 Wht	X22, K13
0.8 940 Wht	X2, X5
0.8 950 Wht	X25, X26

W2 Engine Wiring Harness Color Codes - X475 and X575

Size/No./Color	Wire Connection Points
0.5 Blk	X2, G2
0.5 Blk	X2, G2
0.5 Wht	X2, T1
0.5 Wht	X2, T2
0.5 Yel/Wht	X2, Y2
0.5 Blu/Red	X2, B1
0.5 Yel	X2, B3

W2 Engine Wiring Harness Color Codes - X485 and X585

Size/No./Color	Wire Connection Points
0.5 Lt Blk	X2, G2
0.5 Lt Blk	X2, G2
0.5 Dk Blk	X2, T1
0.5 Dk Blk	X2, T2
0.5 Dk Blk	X2, X24
0.5 Blu/Red	X2, B1
0.5 Yel	X2, B3
0.5 Yel/Blu	X25, Y4
0.5 Blk/Wht	X25, Splice
0.5 Blk/Wht	Splice, W2
0.5 Blk/Wht	X25, Splice
0.5 Brn	X25, X22
0.5 Blk/Wht	X25, Splice
0.5 Blk/Yel	Splice, B4
0.5 Blk/Yel	Splice, B5
0.5 Blk/Yel	Splice, B6
0.5 Yel/Red	X25, Y5
0.5 Org	X25, X22
0.5 Red	X25, X22
0.5 Blk	X25, X24
0.5 Red/Blu	X25, B6
0.5 Wht	X25, Splice
0.5 Wht	Splice, X25
0.5 Wht	Splice, Y4
0.5 Wht	Splice, Y5
0.5 Tan	X25, B6
0.5 Grn	X25, B5
0.5 Blu	X25, B4

W3 Headlight Wiring Harness Color Codes

Size/Color	Wire Connection Points
0.8 Blk	X14, E4
0.8 Blk	X14, E4
0.8 Blk	E3, E4
0.8 Blk	E3, E4

W4 Rear Wiring Harness Color Codes

Size/No./Color	Wire Connection Points
0.8 100 Blk	Splice, E7
0.8 110 Blk	Splice, E8
0.8 120 Blk	Splice, X12
0.8 130 Blk	Splice, E5
2.0 140 Blk	X12, X23
0.5 150 Blk	Splice, E6
2.0 400 Yel	X12, E7
0.8 420 Yel	X12, E5
2.0 592 Pnk	X12, S10
2.0 593 Pnk	S10, X23
0.8 800 Pur	X12, E8
0.8 820 Pur	X12, E6

W5 PTO Option Wiring Harness Color Codes

Size/No./Color	Wire Connection Points
0.8 520 Yel	X6, S8
0.8 740 Blu	X6, S7
0.8 742 Blu	X6, S8
0.8 790 Tan	X6, S7

Operation and Diagnostics

Power Circuit Operation - X465

Function:

To provide unswitched and switched power to the primary electrical components whenever the battery is properly connected. The power circuits are divided among the unswitched power circuit, switched power circuits (key switch in run position), and secondary power circuits. The secondary power circuits become energized when switched power circuits energize relays, providing current paths to the secondary circuits. The secondary power circuits will not be energized if the relays controlling the current path(s) fail.

Operating Conditions, Unswitched Circuits:

Voltage must be present at the following components with the key switch in the off position.

- G1 battery positive terminal
- M1 starter - B terminal
- S1 key switch - B terminal
- G2 Stator
- F2 and F3 fusible links
- F1 fuse
- S10 accessory switch
- N1 voltage regulator/rectifier

The positive battery cable connects the battery to the starting motor. The starting motor bolt is used as the 12 volt DC tie point for the rest of the electrical system. These circuits are protected by the F2 and F3 fusible links and the F1 fuse. The battery cables and the starting motor tie point connections must be good for the machine's electrical system to work properly. Proper starting motor operation depends on the battery cables to carry high current. The ground cable connection is equally as important as the positive cable connection in maintaining electrical system integrity.

Operating Conditions, Switched Circuits:

With the key switch in run position, PTO switch in off position, brake locked, and operator off the seat, switched voltage should be present at the following components:

- S1 key switch - terminal "A"
- A1 ignition interlock module - X4 connector F-2, F-3 and F-6 terminals
- A1 interlock module components - F4 and F5 fuse, start relay common terminal and coil, ignition relay normally open terminal and coil, magneto relay coil, and E1 Grn seat LED.
- S9 light switch - 400 Yel/Blu
- S6 brake switch - 501 Yel and (767 SN 30001-)/765 Pur
- S5 PTO/RIP switch - 765 and 789 Pur, and 705 Pnk/Blk
- S3 seat switch - 502 Yel
- A3 PTO interlock module (option) - X8 connector J4-C terminal, 789 Pur
- S4 RIO switch - 505 Yel
- Instrument panel - X3 connector 17 terminal, 503 Yel
- S8 mid PTO switch (option) - 520 Yel
- H8 brake light (pin 6 at X3 instrument panel connector)

These circuits are controlled by the key switch and are protected by the fusible links F2 - F3 and the F4 and F5 fuses.

With power now available at various locations on the machine, the electrical system is ready to perform the different functions of starting and running the engine, engine monitoring lights, the PTO clutch and the PTO system interlocks, and the headlights.

The ground circuit is equally important as the power circuit connections. Proper systems operation depends on good wires and connections in order to carry the current needed to operate the various relays, solenoids and lights.

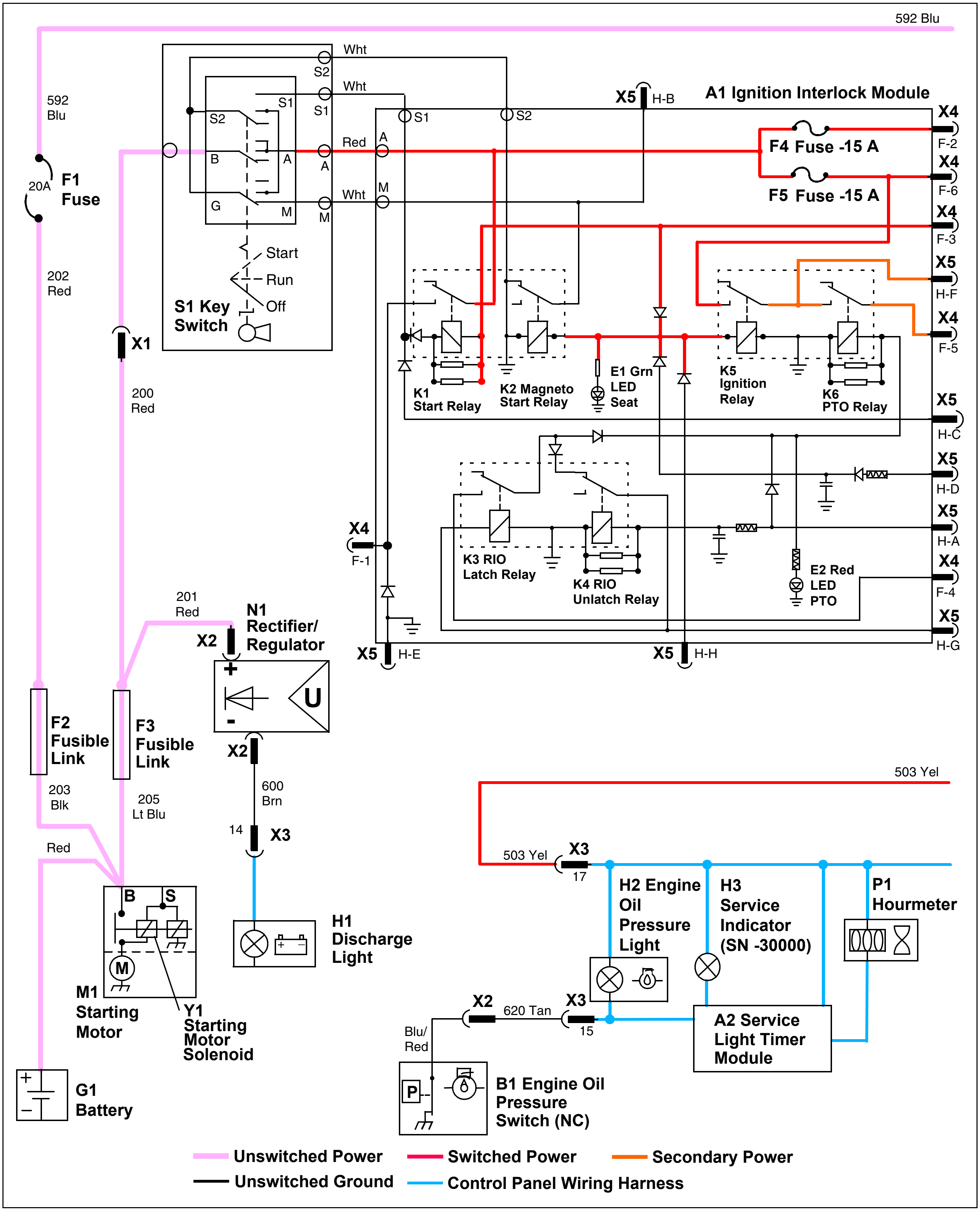
Operating Conditions, Secondary Switched Circuits:

Secondary switched voltage must be present at the following components during the following conditions: Key switch in run position, transmission in neutral, PTO in off position, brake locked, and operator off seat:

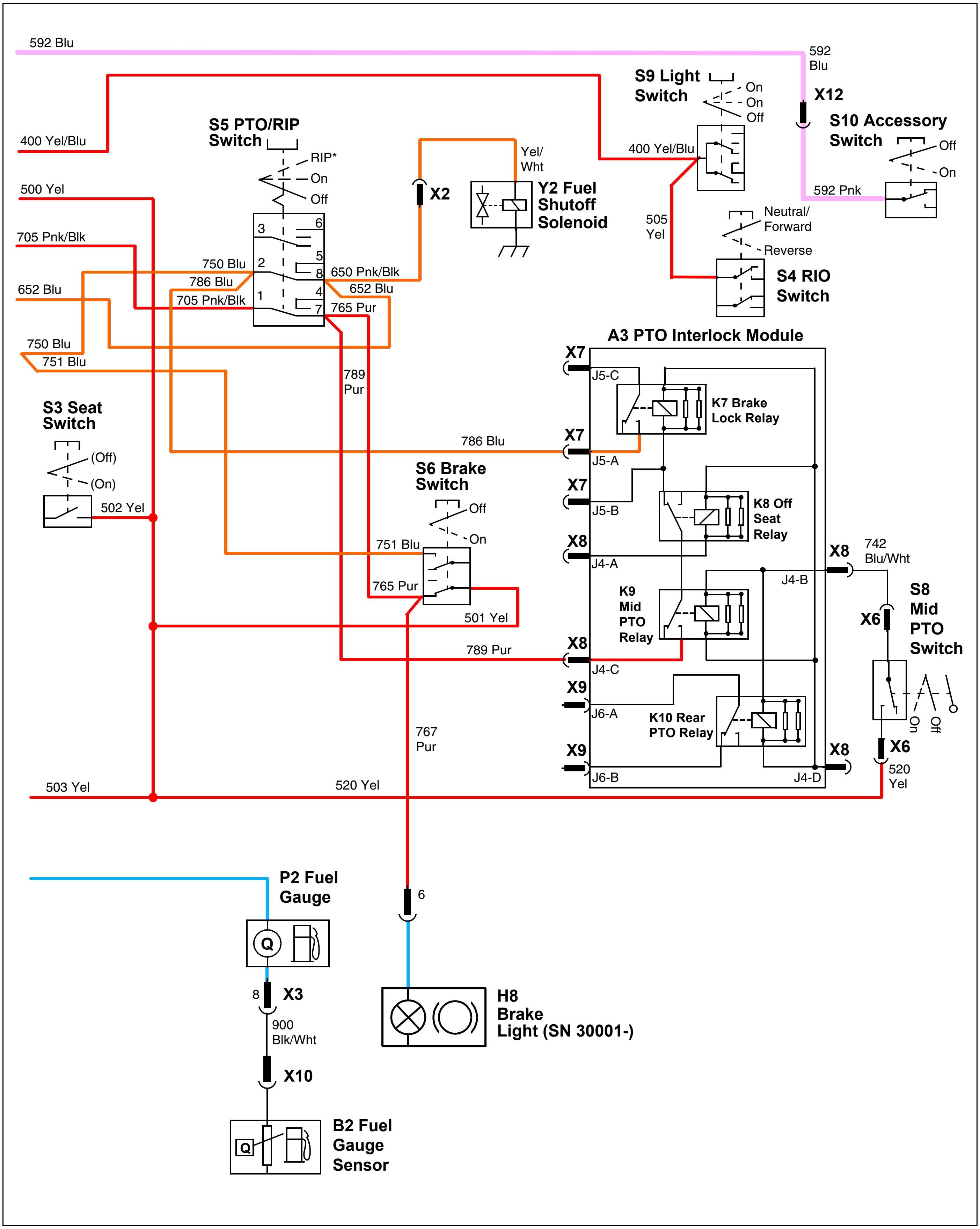
- A1 ignition interlock module - X5 connector H-F terminal, 652 Blu
- S5 PTO/RIP switch - 650 Pnk/Blk and 652, 750, and 786 Blu
- Y2 fuel shutoff solenoid - 650 Pnk/Blk and Yel/Wht
- A1 ignition interlock module - X4 connector F-5 terminal, 750 and 751 Blu
- A1 interlock module components - PTO relay common and normally open terminals and ignition relay common terminal
- S6 brake switch - 751 Blu
- A3 PTO interlock module (option) - X7 connector J5-A terminal, 786 Blu
- A3 PTO interlock module (option) components - brake lock relay normally open terminal

These circuits are controlled by the key switch and are protected by the F3 fusible link and the F4 and F5 fuses.

Power Circuit Wiring Schematic - X465



MIF


592 Blu
592 Blu
400 Yel/Blu
S9 Light Switch
On
On
Off
X12
S10 Accessory Switch
Off
On
592 Pnk
S5 PTO/RIP Switch
RIP*
On
Off
500 Yel
Yel/ Wht
X2
Y2 Fuel Shutoff Solenoid
400 Yel/Blu
505 Yel
Neutral/ Forward
Reverse
S4 RIO Switch
705 Pnk/Blk
652 Blu
750 Blu
786 Blu
705 Pnk/Blk
650 Pnk/Blk
652 Blu
765 Pur
750 Blu
751 Blu
789 Pur
A3 PTO Interlock Module
X7
J5-C
K7 Brake Lock Relay
S3 Seat Switch
(Off)
(On)
502 Yel
786 Blu
X7
J5-A
S6 Brake Switch
Off
On
X7
J5-B
K8 Off Seat Relay
X8
J4-A
751 Blu
742 Blu/Wht
X8
J4-B
K9 Mid PTO Relay
S8 Mid PTO Switch
X6
765 Pur
501 Yel
X8
789 Pur
J4-C
X9
J6-A
K10 Rear PTO Relay
On
Off
767 Pur
X9
J6-B
J4-D
X8
X6
520 Yel
503 Yel
520 Yel
P2 Fuel Gauge
6
X3
900 Blk/Wht
H8 Brake Light (SN 30001-)
X10
B2 Fuel Gauge Sensor

MIF

Power Circuit Operation - X475 and X575

Function:

To provide unswitched and switched power to the primary electrical components whenever the battery is properly connected. The power circuits are divided among the unswitched power circuit, switched power circuits (key switch in run position), and secondary power circuits. The secondary power circuits become energized when switched power circuits energize relays, providing current paths to the secondary circuits. The secondary power circuits will not be energized if the relays controlling the current path(s) fail.

Operating Conditions, Unswitched Circuits:

Voltage must be present at the following components with the key switch in the off position.

- G1 battery positive terminal
- M1 starter - B terminal
- S1 key switch - B terminal
- N1 voltage regulator/rectifier
- F1 fuse
- S10 accessory switch

The positive battery cable connects the battery to the starting motor. The starting motor bolt is used as the 12 volt DC tie point for the rest of the electrical system. These circuits are protected by the F2 and F3 fusible links and the F1 fuse. The battery cables and the starting motor tie point connections must be good for the machine's electrical system to work properly. Proper starting motor operation depends on the battery cables to carry high current. The ground cable connection is equally as important as the positive cable connection in maintaining electrical system integrity

Operating Conditions, Switched Circuits:

With the key switch in run position, PTO switch in off position, brake locked, and operator off the seat, switched voltage should be present at the following components:

- S1 key switch - terminals "A"
- A1 ignition interlock module - X4 connector F-2, F-3 and F-6 terminals
- A1 interlock module components - F4 and F5 fuse, start relay common terminal and coil, ignition relay normally open terminal and coil, magneto relay coil, and E1 Grn seat LED.
- S9 light switch - 400 Yel/Blu
- S6 brake switch - 501 Yel, 765 and (767 SN 30001-)Pur
- S5 PTO/RIP switch - 765 and 789 Pur, and 705 Pnk/Blk
- S3 seat switch - 502 Yel
- A3 PTO interlock module (option) - X8 connector J4-C terminal, 789 Pur
- S4 RIO switch - 505 Yel
- N1 voltage rectifier/regulator - 504 Yel
- Instrument panel - X3 connector 17 terminal, 503 Yel
- S8 mid PTO switch (option) - 520 Yel

These circuits are controlled by the key switch and are protected by the fusible links F2 - F3 and the F4 and F5 fuses.

With power now available at various locations on the machine, the electrical system is ready to perform the different functions of starting and running the engine, engine monitoring lights, the PTO clutch and the PTO system interlocks, and the headlights.

The ground circuit is equally important as the power circuit connections. Proper systems operation depends on good wires and connections in order to carry the current needed to operate the various relays, solenoids and lights.

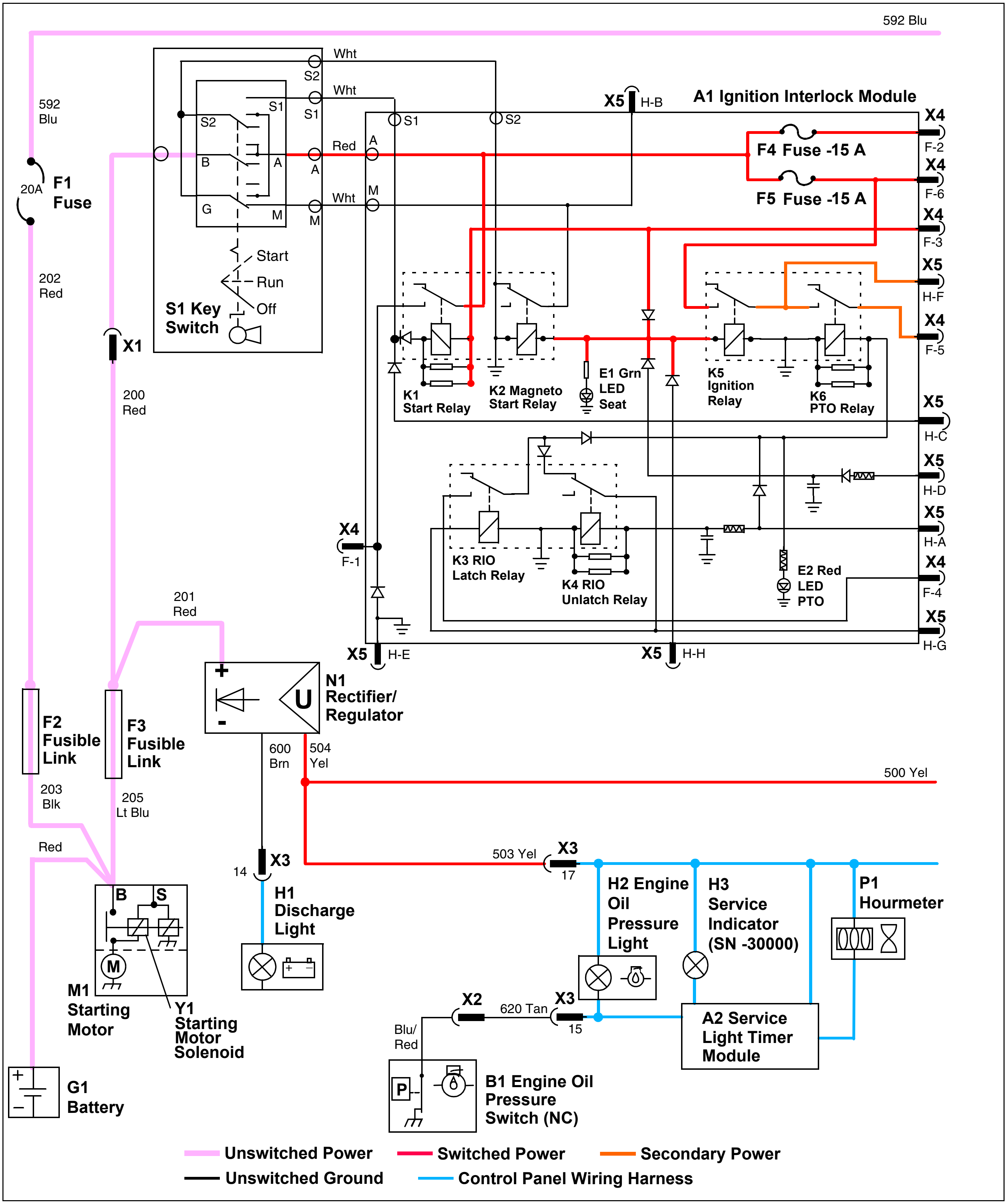
Operating Conditions, Secondary Switched Circuits:

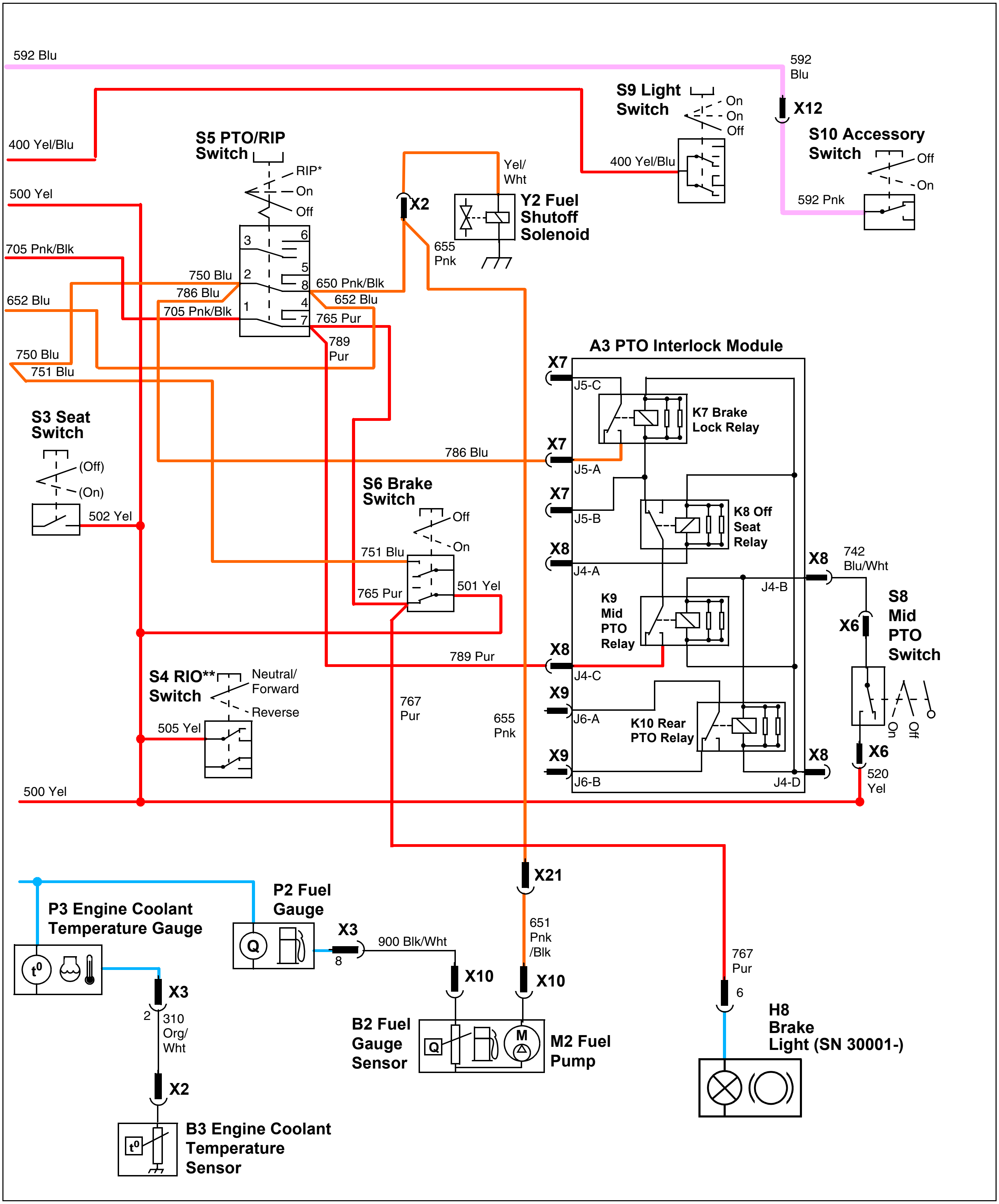
Secondary switched voltage must be present at the following components during the following conditions: Key switch in run position, transmission in neutral, PTO in off position, brake locked, and operator off seat:

- A1 ignition interlock module - X5 connector H-F terminal, 652 Blu
- S5 PTO/RIP switch - 650 Pnk/Blk and 652, 750, and 786 Blu
- Y2 fuel shutoff solenoid - 650 Pnk/Blk and Yel/Wht
- M2 fuel pump - 651 Pnk/Blk
- A1 ignition interlock module - X4 connector F-5 terminal, 750 and 751 Blu
- A1 interlock module components - PTO relay common and normally open terminals and ignition relay common terminal
- S6 brake switch - 751 Blu
- A3 PTO interlock module (option) - X7 connector J5-A terminal, 786 Blu
- A3 PTO interlock module (option) components - brake lock relay normally open terminal

These circuits are controlled by the key switch and are protected by the F3 fusible link and the F4 and F5 fuses.

Power Circuit Electrical Schematic - X475 and X575




592 Blu
592 Blu
X12
400 Yel/Blu
S9 Light Switch
On
On
Off
400 Yel/Blu
S10 Accessory Switch
Off
On
592 Pnk
S5 PTO/RIP Switch
RIP*
On
Off
500 Yel
Yel/ Wht
X2
Y2 Fuel Shutoff Solenoid
655 Pnk
705 Pnk/Blk
750 Blu
786 Blu
705 Pnk/Blk
650 Pnk/Blk
652 Blu
652 Blu
765 Pur
789 Pur
750 Blu
751 Blu
A3 PTO Interlock Module
X7
J5-C
K7 Brake Lock Relay
S3 Seat Switch
(Off)
(On)
502 Yel
786 Blu
X7
J5-A
S6 Brake Switch
Off
On
X7
J5-B
K8 Off Seat Relay
751 Blu
X8
J4-A
X8
742 Blu/Wht
765 Pur
501 Yel
K9 Mid PTO Relay
J4-B
S8 Mid PTO Switch
X6
789 Pur
X8
J4-C
S4 RIO** Switch
Neutral/ Forward
Reverse
505 Yel
767 Pur
655 Pnk
X9
J6-A
K10 Rear PTO Relay
On
Off
X9
J6-B
J4-D
X8
X6
520 Yel
500 Yel
X21
P3 Engine Coolant Temperature Gauge
P2 Fuel Gauge
X3
900 Blk/Wht
8
651 Pnk /Blk
767 Pur
X10
X10
X3
2
310 Org/ Wht
6
H8 Brake Light (SN 30001-)
B2 Fuel Gauge Sensor
M2 Fuel Pump
X2
B3 Engine Coolant Temperature Sensor

Power Circuit Diagnosis - X465, X475 and X575

Test Conditions:

- Key switch off.

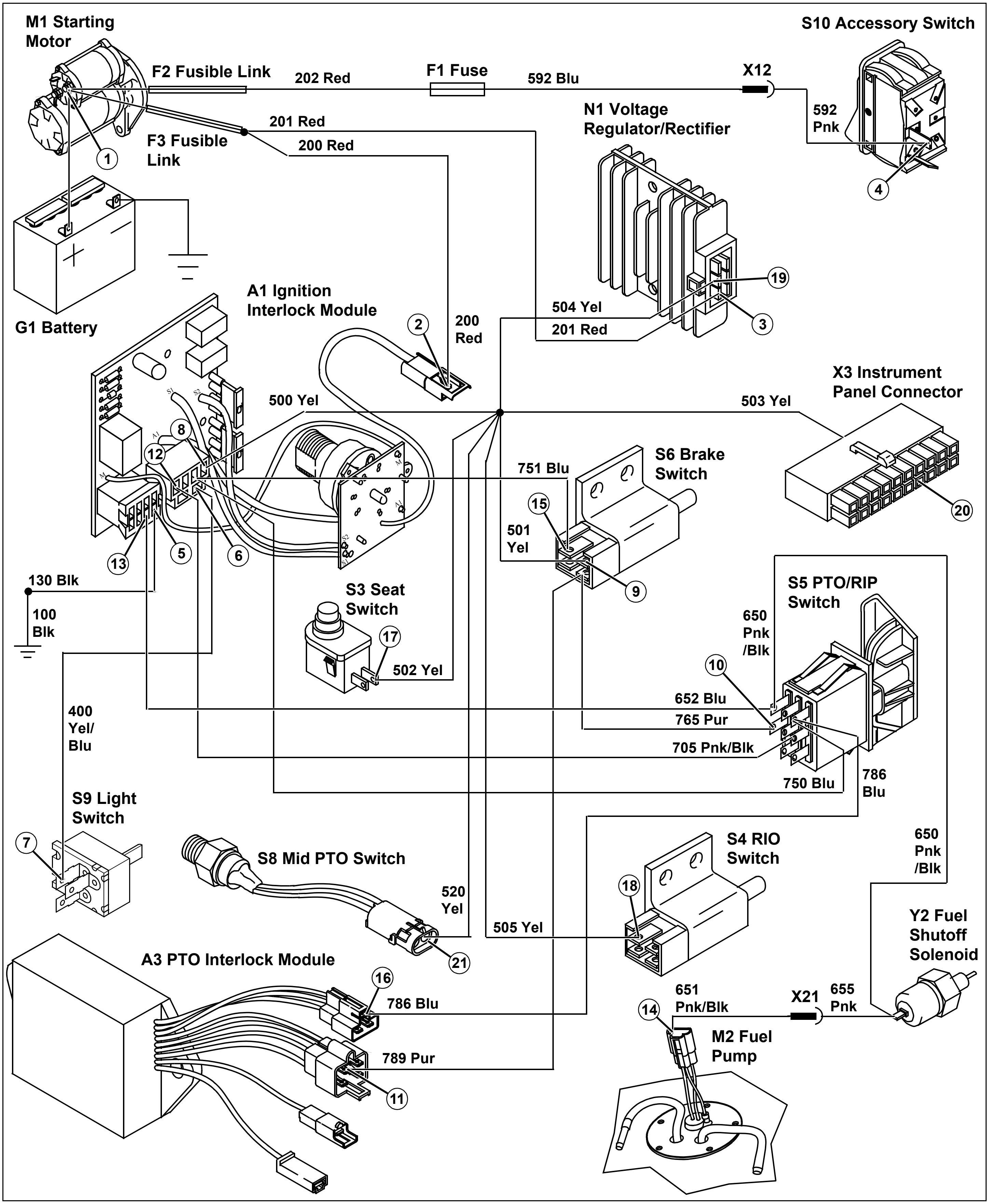
Test/Check Point	Normal	If Not Normal
1. Starting motor solenoid	Battery voltage	Test battery and cable connections.
2. Key switch	Battery voltage	Test F3 fusible link. Check 200 Red wire and connections.
3. Voltage rectifier/regulator	Battery voltage	Test F3 fusible link. Check 201 Red wire and connections.
4. Accessory switch	Battery voltage	Test F1 fuse and F2 fusible link. Check 202 Red, 592 Blu and 592 Pnk wires and connections.

Test Conditions:

- Key switch in run position, engine not running.
- Brake locked.
- PTO off.
- Transmission in neutral.
- Operator off seat.

Test/Check Point	Normal	If Not Normal
5. Ignition interlock module	Continuity to ground	Check 130 and 100 Blk wires and connections.
6. Ignition interlock module	Battery voltage	Test F4 fuse. If ok, replace ignition interlock module.
7. Light switch	Battery voltage	Check 400 Yel/Blu wire and connections.
8. Ignition interlock module	Battery voltage	Test F5 fuse. If ok, replace ignition interlock module.
9. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
10. PTO/RIP switch	Battery voltage	Check 765 Pur wire and connections. If ok, replace brake switch.
11. PTO interlock module (option)	Battery voltage	Check 789 Pur wire and connections. If ok, replace brake switch.
12. Ignition interlock module	Battery voltage	Check 705 Pnk/Blk wire and connections. If ok, replace PTO/RIP switch.
13. Ignition interlock module	Battery voltage, green LED illuminated	Replace ignition interlock module.
14. Fuel pump	Battery voltage	Check 652 Blu, 650, and 651 Pnk/Blk wires and connections.
15. Brake switch	Battery voltage	Check 750 and 751 Blu wires and connections. If ok, replace PTO/RIP switch.
16. PTO interlock module (option)	Battery voltage	Check 786 Blu wires and connections. If ok, replace PTO/ RIP switch.
17. Seat switch	Battery voltage	Check 500 and 502 Yel wires and connections.

Test/Check Point	Normal	If Not Normal
18. RIO switch	Battery voltage	Check 500 and 505 Yel wires and connections.
19. Voltage rectifier/regulator (not on X465)	Battery voltage	Check 500 and 504 Yel wires and connections.
20. X3 instrument panel connector	Battery voltage	Check 500 and 503 Yel wires and connections.
21. Mid PTO switch (option)	Battery voltage	Check 500 and 520 Yel wires and connections.


M1 Starting Motor
F2 Fusible Link
202 Red
F1 Fuse
592 Blu
X12
S10 Accessory Switch
592 Pnk
201 Red
F3 Fusible Link
200 Red
N1 Voltage Regulator/Rectifier
G1 Battery
A1 Ignition Interlock Module
200 Red
504 Yel
201 Red
500 Yel
503 Yel
X3 Instrument Panel Connector
751 Blu
S6 Brake Switch
501 Yel
130 Blk
100 Blk
S3 Seat Switch
502 Yel
S5 PTO/RIP Switch
650 Pnk /Blk
400 Yel/ Blu
652 Blu
765 Pur
705 Pnk/Blk
750 Blu
786 Blu
S9 Light Switch
S8 Mid PTO Switch
S4 RIO Switch
650 Pnk /Blk
520 Yel
505 Yel
Y2 Fuel Shutoff Solenoid
A3 PTO Interlock Module
786 Blu
651 Pnk/Blk
X21
655 Pnk
M2 Fuel Pump
789 Pur
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21

Power Circuit Operation - X485 and X585

Function:

To provide unswitched and switched power to the primary electrical components whenever the battery is properly connected.

The power circuits are divided among the unswitched power circuit, switched power circuits (key switch in run position), and secondary power circuits. The secondary power circuits become energized when switched power circuits energize relays, providing current paths to the secondary circuits. The secondary power circuits will not be energized if the relays controlling the current path(s) fail.

Operating Conditions, Unswitched Circuits:

Voltage must be present at the following components with the key switch in off position.

- G1 battery positive terminal
- M1 starter - B terminal
- S1 key switch - B terminal
- N1 voltage regulator/rectifier
- F1 fuse
- S10 accessory switch
- K12 EFI module power relay - 87 terminal, 221 Red

The positive battery cable connects the battery to the starting motor. The starting motor bolt is used as the 12 volt DC tie point for the rest of the electrical system. These circuits are protected by the F2, F3 and F6 fusible links and the F1 and F7 fuses.

The battery cables and the starting motor tie point connections must be good for the machine's electrical system to work properly. Proper starting motor operation depends on the battery cables to carry high current. The ground cable connection is equally as important as the positive cable connection in maintaining electrical system integrity.

Operating Conditions, Switched Circuits:

With the key switch in run position, PTO switch in off position, brake locked, and operator off the seat, switched voltage should be present at the following components:

- S1 key switch - terminals "A"
- A1 ignition interlock module - X4 connector F-2, F-3 and F-6 terminals
- A1 interlock module components - F4 and F5 fuse, start relay common terminal and coil, ignition relay normally open terminal and coil, magneto relay coil, and E1 Grn seat LED.
- S9 light switch - 400 Yel/Blu
- S6 brake switch - 501 Yel, 765 (767 SN 30001-) Pur
- S5 PTO/RIP switch - 765 and 789 Pur, and 705 Pnk/Blk
- S3 seat switch - 502 Yel
- A3 PTO interlock module (option) - X8 connector J4-C terminal, 789 Pur
- S4 RIO switch - 505 Yel
- N1 voltage rectifier/regulator - 504 Yel
- Instrument panel - X3 connector 17 terminal, 503 Yel
- K12 EFI module power relay - 86 terminal, 560 Red/Yel
- S8 mid PTO switch (option) - 520 Yel

These circuits are controlled by the key switch and are protected by the fusible links F2, F3 and F6, and the F4, F5, and F7 fuses.

With power now available at various locations on the machine, the electrical system is ready to perform the different functions of starting and running the engine, engine monitoring lights, the PTO clutch and the PTO system interlocks, and the headlights.

The ground circuit is equally important as the power circuit connections. Proper systems operation depends on good wires and connections in order to carry the current needed to operate the various relays, solenoids and lights.

Operating Conditions, Secondary Switched Circuits:

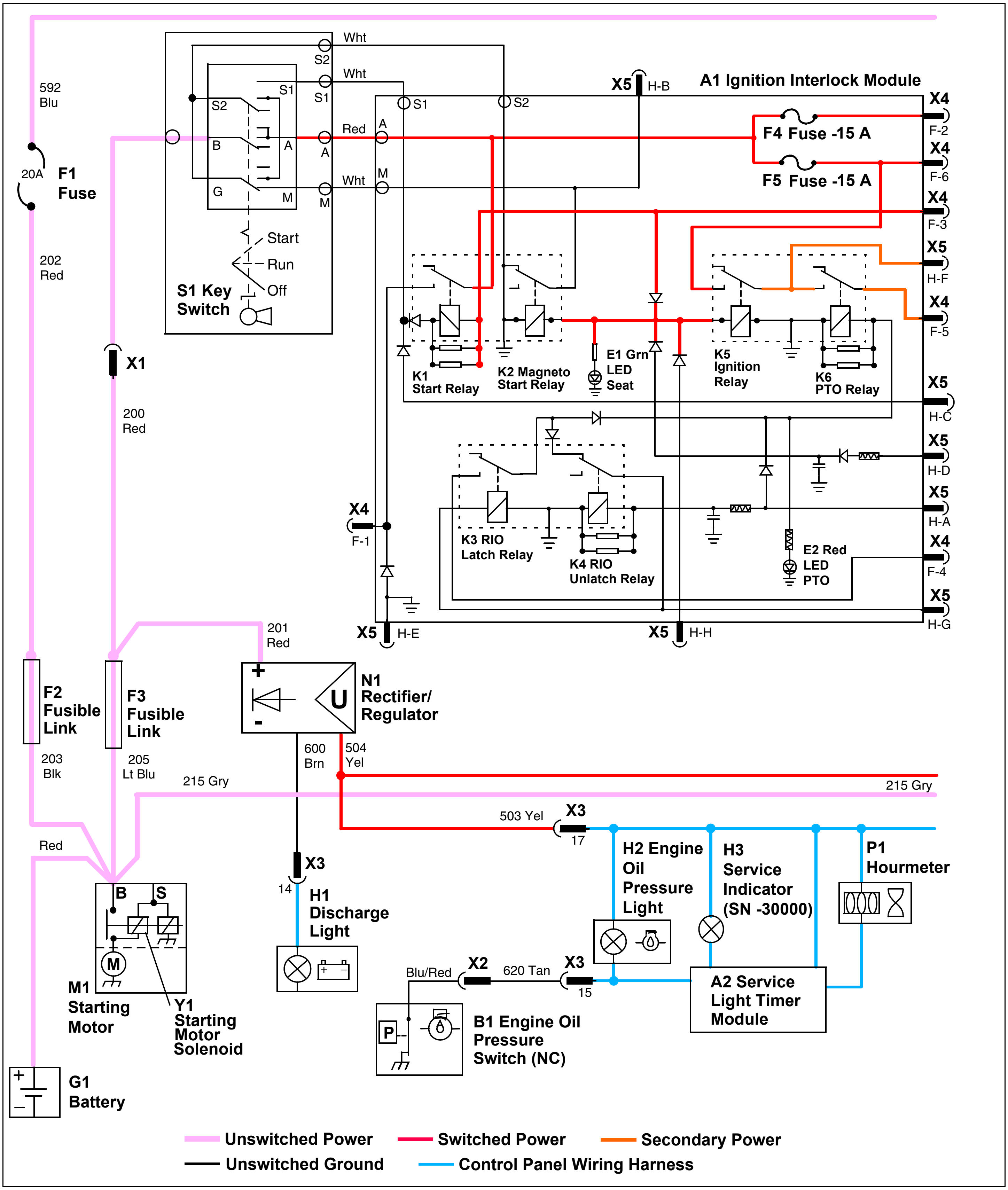
Secondary switched voltage must be present at the following components during the following conditions: Key switch in run position, transmission in neutral, PTO in off position, brake locked, and operator off seat:

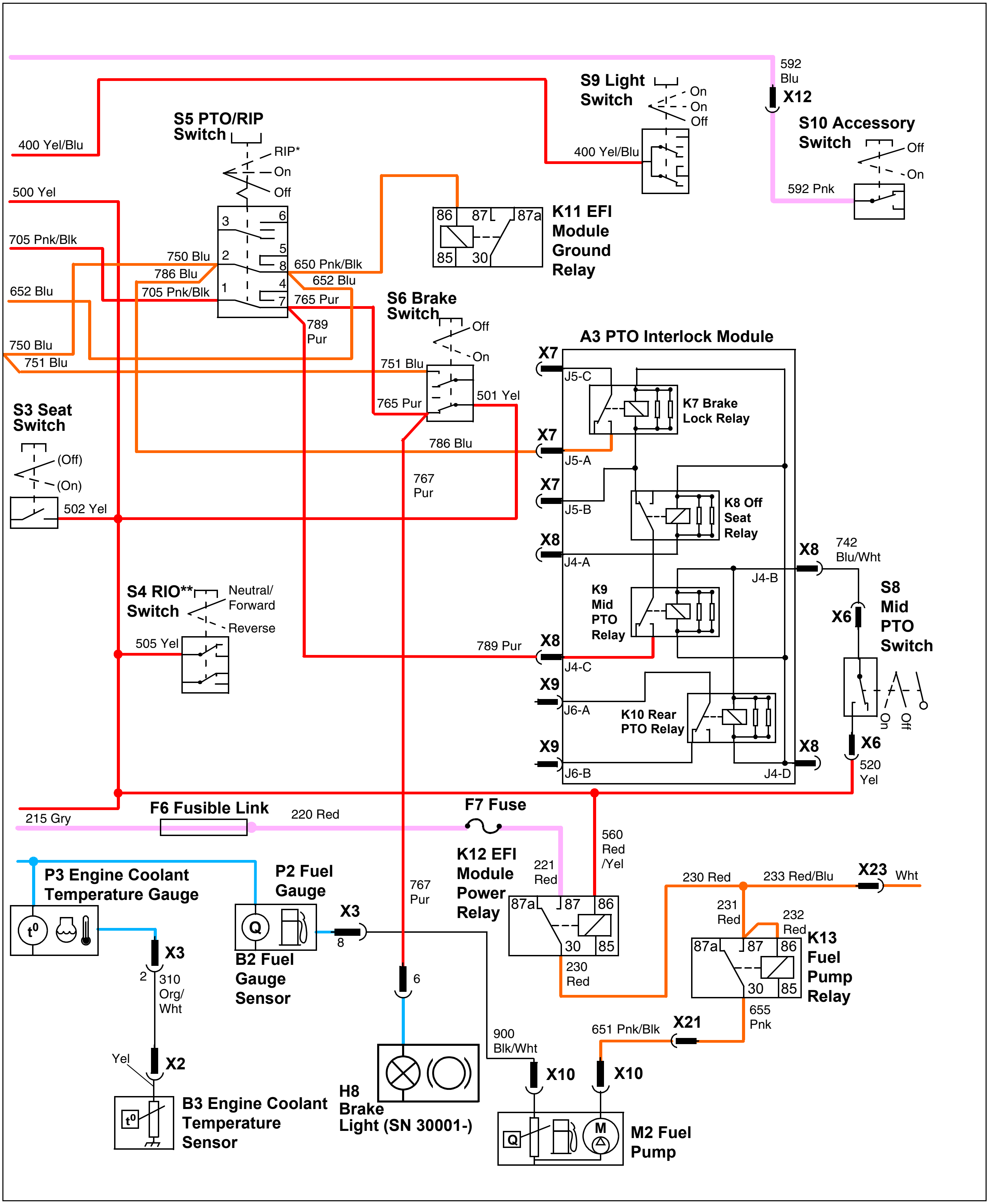
- A1 ignition interlock module - X5 connector H-F terminal, 652 Blu
- S5 PTO/RIP switch - 650 Pnk/Blk and 652, 750, and 786 Blu
- Y2 fuel shutoff solenoid - 650 Pnk/Blk and Yel/Wht
- K11 EFI module ground relay - 86 terminal 650 Pnk/Blk
- K12 EFI module power relay - 30 terminal, 230 Red
- K13 Fuel pump relay - 87, 86, and 30 terminals, 231, 232 Red and 655 Pnk
- M2 fuel pump - 651 Pnk/Blk
- A1 ignition interlock module - X4 connector F-5 terminal, 750 and 751 Blu
- A1 interlock module components - PTO relay common and normally open terminals and ignition relay common terminal
- S6 brake switch - 751 Blu
- Y4 fuel injector - Wht
- Y5 fuel injector - Wht

- A4 electric fuel injector module - P and M terminals, 240 Red and 950 Wht
- Instrument panel - X3 connector 16 terminal, 785 Grn
- A3 PTO interlock module (option) - X7 connector J5-A terminal, 786 Blu
- A3 PTO interlock module (option) components - brake lock relay normally open terminal

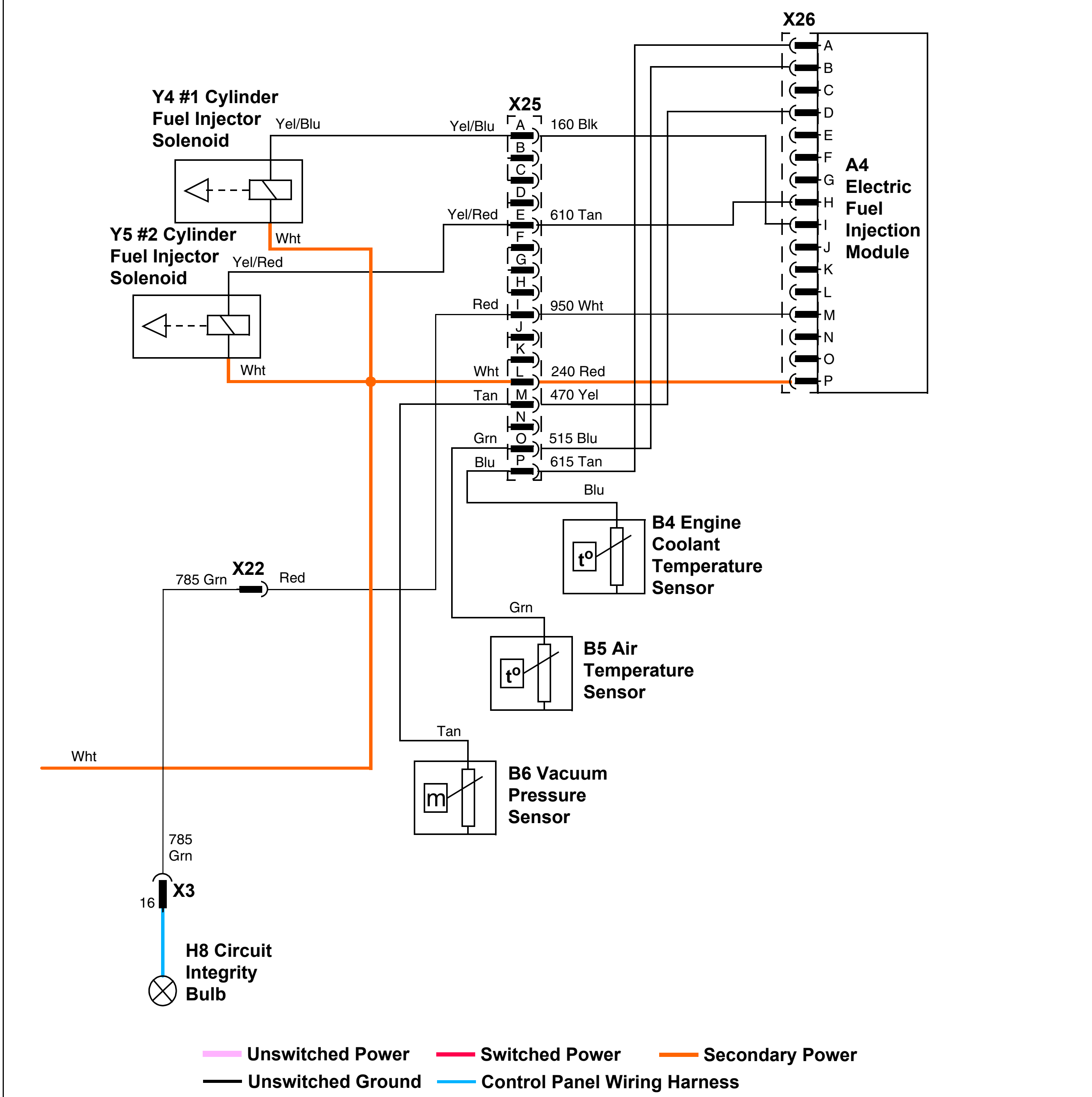
These circuits are controlled by the key switch and are protected by the F3 and F6 fusible links and the F4, F5 and F7 fuses.

Power Circuit Electrical Schematic - X485 and X585





S9 Light Switch
On
On
Off
592 Blu
X12
S10 Accessory Switch
Off
On
592 Pnk
400 Yel/Blu
S5 PTO/RIP Switch
RIP*
On
Off
500 Yel
705 Pnk/Blk
750 Blu
786 Blu
705 Pnk/Blk
652 Blu
750 Blu
751 Blu
650 Pnk/Blk
652 Blu
765 Pur
789 Pur
K11 EFI Module Ground Relay
86
87
87a
85
30
S6 Brake Switch
Off
On
751 Blu
765 Pur
501 Yel
786 Blu
767 Pur
A3 PTO Interlock Module
X7
J5-C
K7 Brake Lock Relay
J5-A
J5-B
K8 Off Seat Relay
X8
J4-A
742 Blu/Wht
J4-B
K9 Mid PTO Relay
S8 Mid PTO Switch
X6
789 Pur
J4-C
X9
J6-A
K10 Rear PTO Relay
J6-B
J4-D
520 Yel
S3 Seat Switch
(Off)
(On)
502 Yel
S4 RIO** Switch
Neutral/ Forward
Reverse
505 Yel
215 Gry
F6 Fusible Link
220 Red
F7 Fuse
560 Red /Yel
K12 EFI Module Power Relay
221 Red
230 Red
233 Red/Blu
X23
Wht
231 Red
232 Red
K13 Fuel Pump Relay
655 Pnk
651 Pnk/Blk
X21
P3 Engine Coolant Temperature Gauge
P2 Fuel Gauge
B2 Fuel Gauge Sensor
X3
8
2
310 Org/ Wht
X2
Yel
B3 Engine Coolant Temperature Sensor
767 Pur
6
H8 Brake Light (SN 30001-)
900 Blk/Wht
X10
M2 Fuel Pump


X26
A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
A4 Electric Fuel Injection Module
Y4 #1 Cylinder Fuel Injector Solenoid
Yel/Blu
X25
Yel/Blu
160 Blk
Yel/Red
610 Tan
Wht
Y5 #2 Cylinder Fuel Injector Solenoid
Yel/Red
Red
950 Wht
Wht
Wht
240 Red
Tan
470 Yel
Grn
515 Blu
Blu
615 Tan
Blu
B4 Engine Coolant Temperature Sensor
785 Grn
X22
Red
Grn
B5 Air Temperature Sensor
Tan
Wht
B6 Vacuum Pressure Sensor
785 Grn
16
X3
H8 Circuit Integrity Bulb
Unswitched Power
Switched Power
Secondary Power
Unswitched Ground
Control Panel Wiring Harness

Power Circuit Diagnosis - X485 and X585

Test Conditions:

- Key switch off.

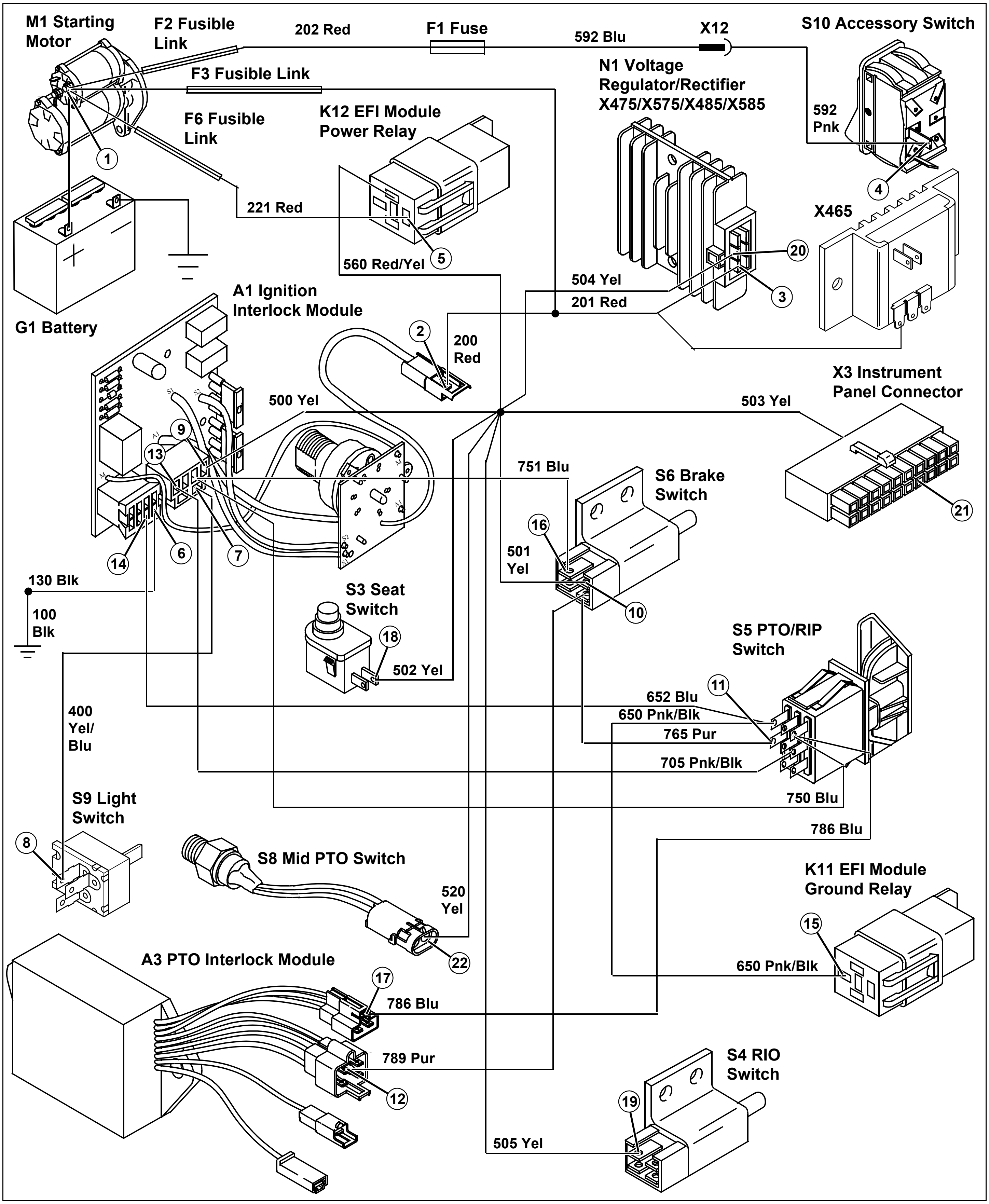
Test/Check Point	Normal	If Not Normal
1. Starting motor solenoid	Battery voltage	Test battery and cable connections.
2. Key switch	Battery voltage	Test F3 fusible link. Check 200 Red wire and connections.
3. Voltage rectifier/regulator	Battery voltage	Test F3 fusible link. Check 201 Red wire and connections.
4. Accessory switch	Battery voltage	Test F1 fuse and F2 fusible link. Check 202 Red, 592 Blu and 592 Pnk wires and connections.
5. K12 EFI module power relay	Battery voltage	Test F7 fuse and F6 fusible link. Check 220 and 221 Red wires and connections.

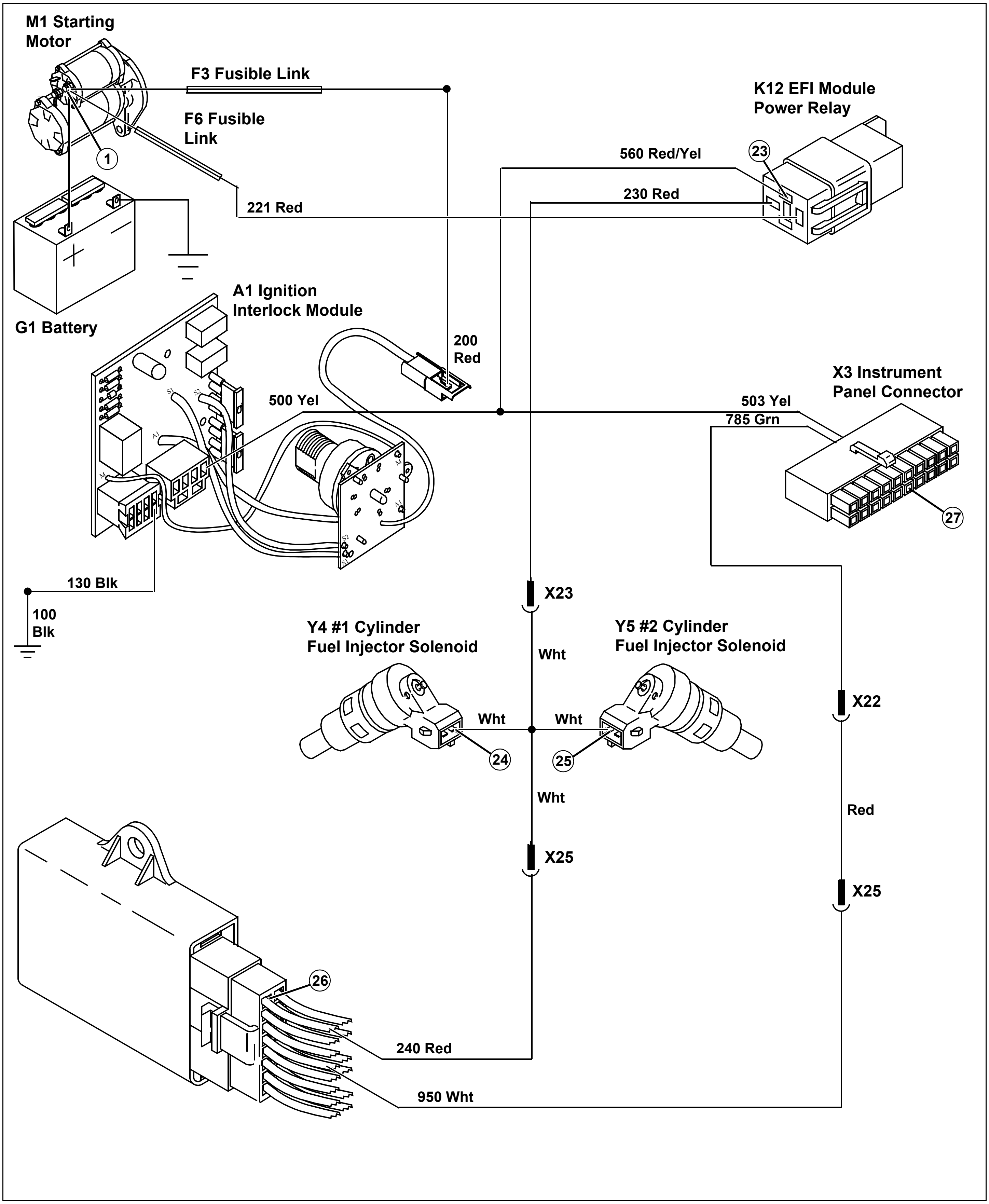
Test Conditions:

- Key switch in run position, engine not running.
- Brake locked.
- PTO off.
- Transmission in neutral.
- Operator off seat.

Test/Check Point	Normal	If Not Normal
6. Ignition interlock module	Continuity to ground	Check 130 and 100 Blk wires and connections.
7. Ignition interlock module	Battery voltage	Test F4 fuse. If OK, replace ignition interlock module.
8. Light switch	Battery voltage	Check 400 Yel/Blu wire and connections.
9. Ignition interlock module	Battery voltage	Test F5 fuse. If OK, replace ignition interlock module.
10. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
11. PTO/RIP switch	Battery voltage	Check 765 Pur wire and connections. If OK, replace brake switch.
12. PTO interlock module (option)	Battery voltage	Check 789 Pur wire and connections. If OK, replace brake switch.
13. Ignition interlock module	Battery voltage	Check 705 Pnk/Blk wire and connections. If OK, replace PTO/RIP switch.
14. Ignition interlock module	Battery voltage, green LED illuminated	Replace ignition interlock module.
15. K11 EFI module ground relay	Battery voltage	Check 652 Blu and 650 Pnk/Blk wires and connections.
16. Brake switch	Battery voltage	Check 750 and 751 Blu wires and connections. If OK, replace PTO/RIP switch.
17. PTO interlock module (option)	Battery voltage	Check 786 Blu wires and connections. If OK, replace PTO/RIP switch.

Test/Check Point	Normal	If Not Normal
18. Seat switch	Battery voltage	Check 500 and 502 Yel wires and connections.
19. RIO switch	Battery voltage	Check 500 and 505 Yel wires and connections.
20. Voltage rectifier/regulator	Battery voltage	Check 500 and 504 Yel wires and connections.
21. X3 instrument panel connector	Battery voltage	Check 500 and 503 Yel wires and connections.
22. Mid PTO switch (option)	Battery voltage	Check 500 and 520 Yel wires and connections.
23. K12 EFI module power relay	Battery voltage	Check 500 Yel and 560 Red/Yel wires and connections.
24. #1 Cylinder fuel injector	Battery voltage	Check 230 Red, 231 Red/Blu, and Wht wires and connections. If OK, replace EFI module power relay.
25. #2 Cylinder fuel injector	Battery voltage	Check 230 Red, 231 Red/Blu, and Wht wires and connections. If OK, replace EFI module power relay.
26. EFI module	Battery voltage	Check 230 Red, 231 Red/Blu, Wht, and 240 Red wires and connections. If OK, replace EFI module power relay.
27. X3 instrument panel connector	Battery voltage	Check 785 Grn, Red, and 950 Wht wires and connections. If OK, replace EFI module.


M1 Starting Motor
F2 Fusible Link
202 Red
F1 Fuse
592 Blu
X12
S10 Accessory Switch
F3 Fusible Link
N1 Voltage Regulator/Rectifier X475/X575/X485/X585
F6 Fusible Link
K12 EFI Module Power Relay
592 Pnk
1
4
X465
221 Red
5
20
560 Red/Yel
504 Yel
3
201 Red
G1 Battery
A1 Ignition Interlock Module
2
200 Red
X3 Instrument Panel Connector
500 Yel
503 Yel
9
13
751 Blu
S6 Brake Switch
21
16
14
6
7
501 Yel
130 Blk
10
100 Blk
S3 Seat Switch
18
502 Yel
S5 PTO/RIP Switch
11
652 Blu
650 Pnk/Blk
400 Yel/ Blu
765 Pur
705 Pnk/Blk
S9 Light Switch
750 Blu
786 Blu
8
S8 Mid PTO Switch
K11 EFI Module Ground Relay
520 Yel
15
A3 PTO Interlock Module
22
650 Pnk/Blk
17
786 Blu
789 Pur
S4 RIO Switch
12
19
505 Yel



M1 Starting Motor
F3 Fusible Link
F6 Fusible Link
1
K12 EFI Module Power Relay
560 Red/Yel
23
230 Red
221 Red
G1 Battery
A1 Ignition Interlock Module
200 Red
X3 Instrument Panel Connector
500 Yel
503 Yel
785 Grn
27
130 Blk
100 Blk
X23
Y4 #1 Cylinder Fuel Injector Solenoid
Y5 #2 Cylinder Fuel Injector Solenoid
Wht
X22
Wht
Wht
24
25
Wht
Red
X25
X25
26
240 Red
950 Wht

Cranking Circuit Operation - All Models

Function:

To energize the starting motor solenoid and engage the starting motor to crank the engine.

Operating Conditions:

- Key switch in start position.
- Brake locked, switch closed.
- PTO/RIP switch in off position.

Theory of Operation:

NOTE: The key switch is connected to the printed circuit board (A1 Interlock Module) by soldered wire connections. Components are solid state and are not serviced separately except for the 15 amp fuses. Lines within border of the Interlock Module (A1) except for key switch connections are circuit paths on the circuit board and are not wires in the harness.

The starting motor is a solenoid shift design. The power circuit provides current to the S1 key switch and protects the cranking circuit with the F3 fusible link. Current flows from the battery positive (+) terminal to the starter solenoid battery terminal, F3 fusible link, 200 Red wire, X1 connector, Red wire and key switch.

With the key switch in the start position, current flows through the Red wire to the A1 interlock module and to the start relay common terminal and the F5 fuse. From the F5 fuse current flows to the 500 and 501 Yel wire, and the S6 brake switch.

With the brake pedal depressed, the brake switch is engaged and current flows across the switch to the 765 pur wire and the S5 PTO/RIP switch. From the PTO/RIP switch current flows to the 705 Pnk/Blk wire back to the A1 interlock module and the start relay coil. The ground path for the start relay is provided through the key switch S1 and S2 contacts.

The start relay is energized and shifts to the normally open contact allowing current to flow to the 720 Pur wire to the Y1 starting motor solenoid and the 721 Pur wire to the H4 RIO defeat light on the instrument panel.

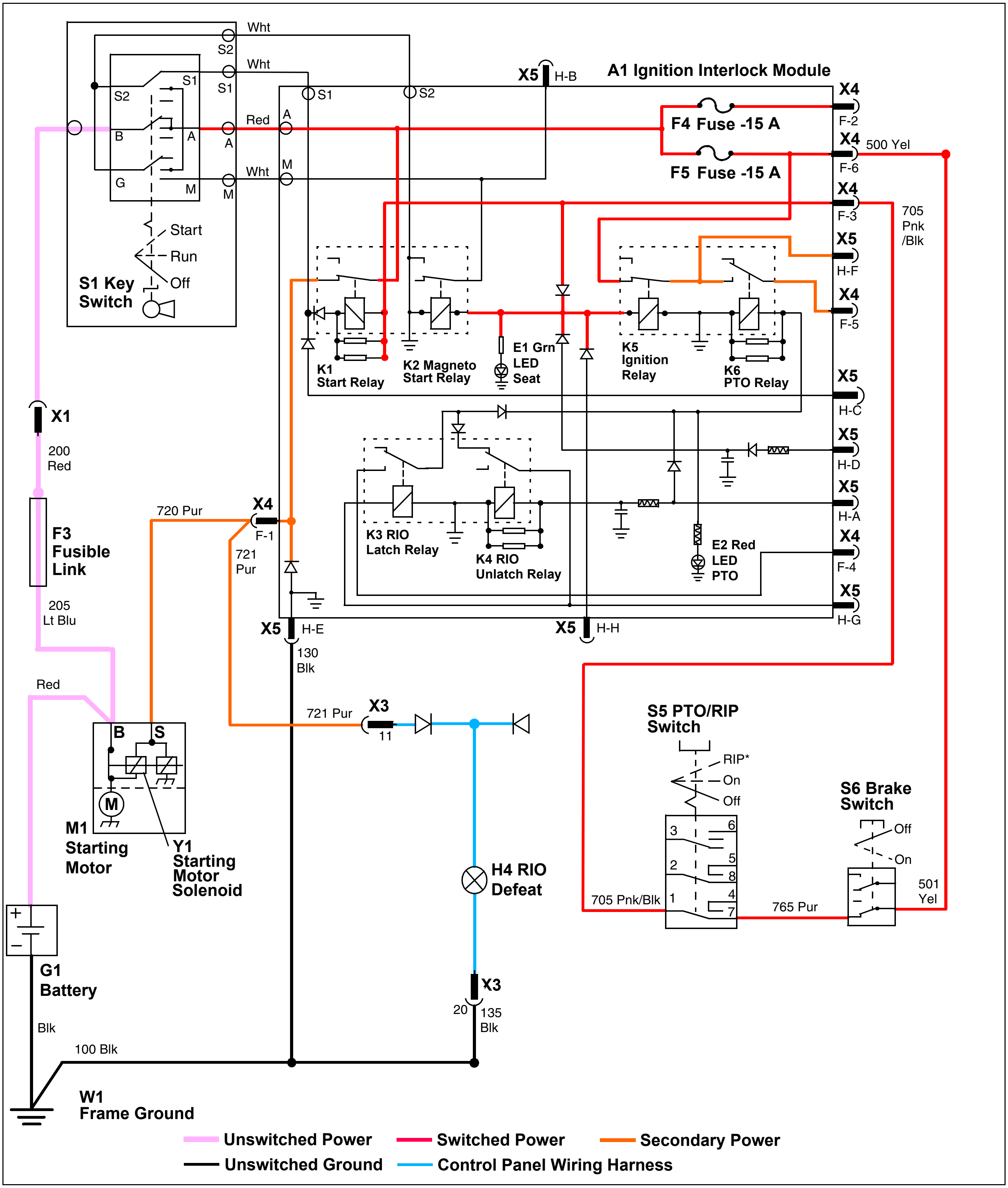
The Y1 starting motor solenoid is engaged by current flowing through the coil windings, pulling the plunger inward. The plunger closes the solenoid main contacts. Current flows through the hold-in windings, keeping the solenoid engaged.

With the solenoid main contacts closed, high current from the battery flows across the main contacts to the M1 starter motor causing it to turn.

Current on the 721 Pur wire is used to illuminate the H4 RIO defeat light and indicate that the starting circuit through the A1 interlock module is functional. If the Y1 starting motor solenoid does not engage but the H4 RIO defeat light is illuminated, then the cranking circuit is functional and the starting motor solenoid may be tested. If the Y1 starting motor solenoid does not engage and the H4 RIO defeat light is not illuminated, then either the operating conditions have not been met, or there is a fault within the cranking circuit.

Additionally, the E1 Grn LED on the A1 ignition interlock module will be illuminated when the cranking circuit operating conditions have been met and are functional.

Cranking Circuit Electrical Schematic - All Models



Cranking Circuit Diagnosis - All Models

Test Conditions:

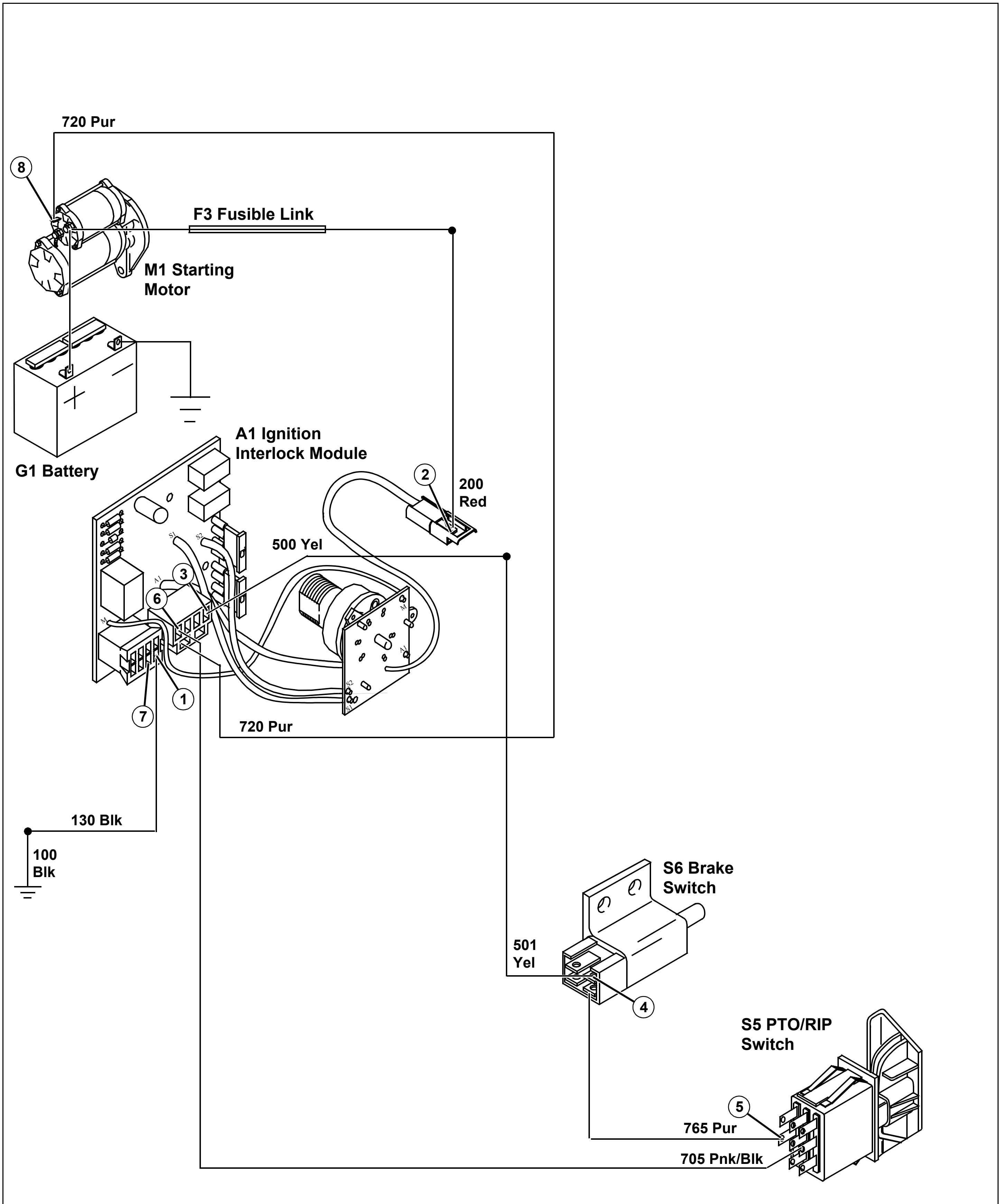
- Key switch in run position, engine not running.
- Brake locked.
- PTO off.

Test/Check Point	Normal	If Not Normal
1. Ignition interlock module	Continuity to ground	Check 130 and 100 Blk wires and connections.
2. Key switch	Battery voltage	Test F3 fusible link. Check 200 Red wire and connections.
3. Ignition interlock module	Battery voltage	Test F5 fuse. If OK, replace ignition interlock module.
4. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
5. PTO/RIP switch	Battery voltage	Check 765 Pur wire and connections. If OK, replace brake switch.
6. Ignition interlock module	Battery voltage	Check 705 Pnk/Blk wire and connections. If OK, replace PTO/RIP switch.
7. Ignition interlock module	Battery voltage, green LED illuminated	Replace ignition interlock module.

Test Conditions:

- Spark plug wires removed and grounded to engine.
- Key switch in start position.
- Brake locked.
- PTO off.

Test/Check Point	Normal	If Not Normal
8. Starting motor solenoid	Battery voltage	Check 720 Pur wire and connections. If OK, replace ignition interlock module.


720 Pur
8
F3 Fusible Link
M1 Starting
Motor
G1 Battery
A1 Ignition
Interlock Module
2
200
Red
500 Yel
3
6
1
7
720 Pur
130 Blk
100
Blk
S6 Brake
Switch
501
Yel
4
S5 PTO/RIP
Switch
5
765 Pur
705 Pnk/Blk

Charging Circuit Operation - All Models

Function:

To maintain battery voltage between 12.4 volts DC or higher.

Operating Conditions:

The key switch must be in the run position, with the engine running, for the charging system to operate.

Theory of Operation:

The main charging system components are the battery, G2 alternator and the N1 voltage regulator/rectifier. Charging output is controlled by the regulator/rectifier. The status of the charge rate is indicated by the H1 discharge light.

X475/X575/X485/X585 - With the key switch in the run position, battery sensing circuit current flows from battery positive terminal to the starting motor terminal, F3 fusible link, 200 Red wire, S1 key switch, F5 fuse, 500 and 504 Yel wire, and the N1 voltage regulator/rectifier. The battery sensing circuit allows the voltage regulator/rectifier to monitor battery voltage.

X465 - With the key switch in the run position, battery sensing current flows from the battery positive terminal to the starting motor terminal, F3 fusible link, 201 Red wire to the N1 voltage regulator/rectifier.

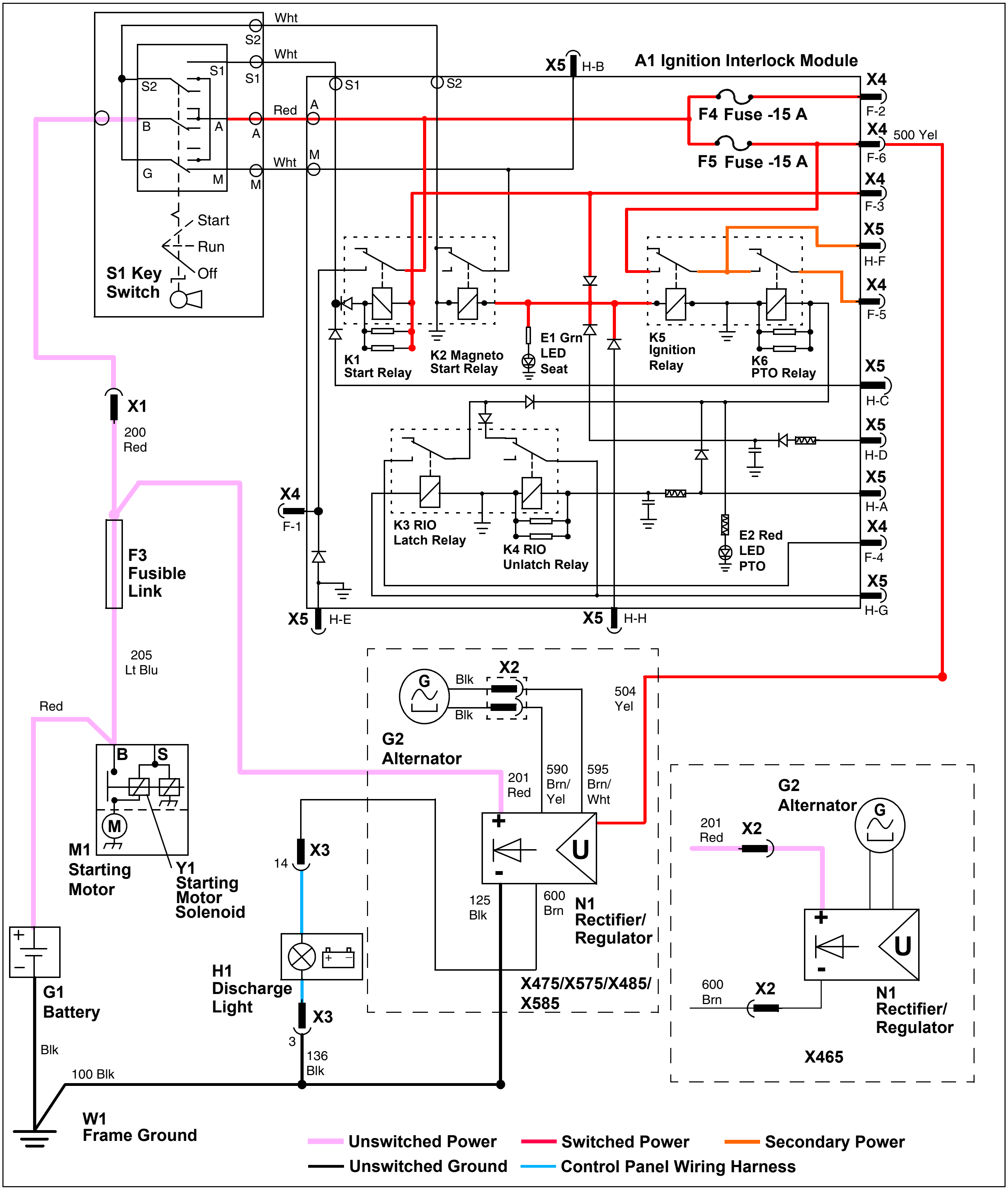
With the engine running, a rotating permanent magnet in the alternator induces AC current in the alternator stator coils. The AC current flows to the voltage regulator/rectifier. The voltage regulator/rectifier converts AC current to DC current needed to charge the battery.

Current flows from the voltage regulator/rectifier to the battery positive (+) terminal through the 201 Red wire and F3 fusible link. When the battery is fully charged, the voltage regulator/rectifier halts current flow to the battery.

If the voltage on the sensing circuit falls below system usage or is insufficient to maintain a preset voltage, the regulator/rectifier provides current to turn on the H1 discharge light.

The ground circuit 125 Blk provides a path to ground for the voltage regulator/rectifier and the 136/100 Blk for the discharge light.

Charging Circuit Electrical Schematic - All Models



Charging Circuit Diagnosis - All Models

Test Conditions:

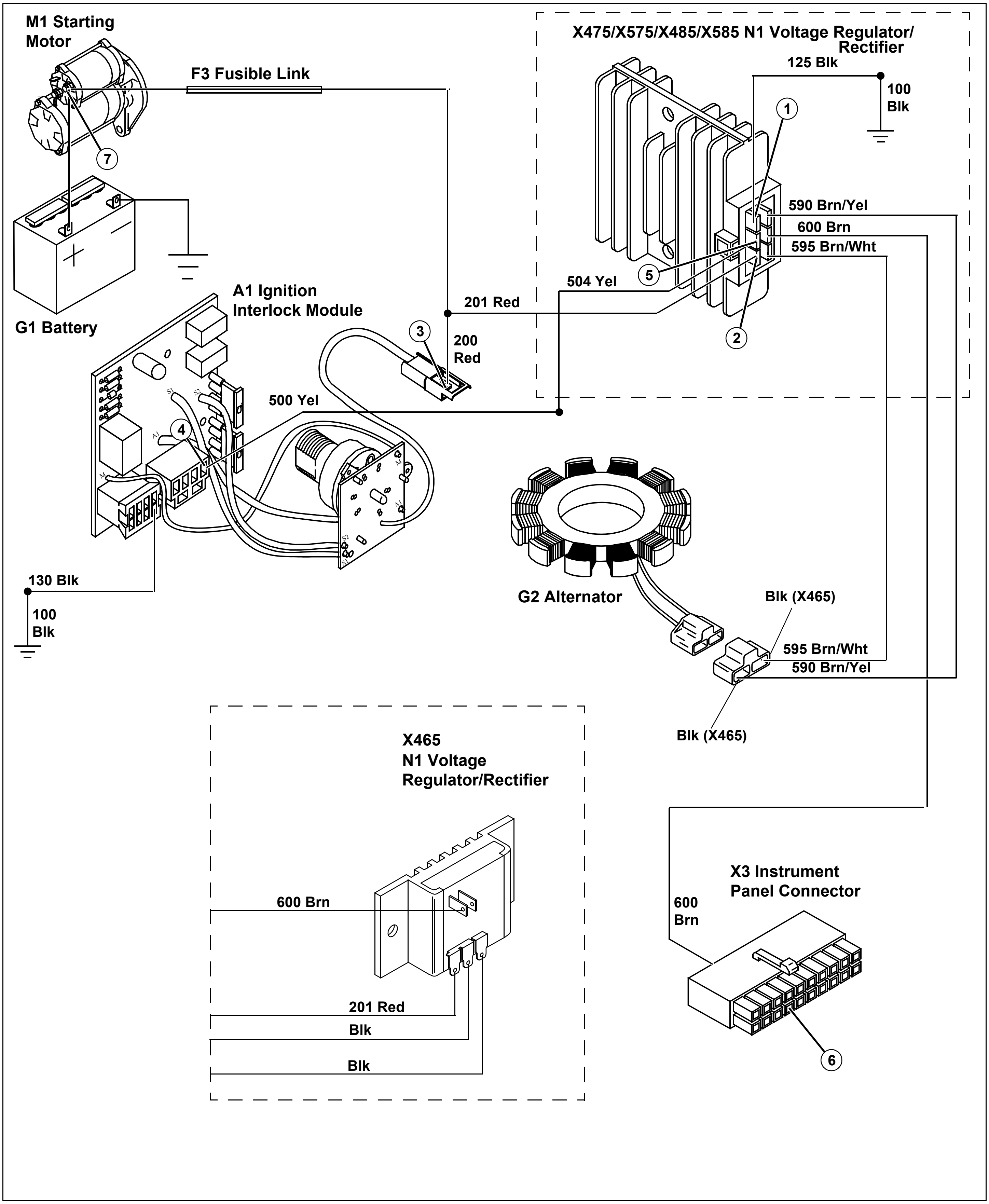
- Key switch in run position, engine not running.
- Brake locked.
- PTO off.

Test/Check Point	Normal	If Not Normal
1. Voltage rectifier/regulator	Continuity to ground	Check 125 (not X465) and 100 Blk wires and connections.
2. Voltage rectifier/regulator	Battery voltage	Test F3 fusible link. Check 201 Red wire and connections.
3. Key switch (X465 - skip to step 6)	Battery voltage	Test F3 fusible link. Check 200 Red wire and connections.
4. Ignition interlock module (X465 - skip to step 6)	Battery voltage	Test F5 fuse. If OK, replace ignition interlock module.
5. Voltage rectifier/regulator (X465 - Skip to step 6)	Battery voltage	Check 500 and 504 Yel wires and connections.
6. X3 Instrument panel connector	Battery voltage	Check 600 Brn wire and connections.

Test Conditions:

- Key switch in run position, engine running at slow idle.
- Brake locked.
- PTO off.
- Engine speed increased from slow idle to fast idle during test.

Test/Check Point	Normal	If Not Normal
7. Battery positive terminal	12.0 volts raising to 15.0 volts	Test voltage rectifier/regulator. Test stator. See appropriate procedures in Tests and Adjustments in this section.


M1 Starting Motor
F3 Fusible Link
7
G1 Battery
A1 Ignition Interlock Module
3
200 Red
201 Red
500 Yel
4
130 Blk
100 Blk
X475/X575/X485/X585 N1 Voltage Regulator/ Rectifier
125 Blk
100 Blk
1
590 Brn/Yel
600 Brn
595 Brn/Wht
504 Yel
5
2
G2 Alternator
Blk (X465)
595 Brn/Wht
590 Brn/Yel
Blk (X465)
X465 N1 Voltage Regulator/Rectifier
600 Brn
201 Red
Blk
Blk
X3 Instrument Panel Connector
600 Brn
6

Ignition Circuit Operation - X465, X475 and X575

Function:

To control the ignition coils ability to create a spark while also controlling the fuel supply to the engine.

Operating Conditions, Stopping Engine:

- Key switch in off position,

or,

- Operator off seat with PTO engaged,

and,

- Park braked unlocked.

Theory of Operation, Stopping Engine

The engine is shut off by grounding the T1 and T2 ignition coils. With the ignition primary coil grounded, a spark cannot be produced.

When the S1 key switch is in the off position, a path to ground is created through the key switch Wht wire (M) solder connection and back to the interlock module through the Wht wire (S2) solder connection. In addition the power circuit current flow is stopped at the key switch. This de-energizes the magneto relay providing a path to ground for ignition coil through the magneto relay.

The magneto relay coil is energized by current from the seat switch circuit or brake circuit. If the operator gets off the seat (seat switch opens) with the PTO engaged without the park brake locked, current flow to the magneto relay coil is stopped.

The seat switch or brake switch also energize the ignition relay. Whenever the ignition relay is de-energized, the fuel shutoff solenoid is also de-energized and fuel flow to the carburetor is stopped.

Operating Conditions, Running Engine:

- Key switch in run or start position,
- Operator on seat,

or,

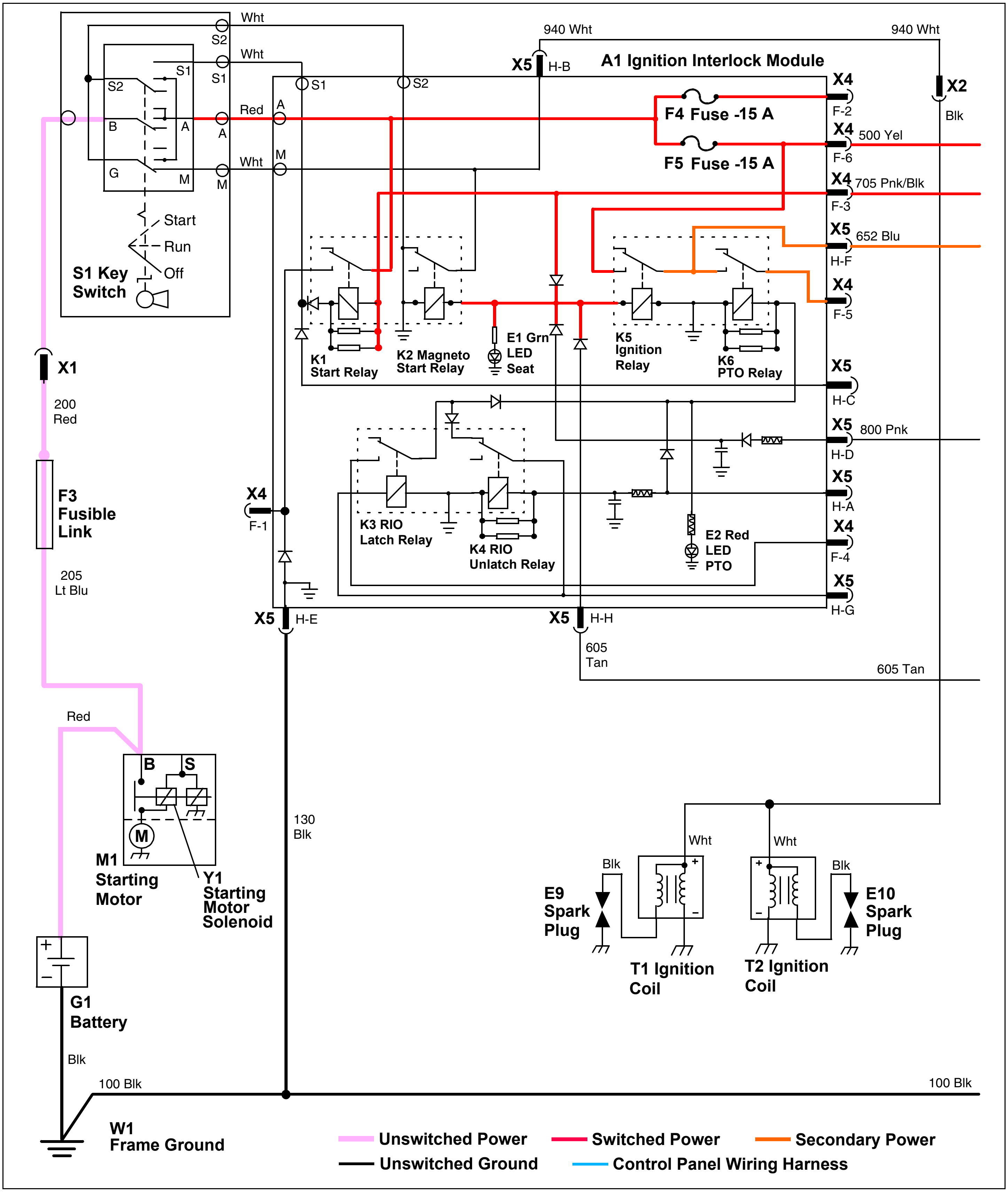
- Key switch in run or start position,
- Operator off seat,
- Park braked locked.

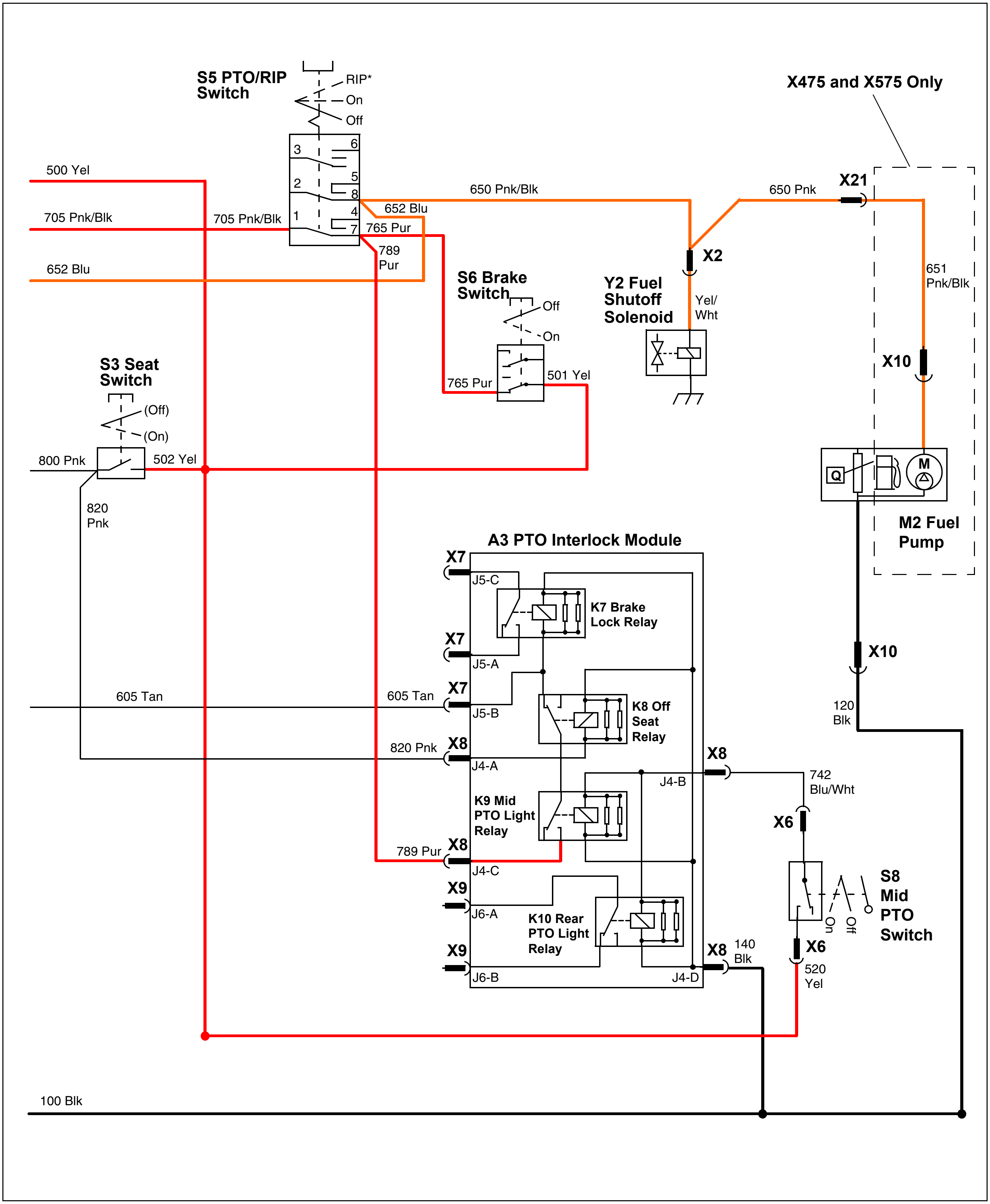
Theory of Operation, Running Engine

The key switch is connected to the A1 ignition interlock module by soldered wire connections. The components are solid state and are not serviced separately except for the 15 amp fuses. Lines within the border of the A1 ignition interlock module except for key switch connections are circuit paths on the circuit board and are not wires in the harness.

The ignition system is a transistor-controlled magneto design. Ignition timing is controlled by the transistor and is not adjustable. As the engine turns over the flywheel magnet induces current into the magneto ignition coil, which in turn produces current high enough to jump the spark plug(s) gap, creating spark to ignite the engine fuel/ air mixture.

Ignition Circuit Electrical Schematic - X465, X475 and X575




S5 PTO/RIP Switch
RIP*
On
Off
X475 and X575 Only
500 Yel
705 Pnk/Blk
705 Pnk/Blk
652 Blu
650 Pnk/Blk
652 Blu
765 Pur
789 Pur
650 Pnk
X21
X2
S6 Brake Switch
Off
On
Y2 Fuel Shutoff Solenoid
Yel/ Wht
651 Pnk/Blk
X10
S3 Seat Switch
(Off)
(On)
765 Pur
501 Yel
800 Pnk
502 Yel
820 Pnk
M
M2 Fuel Pump
A3 PTO Interlock Module
X7
J5-C
K7 Brake Lock Relay
X7
J5-A
X10
605 Tan
605 Tan
X7
J5-B
K8 Off Seat Relay
120 Blk
820 Pnk
X8
J4-A
X8
J4-B
742 Blu/Wht
K9 Mid PTO Light Relay
X6
789 Pur
X8
J4-C
X9
J6-A
K10 Rear PTO Light Relay
S8 Mid PTO Switch
On
Off
X9
J6-B
J4-D
X8
140 Blk
X6
520 Yel
100 Blk

Ignition Circuit Diagnosis - X465, X475 and X575

Test Conditions:

- Key switch off.

Test/Check Point	Normal	If Not Normal
1. Ignition interlock module	Continuity to ground	Check 130 and 100 Blk wires and connections.
2. Ignition interlock module	Continuity to ground	Replace ignition interlock module.
3. X2 connector	Continuity to ground	Check 940 Wht wire and connections.

Test Conditions:

- Key switch in run position, engine not running.
- Park brake locked.
- PTO off.
- Operator off seat.

Test/Check Point	Normal	If Not Normal
4. Key switch	Continuity to ground	Replace ignition interlock module.
5. Ignition interlock module	No continuity to ground	Replace ignition interlock module.
6. Ignition interlock module	Battery voltage	Test F3 fusible link and F5 fuse. Check 200 Red wire and connections. If OK, replace ignition interlock module.
7. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
8. PTO/RIP switch	Battery voltage	Check 765 Pur wire and connections. If OK, replace brake switch.
9. Ignition interlock module	Battery voltage	Check 705 Pnk/Blk wire and connections. If OK, replace PTO/RIP switch.
10. Ignition interlock module	Battery voltage	Replace ignition interlock module.
11. Fuel pump (X475 and X575 only)	Battery voltage	Check 652 Blu, 650 Pnk/Blk, 655 Pnk, and 651 Pnk/Blk wires and connections.

Test Conditions:

- Key switch in run position, engine not running.
- Park brake unlocked.
- PTO off.
- Operator on seat.

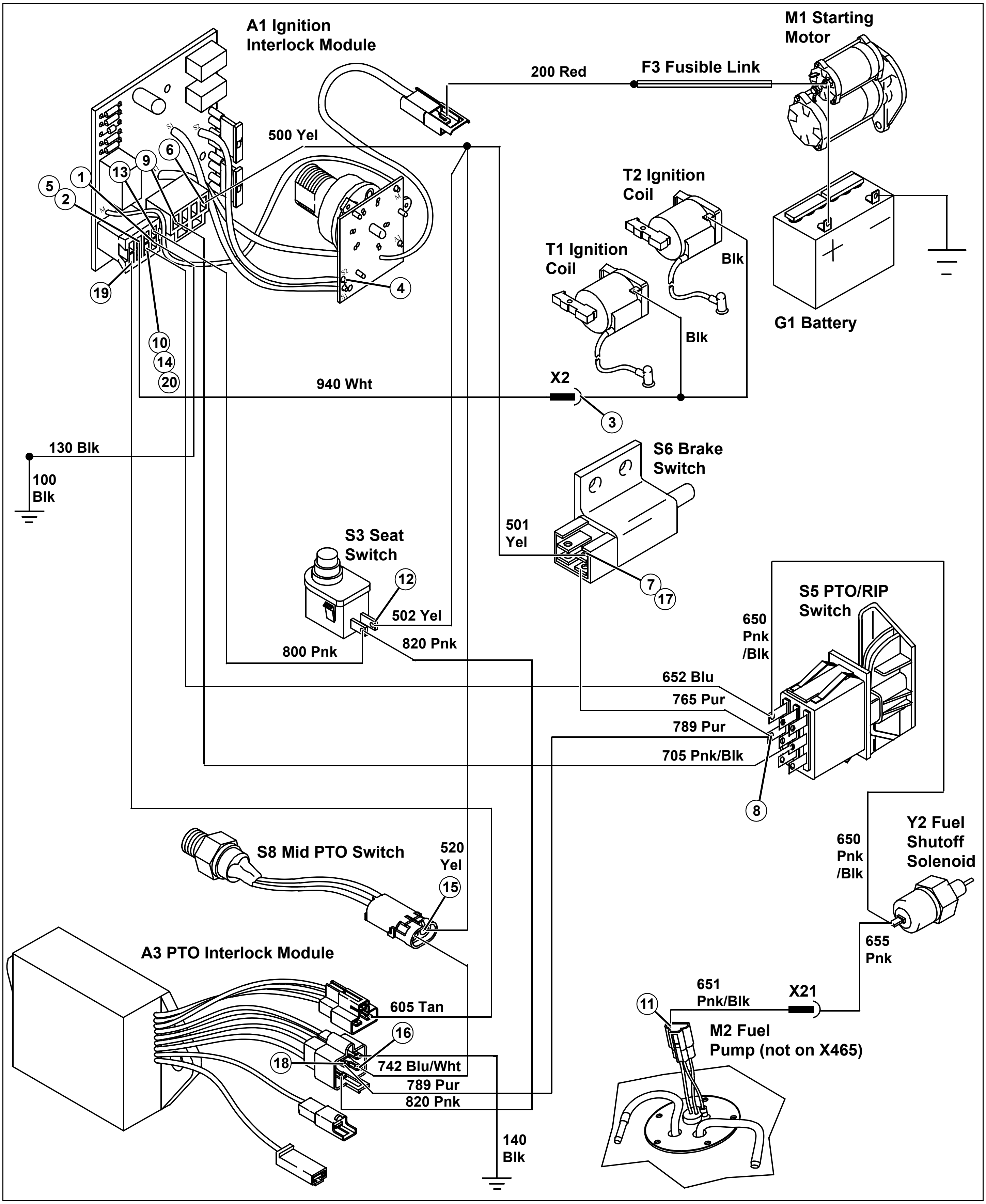
Test/Check Point	Normal	If Not Normal
12. Seat switch	Battery voltage	Check 500 and 502 Yel wires and connections.
13. Ignition interlock module	Battery voltage	Check 800 Pnk wire and connections. If OK, replace seat switch.

Test/Check Point	Normal	If Not Normal
14. Ignition interlock module	Battery voltage	Replace ignition interlock module.

Test Conditions:

- Key switch in run position, engine not running.
- Park brake locked.
- PTO on.
- Operator off seat.

Test/Check Point	Normal	If Not Normal
15. Mid PTO switch	Battery voltage	Check 500 and 520 Yel wires and connections.
16. PTO interlock module (option)	Battery voltage	Check 742 Blu/Wht wire and connections. If OK, replace mid PTO switch.
17. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
18. PTO interlock module (option)	Battery voltage	Check 765 and 789 Pur wires and connections. If OK, replace brake switch.
19. Ignition interlock module	Battery voltage	Check 605 Tan wire and connections. If OK, replace PTO interlock module.
20. Ignition interlock module	Battery voltage	Replace ignition interlock module.


A1 Ignition Interlock Module
M1 Starting Motor
200 Red
F3 Fusible Link
500 Yel
T2 Ignition Coil
T1 Ignition Coil
Blk
Blk
G1 Battery
940 Wht
X2
130 Blk
100 Blk
S6 Brake Switch
501 Yel
S3 Seat Switch
502 Yel
800 Pnk
820 Pnk
S5 PTO/RIP Switch
650 Pnk /Blk
652 Blu
765 Pur
789 Pur
705 Pnk/Blk
S8 Mid PTO Switch
520 Yel
Y2 Fuel Shutoff Solenoid
650 Pnk /Blk
655 Pnk
A3 PTO Interlock Module
605 Tan
651 Pnk/Blk
X21
M2 Fuel Pump (not on X465)
742 Blu/Wht
789 Pur
820 Pnk
140 Blk
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Ignition Circuit Operation - X485 and X585

Function:

To control the ignition coils ability to create a spark while also controlling the fuel supply to the engine.

Operating Conditions, Stopping Engine:

- Key switch in off position,

or,

- Operator off seat with PTO engaged,

and,

- Park braked unlocked.

Theory of Operation, Stopping Engine

The engine is shut off by grounding the T1 and T2 ignition coils. With the ignition primary coil grounded, a spark cannot be produced.

When the S1 key switch is in the off position, a path to ground is created through the key switch Wht wire (M) solder connection and back to the interlock module through the Wht wire (S2) solder connection. In addition the power circuit current flow is stopped at the key switch. This de-energizes the magneto relay providing a path to ground for ignition coil through the magneto relay.

The magneto relay coil is energized by current from the seat switch circuit or brake circuit. If the operator gets off the seat (seat switch opens) with the PTO engaged without the park brake locked, current flow to the magneto relay coil is stopped.

The seat switch or brake switch also energize the ignition relay. Whenever the ignition relay is de-energized, the fuel shutoff solenoid is also de-energized and fuel flow to the carburetor is stopped.

Operating Conditions, Running Engine:

- Key switch in run or start position,
- Operator on seat,

or,

- Key switch in run or start position,
- Operator off seat,
- Park braked locked.

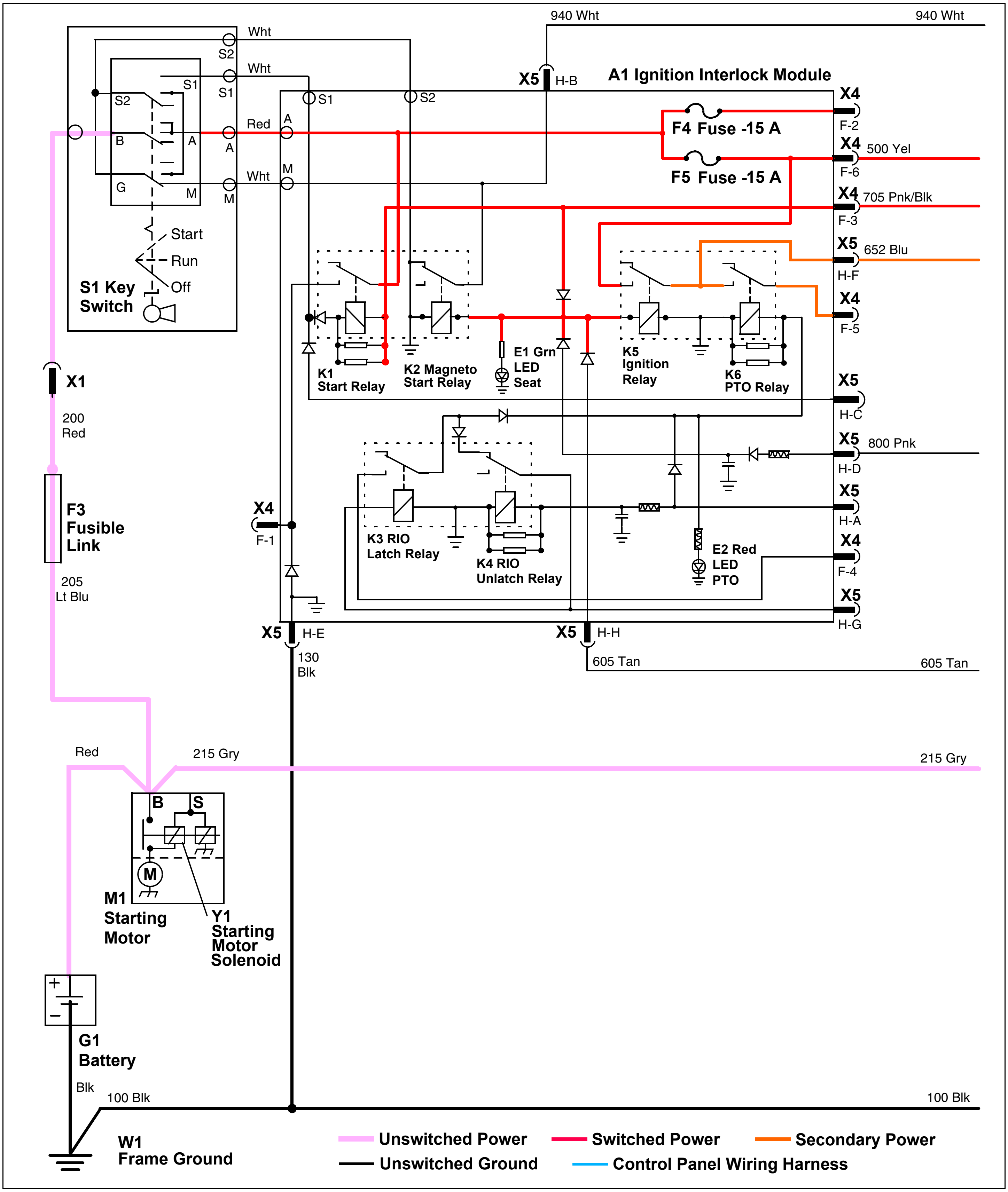
Theory of Operation, Running Engine

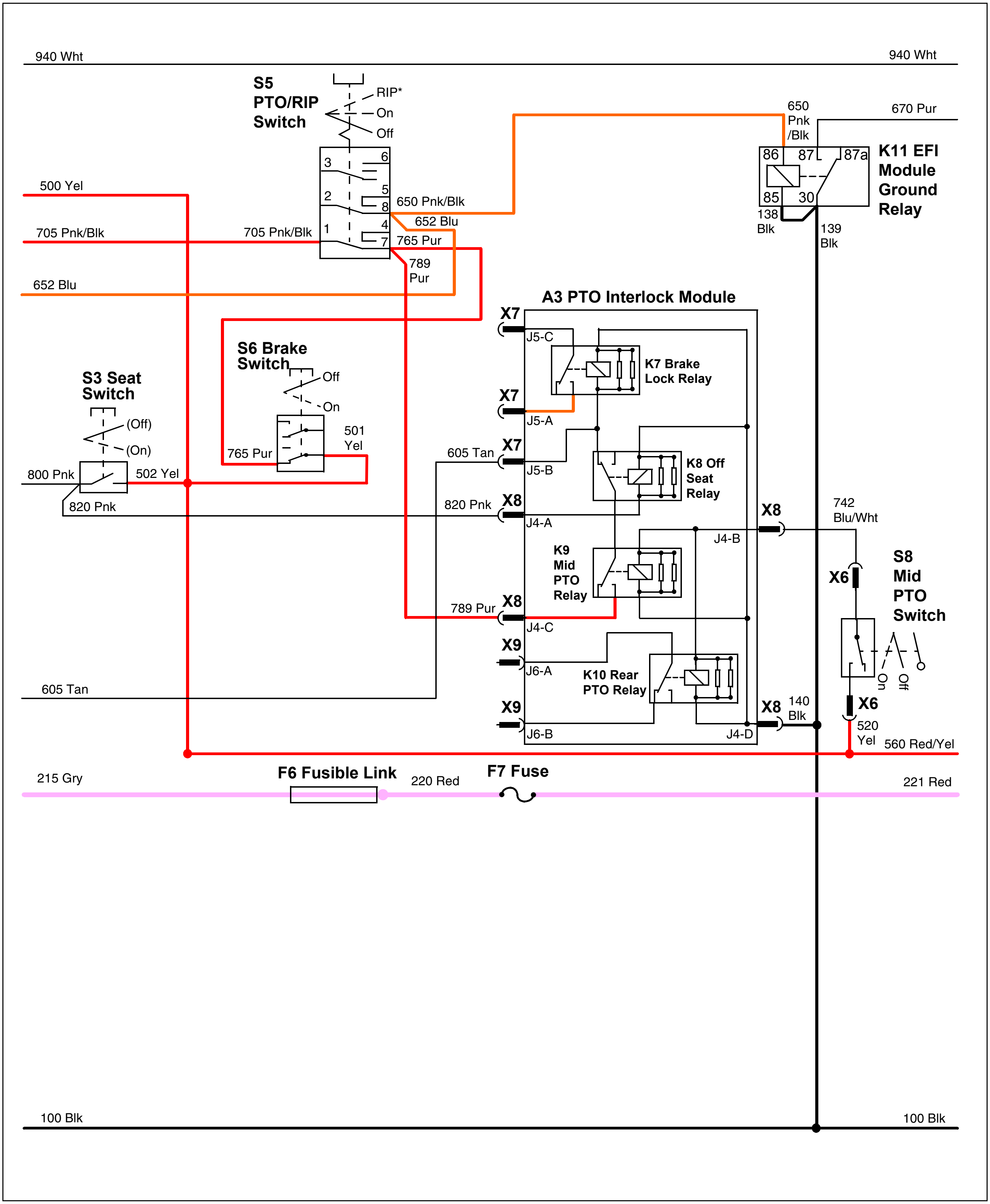
The key switch is connected to the A1 ignition interlock module by soldered wire connections. The components are solid state and are not serviced separately except for the 15 amp fuses. Lines within the border of the A1 ignition interlock module except for key switch connections are circuit paths on the circuit board and are not wires in the harness.

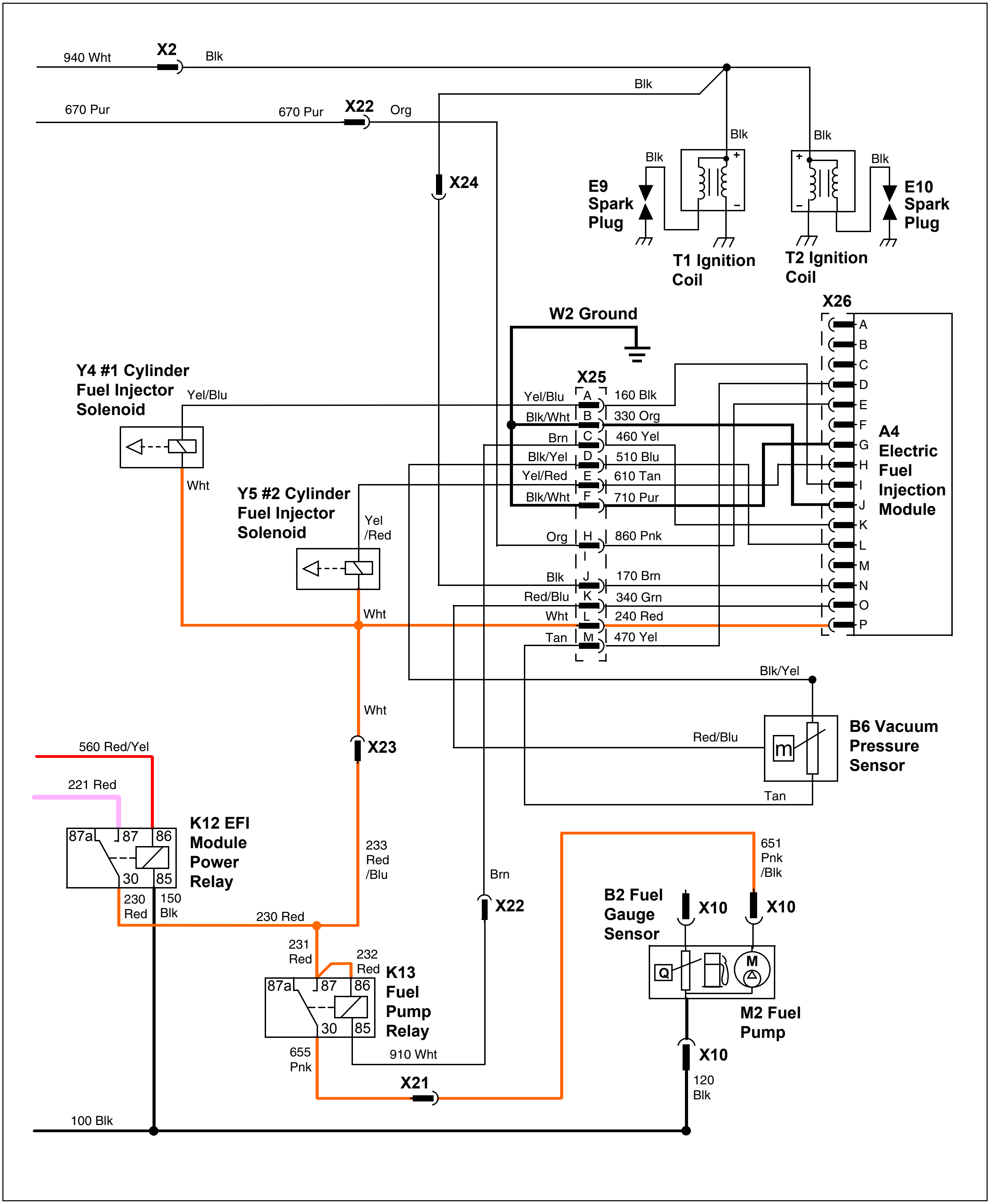
The ignition system is a transistor-controlled magneto design. Ignition timing is controlled by the transistor and is not adjustable. As the engine turns over the flywheel magnet induces current into the magneto ignition coil, which in turn produces current high enough to jump the spark plug(s) gap, creating spark to ignite the engine fuel/air mixture.

Fuel is supplied to the intake manifold through the fuel injectors. The fuel injectors are controlled by the electric fuel injector (EFI) module. The ground circuit for the EFI module is controlled through the ignition interlock module. If the operating conditions are not met, the K11 EFI module ground relay will not be energized and the EFI module will not receive a ground path. This will stop the fuel supply, thereby stopping the engine.

Ignition Circuit Electrical Schematic - X485 and X585




940 Wht
940 Wht
S5
PTO/RIP
Switch
RIP*
On
Off
650 Pnk/Blk
670 Pur
K11 EFI Module Ground Relay
86
87
87a
85
30
138 Blk
139 Blk
500 Yel
705 Pnk/Blk
705 Pnk/Blk
650 Pnk/Blk
652 Blu
765 Pur
789 Pur
652 Blu
A3 PTO Interlock Module
X7
J5-C
K7 Brake Lock Relay
S6 Brake Switch
Off
On
S3 Seat Switch
(Off)
(On)
X7
J5-A
501 Yel
765 Pur
605 Tan
X7
J5-B
K8 Off Seat Relay
800 Pnk
502 Yel
820 Pnk
820 Pnk
X8
J4-A
X8
742 Blu/Wht
J4-B
K9 Mid PTO Relay
S8 Mid PTO Switch
X6
789 Pur
X8
J4-C
X9
J6-A
K10 Rear PTO Relay
On
Off
605 Tan
X9
J6-B
J4-D
X8
140 Blk
X6
520 Yel
560 Red/Yel
215 Gry
F6 Fusible Link
220 Red
F7 Fuse
221 Red
100 Blk
100 Blk



940 Wht
X2
Blk
670 Pur
670 Pur
X22
Org
Blk
Blk
Blk
Blk
Blk
X24
E9 Spark Plug
E10 Spark Plug
T1 Ignition Coil
T2 Ignition Coil
X26
W2 Ground
X25
Y4 #1 Cylinder Fuel Injector Solenoid
Yel/Blu
Yel/Blu
A
160 Blk
Blk/Wht
B
330 Org
Brn
C
460 Yel
Blk/Yel
D
510 Blu
Yel/Red
E
610 Tan
Blk/Wht
F
710 Pur
Org
H
860 Pnk
Blk
J
170 Brn
Red/Blu
K
340 Grn
Wht
L
240 Red
Tan
M
470 Yel
A B C D E F G H I J K L M N O P
A4 Electric Fuel Injection Module
Wht
Y5 #2 Cylinder Fuel Injector Solenoid
Yel /Red
Wht
Blk/Yel
Wht
Red/Blu
B6 Vacuum Pressure Sensor
Tan
X23
560 Red/Yel
221 Red
87a 87 86 30 85
K12 EFI Module Power Relay
230 Red
150 Blk
230 Red
233 Red /Blu
Brn
X22
651 Pnk /Blk
B2 Fuel Gauge Sensor
X10
X10
231 Red
232 Red
87a 87 86 30 85
K13 Fuel Pump Relay
Q
M
M2 Fuel Pump
655 Pnk
910 Wht
X10
X21
120 Blk
100 Blk

Ignition Circuit Diagnosis - X485 and X585

Test Conditions:

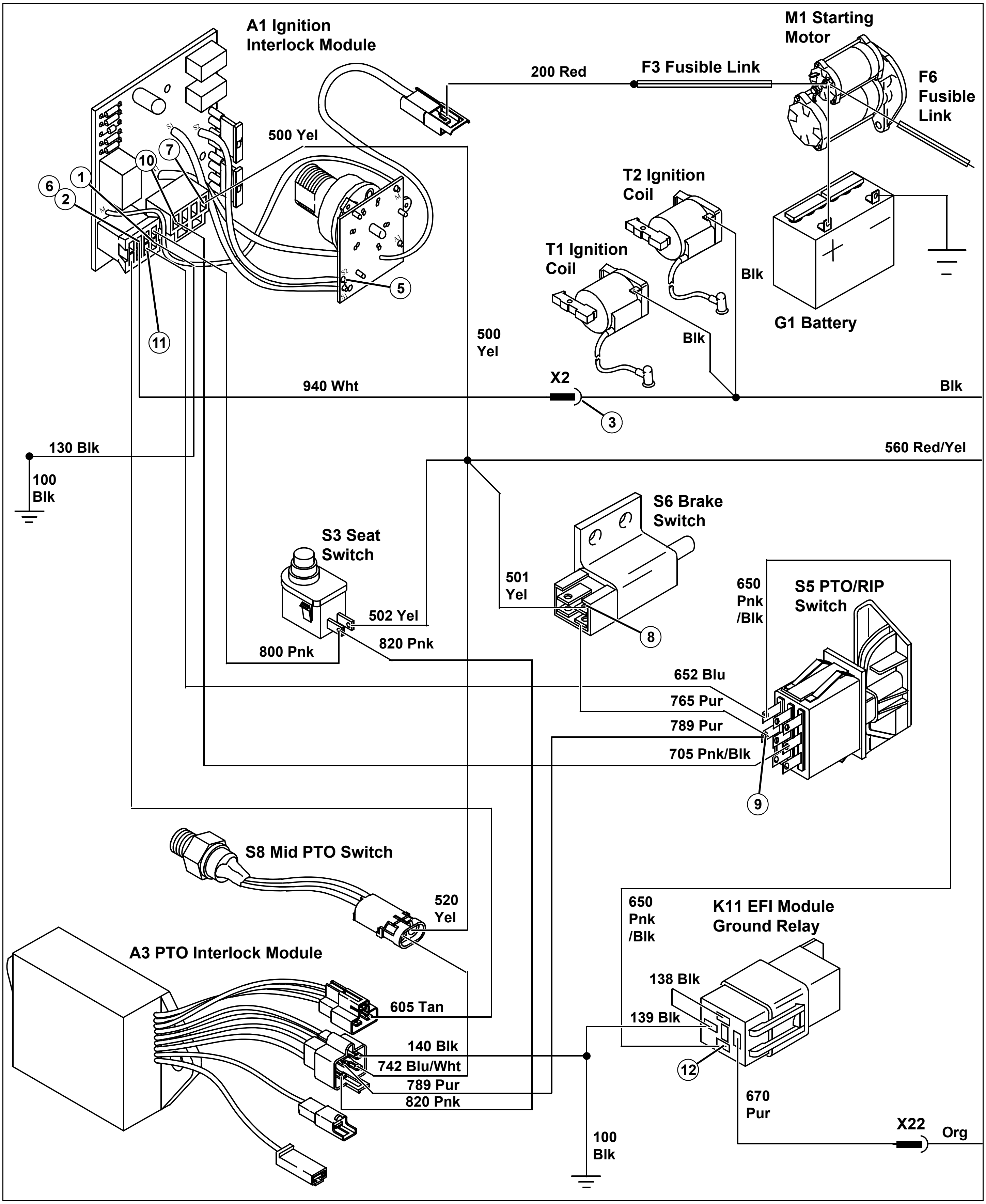
- Key switch off.

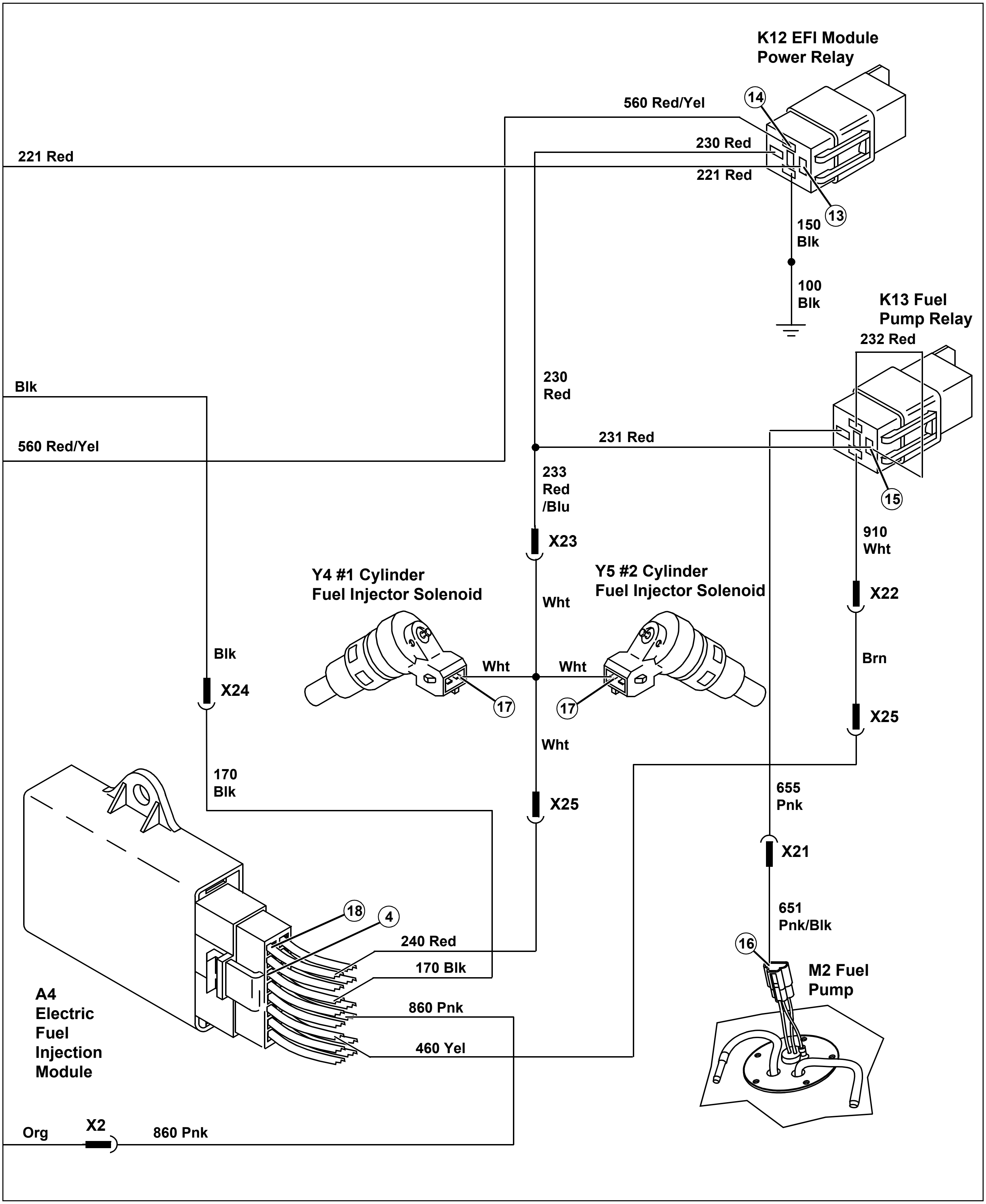
Test/Check Point	Normal	If Not Normal
1. Ignition interlock module	Continuity to ground	Check 130 and 100 Blk wires and connections.
2. Ignition interlock module	Continuity to ground	Replace ignition interlock module.
3. X2 connector	Continuity to ground	Check 940 Wht wire and connections.
4. EFI module	Continuity to ground	Check Blk and 170 Brn wires and connections.

Test Conditions:

- Key switch in run position, engine not running.
- Park brake locked.
- PTO off.
- Operator off seat.

Test/Check Point	Normal	If Not Normal
5. Key switch	Continuity to ground	Replace ignition interlock module.
6. Ignition interlock module	No continuity to ground	Replace ignition interlock module.
7. Ignition interlock module	Battery voltage	Test F3 fusible link and F5 fuse. Check 200 Red wire and connections. If OK, replace ignition interlock module.
8. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
9. PTO/RIP switch	Battery voltage	Check 765 Pur wire and connections. If OK, replace brake switch.
10. Ignition interlock module	Battery voltage	Check 705 Pnk/Blk wire and connections. If OK, replace PTO/RIP switch.
11. Ignition interlock module	Battery voltage	Replace ignition interlock module.
12. K11 EFI module ground relay	Battery voltage	Check 652 Blu and 650 Pnk/Blk wires and connections.
13. K12 EFI module power relay	Battery voltage	Test F6 fusible link and F7 fuse. Check 220 and 221 Red wires and connections.
14. K12 EFI module power relay	Battery voltage	Check 500 and 560 Red/Yel wires and connections.
15. K13 Fuel pump relay	Battery voltage	Check 230 and 231 Red wires and connections. If OK, replace K12 EFI module power relay.
16. Fuel pump	Battery voltage	Check 655 Pnk and 651 Pnk/Blk Red wires and connections. If OK, replace K13 fuel pump relay.
17. Fuel injector(s)	Battery voltage	Check 230 Red, 233 Red/Blu and Wht wires and connections. If OK, replace K12 EFI module power relay.
18. Electric fuel injection module	Battery voltage	Check 230 Red, 233 Red/Blu, Wht, and 240 Red wires and connections. If OK, replace K12 EFI module power relay.


A1 Ignition Interlock Module
M1 Starting Motor
200 Red
F3 Fusible Link
F6 Fusible Link
500 Yel
T2 Ignition Coil
T1 Ignition Coil
Blk
G1 Battery
500 Yel
940 Wht
X2
Blk
130 Blk
100 Blk
560 Red/Yel
S6 Brake Switch
S3 Seat Switch
501 Yel
650 Pnk /Blk
S5 PTO/RIP Switch
502 Yel
800 Pnk
820 Pnk
652 Blu
765 Pur
789 Pur
705 Pnk/Blk
S8 Mid PTO Switch
520 Yel
A3 PTO Interlock Module
650 Pnk /Blk
K11 EFI Module Ground Relay
138 Blk
605 Tan
139 Blk
140 Blk
742 Blu/Wht
789 Pur
820 Pnk
670 Pur
X22
Org
100 Blk
1
2
3
5
6
7
8
9
10
11
12



K12 EFI Module
Power Relay
560 Red/Yel
14
230 Red
221 Red
221 Red
13
150
Blk
100
Blk
K13 Fuel
Pump Relay
232 Red
230
Red
Blk
560 Red/Yel
231 Red
15
233
Red
/Blu
X23
910
Wht
Y4 #1 Cylinder
Fuel Injector Solenoid
Y5 #2 Cylinder
Fuel Injector Solenoid
X22
Wht
Blk
Wht
Wht
Brn
X24
17
17
X25
Wht
170
Blk
655
Pnk
X25
X21
651
Pnk/Blk
18
4
240 Red
16
170 Blk
M2 Fuel
Pump
A4
Electric
Fuel
Injection
Module
860 Pnk
460 Yel
Org
X2
860 Pnk

Test Conditions:

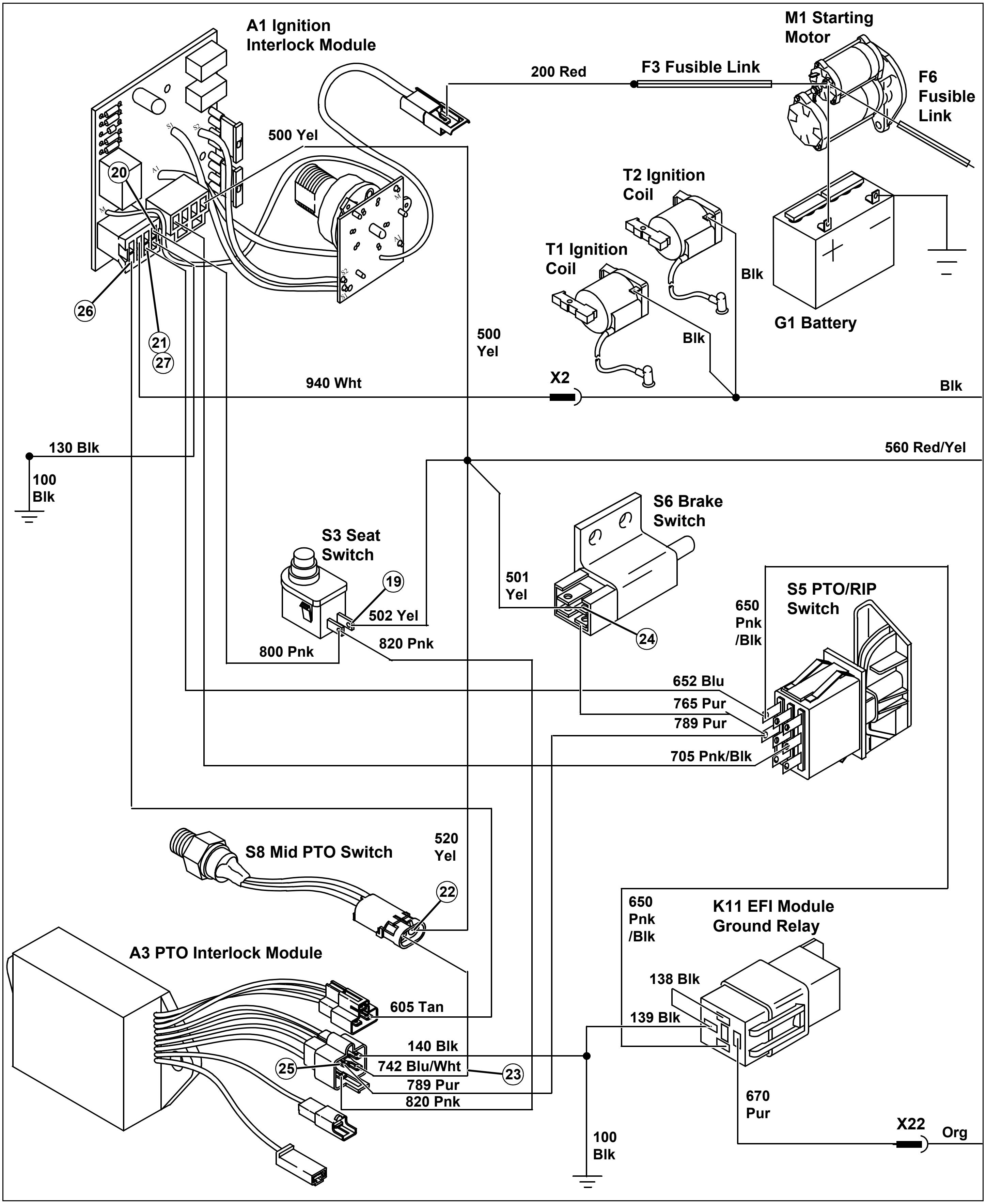
- Key switch in run position, engine not running.
- Park brake unlocked.
- PTO off.
- Operator on seat.

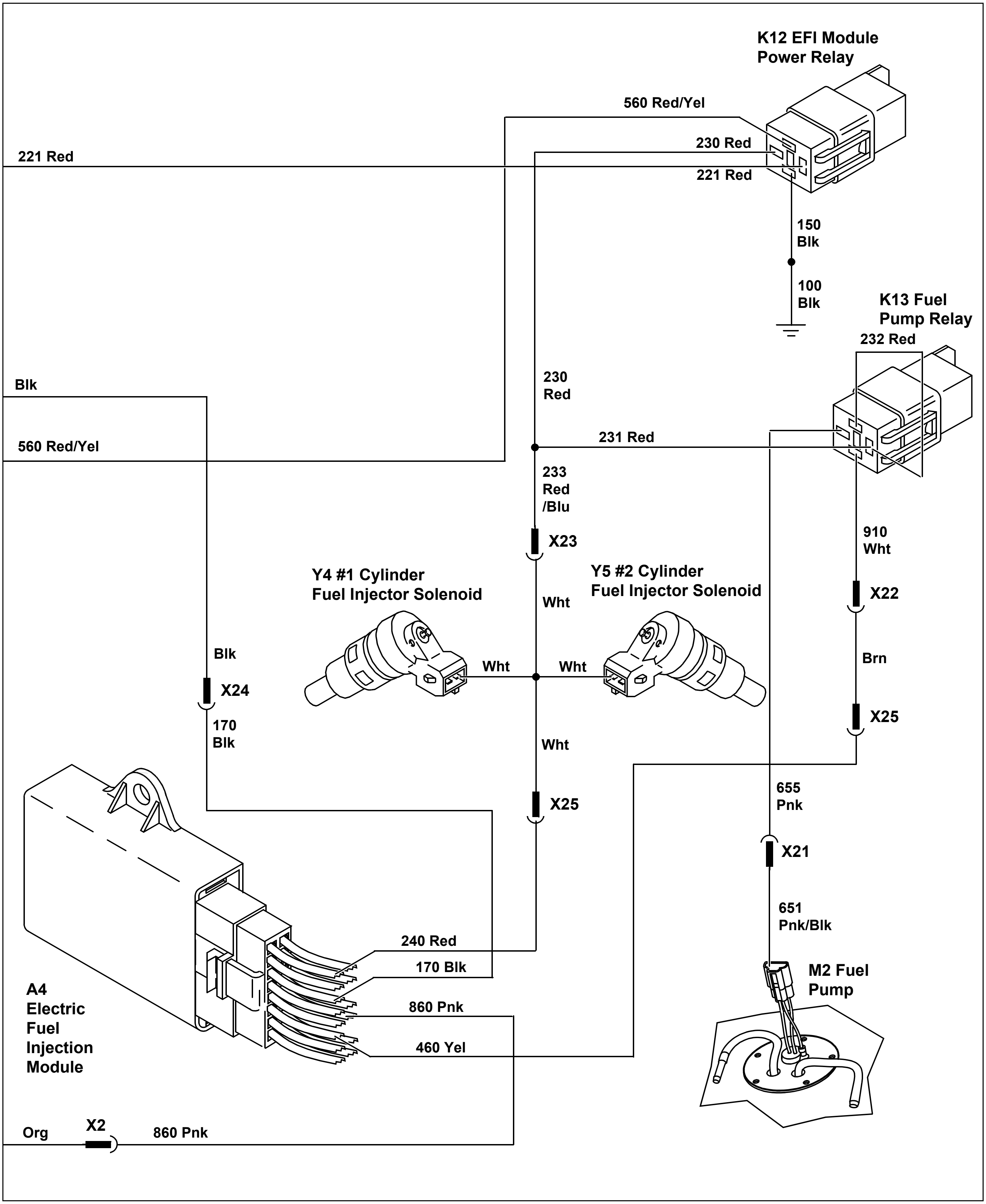
Test/Check Point	Normal	If Not Normal
19. Seat switch	Battery voltage	Check 500 and 502 Yel wires and connections.
20. Ignition interlock module	Battery voltage	Check 800 Pnk wire and connections. If OK, replace seat switch.
21. Ignition interlock module	Battery voltage	Replace ignition interlock module.

Test Conditions:

- Key switch in run position, engine not running.
- Park brake locked.
- PTO on.
- Operator off seat.

Test/Check Point	Normal	If Not Normal
22. Mid PTO switch	Battery voltage	Check 500 and 520 Yel wires and connections.
23. PTO interlock module (option)	Battery voltage	Check 742 Blu/Wht wire and connections. If OK, replace mid PTO switch.
24. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
25. PTO interlock module (option)	Battery voltage	Check 765 and 789 Pur wires and connections. If OK, replace brake switch.
26. Ignition interlock module	Battery voltage	Check 605 Tan wire and connections. If OK, replace PTO interlock module.
27. Ignition interlock module	Battery voltage	Replace ignition interlock module.


A1 Ignition Interlock Module
M1 Starting Motor
200 Red
F3 Fusible Link
F6 Fusible Link
500 Yel
T2 Ignition Coil
T1 Ignition Coil
Blk
G1 Battery
20
26
21
27
500 Yel
Blk
940 Wht
X2
Blk
130 Blk
560 Red/Yel
100 Blk
S6 Brake Switch
S3 Seat Switch
19
501 Yel
502 Yel
S5 PTO/RIP Switch
650 Pnk /Blk
24
800 Pnk
820 Pnk
652 Blu
765 Pur
789 Pur
705 Pnk/Blk
S8 Mid PTO Switch
520 Yel
22
650 Pnk /Blk
K11 EFI Module Ground Relay
A3 PTO Interlock Module
138 Blk
605 Tan
139 Blk
140 Blk
25
742 Blu/Wht
23
789 Pur
820 Pnk
670 Pur
X22
Org
100 Blk


K12 EFI Module Power Relay
560 Red/Yel
230 Red
221 Red
221 Red
150 Blk
100 Blk
K13 Fuel Pump Relay
232 Red
Blk
230 Red
231 Red
560 Red/Yel
233 Red /Blu
X23
910 Wht
Y4 #1 Cylinder Fuel Injector Solenoid
Y5 #2 Cylinder Fuel Injector Solenoid
Wht
X22
Brn
Blk
Wht
Wht
X24
X25
170 Blk
Wht
X25
655 Pnk
X21
651 Pnk/Blk
240 Red
170 Blk
M2 Fuel Pump
A4 Electric Fuel Injection Module
860 Pnk
460 Yel
Org
X2
860 Pnk

PTO/RIP Circuit Operation - All Models

Function:

To provide power to energize the PTO clutch solenoid.

Operating Conditions:

- Key switch in run position,
- PTO/RIP switch on,
- Operator on seat,

or,

- Park brake locked.

Theory of Operation:

The key switch is connected to the A1 interlock module by soldered wire connections. The components are solid state and are not serviced separately except for the 15 amp fuses. Lines within border of the interlock module except for key switch connections are circuit paths on the circuit board and are not wires in the harness.

With the PTO/RIP switch off, current flows from the A1 ignition interlock module to the 652 Blu wire, S5 PTO/RIP switch (PTO disengaged), 750 and 751 Blu wire, S6 brake switch (unlocked position), 770 Pur and 772 Blu wires, S4 RIO switch (neutral/forward position), 735 Org wire into the interlock module to energize the K6 PTO relay coil.

The PTO relay contacts close, supplying power to the 750 and 751 Blu wires. The 751 Blu wire connects back to the brake switch to the 770 Pur wire. This creates a loop to energize the PTO relay. Power is now available at the PTO/RIP switch to energize the RIO circuit and the PTO clutch solenoid when the PTO/RIP switch is moved to the on position.

With the S5 PTO/RIP switch in the on position, power is supplied from the 750 Blu wire across the S5 PTO/RIP switch contacts to the 754 and 755 Blu wires. The 755 Blu wire supplies power to the Y3 PTO clutch solenoid and is grounded through the 115 and 100 Blk wires.

At the same time the 755 Blu wire supplies power to the Y3 PTO clutch solenoid the 754 Blu wire supplies power to the momentary contacts of the S5 PTO/RIP switch. This circuit is part of the RIO circuit to allow for repositioning of the machine in reverse with the PTO engaged.

If the machine is repositioned in reverse without pulling the PTO/RIP switch into the Reverse Implement Position (RIP) the PTO will be disengaged. The PTO/RIP switch will then need to be placed back to the off position to energize the PTO relay and then pulled back to the on position if desired.

With the S5 PTO/RIP switch on, current flow from the S6 brake switch to the S5 PTO/RIP switch and into the A1 ignition interlock module to energize the K5 ignition relay is stopped. The S3 seat switch circuit must keep the K5 ignition relay energized. If the operator gets off the seat with the S5 PTO/RIP switch on, the S3 seat switch de-energizes the K5 ignition relay and K2 magneto relay, which grounds the ignition system and stops current flow to the fuel supply circuit and the K6 PTO relay. The result is that the engine will stop and the PTO clutch solenoid will be de-energized.

Circuit Function - RIO:

To provide a safety interlock with the machine being repositioned in reverse while the PTO is engaged, by disengaging the PTO circuit. An operator who deems it necessary, and chooses to keep the PTO engaged while backing up, must take a deliberate action to actuate the PTO/RIP switch, by pulling up on the PTO/RIP switch momentarily while pressing the reverse pedal.

Operating Conditions - RIO:

- Key switch in the run position,
- Operator on the seat,
- PTO/RIP switch in the RIO (momentary) position while placing the machine in reverse.

Theory of Operation - RIO:

The RIO circuit is powered by either the S4 RIO switch or the S5 PTO/RIP switch. With the machine in neutral or forward, the S4 RIO switch supplies power to the K4 RIO unlatch relay and the K6 PTO relay which in turn supplies power to the S5 PTO/RIP switch to be used when the Y3 PTO clutch solenoid is engaged.

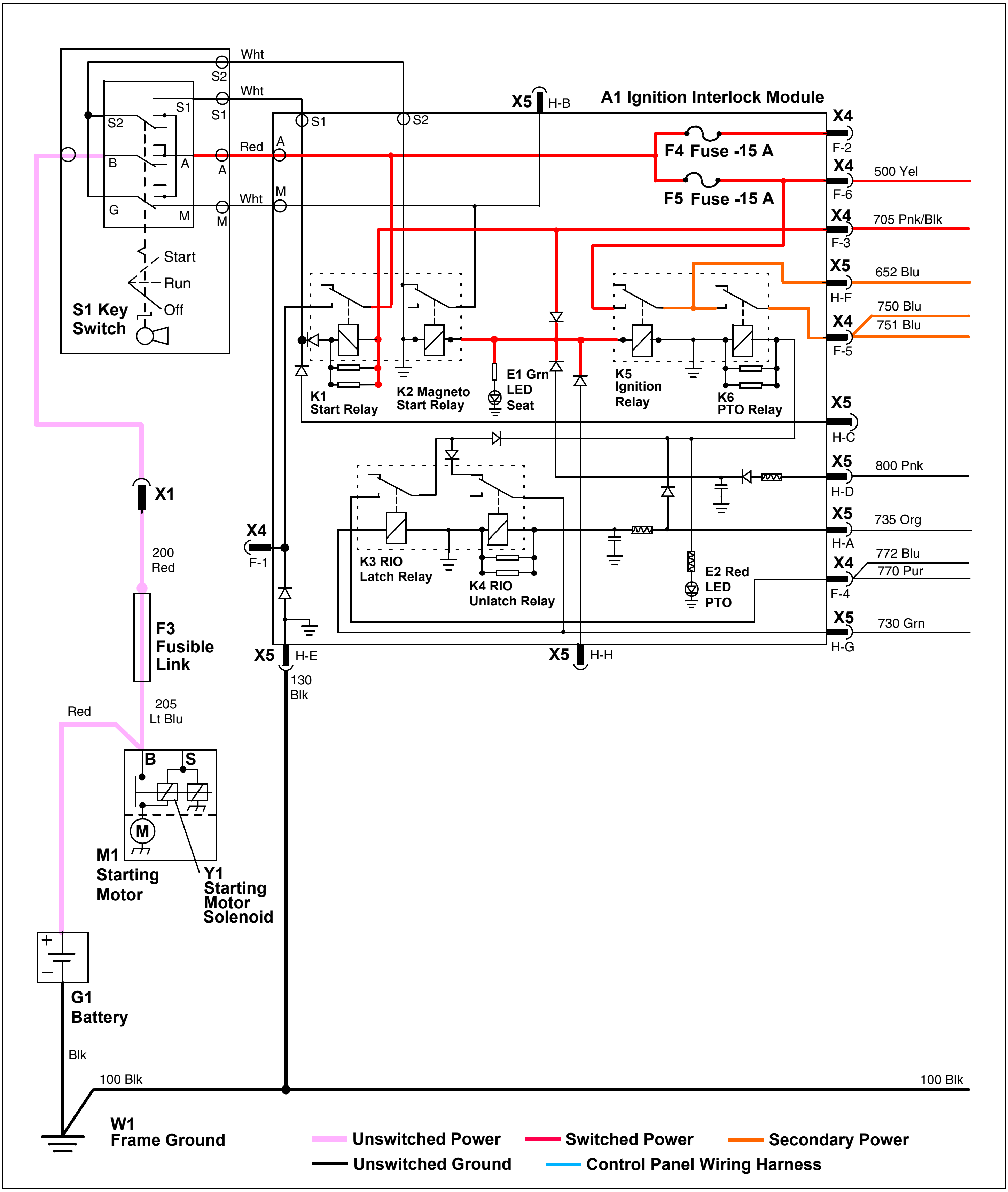
If the machine is placed in reverse the S4 RIO switch is released and its contacts are opened, breaking the circuit to the K6 PTO relay and the K4 RIO unlatch relay. This in turn breaks the circuit to the 750 Blu wire back to the S5 PTO/RIP switch and the 751 Blu wire back to the S6 brake switch where it connects to the 770 Pur wire to continue the loop back to the S4 RIO switch. Because the 750 Pur wire is now de-energized, there is no power for the PTO/RIP switch to energize the PTO clutch solenoid.

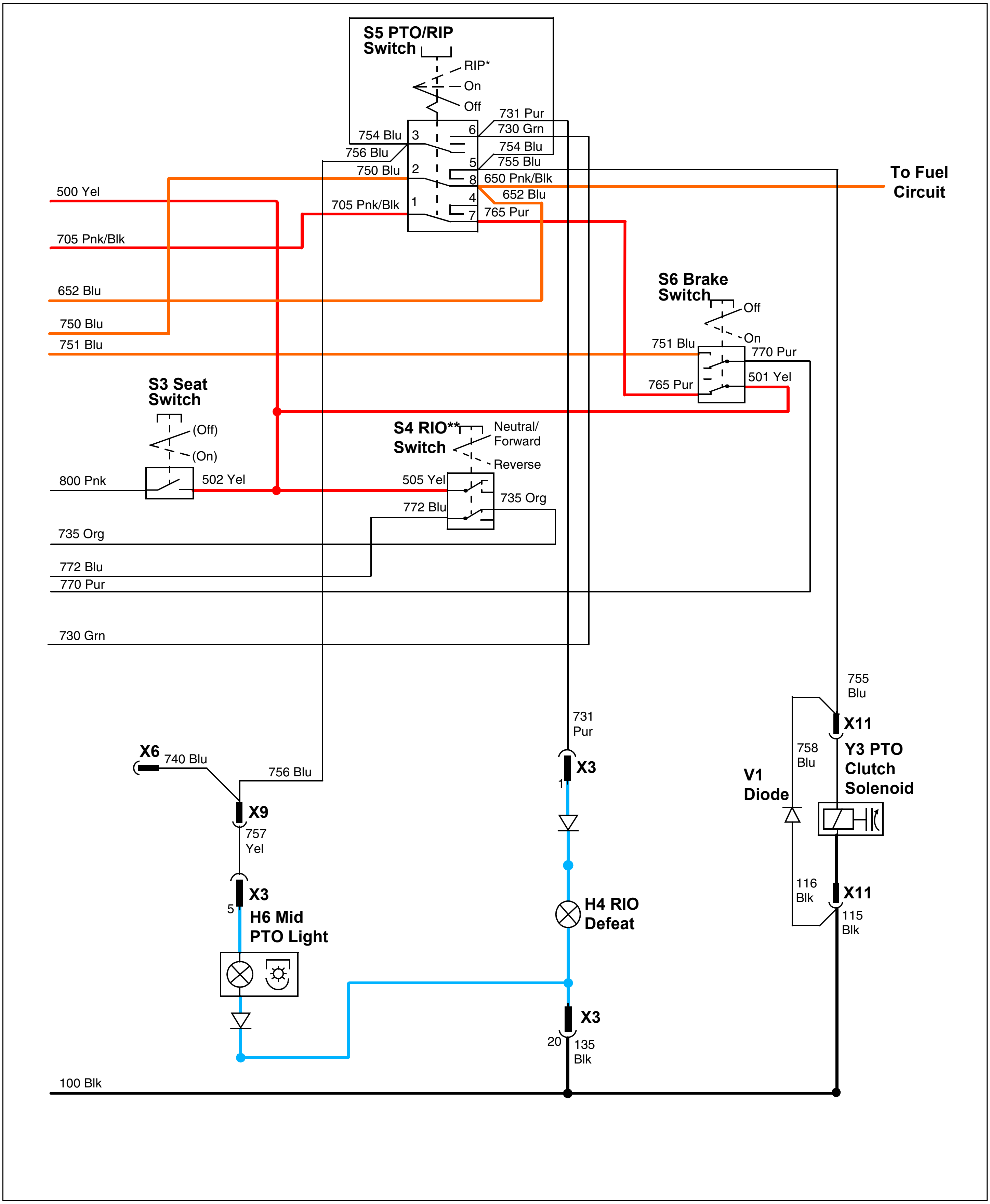
To reenergize these circuits, the machine must be placed in neutral or forward, and the PTO/RIP switch must be placed in the OFF position. The operator can then energize the PTO again if desired.

If the operator chooses to reposition the machine in reverse while the PTO is engaged, the operator must pull up on the PTO/RIP switch while placing the machine in reverse. Once the machine is traveling in reverse the operator can release the PTO/RIP switch. The K3 RIO latch relay feeds itself across the K4 RIO unlatch relay contacts as well as supplying power to the K6 PTO relay and the PTO loop. Power to create this condition comes from the S5 PTO/RIP switch momentary contacts. Wire 754 Blu is connected to the 730 Grn and 731 Pur wires. The 730 Grn wire connects to the interlock module to energize the K3 RIO latch relay coil. The K3 RIO latch relay takes the place of the S4 RIO switch during reverse movement with the RIP circuit activated. The 731 Pur wire connects to the instrument panel to illuminate the H4 RIO defeat light to warn the operator that the RIO circuit is activated.

When the machine is placed back into neutral or forward, the K4 RIO unlatch relay is energized and breaks the loop to the K3 RIO latch relay. This prepares the machine for the next reverse operation with the PTO engaged.

PTO Circuit Electrical Schematic - All Models




S5 PTO/RIP Switch
RIP*
On
Off
731 Pur
730 Grn
754 Blu
755 Blu
650 Pnk/Blk
652 Blu
765 Pur
754 Blu
756 Blu
750 Blu
705 Pnk/Blk
To Fuel Circuit
500 Yel
705 Pnk/Blk
652 Blu
750 Blu
751 Blu
S6 Brake Switch
Off
On
751 Blu
770 Pur
501 Yel
765 Pur
S3 Seat Switch
(Off)
(On)
800 Pnk
502 Yel
S4 RIO** Switch
Neutral/ Forward
Reverse
505 Yel
735 Org
772 Blu
735 Org
772 Blu
770 Pur
730 Grn
755 Blu
X11
Y3 PTO Clutch Solenoid
758 Blu
V1 Diode
731 Pur
X3
X6
740 Blu
756 Blu
X9
757 Yel
X3
H6 Mid PTO Light
H4 RIO Defeat
116 Blk
X11
115 Blk
X3
135 Blk
100 Blk

PTO Circuit Diagnosis - All Models

Test Conditions:

- Key switch in run position, engine not running.
- Operator on seat.
- PTO off.
- Park brake unlocked.
- Transmission in neutral.

Test/Check Point	Normal	If Not Normal
1. Ignition interlock module	Continuity to ground	Check 130 and 100 Blk wires and connections.
2. Ignition interlock module	Battery voltage	Test F3 fusible link and F5 fuse. Check 200 Red wire and connections. If OK, replace ignition interlock module.
3. Seat switch	Battery voltage	Check 500 and 502 Yel wires and connections.
4. Ignition interlock module	Battery voltage	Check 800 Pnk wire and connections. If OK, replace seat switch.
5. Ignition interlock module	Battery voltage	Replace ignition interlock module.
6. PTO/RIP switch	Battery voltage	Check 652 Blu wire and connections.
7. Brake switch	Battery voltage	Check 750 and 751 Blu wires and connections. If OK, replace PTO/RIP switch.
8. RIO switch	Battery voltage	Check 770 Pur and 772 Blu wires and connections. If OK, replace brake switch.
9. Ignition interlock module	Battery voltage	Check 735 Org wire and connections. If OK, replace RIO switch.

Test Conditions:

- Key switch in run position, engine not running.
- Operator on seat.
- PTO on.
- Park brake unlocked.
- Transmission in neutral.

Test/Check Point	Normal	If Not Normal
10. Ignition interlock module	Battery voltage. Red LED illuminated	Check that test conditions have been set properly. If OK, replace ignition interlock module.
11. PTO/RIP switch	Battery voltage	Check 750 Blu wire and connections.
12. PTO clutch solenoid	Battery voltage	Check 755 Blu wire and connections. If OK, replace PTO/RIP switch.
13. X3 Instrument panel connector	Battery voltage. Rear PTO light illuminated	Check 754, 756 Blu, and 757 Yel wires and connections. If OK, replace PTO/RIP switch. Test rear PTO light bulb.

Test Conditions:

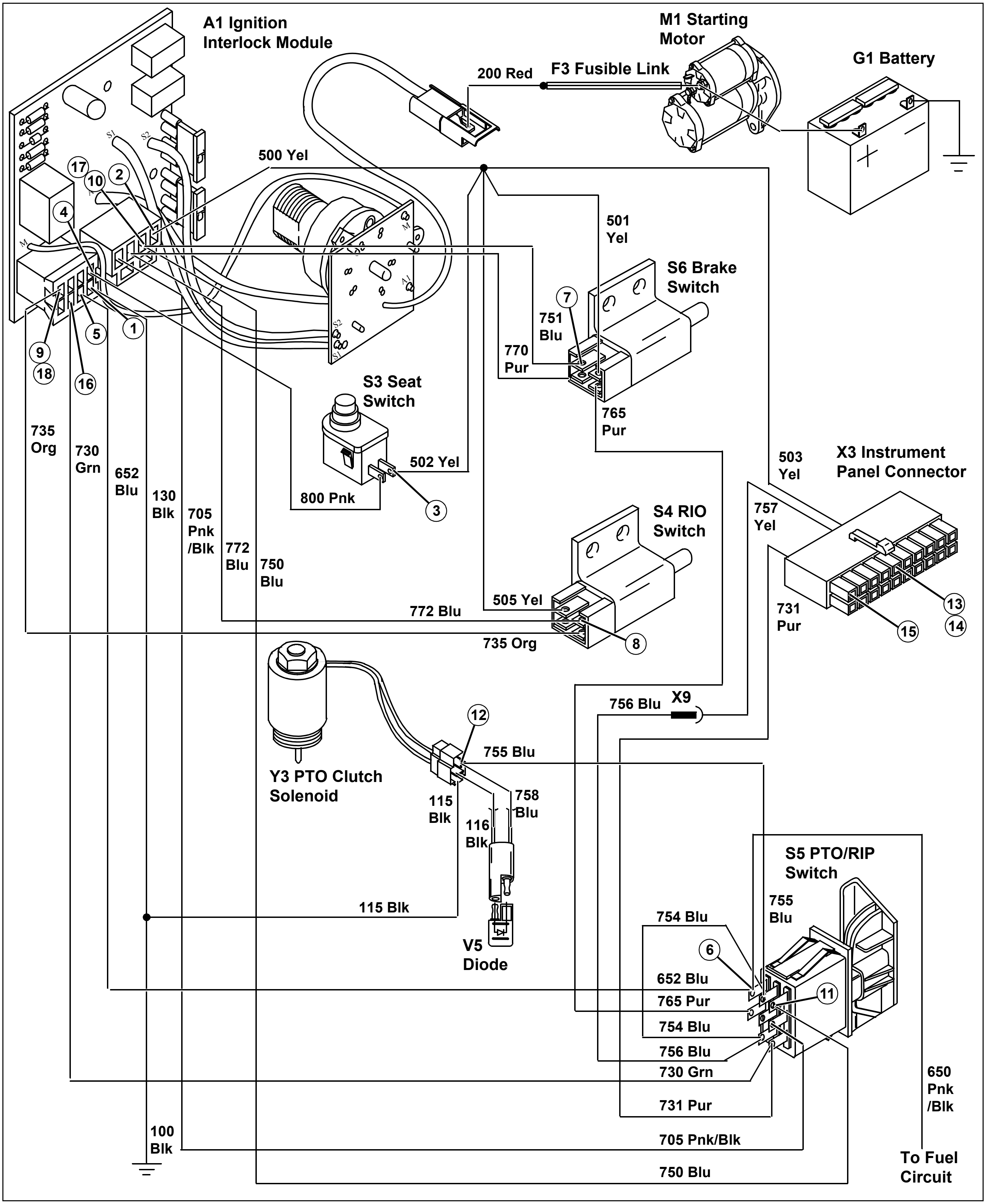
- Key switch in run position, engine not running.
- Operator off seat.
- PTO in RIP position.
- Park brake unlocked.
- Transmission in neutral.

Test/Check Point	Normal	If Not Normal
14. X3 Instrument panel connector	Battery voltage	Check 754, 756 Blu, and 757 Yel wires and connections. If OK, replace PTO/RIP switch.
15. X3 Instrument panel connector	Battery voltage. RIO defeat light illuminated	Check 731 Pur wire and connections. If OK, replace PTO/RIP switch. Test RIO defeat light bulb.
16. Ignition interlock module	Battery voltage	Check 730 Grn wire and connections. If OK, replace PTO/RIP switch.

Test Conditions:

- Key switch in run position, engine not running.
- Operator off seat.
- PTO in RIP position.
- Park brake unlocked.
- Transmission in reverse.

Test/Check Point	Normal	If Not Normal
17. Ignition interlock module	Battery voltage. Red LED illuminated	Check that test conditions have been set properly. If OK, replace ignition interlock module.
18. Ignition interlock module	Less than 0.2 volts	Test RIO switch. See “RIO Switch Test” on page 314. If OK, replace ignition interlock module.


A1 Ignition Interlock Module
M1 Starting Motor
G1 Battery
200 Red
F3 Fusible Link
500 Yel
501 Yel
S6 Brake Switch
751 Blu
770 Pur
765 Pur
S3 Seat Switch
502 Yel
800 Pnk
735 Org
730 Grn
652 Blu
130 Blk
705 Pnk /Blk
772 Blu
750 Blu
503 Yel
X3 Instrument Panel Connector
757 Yel
S4 RIO Switch
505 Yel
772 Blu
735 Org
731 Pur
756 Blu
X9
Y3 PTO Clutch Solenoid
755 Blu
115 Blk
758 Blu
116 Blk
115 Blk
V5 Diode
S5 PTO/RIP Switch
755 Blu
754 Blu
652 Blu
765 Pur
754 Blu
756 Blu
730 Grn
731 Pur
705 Pnk/Blk
750 Blu
650 Pnk /Blk
To Fuel Circuit
100 Blk
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18

Instrument Panel Circuit Operation - All Models

Function - Instrument Panel:

To inform the operator of various machine operating conditions.

Operating Condition - Instrument Panel:

The key switch must be in run or start position.

Theory of Operation - Instrument Panel:

The instrument panel receives current from the power circuit. Power is supplied to the instrument panel from the battery positive (+) terminal to the F3 fusible link, S1 key switch, A1 ignition interlock module F5 fuse, and 500 and 503 Yel wires. The circuit traces on the instrument panel then supply current to the individual lights and gauges.

The ground path for the instrument panel is completed by the 135 and 100 Blk wires.

Function - Engine Oil Pressure Light:

To alert operator of low engine oil pressure.

Theory of Operation - Engine Oil Pressure Light:

The engine oil pressure light circuit uses a pressure activated switch to provide a path to ground for the engine oil pressure light. The switch is closed when engine oil pressure is at or below 98 kPa (14.2 psi).

If the engine is not running or engine oil pressure is at or below 98 kPa (14.2 psi), the B1 engine oil pressure switch will be closed. The engine oil pressure switch completes the path to ground and the engine oil pressure light comes on.

When the engine starts and engine oil pressure increases above 98 kPa (14.2 psi), the engine oil pressure switch opens, breaking the path to ground and the light goes out.

Function - Engine Coolant Temperature Gauge (not on X465):

Inform the operator of engine and coolant operating temperature.

Theory of Operation - Engine Coolant Temperature Gauge (not on X465):

The engine coolant temperature sensor is a variable resistor, providing a ground circuit path for the temperature gauge. As the engine coolant heats, the resistance decreases. As the resistance decreases, more current is allowed to flow across the engine coolant temperature gauge causing the needle raise and indicate a higher temperature. The temperature gauge is part of the instrument panel.

The engine coolant temperature sensor resistance is approximately 40 - 700 ohms.

Function - Fuel Gauge:

Inform the operator of the fuel level in the fuel tank.

Theory of Operation - Fuel Gauge:

The fuel level in the fuel tank is measured by the B2 fuel gauge sensor. The sensor is a variable resistor. The resistance is set by movement of a mechanical linkage connected to a float in the fuel tank. The 5 to 95 ohm variable resistance creates a variable voltage difference across the P2 fuel gauge. The voltage difference ranges from approximately 5.72 VDC (fuel tank FULL) to approximately 0.87 VDC (fuel tank EMPTY). The fuel gauge is part of the instrument panel.

Function - Hourmeter and Service Timer Light:

Hourmeter and Service Timer Light Circuit (SN -30000)

To record the number of hours the key switch is in the run position. To inform the operator that the machine is within the service time limit.

Theory of Operation - Hourmeter and Service Timer Light:

The P1 hourmeter not only counts the total number of hours the key switch is in the run position, it is also tied into the A2 service light timer module. The service light timer module is also tied into the engine oil pressure light circuit.

While the hourmeter counts time when the key is in the run position, the service light timer module is activated when the engine has enough oil pressure to open the engine oil pressure switch.

The H3 service light will illuminate automatically 2 hours before scheduled maintenance. The light will stay illuminated until the S2 service timer reset switch has been pressed. The service light circuit is automatic and nonadjustable. When the scheduled maintenance has been preformed, the service technician will press and release the S2 service timer reset switch to start the next service interval cycle. The key must be in the run or start position for the A2 service light timer module to be reset.

The P1 hourmeter will continue to count the total hours of operation and will not reset to zero with the reset switch. The P1 hourmeter, A2 service light timer module and H3 service light receive power from the switched power circuit.

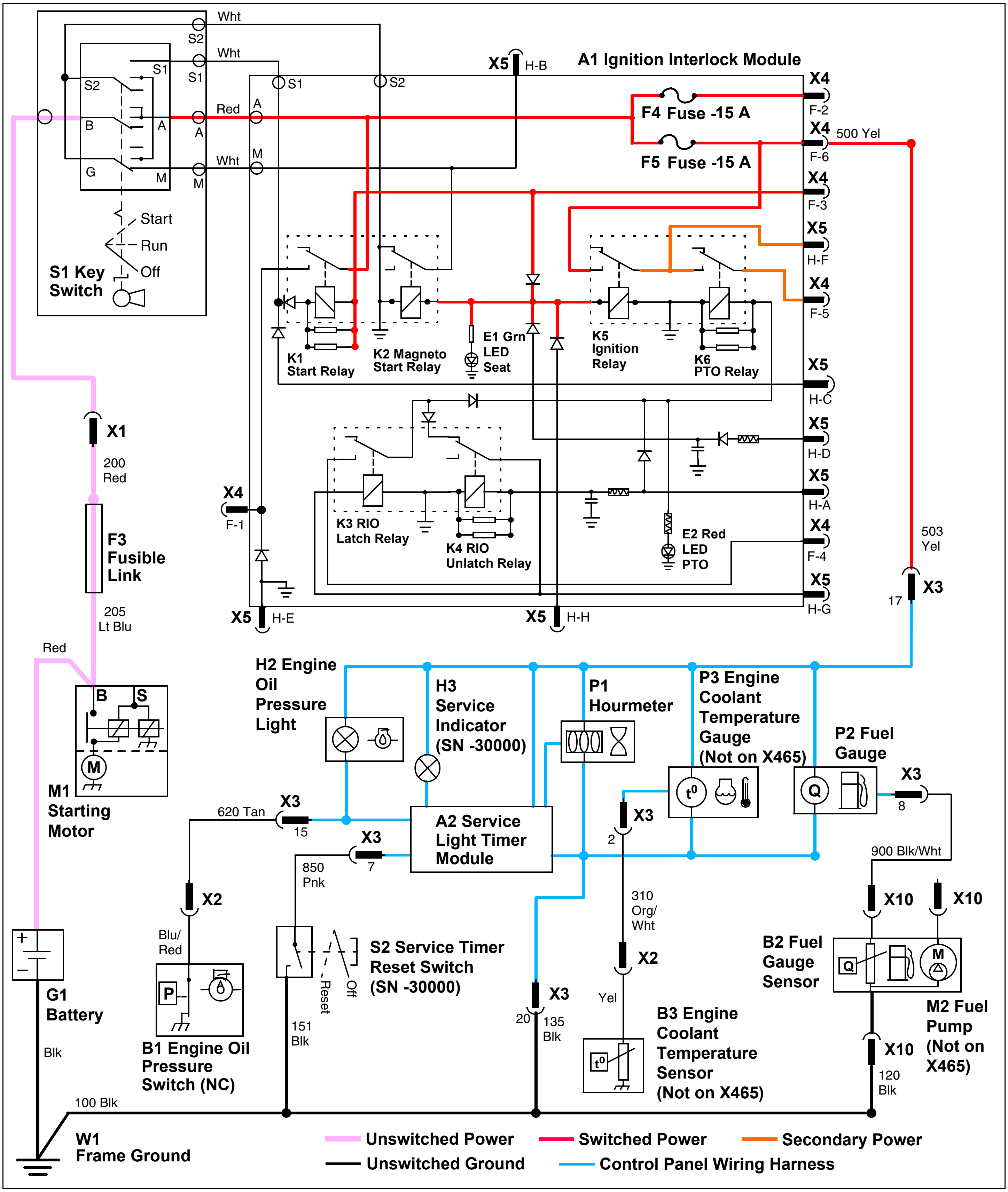
Function - Brake Light (SN 30001-)

Inform the operator that the brake pedal is depressed or the park brake is engaged.

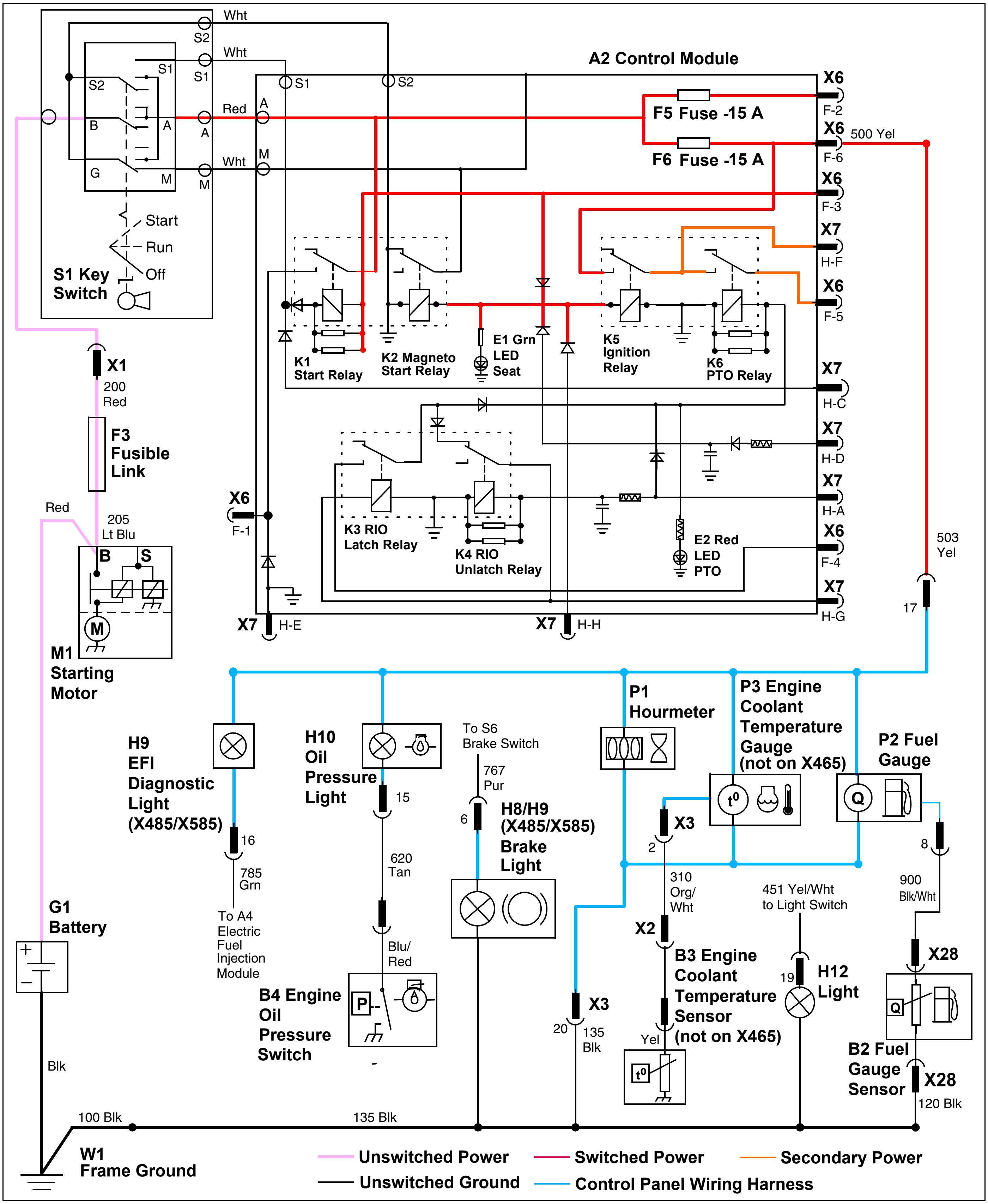
Theory of Operation - Brake Light

The brake light will illuminate whenever the brake pedal is depressed or the park brake is engaged with the key switch in the on position. It is activated by the S6 brake switch and receives its power from the 767 Pur wire to the H13 brake light on the instrument panel. A path to ground is provided by the 135 Blk wire on the instrument panel and to frame ground.

Instrument Panel Circuit Electrical Schematic - All Models (SN -30000)



Instrument Panel Circuit Schematic - All Models (SN 30001-)



Instrument Panel Circuit Diagnosis - All Models

Test Conditions:

- Key switch in off position.
- X3 connector to instrument panel disconnected.
- Service timer reset switch depressed (SN -30000).

Test/Check Point	Normal	If Not Normal
1. X3 Instrument panel connector	Continuity to ground	Check 135 and 100 Blk wires and connections.
2. Service timer reset switch (SN -30000)	Continuity to ground	Check 135 and 100 Blk wires and connections.
3. X3 Instrument panel connector (SN -30000)	Continuity to ground	Check 850 Pnk wire and connections. If OK, replace service timer reset switch.

Test Conditions:

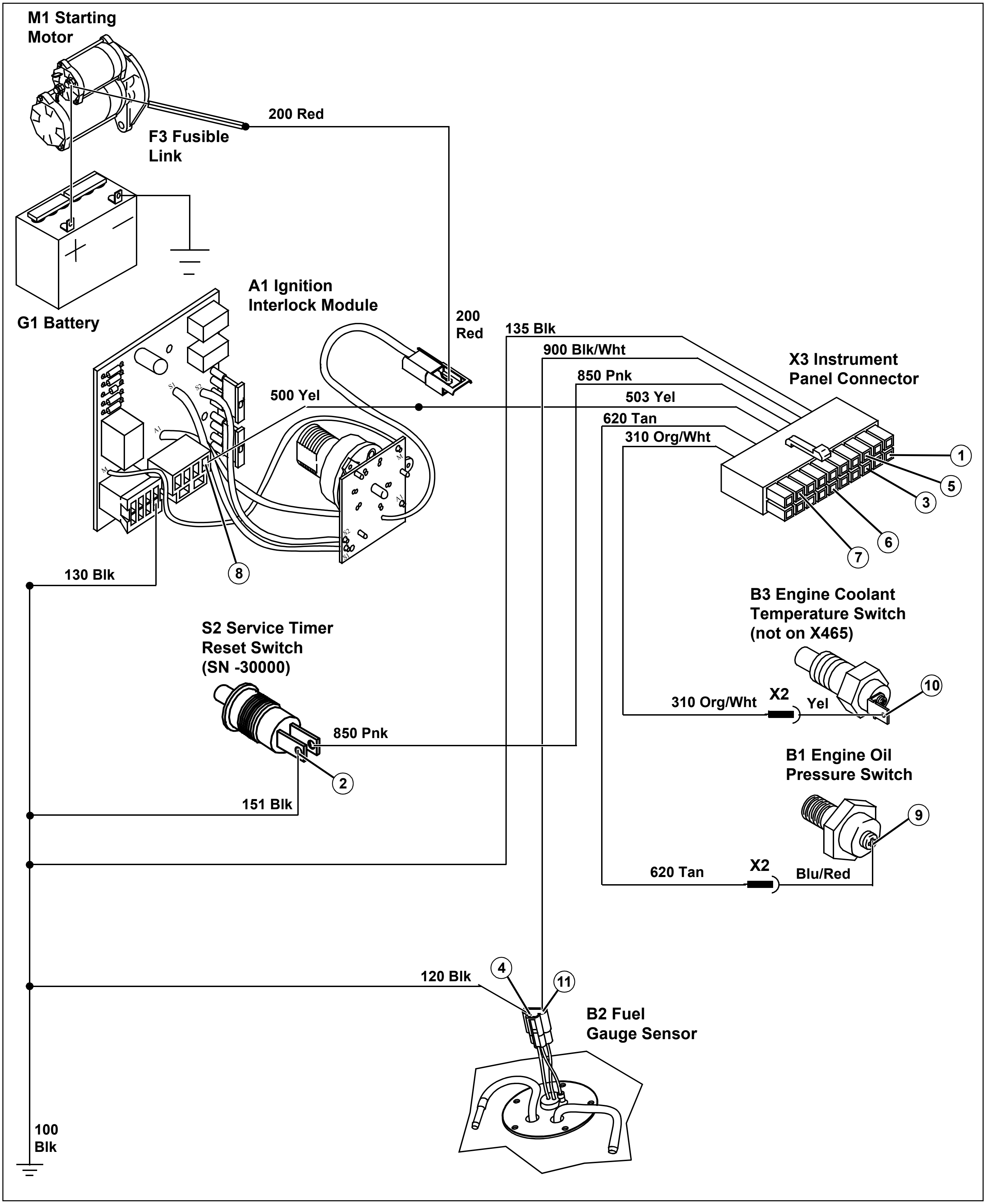
- Key switch in off position.
- X3 connector to instrument panel disconnected.

Test/Check Point	Normal	If Not Normal
4. Fuel gauge sensor	Continuity to ground	Check 120 and 100 Blk wires and connections.
5. X3 Instrument panel connector	Resistance between 6 ohms - empty to 200 ohms - full	Check 900 Blk/Wht wire and connections. Test fuel gauge sensor.
6. X3 Instrument panel connector	Continuity to ground	Check 620 Tan wire and connections. If OK, replace engine oil pressure switch.
7. X3 Instrument panel connector (not on X465)	Resistance between 700 ohms - cold 40 ohms - hot	Check 310 Org/Wht wire and connections. If OK, replace engine coolant temperature sensor.

Test Conditions:

- Key switch in run position, engine not running.
- X3 connector to instrument panel connected.

Test/Check Point	Normal	If Not Normal
8. Ignition interlock module	Battery voltage.	Test F3 fusible link and F5 fuse. If OK, replace ignition interlock module.
9. Engine oil pressure switch	Less than 0.2 volts. Engine oil pressure light illuminated.	Test engine oil pressure light bulb. Check 200 Red and 500 Yel wires and connections. If OK, replace instrument panel.
10. Engine coolant temperature sensor (not on X465)	7.0 - 6.0 volts	Test engine coolant temperature sensor. If OK, replace instrument panel.
11. Fuel gauge sensor	0.87 - 5.72 volts	Test fuel gauge sensor. If OK, replace instrument panel.


M1 Starting Motor
200 Red
F3 Fusible Link
G1 Battery
A1 Ignition Interlock Module
200 Red
135 Blk
900 Blk/Wht
850 Pnk
500 Yel
503 Yel
620 Tan
310 Org/Wht
X3 Instrument Panel Connector
1
5
3
6
7
8
130 Blk
B3 Engine Coolant Temperature Switch (not on X465)
S2 Service Timer Reset Switch (SN -30000)
10
310 Org/Wht
X2
Yel
850 Pnk
2
B1 Engine Oil Pressure Switch
151 Blk
9
620 Tan
X2
Blu/Red
120 Blk
4
11
B2 Fuel Gauge Sensor
100 Blk

Lights Circuit Operation - All Models

Function:

Provides current to the headlights, tail lights, backup lights, and instrument panel lights, in combinations depending upon the position the light switch or transmission (RIO switch).

Operating Conditions:

- Light switch in on or on with backup lights position,

or,

- Transmission in reverse.

Theory of Operation:

Power is provide to the S9 light switch through the A1 ignition interlock module and is protected by the F3 fusible link and the F4 fuse. The 400 Yel/Blu wire connects to the S9 light switch.

With the S9 light switch in the on position, current is provided to the 450 Yel/Wht wire which splices into the 451, 455, and 459 Yel/Wht wires.

The 451 Yel/Wht wire supplies current to the X3 connector to illuminate the H7 dash lights. The 135 and 100 Blk wires provide the ground from the instrument panel.

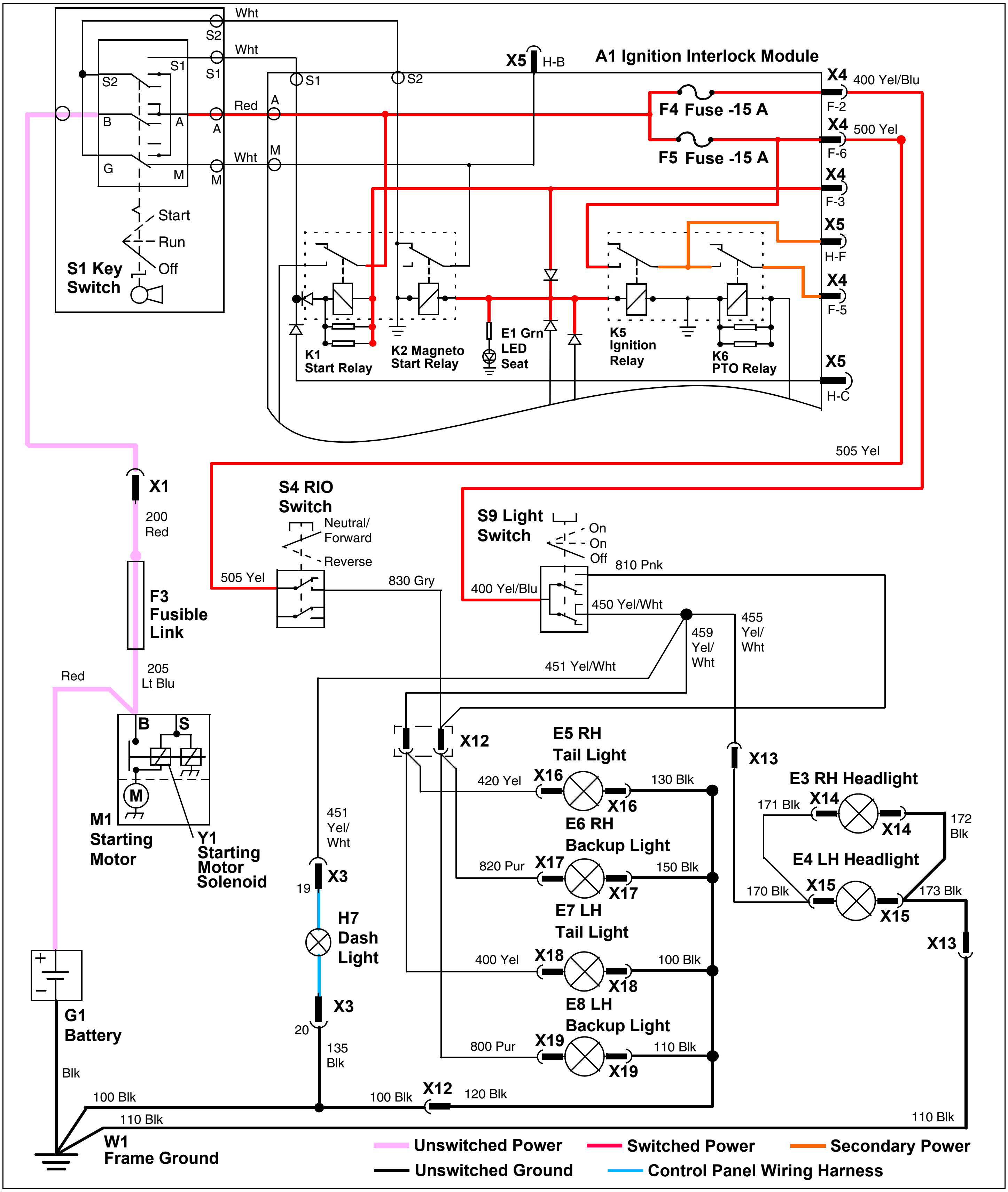
The 455 Yel/Wht wire supplies current to the X13 connector and the 170 and 171 Blk wires to illuminate the E3 and E4 headlights. The ground is provided by the 172 and 173 Blk wires, X13 connector to the 110 Blk wire.

The 459 Yel/Wht wire supplies current to the X12 connector and the 400 and 420 Yel wires to illuminate the E5 and E7 tail lights. The ground is provided by the 100, 130 and 120 Blk wires, X12 connector to the 122 and 100 Blk wires.

With the light switch in the on with backup lights position, in addition to the H7 dash light, E3 and E4 headlights, and the E5 and E7 tail lights, current is provided to the to the E6 and E8 backup lights. The 810 Pnk with receives current and connects through the X12 connector to the 800 and 820 Pur wires and the E6 and E8 backup lights. The ground is provided by the 110, 150, and 120 Blk wires, X12 connector to the 122 and 100 Blk wires.

The backup light can also be powered by the S4 RIO switch. Power is provide to the S4 RIO switch through the A1 ignition interlock module and is protected by the F3 fusible link and the F5 fuse. The 500 and 505 Yel wires connect to the S4 RIO switch. When the transmission is in reverse, the switch is released and the 505 Yel wire is connected to the 830 Gry wire. The 830 Gry wire supplies current to the X12 connector to the 800 and 820 Pur wires and the E6 and E8 backup lights. The ground is provided by the 110, 150, and 120 Blk wires, X12 connector to the 122 and 100 Blk wires.

Lights Circuit Electrical Schematic - All Models



Lights Circuit Diagnosis - All Models

Test Conditions:

- Key switch in run position, engine not running.
- Light switch in on position.
- Transmission in neutral.
- X3 connector to instrument panel disconnected.

Test/Check Point	Normal	If Not Normal
1. Light switch	Battery voltage	Test F3 fusible link and F4 fuse. Check 400 Yel/Blu wires and connections. If OK, replace ignition interlock module.
2. X3 Instrument panel connector	Battery voltage	Check 450 and 451 Yel/Wht wires and connections. If OK, replace light switch.
3. E5 and E7 tail lights	Battery voltage	Check 450, 459 Yel/Wht wires, X12 connector, and 400 and 420 Yel wires and connections. If OK, replace light switch.
4. E3 and E4 headlights	Battery voltage	Check 450, 455 Yel/Wht wires, X13 connector, and 170 and 171 Blk wires and connections. If OK, replace light switch.

Test Conditions:

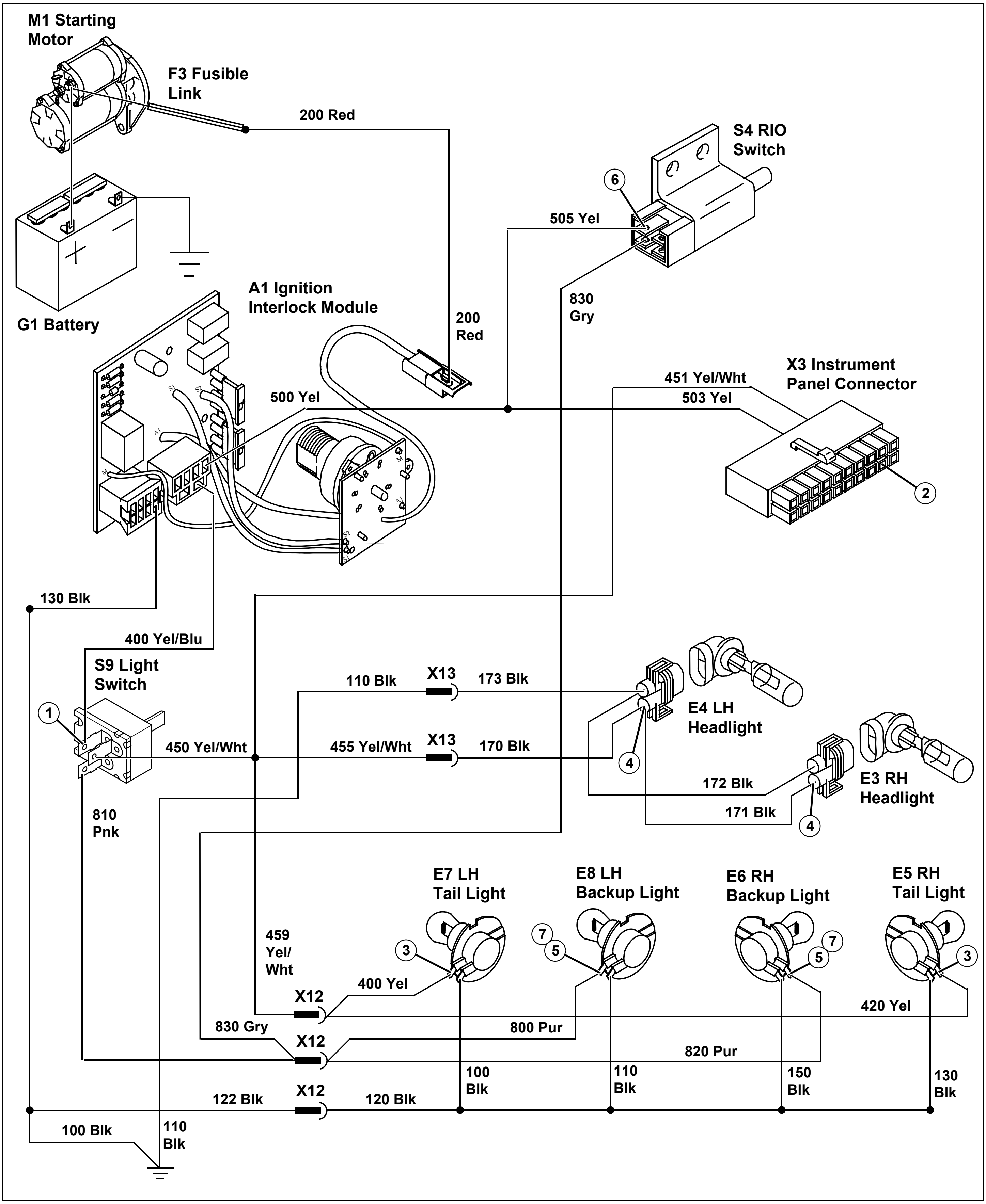
- Key switch in run position, engine not running.
- Light switch in on with backup lights position.
- Transmission in neutral.

Test/Check Point	Normal	If Not Normal
5. E6 and E8 backup lights	Battery voltage	Check 810 Pnk wire, X12 connector, and 800 and 820 Pur wires and connections. If OK, replace light switch.

Test Conditions:

- Key switch in run position, engine not running.
- Light switch off.
- Transmission in reverse.

Test/Check Point	Normal	If Not Normal
6. RIO switch	Battery voltage	Test F3 fusible link and F5 fuse. Check 500 and 501 Yel wires and connections. If OK, replace ignition interlock module.
7. E6 and E8 backup lights	Battery voltage	Check 810 Pnk wire, X12 connector, and 800 and 820 Pur wires and connections. If OK, replace light switch.



M1 Starting Motor
F3 Fusible Link
200 Red
S4 RIO Switch
6
505 Yel
G1 Battery
A1 Ignition Interlock Module
200 Red
830 Gry
X3 Instrument Panel Connector
451 Yel/Wht
503 Yel
500 Yel
2
130 Blk
400 Yel/Blu
S9 Light Switch
1
110 Blk
X13
173 Blk
E4 LH Headlight
450 Yel/Wht
455 Yel/Wht
X13
170 Blk
4
172 Blk
E3 RH Headlight
171 Blk
4
810 Pnk
E7 LH Tail Light
E8 LH Backup Light
E6 RH Backup Light
E5 RH Tail Light
459 Yel/ Wht
3
7
5
7
5
3
400 Yel
X12
420 Yel
830 Gry
800 Pur
X12
820 Pur
100 Blk
110 Blk
150 Blk
130 Blk
122 Blk
X12
120 Blk
100 Blk
110 Blk

Fuel Pump and Fuel Shutoff Circuit Operation - X465, X475 and X575

Function:

To provide power to the fuel shutoff solenoid and the fuel pump to supply fuel to the engine.

Operating Conditions:

- Key switch in run or start position,
- Operator on seat,

or,

- Key switch in run or start position,
- Operator off seat,
- Park braked locked.

Theory of Operation:

NOTE: The X465 model circuit does not have the M2 electric fuel pump (655 Pnk wire and M2 fuel pump). See Engine section.

The fuel shutoff solenoid and fuel pump circuit is powered by the K5 ignition relay on the A1 ignition interlock module.

Current from the switched power circuit is supplied to the S3 seat switch from the 500 and 502 Yel wires. When the operator is on the seat, current flows across the seat switch to the 800 Pnk wire and into the A1 ignition interlock module to energize the K5 ignition relay. With the relay energized, current flows across the ignition relays normally open contacts to the 652 Blu wire. The 652 Blu wire connects to the 650 Pnk/Blk and 655 Pnk wires. The 650 Pnk/Blk wire supplies current to the Y2 fuel shutoff solenoid while the 655 Pnk wire supplies current to the M2 fuel pump.

With both the fuel pump and fuel shutoff solenoid energized, fuel is now able to flow to the carburetor and into the combustion chamber.

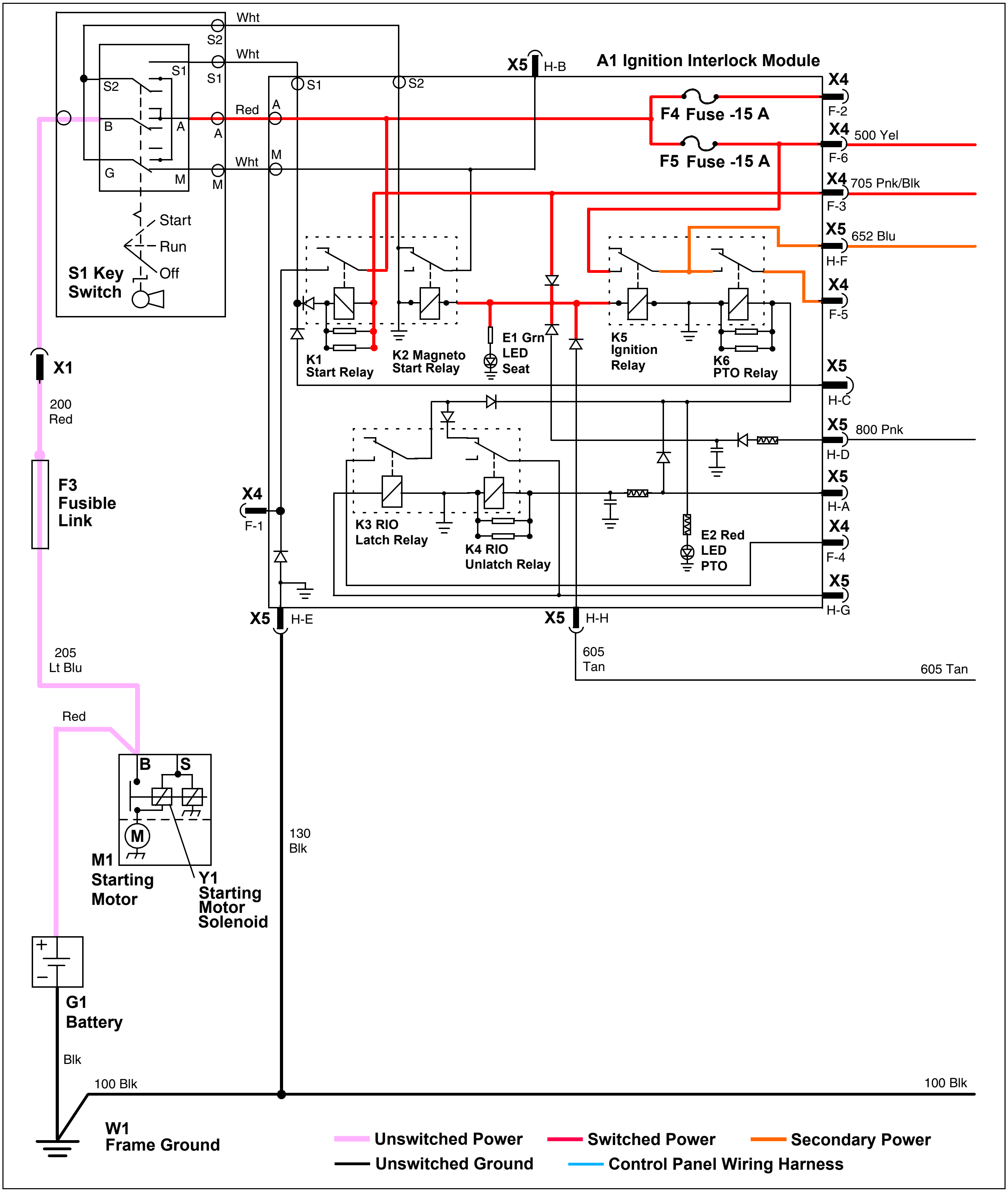
Current can also be supplied to the K5 ignition relay through the brake switch.

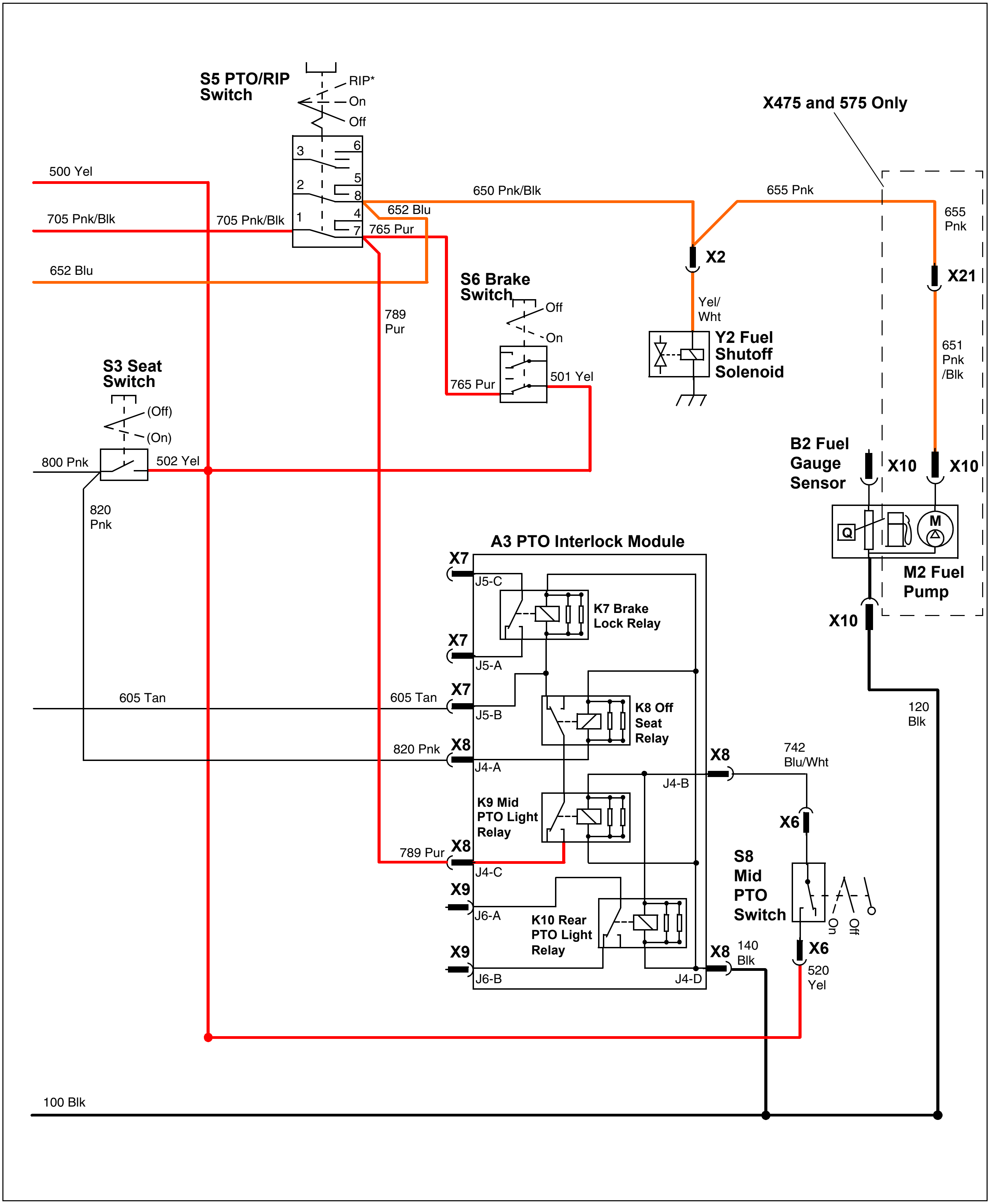
Current from the switched power circuit is supplied to the S6 brake switch from the 500 and 501 Yel wires. When the brake is on, current flows across the brake switch to the 765 and 789 Pur wires. The 765 Pur wire supplies current to the S5 PTO/RIP switch. With the PTO/RIP switch in the off position current flows across the PTO/RIP switch to the 705 Pnk/Blk wire and into the A1 ignition interlock module to energize the K5 ignition relay. With the relay energized, current flows across the ignition relays normally open contacts to the 652 Blu wire. The 652 Blu wire connects to the 650 Pnk/Blk and 655 Pnk wires. The 650 Pnk/Blk wire supplies current to the Y2 fuel shutoff solenoid while the 655 Pnk wire supplies current to the M2 fuel pump.

If the PTO/RIP switch is in the on position and the operator is off the seat then the 789 Pur wire which receives power from the brake switch will supply current to the A3 PTO interlock module.

With the operator off the seat, the K8 off seat relay on the A3 PTO interlock module is deenergized and its normally closed and common terminals and connected. The S8 mid PTO switch has energized the K9 mid PTO relay and current flows from the 789 Pur wire across the K9 mid PTO relay and the K8 off seat relay and to the 605 Tan wire. The 605 Tan wire supplies current into the A1 ignition interlock module to energize the K5 ignition relay. With the relay energized, current flows across the ignition relays normally open contacts to the 652 Blu wire. The 652 Blu wire connects to the 650 Pnk/Blk and 655 Pnk wires. The 650 Pnk/Blk wire supplies current to the Y2 fuel shutoff solenoid while the 655 Pnk wire supplies current to the M2 fuel pump.

Fuel Pump and Fuel Shutoff Circuit Electrical Schematic - X465, X475 and X575




S5 PTO/RIP Switch
RIP*
On
Off
X475 and 575 Only
500 Yel
705 Pnk/Blk
705 Pnk/Blk
652 Blu
650 Pnk/Blk
655 Pnk
652 Blu
765 Pur
655 Pnk
X2
X21
S6 Brake Switch
Off
On
789 Pur
Yel/ Wht
Y2 Fuel Shutoff Solenoid
651 Pnk /Blk
S3 Seat Switch
(Off)
(On)
765 Pur
501 Yel
800 Pnk
502 Yel
B2 Fuel Gauge Sensor
X10
X10
M
820 Pnk
A3 PTO Interlock Module
M2 Fuel Pump
X7
J5-C
K7 Brake Lock Relay
X10
X7
J5-A
605 Tan
605 Tan
X7
J5-B
K8 Off Seat Relay
120 Blk
820 Pnk
X8
J4-A
X8
742 Blu/Wht
J4-B
K9 Mid PTO Light Relay
X6
789 Pur
X8
J4-C
S8 Mid PTO Switch
X9
J6-A
K10 Rear PTO Light Relay
On
Off
X9
J6-B
J4-D
X8
140 Blk
X6
520 Yel
100 Blk

Fuel Pump and Fuel Shutoff Circuit Diagnosis - X465, X475 and X575

NOTE: The X465 model circuit does not have the M2 electric fuel pump (655 Pnk wire and M2 fuel pump). See Engine section.

Test Conditions:

- Key switch in run position, engine not running.
- Park brake locked.
- PTO off.
- Operator off seat.

Test/Check Point	Normal	If Not Normal
1. Ignition interlock module	Battery voltage	Test F3 fusible link and F5 fuse. Check 200 Red wire and connections. If OK, replace ignition interlock module.
2. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
3. PTO/RIP switch	Battery voltage	Check 765 Pur wire and connections. If OK, replace brake switch.
4. Ignition interlock module	Battery voltage	Check 705 Pnk/Blk wire and connections. If OK, replace PTO/RIP switch.
5. Ignition interlock module	Battery voltage	Replace ignition interlock module.
6. Fuel pump (not X465)	Battery voltage	Check 652 Blu, 650 Pnk/Blk, 655 Pnk, and 651 Pnk/Blk wires and connections.

Test Conditions:

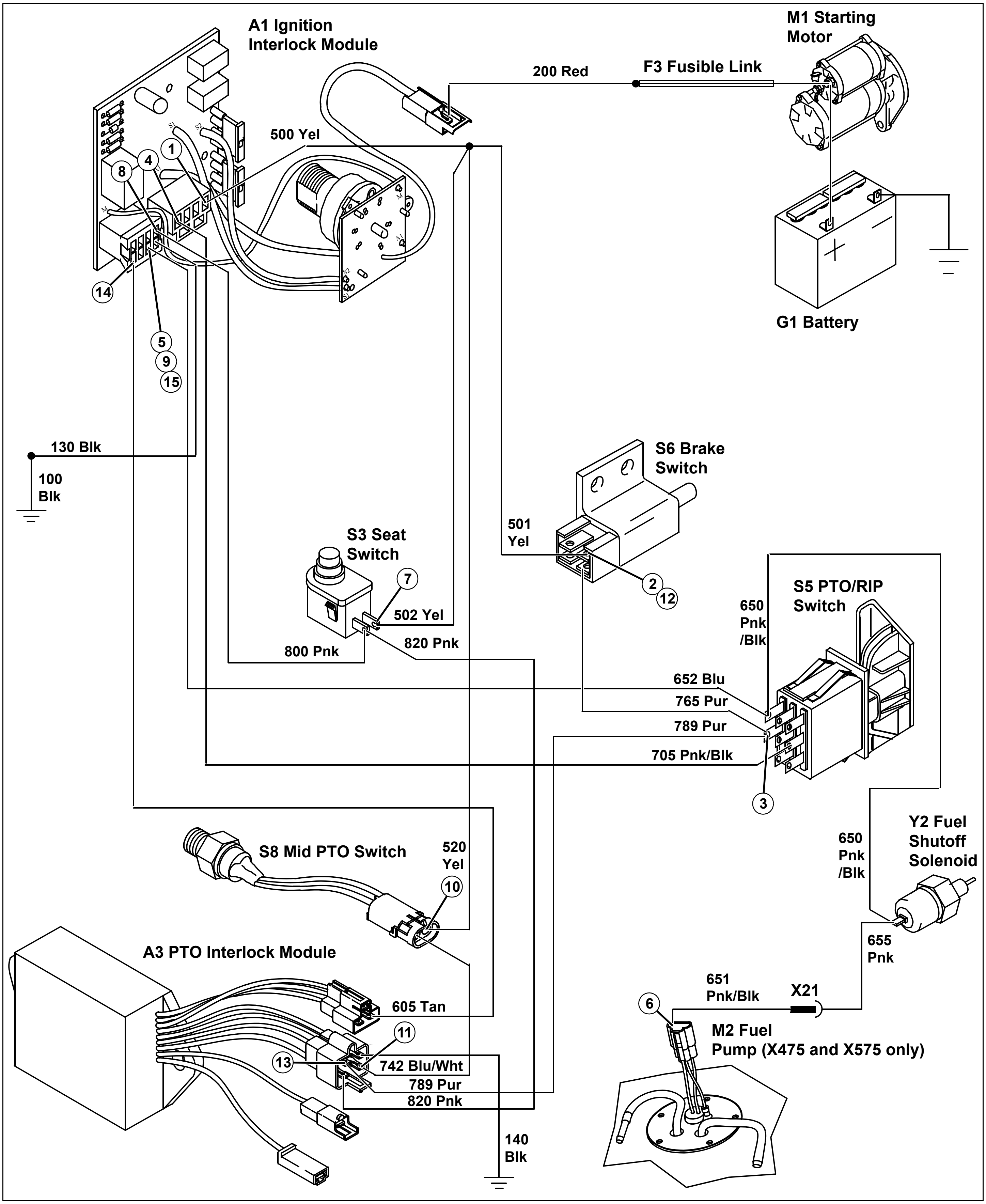
- Key switch in run position, engine not running.
- Park brake unlocked.
- PTO off.
- Operator on seat.

Test/Check Point	Normal	If Not Normal
7. Seat switch	Battery voltage	Check 500 and 502 Yel wires and connections.
8. Ignition interlock module	Battery voltage	Check 800 Pnk wire and connections. If OK, replace seat switch.
9. Ignition interlock module	Battery voltage	Replace ignition interlock module.

Test Conditions:

- Key switch in run position, engine not running.
- Park brake locked.
- PTO on.
- Operator off seat.

Test/Check Point	Normal	If Not Normal
10. Mid PTO switch	Battery voltage	Check 500 and 520 Yel wires and connections.
11. PTO interlock module (option)	Battery voltage	Check 742 Blu/Wht wire and connections. If OK, replace mid PTO switch.
12. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
13. PTO interlock module (option)	Battery voltage	Check 765 and 789 Pur wires and connections. If OK, replace brake switch.
14. Ignition interlock module	Battery voltage	Check 605 Tan wire and connections. If OK, replace PTO interlock module.
15. Ignition interlock module	Battery voltage	Replace ignition interlock module.


A1 Ignition
Interlock Module
M1 Starting
Motor
200 Red
F3 Fusible Link
500 Yel
G1 Battery
130 Blk
100
Blk
S6 Brake
Switch
501
Yel
S3 Seat
Switch
502 Yel
800 Pnk
820 Pnk
S5 PTO/RIP
Switch
650
Pnk
/Blk
652 Blu
765 Pur
789 Pur
705 Pnk/Blk
S8 Mid PTO Switch
520
Yel
Y2 Fuel
Shutoff
Solenoid
A3 PTO Interlock Module
655
Pnk
651
Pnk/Blk
X21
605 Tan
M2 Fuel
Pump (X475 and X575 only)
742 Blu/Wht
789 Pur
820 Pnk
140
Blk
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

Fuel Pump and Fuel Injector Circuit Operation - X485 and X585

Function:

To provide power to the fuel pump and fuel injectors to supply fuel to the engine.

Operating Conditions:

- Key switch in run or start position,
- Operator on seat,

or,

- Key switch in run or start position,
- Operator off seat,
- Park braked locked.

Theory of Operation:

The fuel pump and fuel injector circuit is powered by the A4 electric fuel injection (EFI) module which is controlled through the K5 ignition relay on the A1 ignition interlock module.

Current from the switched power circuit is supplied to the S3 seat switch from the 500 and 502 Yel wires. When the operator is on the seat, current flows across the seat switch to the 800 Pnk wire and into the A1 ignition interlock module to energize the K5 ignition relay. With the relay energized, current flows across the ignition relays normally open contacts to the 652 Blu wire. The 652 Blu wire connects to the 650 Pnk/Blk wire. The 650 Pnk/Blk wire supplies current to the K11 EFI module ground relay. The K11 EFI module ground relay closes and provides a path to ground for the A4 EFI module through the 860 Pnk, Org, and 670 Pur wires, K11 EFI module ground relay, 139, and 100 Blk wires.

At the same time power is being supplied to the A4 EFI module through the K12 EFI power relay. With the key in the run position, switched power is provided to the K12 EFI module power relay from the 500 Yel and 560 Red/Yel wires. This energizes the K12 EFI module power relay to supply unswitched power from the F6 fusible link, 220 and 221 Red wires to the 230, 231 Red and 233 Red/Blu wires to the X23 connector to the Wht wire in the W2 engine wiring harness (X485 and X585). The Wht wire provides power to the Y4 and Y5 fuel injector solenoids and the 240 Red wire to the A4 EFI module. With power to the A4 EFI module, internal circuitry provides a ground circuit for the K13 fuel pump relay. The power input for the for the K13 fuel pump relay is provided from the K12 EFI module relay and 230 and 231 and 232 Red wires. With the K13 fuel pump relay energized, current flows from the 231 Red wire to the 655 Pnk and 651 Pnk/Blk wires to the M2 fuel pump.

With the fuel pump, fuel injectors, and the EFI module energized, fuel is now able to be delivered into the combustion chamber.

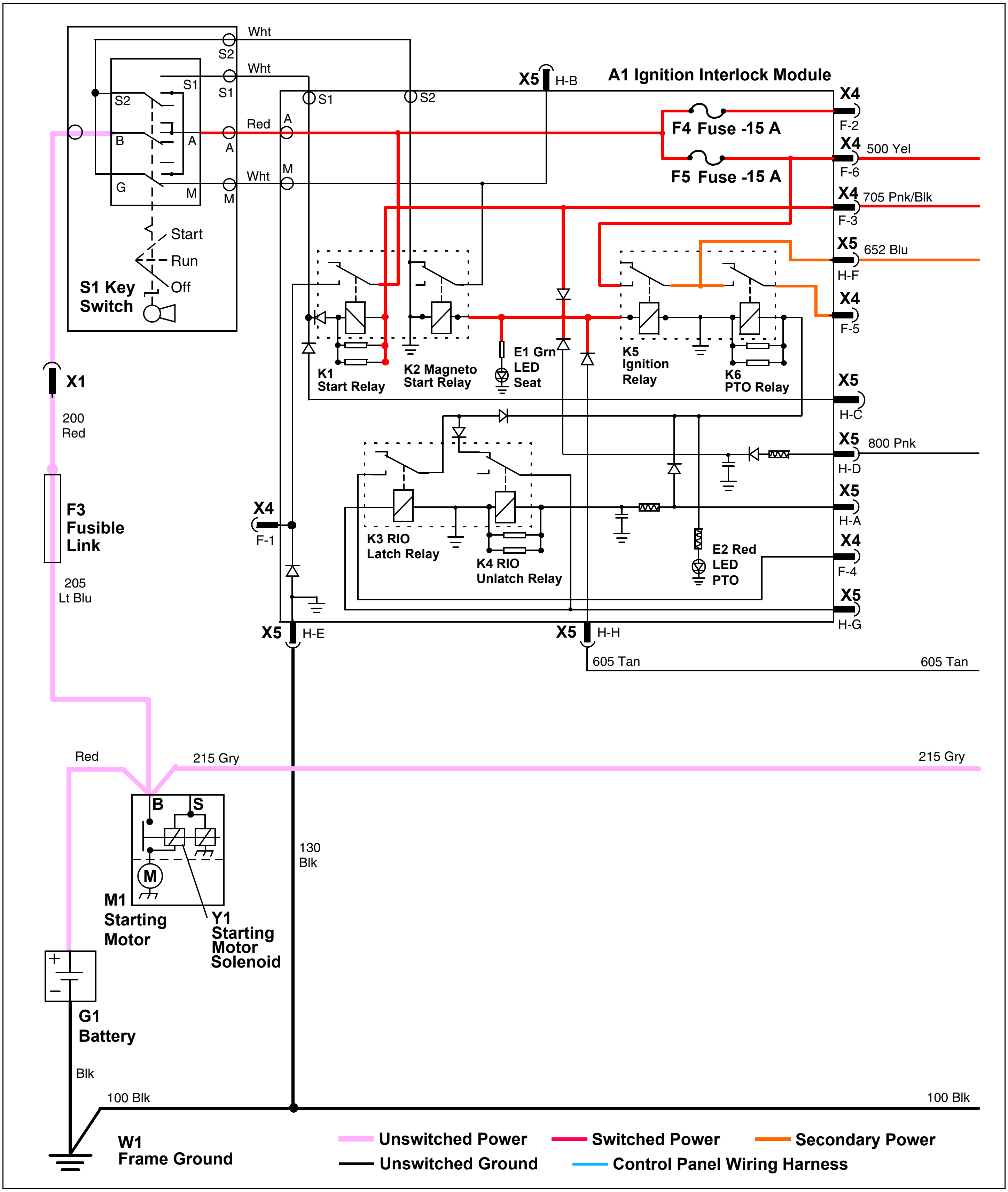
Current can also be supplied to the K5 ignition relay through the S6 brake switch.

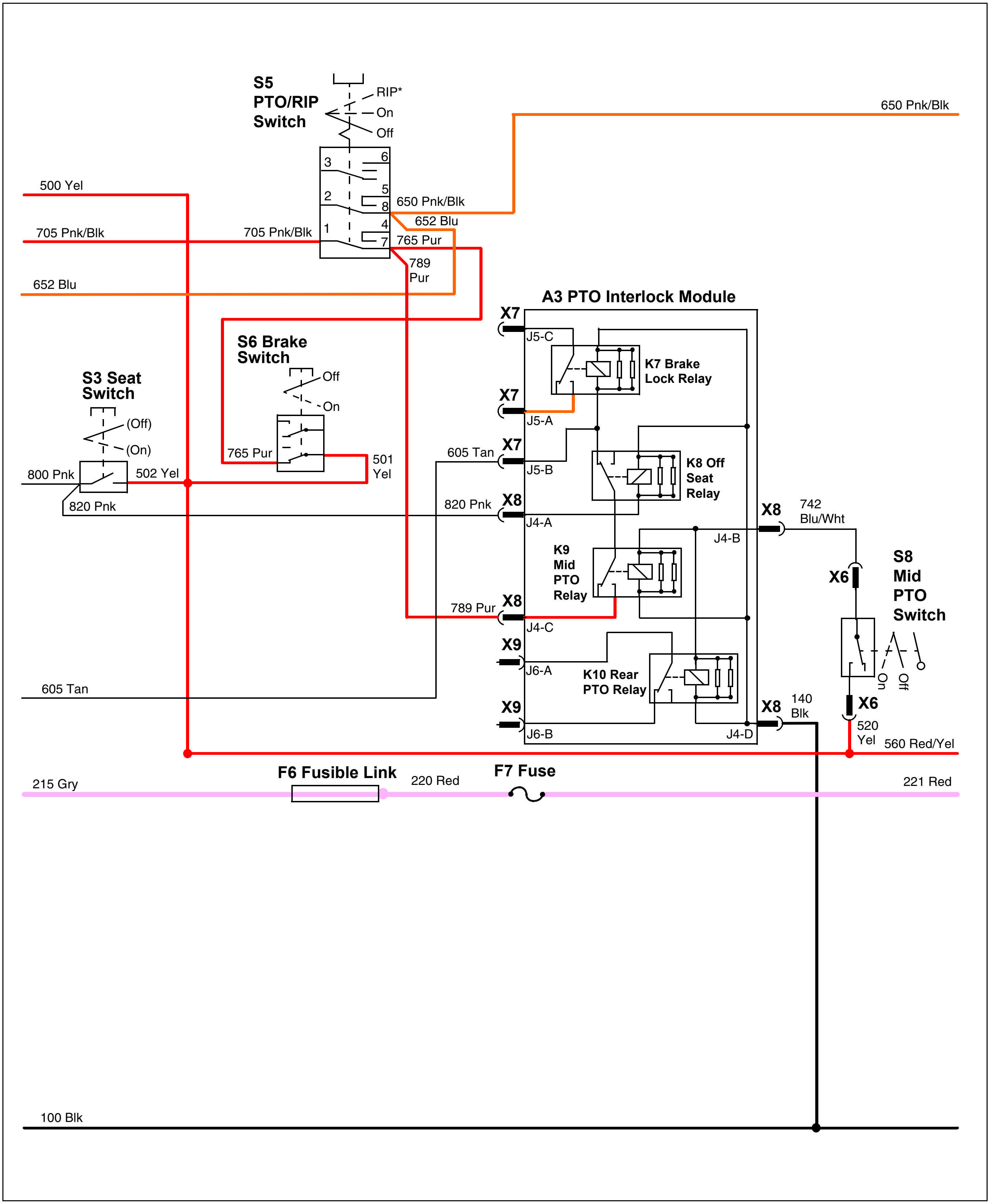
Current from the switched power circuit is supplied to the S6 brake switch from the 500 and 501 Yel wires. When the brake is on, current flows across the brake switch to the 765 and 789 Pur wires. The 765 Pur wire supplies current to the S5 PTO/RIP switch. With the PTO/RIP switch in the off position current flows across the PTO/RIP switch to the 705 Pnk/Blk wire and into the A1 ignition interlock module to energize the K5 ignition relay. With the relay energized, current flows across the ignition relay's contacts to the 652 Blu wire. The 652 Blu wire connects to the 650 Pnk/Blk wire. The 650 Pnk/Blk wire supplies current to the K11 EFI module ground relay.

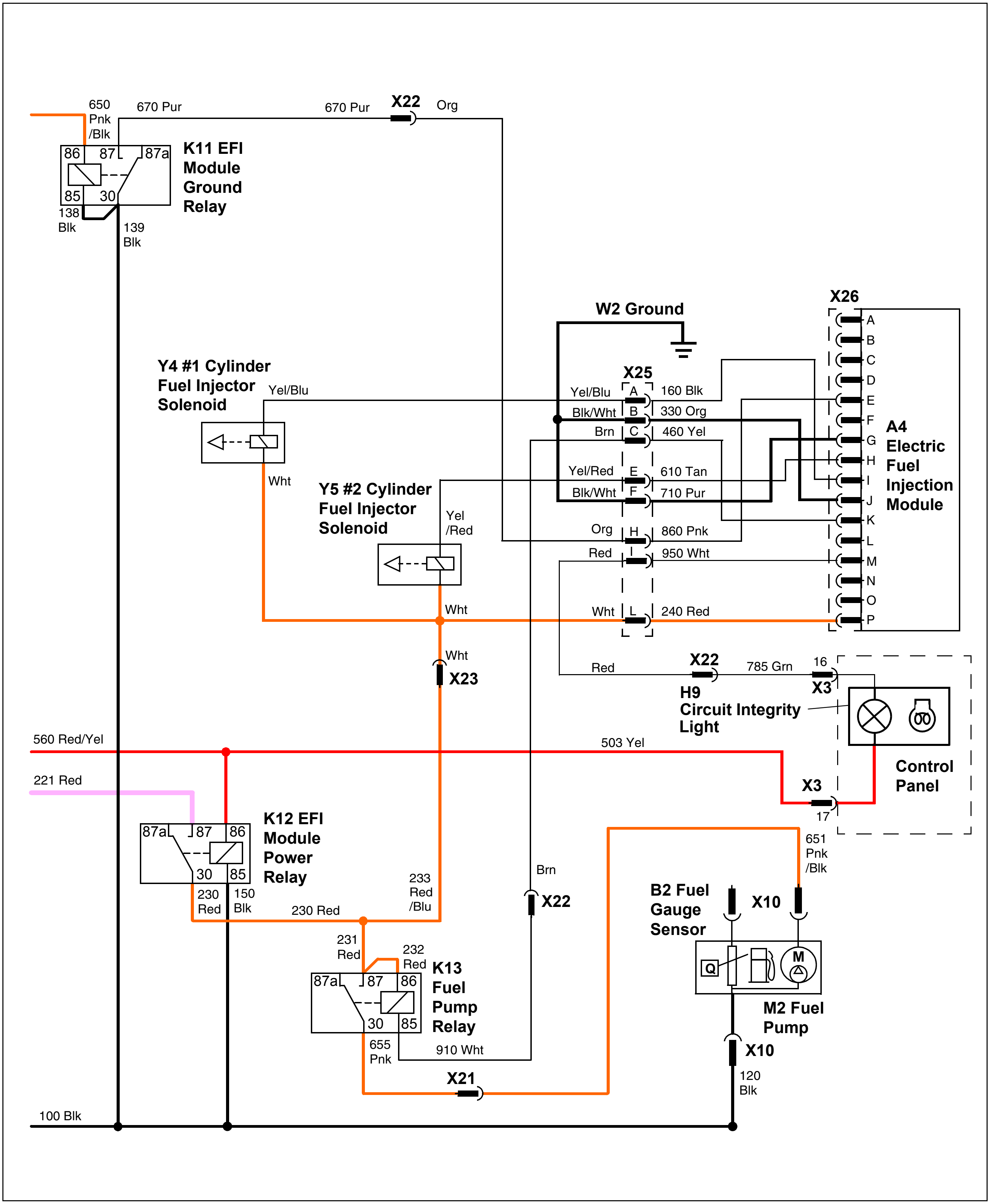
If the PTO/RIP switch is in the on position and the operator is off the seat then the 789 Pur wire, which receives power from the brake switch, will supply current to the A3 PTO interlock module.

With the operator off the seat, the K8 off seat relay on the A3 PTO interlock module is deenergized and its normally closed and common terminals are connected. The S8 mid PTO switch has energized the K9 mid PTO relay and current flows from the 789 Pur wire across the K9 mid PTO relay and the K8 off seat relay and to the 605 Tan wire. The 605 Tan wire supplies current into the A1 ignition interlock module to energize the K5 ignition relay. With the relay energized, current flows across the ignition relays normally open contacts to the 652 Blu wire. The 652 Blu wire connects to the 650 Pnk/Blk wire. The 650 Pnk/Blk wire supplies current to the K11 EFI module ground relay.

Fuel Pump and Fuel Injector Circuit Electrical Schematic - X485 and X585




S5
PTO/RIP
Switch
RIP*
On
Off
650 Pnk/Blk
500 Yel
705 Pnk/Blk
705 Pnk/Blk
652 Blu
650 Pnk/Blk
652 Blu
765 Pur
789
Pur
A3 PTO Interlock Module
X7
J5-C
K7 Brake
Lock Relay
S6 Brake
Switch
Off
On
S3 Seat
Switch
(Off)
(On)
800 Pnk
502 Yel
820 Pnk
765 Pur
501
Yel
605 Tan
J5-A
J5-B
K8 Off
Seat
Relay
820 Pnk
X8
J4-A
J4-B
742
Blu/Wht
K9
Mid
PTO
Relay
X6
S8
Mid
PTO
Switch
789 Pur
J4-C
X9
J6-A
K10 Rear
PTO Relay
On
Off
605 Tan
J6-B
J4-D
140
Blk
520
Yel
560 Red/Yel
215 Gry
F6 Fusible Link
220 Red
F7 Fuse
221 Red
100 Blk


650 Pnk /Blk
670 Pur
670 Pur
X22
Org
86
87
87a
85
30
K11 EFI Module Ground Relay
138 Blk
139 Blk
W2 Ground
X26
Y4 #1 Cylinder Fuel Injector Solenoid
Yel/Blu
X25
Yel/Blu
160 Blk
Blk/Wht
330 Org
Brn
460 Yel
Yel/Red
610 Tan
Blk/Wht
710 Pur
Org
860 Pnk
Red
950 Wht
Wht
240 Red
A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
A4 Electric Fuel Injection Module
Wht
Y5 #2 Cylinder Fuel Injector Solenoid
Yel /Red
Wht
Wht
X23
Red
X22
785 Grn
16
X3
H9 Circuit Integrity Light
560 Red/Yel
503 Yel
221 Red
X3
17
Control Panel
87a
87
86
30
85
K12 EFI Module Power Relay
230 Red
150 Blk
230 Red
233 Red /Blu
Brn
X22
651 Pnk /Blk
B2 Fuel Gauge Sensor
X10
231 Red
232 Red
K13 Fuel Pump Relay
87a
87
86
30
85
655 Pnk
910 Wht
M2 Fuel Pump
X10
120 Blk
X21
100 Blk

Fuel Pump and Fuel Injector Circuit Diagnosis - X485 and X585

Test Conditions:

- Key switch in run position, engine not running.
- Park brake locked.
- PTO off.
- Operator off seat.
- Confirm that the EFI fault light illuminates for approximately two seconds when the key is first turned to the run position.

Test/Check Point	Normal	If Not Normal
1. Ignition interlock module	Battery voltage	Test F3 fusible link and F5 fuse. Check 200 Red wire and connections. If OK, replace ignition interlock module.
2. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
3. PTO/RIP switch	Battery voltage	Check 765 Pur wire and connections. If OK, replace brake switch.
4. Ignition interlock module	Battery voltage	Check 705 Pnk/Blk wire and connections. If OK, replace PTO/RIP switch.
5. Ignition interlock module	Battery voltage	Replace ignition interlock module.
6. K11 EFI module ground relay	Battery voltage	Check 652 Blu and 650 Pnk/Blk wires and connections.
7. K12 EFI module power relay	Battery voltage	Test F6 fusible link and F7 fuse. Check 220 and 221 Red wires and connections.
8. K12 EFI module power relay	Battery voltage	Check 500 and 560 Red/Yel wires and connections.
9. K13 Fuel pump relay	Battery voltage	Check 230 and 231 Red wires and connections. If OK, replace K12 EFI module power relay.
10. Fuel pump	Battery voltage	Check 655 Pnk and 651 Pnk/Blk wires and connections. If OK, replace K13 fuel pump relay.
11. Fuel injector(s)	Battery voltage	Check 230 Red, 233 Red/Blu and Wht wires and connections. If OK, replace K12 EFI module power relay.
12. Electric fuel injection module	Battery voltage	Check 230 Red, 233 Red/Blu, Wht, and 240 Red wires and connections. If OK, replace K12 EFI module power relay.

Test Conditions:

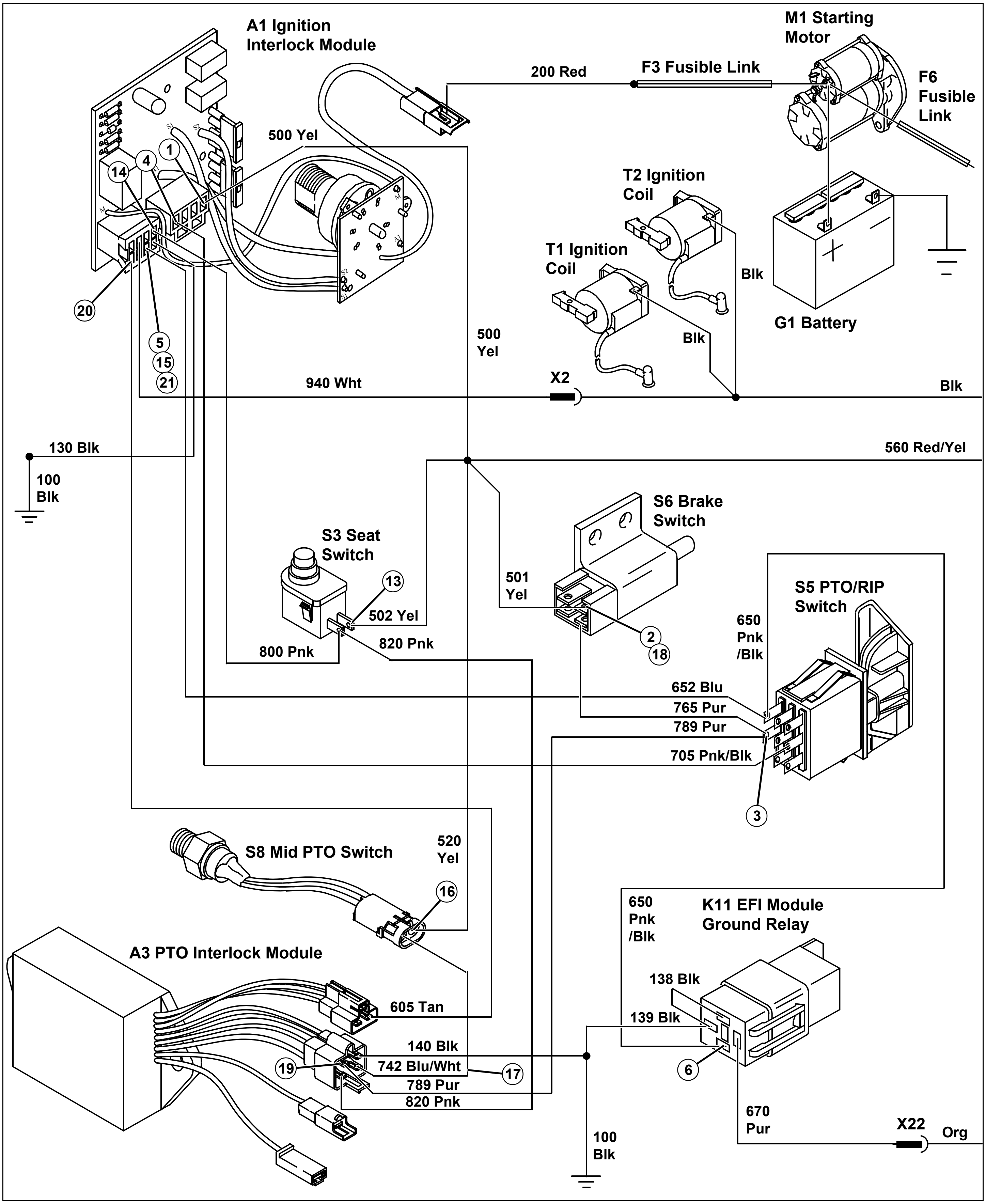
- Key switch in run position, engine not running.
- Park brake unlocked.
- PTO off.
- Operator on seat.
- Confirm that the EFI fault light illuminates for approximately two seconds when the key is first turned to the run position.

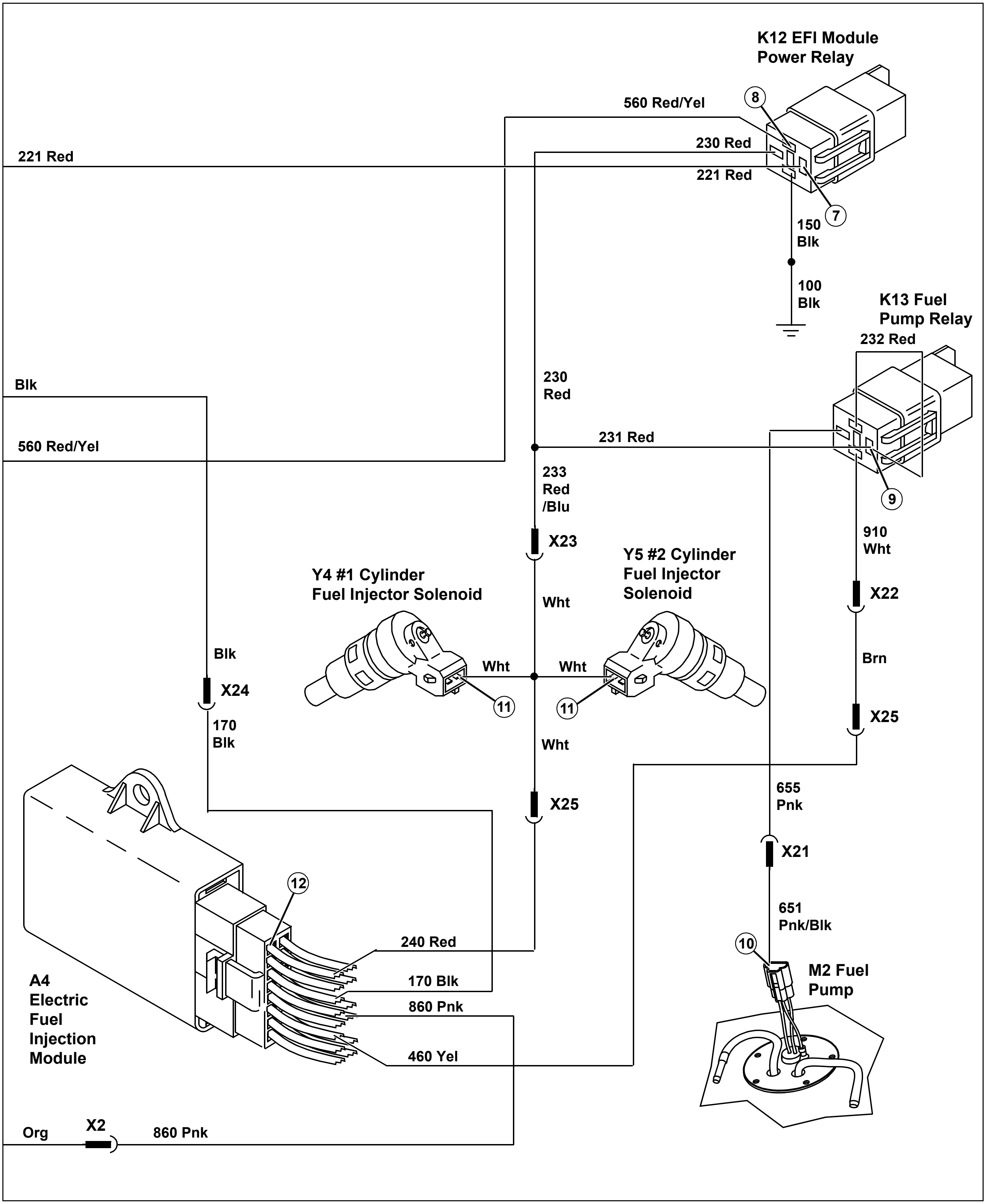
Test/Check Point	Normal	If Not Normal
13. Seat switch	Battery voltage	Check 500 and 502 Yel wires and connections.
14. Ignition interlock module	Battery voltage	Check 800 Pnk wire and connections. If OK, replace seat switch.
15. Ignition interlock module	Battery voltage	Replace ignition interlock module.

Test Conditions:

- Key switch in run position, engine not running.
- Park brake locked.
- PTO on.
- Operator off seat.
- Confirm that the EFI fault light illuminates for approximately two seconds when the key is first turned to the run position.

Test/Check Point	Normal	If Not Normal
16. Mid PTO switch	Battery voltage	Check 500 and 520 Yel wires and connections.
17. PTO interlock module (option)	Battery voltage	Check 742 Blu/Wht wire and connections. If OK, replace mid PTO switch.
18. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
19. PTO interlock module (option)	Battery voltage	Check 765 and 789 Pur wires and connections. If OK, replace brake switch.
20. Ignition interlock module	Battery voltage	Check 605 Tan wire and connections. If OK, replace PTO interlock module.
21. Ignition interlock module	Battery voltage	Replace ignition interlock module.


A1 Ignition Interlock Module
M1 Starting Motor
200 Red
F3 Fusible Link
F6 Fusible Link
500 Yel
T2 Ignition Coil
T1 Ignition Coil
Blk
G1 Battery
500 Yel
940 Wht
X2
Blk
130 Blk
100 Blk
560 Red/Yel
S6 Brake Switch
S3 Seat Switch
501 Yel
502 Yel
800 Pnk
820 Pnk
S5 PTO/RIP Switch
650 Pnk /Blk
652 Blu
765 Pur
789 Pur
705 Pnk/Blk
S8 Mid PTO Switch
520 Yel
A3 PTO Interlock Module
K11 EFI Module Ground Relay
605 Tan
138 Blk
139 Blk
140 Blk
742 Blu/Wht
789 Pur
820 Pnk
670 Pur
X22
Org
100 Blk
1
2
3
4
5
6
13
14
15
16
17
18
19
20
21


K12 EFI Module Power Relay
560 Red/Yel
8
230 Red
221 Red
221 Red
7
150 Blk
100 Blk
K13 Fuel Pump Relay
232 Red
Blk
230 Red
231 Red
560 Red/Yel
233 Red /Blu
9
X23
910 Wht
Y5 #2 Cylinder Fuel Injector Solenoid
Y4 #1 Cylinder Fuel Injector Solenoid
X22
Wht
Brn
Blk
Wht
Wht
X24
11
11
X25
170 Blk
Wht
655 Pnk
X25
X21
12
651 Pnk/Blk
10
240 Red
M2 Fuel Pump
A4 Electric Fuel Injection Module
170 Blk
860 Pnk
460 Yel
Org
X2
860 Pnk

Fuel Injection Sensor and Diagnostic Circuit Operation - X485 and X585

Function:

Provides the inputs to the EFI module which in turn provides the proper output to the injectors.

The H9 circuit integrity light flashes in specific sequences of flashes to provide a visual indication of fault diagnostic codes. These fault codes will aid in the diagnosis of operational problems that may occur.

Operating Conditions:

- Key in run or start position,
- Engine running.

Theory of Operation:

Power is supplied to the electric fuel injection module when the key is in the run or start positions and the K12 EFI module power relay has been energized. At the same time the ground circuit is controlled by the K11 EFI module ground relay which is controlled by the ignition interlock circuit.

When the operating conditions have been met, the fuel injector sensors provide the input to the EFI module so that the EFI module will provide the ground path for the injectors at the correct time and duration.

The EFI module also has a self diagnostic light that will flash a fault code if any of the three input sensors is not operating properly. The fault codes are shown by a series of long and short flashes of the H8 circuit integrity light. When the circuit is operating normally, the light will be off. If a fault exists, the light will flash all the fault codes that are detected by the EFI module starting with the lowest numbered code. When all the fault codes have been displayed, the light will be off for a long period of time and then the fault codes will repeat.

The B4 coolant temperature, B5 Air Temperature and B6 vacuum pressure sensors are all resistance operated components that change the voltage to the EFI module. The fault codes for these components are based on the EFI module reading either a open or shorted circuit. This means that the EFI is receiving either battery voltage or no voltage from the sensors.

Diagnostic Circuit Operation:

During normal operation the circuit integrity light will illuminated for approximately 2 seconds when the key is turned ON as part of a self test, then the light will not be illuminated. The status indicator light will begin to flash a specific fault code when the input from any one or more sensors is not operating properly. If more than one fault exists, each fault will be displayed before the controller repeats the fault codes.

The EFI module has 4 different fault codes available to assist with diagnostics. Each fault code is a combination of long and short flashes. A fault code of 1 long and 2 short flashes would be signaling that the air temperature sensor circuit is malfunctioning. This means that the EFI is receiving either battery voltage or no voltage from the sensor.

The long flash is approximately 1.0 second, the short flash is approximately 0.5 second and the OFF period is approximately 4 seconds. The flash code begins after the off period. Whenever reading a fault flash code be sure to let the code cycle several times while reading the code to be sure you are at the beginning of the code. Once a code has been read, it can be matched to the fault code chart.

EFI Module Fault Code Chart

The long flash “ **-** ” = 1.0 second

The short flash “•” = 0.5 second

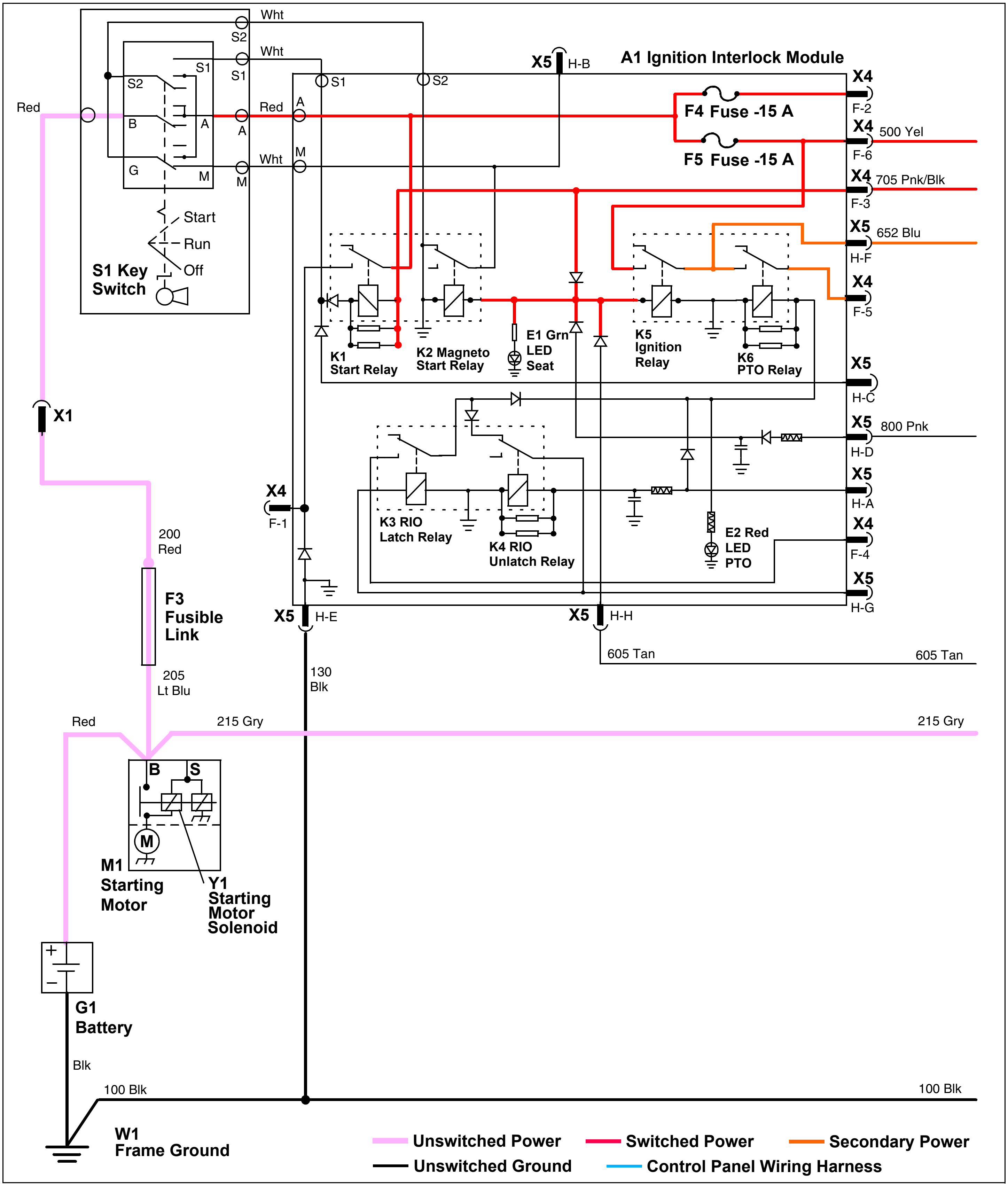
OFF period = 3 to 4 seconds.

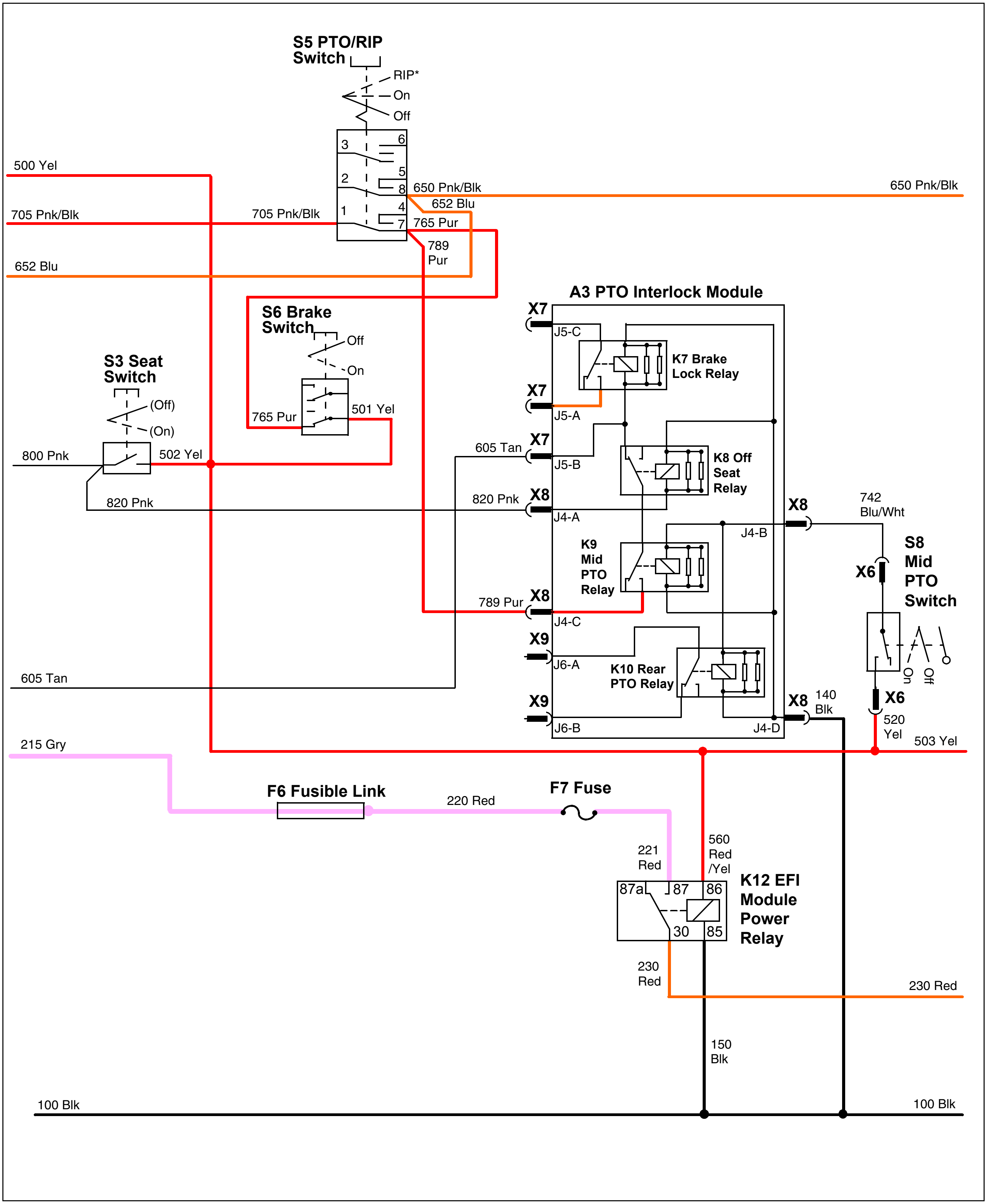
The flash code begins after the off period.

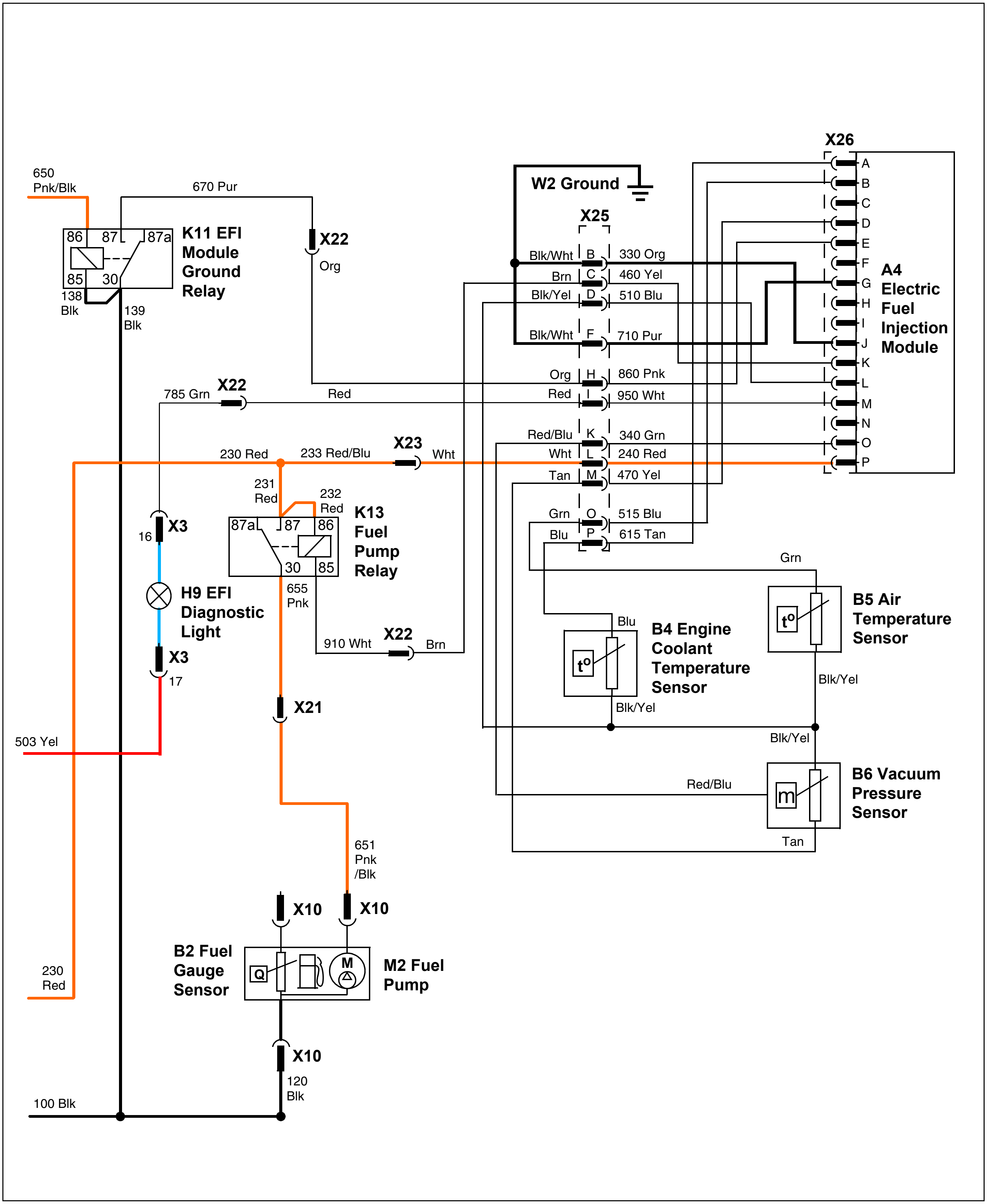
Let the code cycle several times while reading the code to be sure you are at the beginning of the code.

FAULT CODE	FLASH SEQUENCE	TEST POINT	FAULT CAUSE
12	- ••	Air temperature sensor	B5 air temperature sensor circuit malfunctioning. Circuit either open or shorted.
13	- •••	Coolant temperature sensor	B4 coolant temperature sensor circuit malfunctioning. Circuit either open or shorted.
21	- - •	Vacuum pressure sensor	B6 vacuum pressure sensor circuit malfunctioning. Circuit either open or shorted.
22	- - ••	N/A	EFI module air/fuel pressure calculation error. To reset: Stop engine, wait for fault indicator light to go off and fuel pump to stop (approximately 0.5 to 1 second). Start engine.

Fuel Injection Sensor and Diagnostic Circuit Electrical Schematic - X485 and X585




S5 PTO/RIP Switch
RIP*
On
Off
3
6
2
5
8
1
4
7
500 Yel
705 Pnk/Blk
705 Pnk/Blk
652 Blu
650 Pnk/Blk
650 Pnk/Blk
652 Blu
765 Pur
789 Pur
S6 Brake Switch
Off
On
765 Pur
501 Yel
S3 Seat Switch
(Off)
(On)
800 Pnk
502 Yel
820 Pnk
A3 PTO Interlock Module
X7
J5-C
K7 Brake Lock Relay
X7
J5-A
605 Tan
X7
J5-B
K8 Off Seat Relay
820 Pnk
X8
J4-A
X8
742 Blu/Wht
J4-B
K9 Mid PTO Relay
S8 Mid PTO Switch
X6
789 Pur
X8
J4-C
X9
J6-A
K10 Rear PTO Relay
On
Off
605 Tan
X9
J6-B
J4-D
X8
140 Blk
X6
520 Yel
503 Yel
215 Gry
F6 Fusible Link
220 Red
F7 Fuse
221 Red
560 Red /Yel
87a
87
86
30
85
K12 EFI Module Power Relay
230 Red
230 Red
150 Blk
100 Blk
100 Blk



X26
650
Pnk/Blk
670 Pur
W2 Ground
X25
86
87
87a
85
30
K11 EFI
Module
Ground
Relay
X22
Org
138
Blk
139
Blk
Blk/Wht
B
330 Org
Brn
C
460 Yel
Blk/Yel
D
510 Blu
Blk/Wht
F
710 Pur
A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
A4
Electric
Fuel
Injection
Module
785 Grn
X22
Red
Org
H
860 Pnk
Red
I
950 Wht
230 Red
233 Red/Blu
X23
Wht
Red/Blu
K
340 Grn
Wht
L
240 Red
Tan
M
470 Yel
231
Red
232
Red
K13
Fuel
Pump
Relay
87a
87
86
30
85
X3
16
Grn
O
515 Blu
Blu
P
615 Tan
Grn
H9 EFI
Diagnostic
Light
655
Pnk
X3
17
910 Wht
X22
Brn
Blu
B4 Engine
Coolant
Temperature
Sensor
B5 Air
Temperature
Sensor
Blk/Yel
Blk/Yel
X21
503 Yel
Blk/Yel
Red/Blu
B6 Vacuum
Pressure
Sensor
Tan
651
Pnk
/Blk
X10
X10
B2 Fuel
Gauge
Sensor
M
M2 Fuel
Pump
230
Red
X10
120
Blk
100 Blk

Fuel Injection Sensor and Diagnostic Circuit Diagnosis - X485 and X585

Test Conditions:

- Key switch in run position, engine not running.
- Park brake locked.
- PTO off.
- Operator off seat.
- Confirm that the EFI fault light illuminates for approximately two seconds when the key is first turned to the run position.

Test/Check Point	Normal	If Not Normal
1. Ignition interlock module	Battery voltage	Test F3 fusible link and F5 fuse. Check 200 Red wire and connections. If OK, replace ignition interlock module.
2. Brake switch	Battery voltage	Check 500 and 501 Yel wires and connections.
3. PTO/RIP switch	Battery voltage	Check 765 Pur wire and connections. If OK, replace brake switch.
4. Ignition interlock module	Battery voltage	Check 705 Pnk/Blk wire and connections. If OK, replace PTO/RIP switch.
5. Ignition interlock module	Battery voltage	Replace ignition interlock module.
6. K11 EFI module ground relay	Battery voltage	Check 652 Blu and 650 Pnk/Blk wires and connections.
7. K12 EFI module power relay	Battery voltage	Test F6 fusible link and F7 fuse. Check 220 and 221 Red wires and connections.
8. K12 EFI module power relay	Battery voltage	Check 500 Yel wire, 560 Red/Yel wire, and connections.
9. Fuel injector(s)	Battery voltage	Check 230 Red, 233 Red/Blu and Wht wires and connections. If OK, replace K12 EFI module power relay.
10. Electric fuel injection module	Battery voltage	Check 230 Red, 233 Red/Blu, Wht, and 240 Red wires and connections. If OK, replace K12 EFI module power relay.

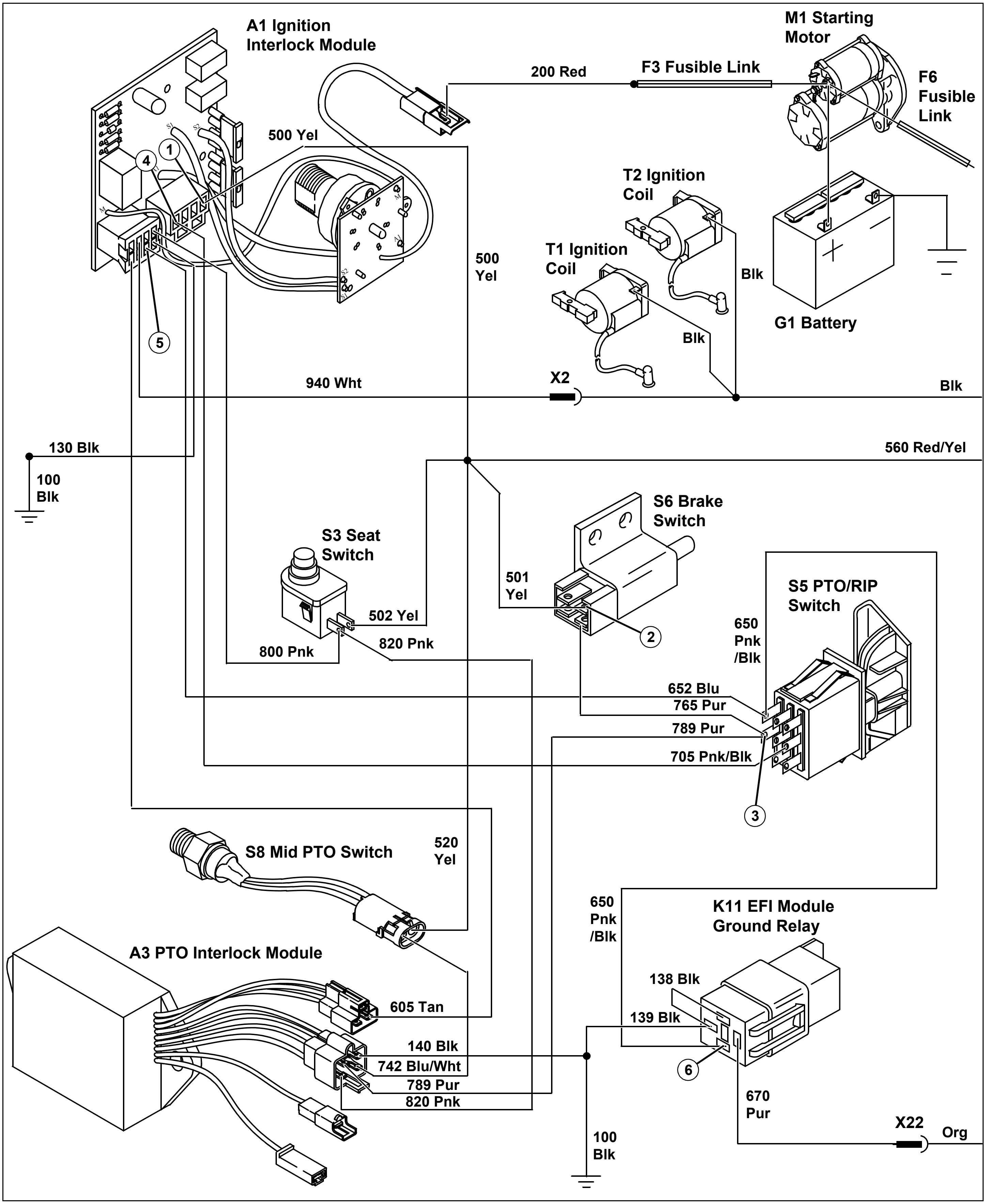
Test Conditions:

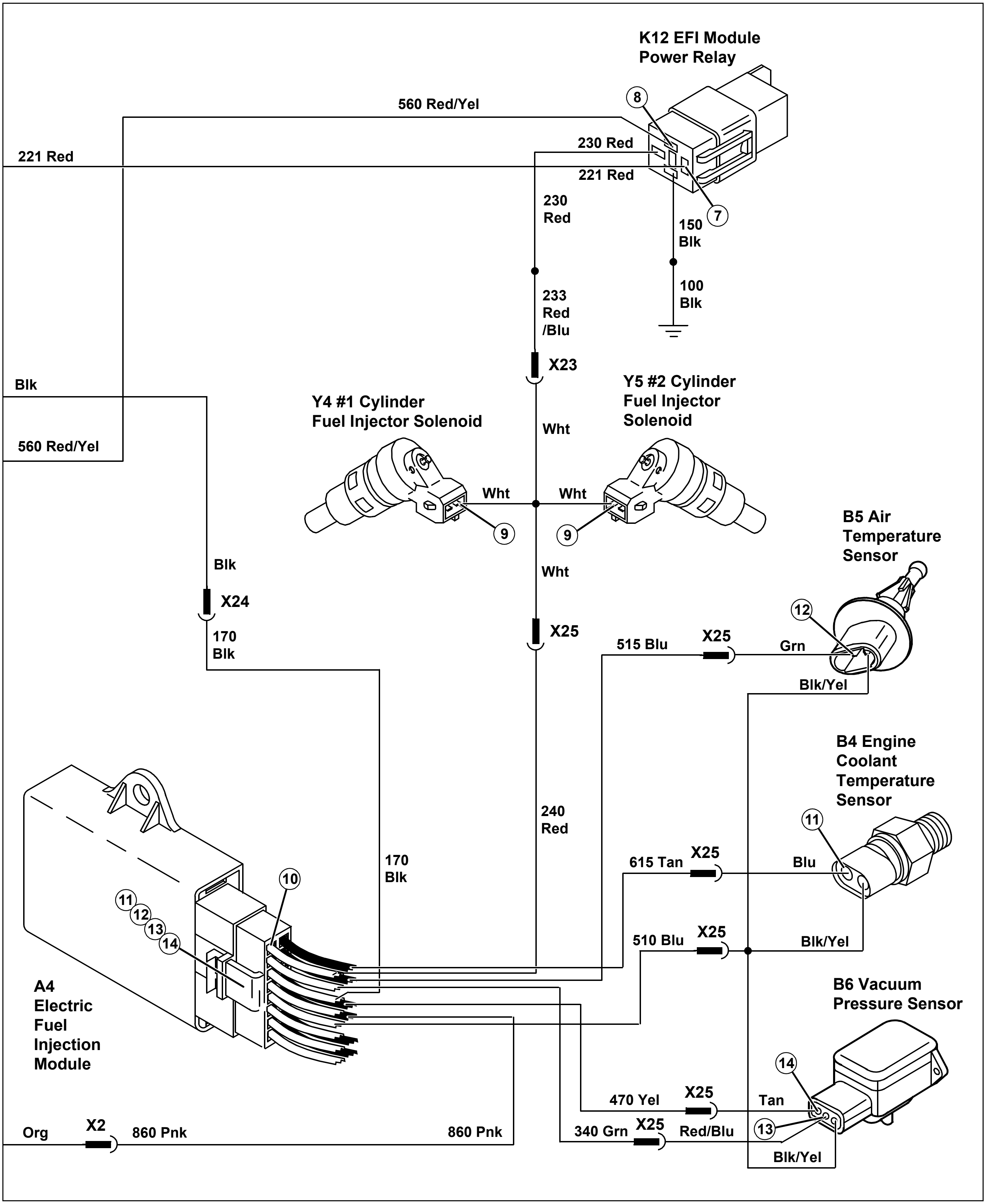
- Key switch in run position, engine not running.
- Park brake locked.
- PTO off.
- Measure voltage between test point and 510 Blu wire (terminal L) at EFI module.
- Confirm that the EFI fault light illuminates for aproximately two seconds when key is first turned to the run position.

Test/Check Point	Normal	If Not Normal
11. Coolant temperature sensor	2.26 - 2.5 volts	Check Blu and 615 Tan wires and connections. Test coolant temperature sensor. See “Coolant Temperature Sensor Test - X485 and X585” on page 322.

Test/Check Point	Normal	If Not Normal
12. Air temperature sensor	2.26 - 2.5 volts	Check Grn and 515 Blu wires and connections. Test air temperature sensor. See “Air Temperature Sensor Test - X485 and X585” on page 322.
13. Vacuum pressure sensor	4.75 - 5.25 volts	Check Red/Blu and 340 Grn wires and connections. Test vacuum pressure sensor. See “Vacuum Pressure Sensor Test - X485 and X585” on page 321.
14. Vacuum pressure sensor	3.5 - 3.8 volts	Check Tan and 470 Yel wires and connections. Test vacuum pressure sensor. See “Vacuum Pressure Sensor Test - X485 and X585” on page 321.

NOTE: If above tests do not solve problem, replace EFI control module with a known good unit.


A1 Ignition Interlock Module
M1 Starting Motor
200 Red
F3 Fusible Link
F6 Fusible Link
500 Yel
1
4
T2 Ignition Coil
G1 Battery
500 Yel
T1 Ignition Coil
Blk
Blk
5
940 Wht
X2
Blk
130 Blk
100 Blk
560 Red/Yel
S6 Brake Switch
S3 Seat Switch
501 Yel
S5 PTO/RIP Switch
502 Yel
2
650 Pnk /Blk
800 Pnk
820 Pnk
652 Blu
765 Pur
789 Pur
705 Pnk/Blk
3
S8 Mid PTO Switch
520 Yel
650 Pnk /Blk
K11 EFI Module Ground Relay
A3 PTO Interlock Module
138 Blk
605 Tan
139 Blk
140 Blk
742 Blu/Wht
789 Pur
820 Pnk
6
670 Pur
X22
Org
100 Blk


K12 EFI Module Power Relay
560 Red/Yel
221 Red
230 Red
221 Red
230 Red
150 Blk
100 Blk
233 Red /Blu
X23
Blk
560 Red/Yel
Y4 #1 Cylinder Fuel Injector Solenoid
Y5 #2 Cylinder Fuel Injector Solenoid
Wht
Wht
Wht
Wht
B5 Air Temperature Sensor
Blk
X24
170 Blk
X25
515 Blu
X25
Grn
Blk/Yel
B4 Engine Coolant Temperature Sensor
240 Red
170 Blk
615 Tan
X25
Blu
510 Blu
X25
Blk/Yel
A4 Electric Fuel Injection Module
B6 Vacuum Pressure Sensor
470 Yel
X25
Tan
Org
X2
860 Pnk
860 Pnk
340 Grn
X25
Red/Blu
Blk/Yel
8
7
9
9
12
11
10
11
12
13
14
14
13

Tests and Adjustments

Common Circuit Tests

Shorted/Grounded Circuit:

A shorted circuit on the ground side of a component (i.e. improper wire-to-wire or wire to ground contact) may result in improper component operation.

A shorted circuit on the power side of a component or contact of two power circuits (i.e. improper wire-to-wire or wire to ground contact) may result in blown fusible link and fuses.

To test for a shorted or improperly wired circuit:

1. Turn component switch ON.

2. Start at the controlling switch of the component that should not be operating.

3. Follow the circuit and disconnect wires at connectors until components stop operating.

4. Shorted or improper connections will be the last two wires disconnected.

HIgh Resistance or Open Circuit:

High resistance or open circuits usually result in slow, dim, or no component operation (i.e. poor, corroded, or severed connections). Voltage at the component will be low when the component is in operation.

To test for high resistance and open circuits:

1. Check all terminals and ground connections of the circuit for corrosion.

2. If terminals are not loose or corroded, the problem is in the component or wiring.

Ground Circuit Tests

Reason:

To check for opens, loose terminal wire crimps, poor connections, or corrosion in the ground circuit. The voltmeter method checks ground connections under load.

Equipment:

- Ohmmeter or Voltmeter

Ohmmeter Procedure:

1. Turn key switch to OFF position. Lock park brake.

2. Connect ohmmeter negative (black) lead to negative (-) terminal of battery. Put meter positive (red) lead on negative terminal of battery and record reading. Reading should be 0.1 ohm or less.

3. Put meter red lead on ground terminal of circuit or component to be tested that is closest to the battery negative terminal. Resistance reading must be very close to or the same as the battery negative terminal reading. Work backwards from the battery on the ground side of the problem circuit until the resistance reading increases above 0.1 ohms. The problem is between the last two test points. If a problem is indicated, disconnect the wiring harness connector to isolate the wire or component and check resistance again. Maximum allowable resistance in the circuit is 0.1 ohms. Check both sides of connectors closely as disconnecting and connecting may temporarily solve problem.

Voltmeter Procedure:

1. Put transmission in neutral. Lock park brake. Put PTO switch in off position. Turn key switch to on position.

2. Connect voltmeter negative (black) lead to negative terminal of battery.

3. Put meter positive (red) lead on ground terminal of component to be tested. Be sure the component circuit is activated (key in run position, switches closed) so voltage will be present at the component. Record voltage. Voltage must be greater than 0 but less than 1 volt. Some components will have a very small voltage reading on the ground side and still be operating correctly.

Results:

- If resistance is above 0.1 ohms, check for open wiring, loose terminal wire crimps, poor connections, or corrosion in the ground circuit.
- If voltage is 0.0, the component is open.
- If voltage is greater than 1 volt, the ground circuit is bad. Check for open wiring, loose terminal wire crimps, poor connections, or corrosion in the ground circuit.

Battery Voltage and Specific Gravity Test

Reason:

To check voltage and determine condition of battery.

Equipment:

- Voltmeter or JTO5685 Battery Tester
- Hydrometer Procedure
- Clean battery terminals and top of battery

Procedure:

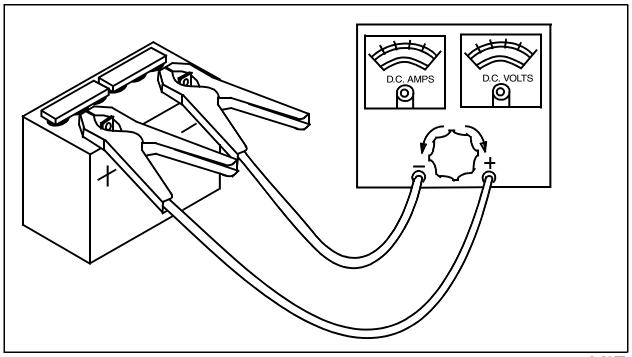
> **CAUTION: Avoid injury! Sulfuric acid in battery electrolyte is poisonous. It is strong enough to burn skin, eat holes in clothing, and cause blindness if splashed into the eyes.**
>
> **Avoid the hazard by:**
> **Filling batteries in a well-ventilated area,**
> **Wearing eye protection and rubber gloves,**
> **Avoiding breathing fumes when electrolyte is added.**
> **Avoid spilling or dripping electrolyte.**
> **Use proper jump-start procedure.**
>
> **If you spill acid on yourself:**
> **Flush your skin with water.**
> **Apply baking soda or lime to help neutralize the acid.**
> **Flush your eyes with water for 10 - 15 minutes. Get medical attention immediately.**
>
> **If acid is swallowed:**
> **Drink large amounts of water or milk.**
> **Then drink milk of magnesia, beaten eggs, or vegetable oil.**
> **Get medical attention immediately.**

1. Park machine safely with park brake locked.

2. Inspect battery terminals and case for breakage or cracks.

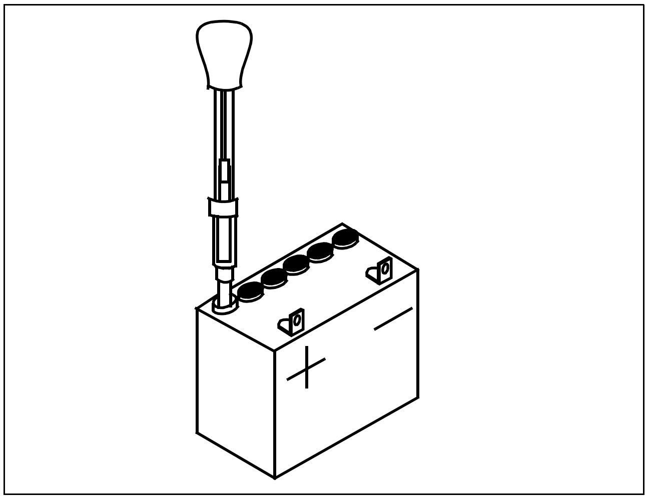
3. Check electrolyte level in each battery cell. Add clean, soft water as needed. If water added, charge battery for 20 minutes at 10 amps.

4. Remove surface charge by placing a small load on the battery for 15 seconds.



MIF

5. Check battery voltage with voltmeter or JTO5685 battery tester.



MIF

6. Use an hydrometer to check for a minimum specific gravity of **1.265** with less than **50** point variation in each cell at full charge at **26.7°C (80°F)**.

Results:

- Battery voltage less than **12.4 VDC**, charge battery.
- Battery voltage more than **12.4 VDC**, test specific gravity.
- All cells less than **1.175**, charge battery at 10 amp rate.
- All cells less than **1.225** with less than 50 point variation, charge battery at 10 amp rate.
- All cells more than **1.225** with less than 50 point variation, load test battery.
- More than **50** point variation: replace battery.

Specifications:

Minimum battery voltage . . . . .12.4 volts
Minimum specific gravity
. . . . 1.225 with less than 50 point variation

Battery - Charge

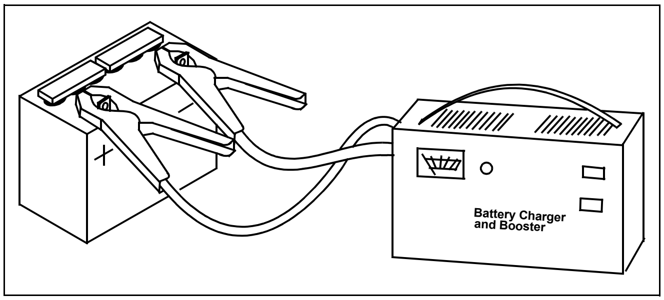
Reason:

To increase battery charge after battery has been discharged.

Equipment:

- Battery charger (variable rate)

Procedure:



MIF

1. Connect variable rate charger to battery.

NOTE: Maximum charge time at boost setting is 10 minutes. Allow an additional 5 minutes for each 10 degrees below 70 degrees F.

2. Start charger at slow rate. Increase charge rate one setting at a time. Check charger ammeter after 1 minute at each setting. Maintain 10 amp charge rate. Use boost setting as necessary.

3. Check if battery is accepting a 10 amp charge after 10 minutes at boost setting.
   - Battery will not accept 10 amp charge after 10 minutes at boost setting: replace battery
   - Battery is accepting 10 amp charge after 10 minutes at boost setting, and battery did not need water: go to steps 6 and 7
   - Battery is accepting 10 amp charge after 10 minutes at boost setting, but battery did need water or all cells were below 1.175: go to steps 4 and 5

IMPORTANT: Avoid damage! Decrease charge rate if battery gases or bubbles excessively or becomes too warm to touch.

4. Set charger at 15 - 25 amps.

5. Check specific gravity after 30 minutes (60 minutes for maintenance-free battery).

NOTE: If battery was discharged at slow or unknown rate, charge at 10 - 15 amps for 6 - 12 hours (Maintenance-free battery: 12 - 24 hours). If battery was discharged at fast rate, charge at 20 - 25 amps for 2 - 4 hours (Maintenance-free battery: 4 - 8 hours).

   - More than 50 point variation between cells: replace battery
   - Less than 50 point variation between cells: go to steps 6 and 7

6. Continue charging battery until specific gravity is **1.230 - 1.265 points**.

7. Load test battery.

Battery - Load Test

Reason:

To check condition of battery under load.

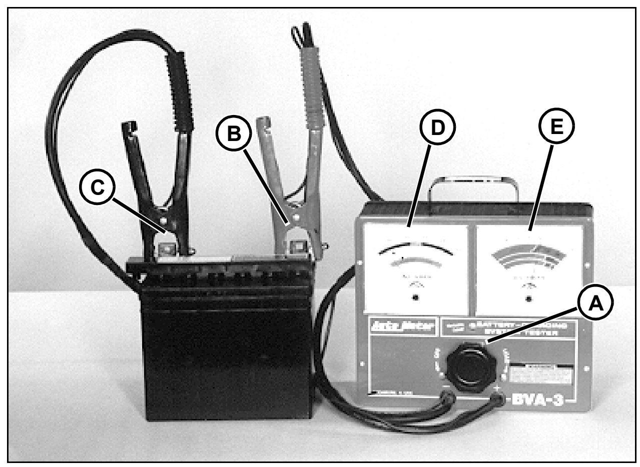
NOTE: Charge battery before performing load test.

Equipment:

- JTO5685 Battery Tester

NOTE: Use the procedures given with the tester.

Connections:



M49597

1. Turn load knob (A) of tester counter-clockwise to off.

2. Connect tester positive cable (B) to battery positive terminal.

3. Connect tester negative cable (C) to battery negative terminal.

Procedure:

1. Turn load knob of tester clockwise until amperage reading is equal to:
   - Cold cranking amperage rating (blue scale).

   or

   - Three times ampere hour rating (black scale).

2. Hold for 15 seconds and turn load knob of tester off.

3. Read battery voltage.

Results:

- If the battery does not indicate 9.6 volts or more, replace battery

Regulated Amperage and Voltage Test

Reason:

To determine the regulated charging output of the alternator.

NOTE: Battery must be in a good state of charge.

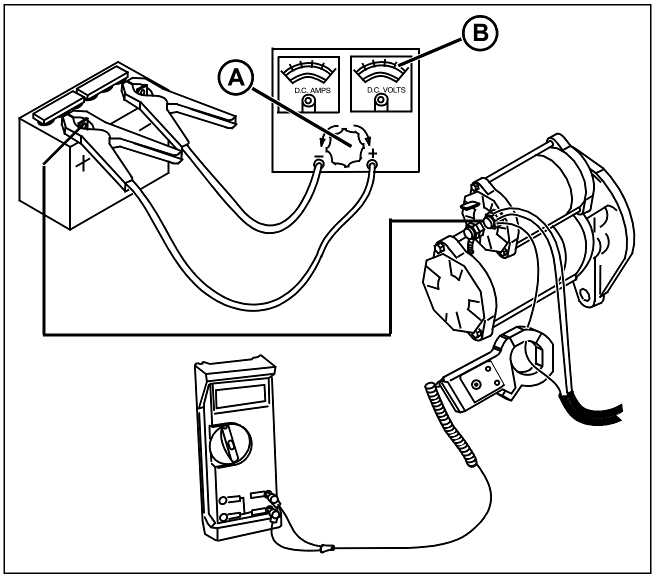
Equipment:

- JTO5712 Current Gun
- Voltmeter
- JTO5685 Battery Tester

Procedure:

1. Park machine safely with park brake locked.
2. Raise hood.

IMPORTANT: Avoid damage! Turn load knob (A) fully counterclockwise (out) into OFF position BEFORE making any test connections.



MIF

3. Connect JTO5712 Current Gun to voltmeter and put around red wire going to voltage rectifier/regulator. Set current gun for DC current.
4. Connect battery tester to battery.

IMPORTANT: Avoid damage! Perform these tests quickly to prevent damage to battery tester. DO NOT apply full load to battery for more than 5 - 10 seconds.

5. Turn load knob clockwise (in) until voltage on voltage tester scale reads 11 volts for 5 seconds only to partially drain battery.
6. Quickly turn load knob completely counter-clockwise (out) to off position.
7. Start and run engine at high idle. Battery voltage should read between **14.2 - 14.8** VDC.
8. Turn load knob (A) clockwise (in) until voltage on tester voltage scale (B) reads 11 volts and look at current gun for a minimum reading of **13.0** amps.
9. Quickly turn load knob (A) completely counter-clockwise (out) to off position.
10. After load test, battery voltage (B) should return to the voltage level prior to test.

Results:

- If current gun amp reading is below specification, test for unregulated voltage output. If unregulated voltage output test meets specifications and you have verified voltage to ground to regulator/rectifier, replace regulator/rectifier.
- If at any time voltage increase exceeds **14.8** VDC, replace regulator/rectifier.

Specifications:

High Idle Voltage . . . . 14.2 - 14.8 VDC
Minimum High Idle Amperage. . . . . 13.5 Amps

Regulated Voltage Test

Reason:

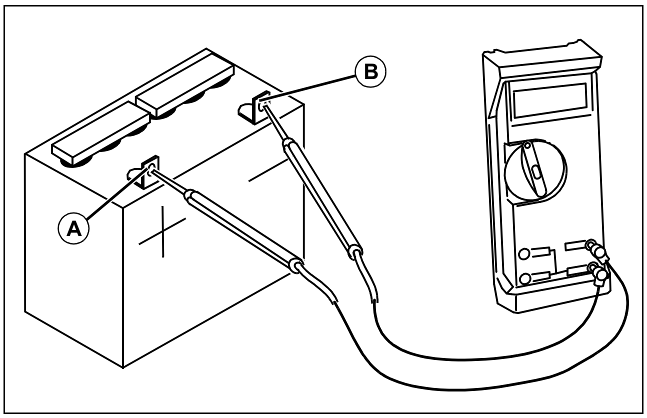
To determine regulated voltage output of the regulator/ rectifier.

Equipment:

- Voltmeter

Procedure:

1. Park machine with park brake locked.
2. Raise hood.
3. Remove surface charge from battery by placing a small load on the battery for 15 seconds.
4. Set voltmeter to DC volts scale.



MIF

5. Connect meter red lead (A) to positive battery terminal.
6. Connect meter black lead (B) to negative battery terminal.
7. Start and run engine at low idle.
8. Read meter while increasing engine speed to fast idle.

Voltage should increase from battery voltage at low idle to maximum regulated voltage at fast idle.

IMPORTANT: Avoid damage! Do not allow the battery voltage to exceed 15.5 volts or the battery and charging system will be damaged.

Specifications:

Regulated voltage (maximum) . . . . .15.0 VDC

Results:

- If the DC voltage stays below the minimum specification, test unregulated voltage output. If ok, replace the voltage rectifier/regulator.
- If the DC voltage goes above the maximum specification, replace the regulator.

Voltage Regulator/Rectifier Test - X465

Reason:

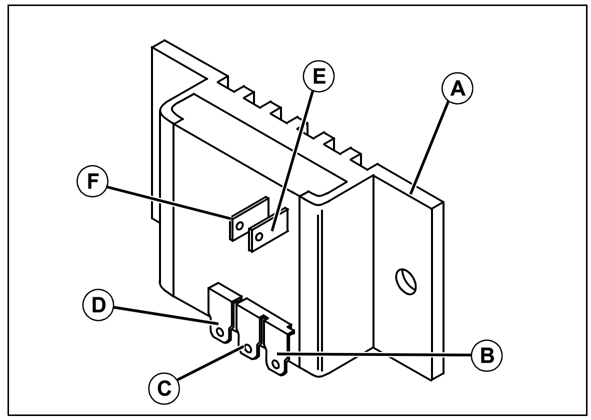
To verify proper operation capability of the voltage regulator/rectifier.

Equipment:

- Multimeter

Procedure:

1. Disconnect stator connector from the voltage regulator/ rectifier.
2. Set the multi-meter to the Ohms scale.



MIF

3. Measure resistance between each point as shown in the chart.

	A	B	C	D	E	F
A		OL	OL	OL	19 - 3k	3k - OL
B	OL		0	500 - OL	OL	OL
C	OL	0		500 - OL	OL	OL
D	7.5k - OL	925 - OL	925 - OL		10k - OL	10k - OL
E	330 - 10k	OL	OL	OL		1.2k - OL
F	14k - OL	OL	OL	OL	110 - OL	

4. If resistance is not within specification, replace voltage regulator/rectifier.

Unregulated Voltage Test - X465

Reason:

To measure AC voltage output of stator and verify correct resistance of stator.

Equipment:

- Multimeter

Procedure:

1. Disconnect connector from stator.
2. Set multimeter for AC volts.
3. Attach red test lead to either pin on stator side of stator connector.
4. Attach black test lead to other pin.
5. Lock parking brake, start engine and run at full throttle.
6. Check output.

Results:

- Output will be approximately **30 - 32 volts**.
- If no or low output is found, replace stator.

Specifications:

AC output at full throttle . . . . . . 26 volts AC (minimum)

Unregulated Voltage Test - X475, X575, X485, and X585

Reason:

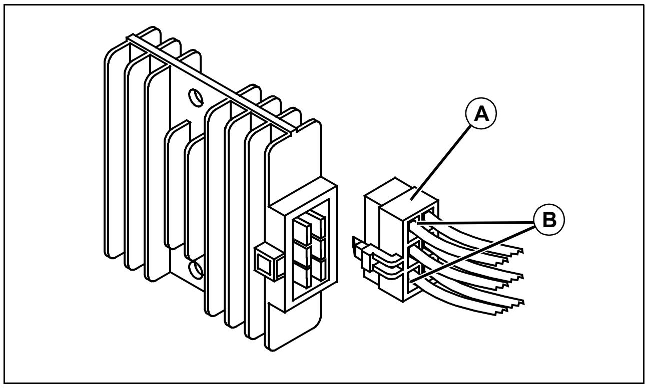
To determine charging output of the alternator stator.

Test Equipment:

- Voltmeter

Procedure:

1. Park machine with park brake locked.
2. Raise hood.



MIF

X475/X575/X485/X585

3. Disconnect connector (A) from voltage rectifier/ regulator.
4. Set voltmeter to AC volts.
5. Connect voltmeter to stator terminals (B).
6. Start and run engine at 3000 rpm.
7. Record stator voltage.

Specifications:

Unregulated voltage (minimum) . . . . 26 VAC

Results:

- If voltage is less than specification, replace stator.

Stator Resistance Test

Reason:

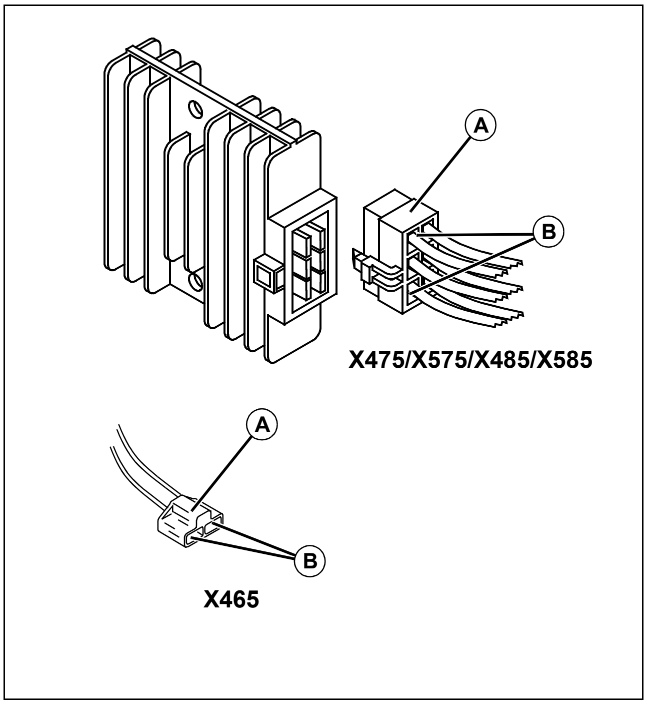
To determine if stator windings are open or grounded.

Equipment:

- Voltmeter

Procedure:

1. Park machine safely with engine off and park brake locked.
2. Raise hood.



MIF

3. Disconnect connector (A) from voltage rectifier/ regulator.
4. Set voltmeter to ohms.
5. Connect meter across terminals (B) of alternator.
6. If reading is not within specification, replace stator.
7. Connect meter to either terminal of alternator connector and to frame ground.
8. If meter does show continuity, replace stator.

Specifications:

Stator Resistance . . . . .0.11 - 0.18 ohms

Starting Motor Condition

1. The starting motor overheats because of:
   - Long cranking.
   - Armature binding.
2. The starting motor operates poorly because of:
   - Armature binding.
   - Dirty or damaged starter drive.
   - Badly worn brushes or weak brush springs.
   - Excessive voltage drop in cranking system.
   - Battery or wiring defective.
   - Shorts, opens, or grounds in armature.

NOTE: Starting motor repair is limited to brushes, end caps, and starter drive. Fields in starting motor are permanent magnets and are not serviceable. If housing or armature is damaged, replace starting motor.

Starter Solenoid Test

Reason:

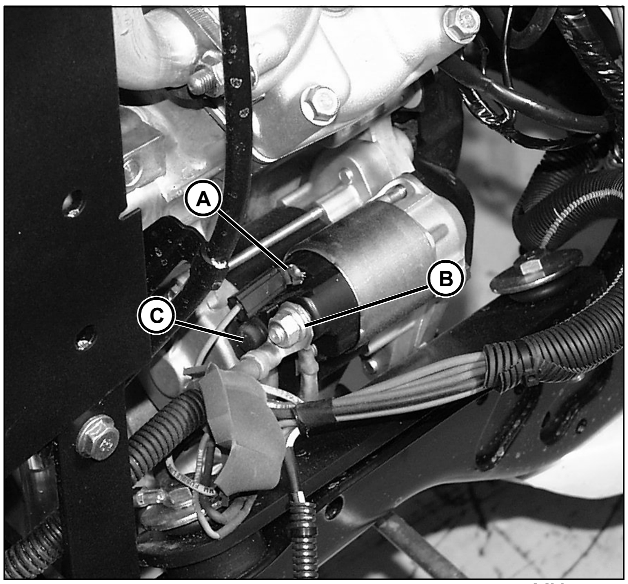
To determine if starter solenoid or starting motor is defective.

Equipment:

- Jumper wire

Procedure:

1. Put transmission in neutral. Lock park brake. Move key switch to off position.
2. Raise hood.
3. Disconnect and ground spark plug leads.



MX14279

4. Disconnect 720 Pur wire from starter solenoid terminal (A).
5. Connect jumper wire to positive battery terminal and briefly jump to starter solenoid terminal (A).
    - Starting motor runs: solenoid is good, check cranking circuit.
    - Starting motor does not run: go to next step.
6. Remove red and black rubber boots from starter solenoid terminals (B and C).
7. Connect jumper wire between starter solenoid large terminals (B and C).
    - Starter runs: Replace solenoid.
    - Starting motor does not run: Check battery cables then replace starting motor.

Starting Motor Amp Draw Test

Reason:

To determine the amperage required to crank the engine and check starter motor operation under load.

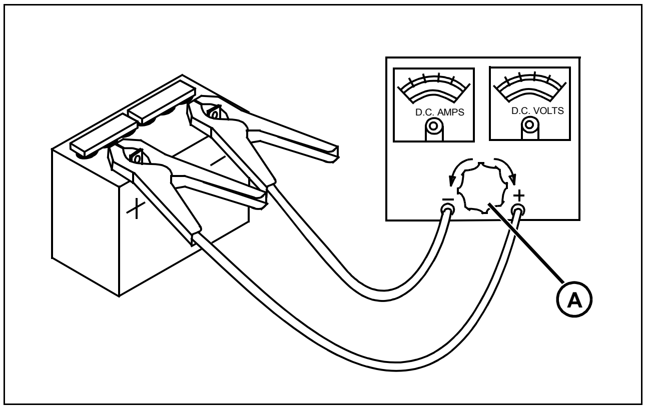
Equipment:

- JTO5685 Battery Tester

Procedure:

1. Put transmission in neutral. Lock park brake. Put key switch in off position.
2. Raise hood.
3. Disconnect and ground spark plug leads.

IMPORTANT: Avoid damage! Turn load knob (A) fully counterclockwise before making any test connections



MIF

4. Connect JTO5685 Battery Tester to battery.
5. Crank engine and read voltage.
6. Turn key switch to off position. Adjust load knob until battery voltage reads the same as when cranking.
7. Read amperage on meter.
8. Turn load knob fully counterclockwise.

Specifications:

Starter amp draw (maximum) . . . . .85 amps

Results:

- If amperage is above specification, check starting motor no load amperage draw to determine if starting motor is binding or damaged.
- If starting motor is good, check internal engine components for binding or damage.

Starting Motor No-load Amperage and RPM Test

Reason:

To determine if starting motor is binding or has excessive amperage draw under no-load.

Equipment:

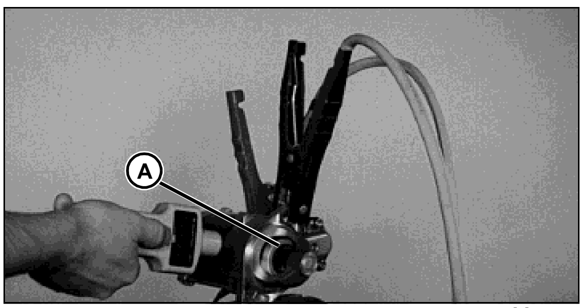
- JTO5712 Current Gun
- JTO5719 Photo Tachometer

Procedure:

1. Remove starter from engine.

NOTE: Check that battery is fully charged and of proper size to ensure accuracy of test.

2. Connect jumper cables to battery.



M45864a

3. Put screwdriver between gear and bendix drum (A) and pull outward. Install reflective tape on drum.
4. Connect negative cable to starter body.
5. Connect positive cable to solenoid battery terminal.
6. Attach current gun to positive jumper cable.

IMPORTANT: Avoid damage! Complete this test in 20 seconds or less to prevent starter damage.

7. Use jumper wire to briefly connect solenoid battery terminal and solenoid engagement terminal.
8. Measure starter amperage and rpm.

Specifications:

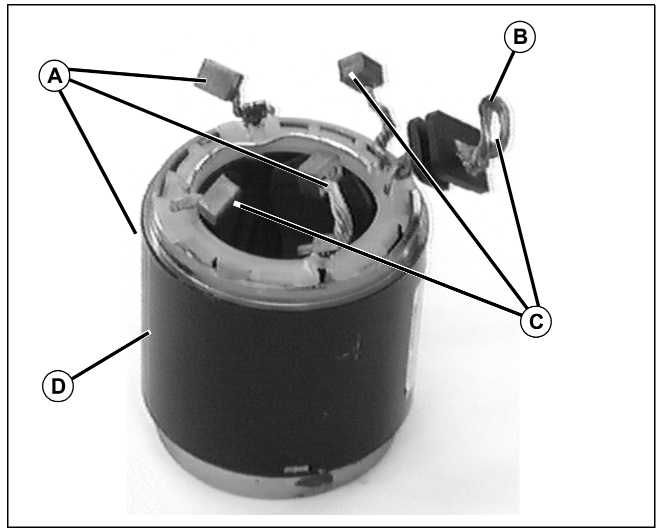
Starter amperage (maximum) . . . . 50 amps at 6000 rpm
Starter rpm (minimum). . . . . .6000 rpm

Results:

- If amperage or rpm is out of specification, check for binding or seizing bearings, sticking brushes, dirty or worn commutator. Repair or replace starter.

Field Windings Test

NOTE: Field winding case is a tie point for many separate field coils. It may be difficult to detect one bad coil. If rpm was slow and armature tests are normal, replace field coil assembly.



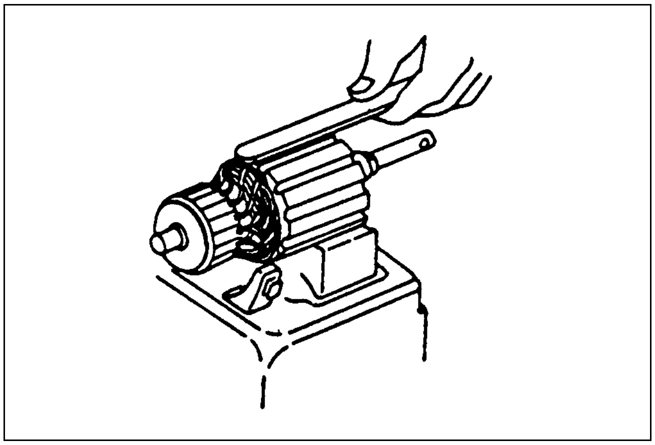
M57356

A - Field Brushes (Continuity between brushed and to case)
B - To Solenoid
C - Armature Brushes (Continuity between brushes and solenoid lead. Open to case.)
D - Starter Case

Replace field coil if not according to specifications.

Starter Armature Test

IMPORTANT: Avoid damage! Do not clean armature with solvent. Solvent can damage insulation on windings. Use only mineral spirits and a brush.

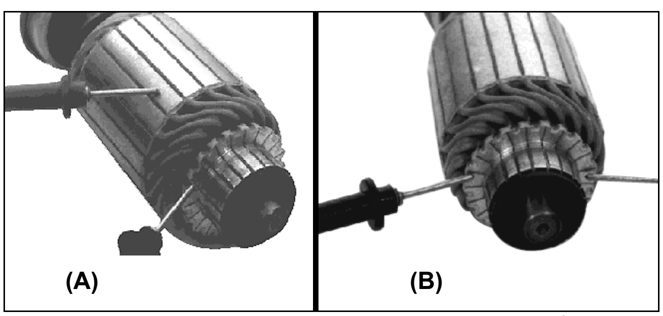


M24861

1. Locate short circuits by rotating armature on a growler while holding a hacksaw blade or steel strip on armature. The hacksaw blade will vibrate in area of short circuit.

2. Shorts between bars are sometimes caused by dirt or copper between bars. Inspect for this condition.

3. If test indicates short circuited windings, clean the commutator of dust and fillings. Check armature again. If test still indicates short circuit, replace armature.



M50112/M50113

A - Test for grounds
B - Test for open

4. Using an ohmmeter, test each individual armature windings for grounded or open circuits.

NOTE: Armature windings are connected in parallel, so each commutator bar needs to be checked.

5. If either test fails, armature must be replaced.

Ignition Interlock Test

Reason:

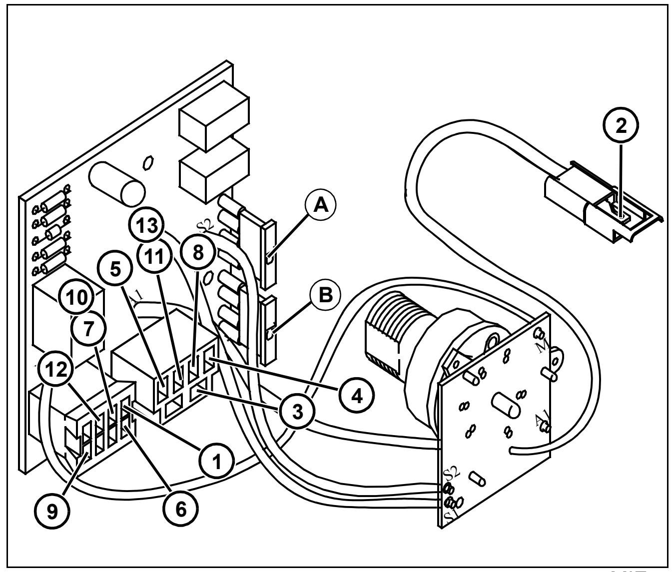
To make sure the ignition interlock module has input and output voltage at the correct terminals for proper operation. This test determines if the ignition interlock module is defective or the input circuit is defective.

Equipment:

- Voltmeter
- Ohmmeter

Procedure:

1. Park machine safely with park brake locked.
2. Set test conditions as described before each test.
3. Connect meter negative (black) lead to battery negative terminal.
4. Check battery voltage.



MIF

5. Check fuses (A) and (B) to be sure they are not open.
6. Put meter positive (red) lead at terminals as indicated in the test table.
7. Check the terminals for input and output battery voltage or good ground.

Test Conditions:

- Key switch in off position.

Test/Check Point	Normal Condition	If Not Normal
1. Terminal E.	Maximum 0.1 ohms resistance.	Check battery negative cable and harness ground connection, and 100, 130 Blk wires.

Test Conditions:

- Key switch in run position.
- Park brake locked.
- PTO off.
- Operator on seat.

Test/Check Point	Normal Condition	If Not Normal
2. Key switch power connector.	Battery voltage - input.	Check 200 Red and 205 Lt Blu wires.
3. Terminal 2.	Battery voltage - output.	Check fuses; if OK, replace ignition interlock module.
4. Terminal 6.	Battery voltage - output.	Check fuses; if OK, replace ignition interlock module.
5. Terminal 3.	Battery voltage - input.	Check cranking circuit test points.
6. Terminal D.	Battery voltage - input.	Check seat switch, 800 Pnk, 502 and 500 Yel wires.
7. Terminal F.	Battery voltage - output.	Replace ignition interlock module.

Test Conditions:

- Key switch in run position.
- Park brake unlocked.
- PTO off.
- Operator on seat.

Test/Check Point	Normal Condition	If Not Normal
8. Terminal 4.	Battery voltage - input.	Check PTO circuit test points.
9. Terminal A.	Battery voltage - input. Red LED on.	Check RIO switch, 735 Org and 772 Blu wires. If OK, replace ignition interlock module.
10. Terminal F.	Battery voltage - output.	Replace ignition interlock module.

Test Conditions:

- Key switch in run position.
- Park brake unlocked.
- Operator on seat.
- PTO switch in on position then put machine in reverse.

Test/Check Point	Normal Condition	If Not Normal
11. Terminal 5.	0.0 volts - output.	Check PTO switch and RIO switch. If switches and wires are good, replace ignition interlock module.

Test Conditions:

- Key switch in run position.
- Machine in forward then cycle the PTO switch to off and then to the RIP position.

Test/Check Point	Normal Condition	If Not Normal
12. Terminal G.	Battery voltage - input.	Check PTO switch, 730 Grn and 754 Blu wires.

Test Conditions:

- Key switch in off position.
- While holding PTO switch in RIP position, place machine in reverse then release PTO switch to the on position.

Test/Check Point	Normal Condition	If Not Normal
13. Terminal 5.	Battery voltage - output.	Check PTO switch, 730 Grn and 754 Blu wires. If PTO switch and wires are good, replace ignition interlock module.

Test Conditions:

- Key switch in off position.
- Place machine in forward and back to reverse.

Test/Check Point	Normal Condition	If Not Normal
14. Terminal 3.	0.0 volts - output.	Replace ignition interlock module.

Test Conditions:

- Key switch in off position.
- Park brake locked.
- PTO switch in off position.
- Starting motor solenoid wire disconnected.
- Key switch in start position.

Test/Check Point	Normal Condition	If Not Normal
15. Terminal 5.	Battery voltage - output.	Replace ignition interlock module.

Results:

- If terminal does not have input voltage or voltage is low, check proper circuit before the ignition interlock module.
- If terminal does not have output voltage, replace the ignition interlock module.

Engine Coolant Temperature Sensor Test - X475, X575, X485, and X585

Reason:

To verify coolant temperature sensor is functioning properly.

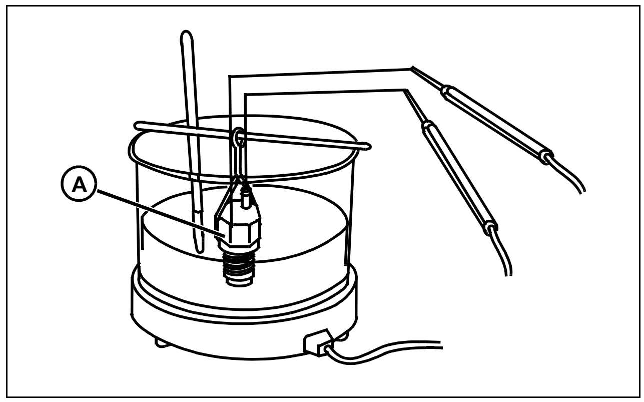
Equipment:

- Ohmmeter

Procedure:

NOTE: Perform test with engine at room temperature.

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Disconnect 310 Org/Wht wire from engine coolant temperature sensor.
4. Measure resistance between terminal and sensor body.
5. If resistance does not meet specification, replace coolant temperature switch.
6. Drain engine coolant and remove coolant temperature switch.



M91295

7. Place sensor (A) in antifreeze solution heated to approximately 110°C (230°F). Measure resistance while sensor is heated.
8. If resistance does not meet specification, replace coolant temperature sensor.

Specifications:

Variable Resistance . . . . 40 - 700 ohms

Engine Oil Pressure Switch Test

Reason:

To determine the proper operation of the oil pressure switch.

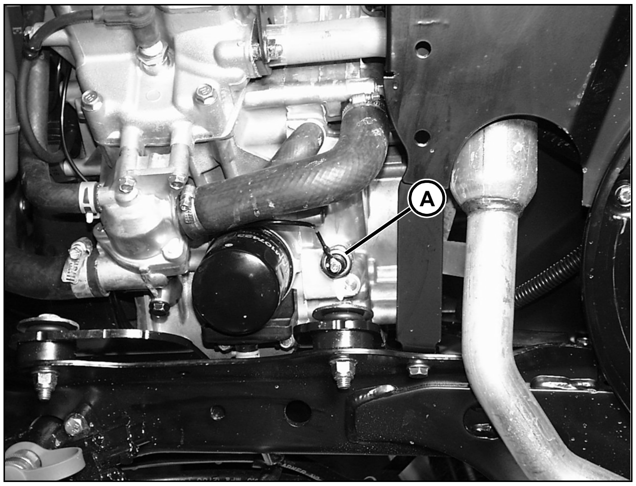
Equipment:

- Ohmmeter

Procedure:

NOTE: Perform test with engine at room temperature.

1. Park machine safely in neutral with park brake locked.
2. Raise hood.



MX14287

Picture Note: X485 shown

3. Disconnect Blu/Red wire from engine oil pressure switch (A) located near oil filter.
4. Connect black lead of ohmmeter to engine block and red lead of ohmmeter to terminal of switch.
5. Measure resistance between terminal and engine block.

Results:

- There should be continuity between terminal and ground.
- If there is no continuity between terminal and ground; replace the oil pressure switch.

NOTE: Be sure to apply Pipe Sealant with TEFLON® to threads of switch anytime it is installed.

TEFLON® is a registered trademark of the DuPont Company.

IMPORTANT: Avoid damage! Do not allow connector of wire to contact the engine or frame as there will be voltage on it during the test.

1. Start and run engine.
2. Measure resistance between terminal and engine block.

Results:

There should be no continuity between switch terminal and ground with the engine running.

- If the switch does have continuity to engine block (ground) with engine running, check engine oil pressure.
- If oil pressure meets specification; replace the oil pressure switch.

Lights Switch Test

Reason:

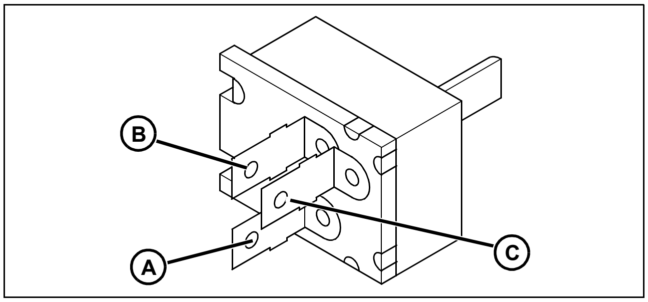
To verify terminal continuity is correct for each switch position.

Equipment:

- Ohmmeter or continuity tester

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Remove left side panel.
4. Disconnect lights switch connector.
5. Put light switch in off position.



MIF

6. Use and ohmmeter to test switch continuity in off, headlights on, and backup lights on positions.

Specifications:

Switch Position	**Terminal Continuity**
Off	**no continuity between terminals**
Headlights On	**A and B**
Backup lights on	**A and B**
	A and C
	B and C

Results:

- If continuity is not correct, replace lights switch.

PTO/RIP Switch Test

Reason:

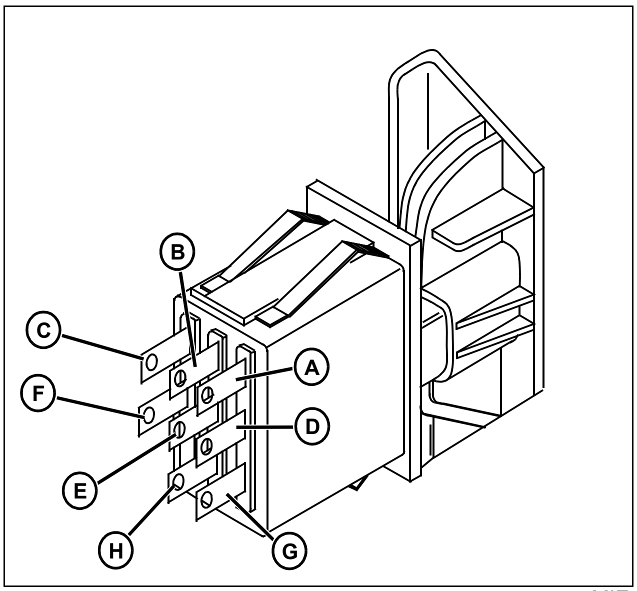
To verify terminal continuity is correct for each switch position.

Equipment:

- Ohmmeter or continuity tester

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Disconnect PTO switch connector.
4. Put PTO switch in the off position.



MIF

5. Use and ohmmeter to test switch continuity in off, on, and momentary (RIP) positions.

Specifications:

Switch Position	Terminal Continuity
Off	A and C
	D and F
On	A and B
	D and E
Momentary (RIP)	A and B
	D and E
	G and H

Results:

All other possible combinations have infinite resistance. If any continuity is NOT correct, replace switch.

Relay Test

Reason:

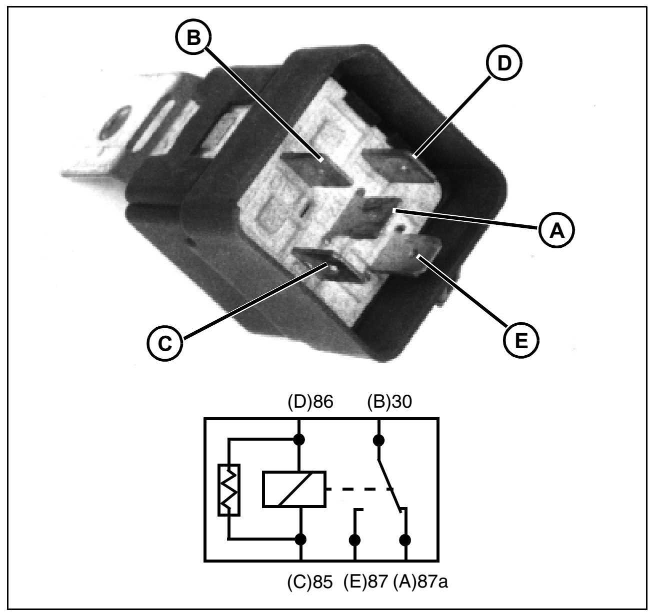
To check relay terminal continuity in the energized and de-energized condition.

Equipment:

- Ohmmeter or continuity tester

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Disconnect relay connector.
4. Check terminal continuity using an ohmmeter or continuity tester.



M56817

5. There should be continuity between terminals 87A (A) and 30 (B), and 85 (C) and 86 (D).
6. There should not be continuity between terminals 87 (E) and 30 (B).
7. Connect a jumper wire from battery positive (+) terminal to relay terminal 85 (C). Connect a jumper wire from relay terminal 86 (D) to ground (-).
8. There should be continuity between terminals 87 (E) and 30 (B).

Results:

- If continuity is not correct, replace relay.

Seat Switch Test

Reason:

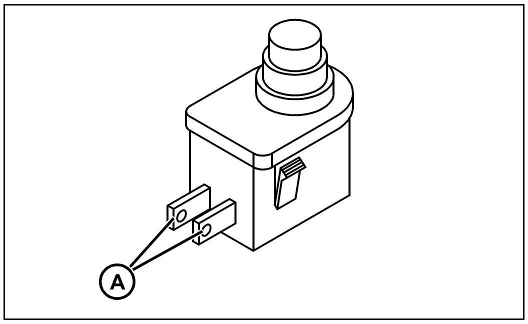
To verify proper operation of seat switch.

Equipment:

- Ohmmeter

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Disconnect seat switch connector.
3. Depress the seat switch plunger.



MIF

4. Check continuity across seat switch connector terminals (A). There should be continuity.
5. Release the seat switch plunger.
6. Check continuity across connector terminals. There should not be continuity.

Specifications:

Seat switch plunger depressed. . . . . continuity
Seat switch plunger released . . . . no continuity

Results:

- If the seat switch does not have continuity with the operator on the seat, check the seat switch bracket and spring for damage.
- If the seat switch does not have continuity with the plunger depressed, replace the switch.

Brake Switch Test and Adjustment

Reason:

To make sure the brake switch has continuity when the brake pedal is depressed.

Equipment:

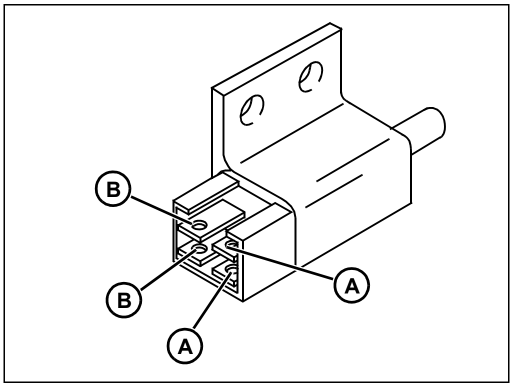
- Ohmmeter

Procedure:

1. Park machine safely in neutral with park brake locked.

NOTE: Brake switch is located on left side of frame under footrest.

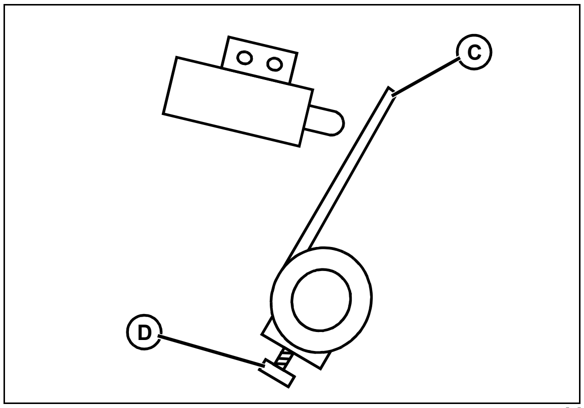
2. Remove footrest.
3. Disconnect brake switch connector.



MIF

4. Check continuity across terminals (A). Check continuity across terminals (B).

NOTE: Be sure actuator (C) contacts switch plunger (plunger depressed) and not switch body.



MIF

NOTE: Side view shown

5. Release park brake pedal.
6. Check continuity across terminals. Be sure actuator does not contact switch plunger (plunger released).

Specifications:

Brake pedal depressed (plunger depressed) continuity between terminals . . . . (A) only

Brake pedal released (plunger released) continuity between terminals . . . . (B) only

Results:

- If continuity is not correct, replace switch.
- If actuator position is not correct, loosen cap screw (D). Depress brake pedal and lock park brake. Rotate actuator (C) until the switch plunger is depressed, but not bottomed out. The actuator must not contact the switch body. Hold actuator in position and tighten cap screw.

RIO Switch Test

Reason:

To make sure the RIO switch has proper continuity.

Equipment:

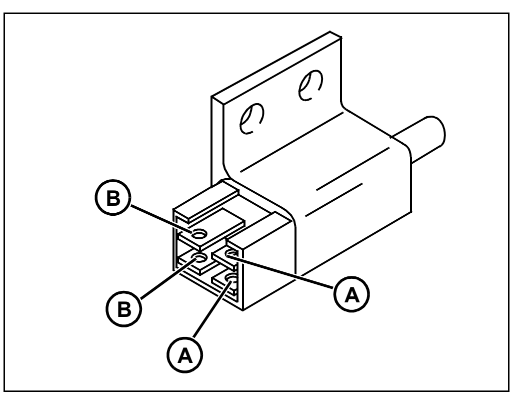
- Ohmmeter

Procedure:

1. Park machine safely in neutral with park brake locked.

NOTE: RIO switch is located on right side of frame under footrest.

2. Remove footrest.
3. Disconnect RIO switch connector.



MIF

4. Press switch plunger and check continuity across terminals (A) and then check continuity across terminals (B).

5. Press switch plunger and check continuity across terminals.

Specifications:

Plunger depressed) continuity between terminals . . . . (A) only

Plunger released) continuity between terminals . . . . (B) only

Results:

- If continuity is not correct, replace switch.
- Adjustment switch to specification if needed.

RIO Switch Adjustment

Reason:

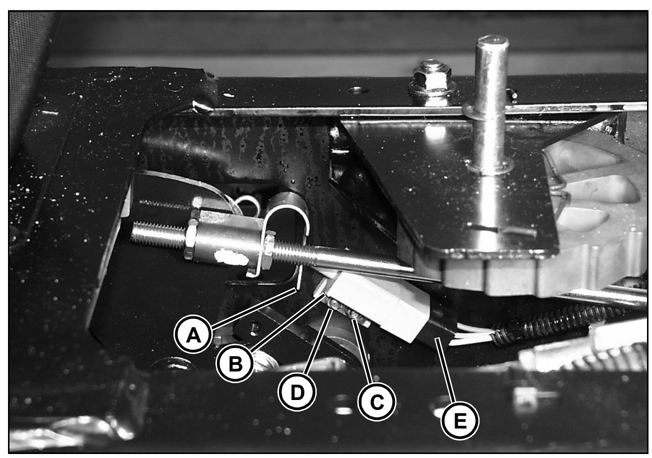
To set the proper air gap so that the RIO switch is activated at the proper time.

Equipment:

- Ruler
- Continuity tester

Procedure:

1. Stop engine and lock park brake.
2. Remove fender deck.



MX16074

3. Measure the gap between the bottom edge (A) of the spring and the bottom edge (B) of the RIO switch body.
   - The gap should be **12 mm (0.47in.)**.

4. The bottom edge of the switch should be in line with the bottom edge of the spring.

5. Slightly loosen the back nut (C) and then the front nut (D).

6. Position the RIO switch at the proper angle with a **12 mm (0.47 in.)** gap.

7. Tighten the nuts to **13.5 N•m (10 lb-ft)** while maintaining the position of the switch.

8. Perform RIO switch test.
9. Install the fender deck.

Specifications:

Air Gap. . . . . 12 mm (0.47 in.)
Torque . . . . 13.5 N•m (10 lb-ft)

Service Timer Reset Switch (SN -30000)

Reason:

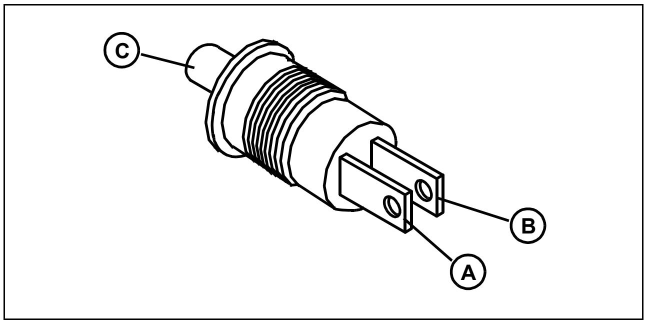
To determine proper operation of service timer reset switch.

Equipment:

- Ohmmeter

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Remove service timer reset switch connectors.



MIF

4. Connect meter leads to switch terminals (A) and (B) and compare to specification.
5. Press and release plunger (C) of switch.

Specifications:

Switch plunger not pressed
. no continuity between A and B
Switch plunger pressed
. . . . . continuity between A and B

Results:

If service timer reset switch does not pass both tests, replace switch.

Fuse Test

Reason:

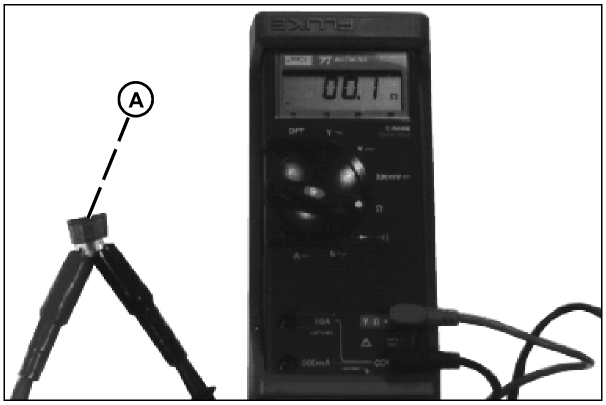
To verify that fuse has continuity.

Equipment:

- Ohmmeter or continuity tester

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Remove fuse from fuse holder.



M49391

4. Check visually for broken filament (A).
5. Connect ohmmeter or continuity tester to each end of fuse.
6. Check for continuity.

Results:

- If continuity is not indicated, replace the fuse.

Instrument Panel Light Test

Reason:

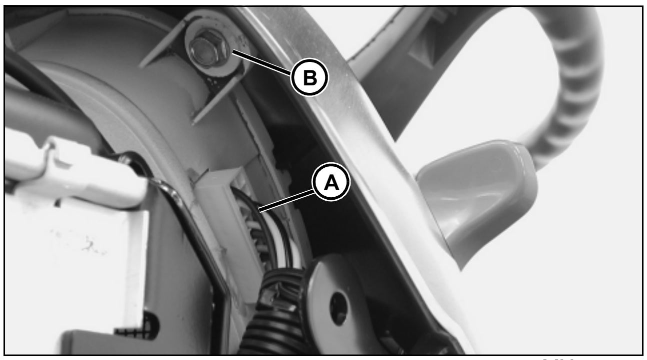
To verify that bulb has continuity.

Equipment:

- Ohmmeter or continuity tester

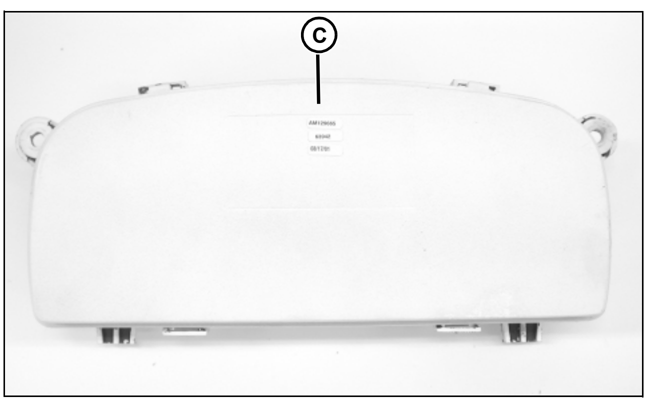
Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Remove side panels.



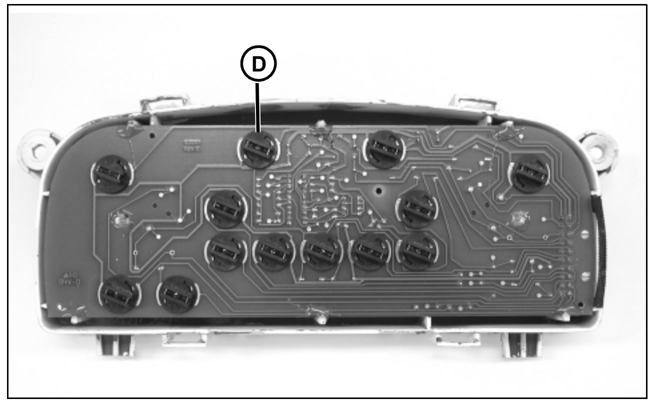
MX13632

4. Disconnect wire harness (A) from instrument panel.
5. Remove capscrews (B) on both sides of instrument panel.
6. Lift instrument panel bottom tabs out of holder, and pivot panel to remove from machine.

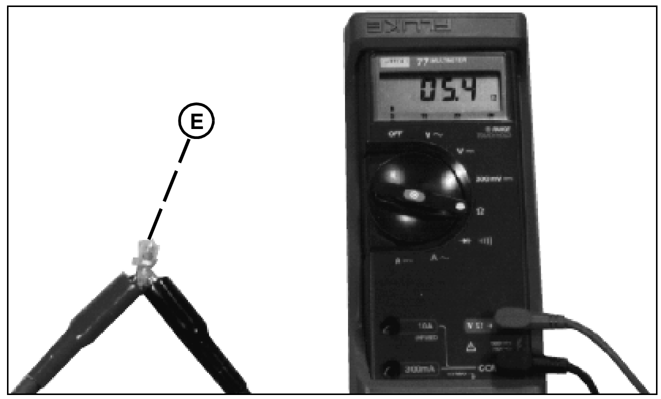


MX13633

7. Push down on tabs and remove panel back (C).



8. Turn bulb sockets (D) counterclockwise to remove.



M49392

9. Check visually for broken filament (E).
10. Connect ohmmeter or continuity tester to each terminal of bulb.
11. Check for continuity.

Results:

- If continuity is not indicated, replace bulb.

PTO Solenoid Test

Reason:

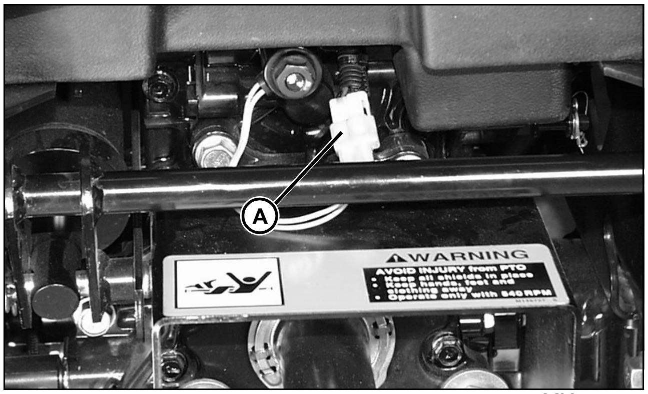
To verify that the solenoid is operating properly.

Equipment:

- Ohmmeter or continuity tester

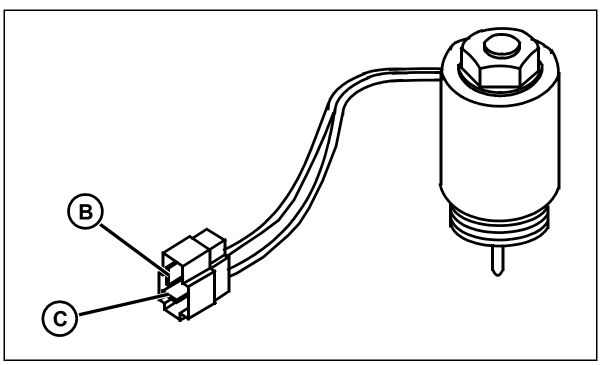
Procedure:

1. Park machine safely in neutral with park brake locked.



MX14285

2. Disconnect PTO solenoid connector (A) at rear of machine.



M91222

3. Using an ohmmeter or continuity tester, check if continuity exists between solenoid terminals.

Results:

- If resistance does not meet specifications, replace solenoid.

PTO Solenoid Continuity:

B to C . . . . 8.0 - 10.0 ohms

Short to Case Test

1. Check for grounds or shorts by connecting tester to one coil terminal and the other to bare metal of coil case or frame.

Results:

- Replace fuel shutoff solenoid if continuity is present from either terminal to coil case.

Audibility Test:

1. Connect PTO solenoid connector (A).
2. Listen for click when PTO solenoid pulls in.

Results:

- If click when PTO solenoid pulls in is not heard, replace solenoid.

PTO Switch Test (optional)

Reason:

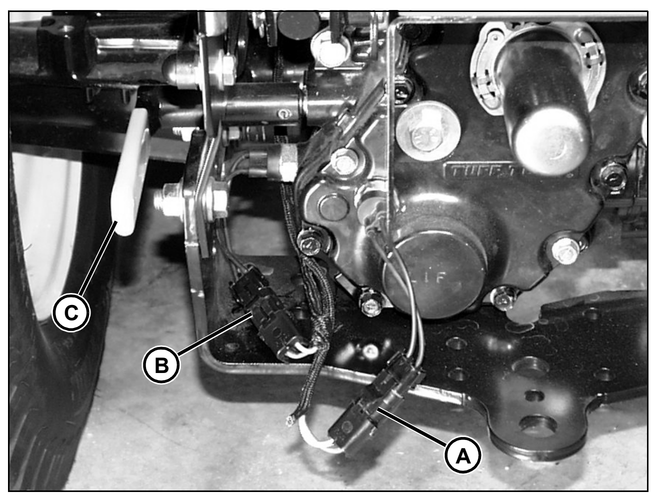
To verify that mid and rear PTO switches function properly.

Equipment:

- Ohmmeter or continuity tester

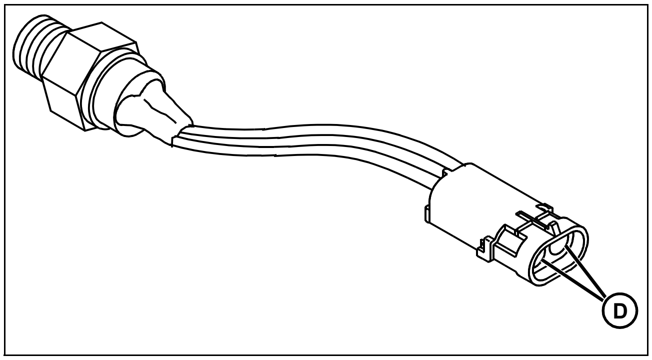
Procedure:

1. Park machine safely in neutral with park brake locked.



MX14286

2. Locate rear PTO switches at rear left side of transaxle and disconnect mid PTO switch connector (A) and rear PTO switch connector (B).
3. Move PTO selector lever (C) to top position.



MIF

4. Check for continuity between switch terminals (D) and compare to specification.

5. Move PTO selector lever (C) to middle position.

6. Check for continuity between switch terminals (D) and compare to specification.

7. Move PTO selector lever (C) to bottom position.

8. Check for continuity between switch terminals (D) and compare to specification.

Replace switch if correct continuity can not be obtained.

Mid PTO Switch Continuity:

PTO Lever Position	Terminal Continuity
Top	No Continuity
Middle	Continuity
Bottom	Continuity

Rear PTO Switch Continuity:

PTO Lever Position	Terminal Continuity
Top	Continuity
Middle	Continuity
Bottom	No Continuity

Fuel Shutoff Solenoid Test - X465, X475 and X575

Reason:

To determine if the fuel shutoff plunger retracts when the solenoid is energized.

Equipment:

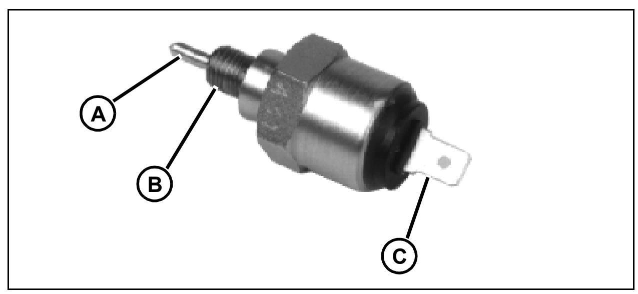
- 2 jumper wires

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.

CAUTION: Avoid Injury! Keep gasoline away from sparks, flame, or hot engine parts or personal injury can result.

3. Remove drain screw and spring to drain gasoline from float bowl.
4. Disconnect fuel shutoff solenoid connector.
5. Remove fuel shutoff solenoid, washer and float bowl.



M82845

6. Connect a jumper wire from the battery positive (+) terminal to solenoid terminal (C). It may be necessary to push plunger (A) inward slightly for plunger to retract.

NOTE: It may be necessary to push plunger (A) inward slightly for plunger to retract.

7. Connect a jumper wire from the battery negative (-) terminal to solenoid threads (B). Plunger should now retract with the solenoid energized.

8. Remove jumper wire from the battery negative (-) terminal. Plunger should extend.

Results:

If plunger does not move, replace solenoid.

Diode Test

Reason:

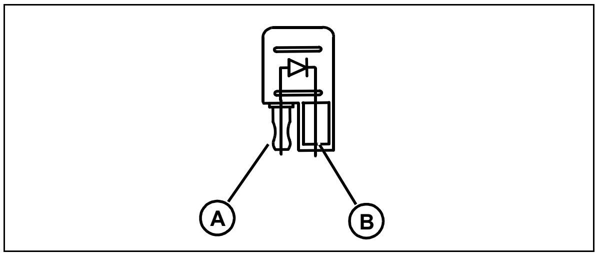
To verify that diode has proper continuity.

Equipment:

- Ohmmeter or continuity tester

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Remove diode from connector.



MIF

4. Connect ohmmeter red (+) lead to pin (A) of diode.
5. Connect ohmmeter black (-) lead to pin (B) of diode. Check for continuity.
6. Reverse test leads. Check for continuity.

Results:

Diode must have continuity in one direction only. Replace defective diode.

Fuel Tank Sensor Test

Reason:

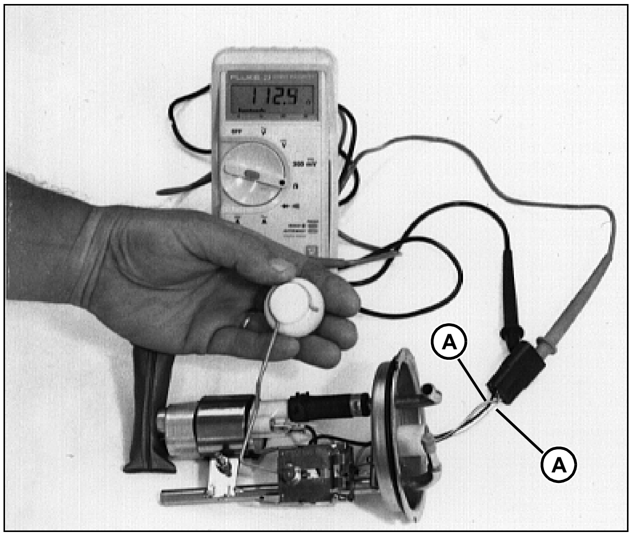
To make sure the fuel tank sensor resistance changes as the float is raised or lowered.

Equipment:

- Ohmmeter

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Remove fender deck.
3. Disconnect fuel tank sensor connector.
4. Remove fuel tank sensor.



M46303a

5. Measure resistance across fuel tank sensor terminals with black and black/white wires (A).
6. Move sensor float arm up and down.
7. Meter must show resistance increase and decrease as float moves up and down. Resistance is the lowest when the float is in the lowest (empty) position.

Specifications:

Sensor resistance (approximately) . . . . . . 6 - 103 ohms

Results:

If the resistance does not change or does not meet the specifications, replace the fuel tank sensor.

Fuel Injector Test - X485 and X585

Reason:

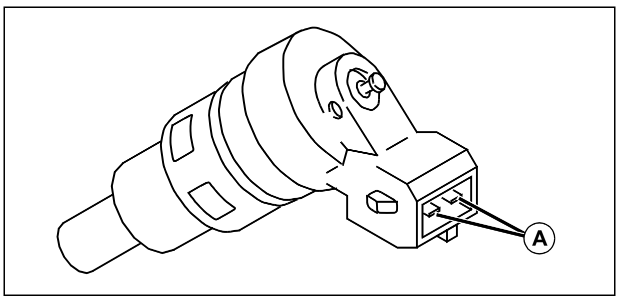
To verify fuel injector continuity and operation.

Equipment:

- Ohmmeter

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Remove air cleaner assembly.
4. Disconnect fuel injector connector.



MIF

5. Measure resistance across fuel injector terminals (A).
6. Connect a jumper wire from battery negative terminal to one of the fuel injector terminals.
7. Briefly and repeatedly connect a jumper wire from battery positive terminal to the other fuel injector terminal.
8. The fuel injector must click each time the jumper wire contacts the terminal.

Specifications:

Resistance at 20° C (68° F) . . . . .13.8 ohms

Results:

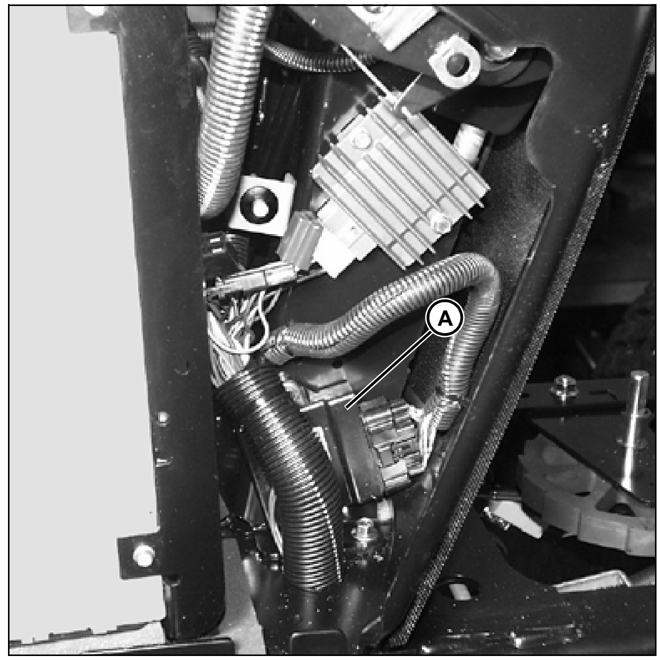
- If resistance does not meet specifications, check fuel injector operation before replacing the fuel injector. The tested resistance values may vary from the specifications due to type of meter used or temperature.
- If the fuel injector does not click, replace the fuel injector.

Fuel Injection Module Test - X485 and X585

Reason:

To determine if fuel injection module is defective.

Procedure:



MX16071

The fuel injection module (A) is very sensitive to the type of meter used to check resistance. Due to variations in the meters, the best way to determine if the fuel injection module is good is to replace the questionable fuel injection module with a known good module.

Results:

- If the new fuel injection module does not solve the problem, check the other fuel injection components.

Vacuum Pressure Sensor Test - X485 and X585

Reason:

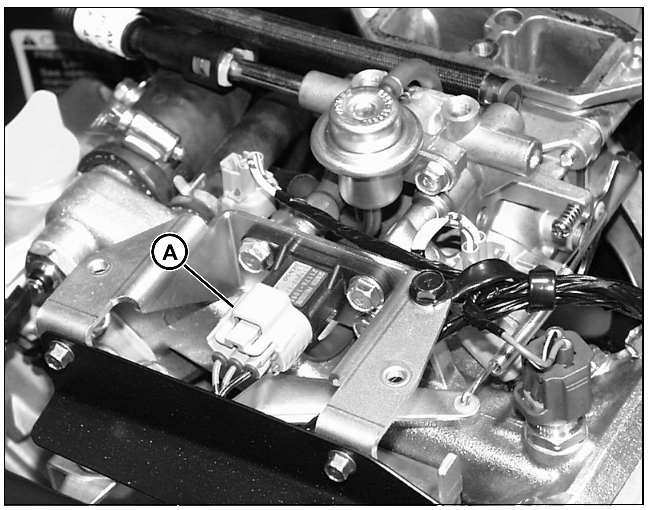
To verify vacuum pressure sensor continuity and operation.

Equipment:

- Ohmmeter
- Voltmeter

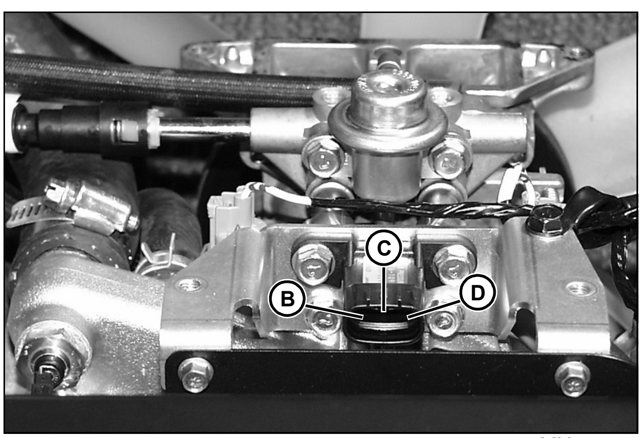
Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Remove air cleaner assembly.



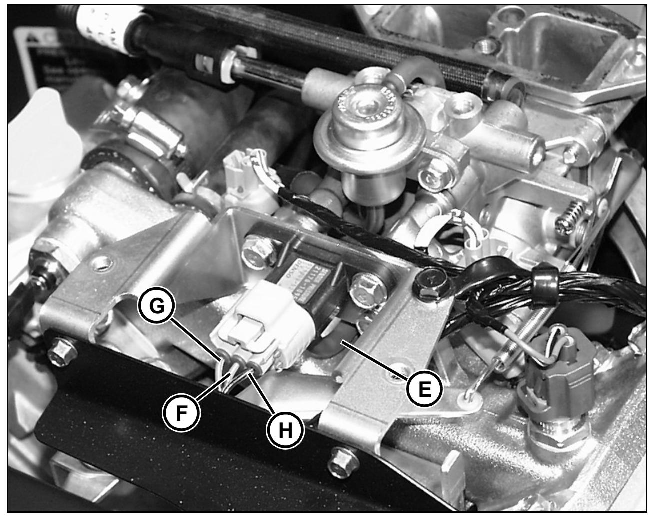
MX14288

4. Disconnect air pressure sensor connector (A).



MX14289

5. Measure resistance between left (B), center (C), and right (D) terminals on vacuum sensor.
6. Connect wire harness to air pressure sensor.



MX14288

7. Disconnect air pressure sensor hose (E).
8. Turn key switch to run position.

NOTE: It will be necessary to remove the rubber seals around the wires to take the voltag e readings.

9. Connect voltmeter negative lead to wire terminal (F).
10. Connect voltmeter positive lead to wire terminal (G).
11. Measure input voltage. Voltage should be about 5 volts.
12. Connect voltmeter positive lead to wire terminal (H).
13. Measure output voltage. Voltage should be something less than 5 volts depending on air pressure.
14. Apply slight air pressure to sensor hose. Voltage should increase.
15. Apply slight vacuum to hose and voltage should decrease.

Specifications:

Terminal resistance (Approximation):

Left (B) and center (C) . . . . 2986 - 3034 ohms
Left (B) and right (D) . . . . 773 - 787 ohms
Center (C) and right (D) . . . . 3774 - 3798 ohms

Input voltage (G). . . . . about 5 volts
Output voltage (H) . . . . 0.5 - 4.9 volts
. . . . . depending on air pressure

Results:

- If resistance does not meet specifications, check output voltage before replacing the air pressure sensor. The tested resistance values may vary from the specifications due to type of meter used or temperature.
- If the output voltage does not increase or decrease, replace the air pressure sensor.

Air Temperature Sensor Test - X485 and X585

Reason:

To verify air temperature sensor continuity.

Equipment:

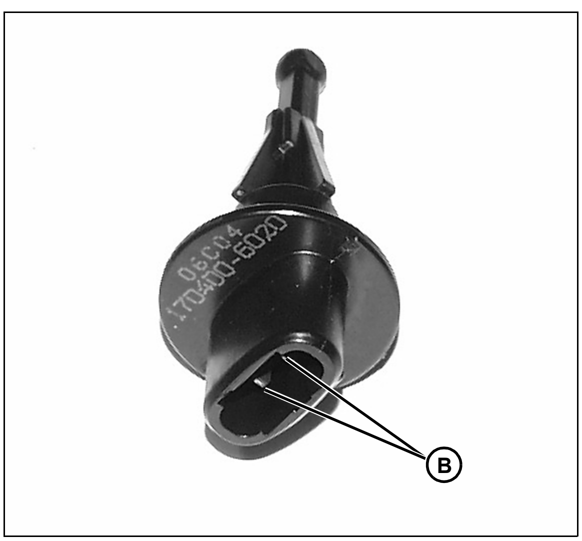
- Ohmmeter

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Remove air cleaner assembly.

MX14290

4. Disconnect air temperature sensor connector (A).



MX14291

5. Measure resistance across air temperature sensor terminals (B).

Specifications:

Resistance at 20°C (68°F) . . . . 2.21 - 2.69 k-ohms
Resistance at 0 - 30°C (32 - 86°F)
. . . . . 5.88 - 1.65 k-ohms

Results:

- If resistance does not meet specifications, replace the air temperature sensor. The tested resistance values may vary from the specifications due to type of meter used or temperature.

Coolant Temperature Sensor Test - X485 and X585

Reason:

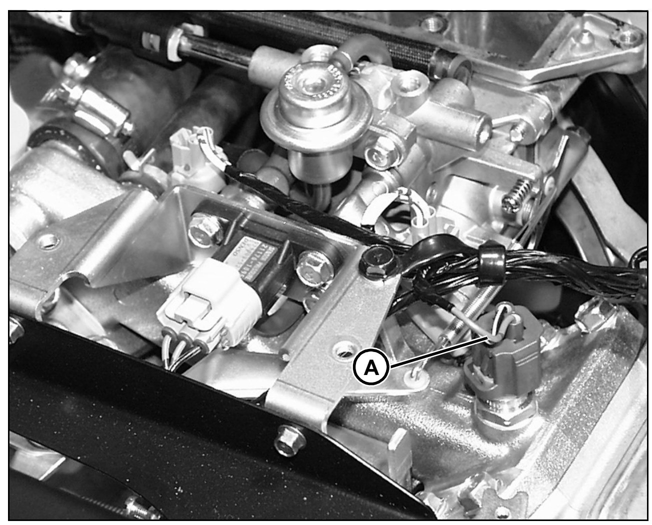
To verify coolant temperature sensor continuity.

Equipment:

- Ohmmeter

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Remove air cleaner assembly.



MX14288

4. Disconnect coolant temperature sensor connector (A).
5. Measure resistance across coolant temperature sensor terminals.

Specifications:

Resistance at 20°C (68°F) . . . . 2.21 - 2.69 k-ohms
Resistance at 0 - 30°C (32 - 86°F)
. . . . . 5.88 - 1.65 k-ohms

Results:

- If resistance does not meet specifications, replace the coolant temperature sensor. The tested resistance values may vary from the specifications due to type of meter used or temperature.

Spark Plug Gap Test

Spark plugs should not be burned, blistered, or have cracked insulator tips or badly eroded electrode.

Make sure the plug is correctly gapped.

Specifications:

X475/X485/X575/X585 - Spark Plug (NGK BPR4ES) . . . . 0.75 mm (0.030 in.)

X465 - Spark Plug (Champion RCJ8Y) . . . . 1.0 mm (0.040 in.)

NOTE: Bending center wire or hitting plug with gapping tool can break insulator.

John Deere recommended plug is the best heat range for the engine application.

If hotter plug is used, go no more than (1) one range hotter.

- For NGK plugs: the lower the number the hotter the plug.
- For Champion Plugs: The higher the number the hotter the plug.

If spark plugs are burning improperly:

- Use known good grade and source of fuel.
- Test fuel for alcohol content. Tester is available at an auto parts stores. Alcohol content should be a maximum of 10%. This rating should be listed on service station pumps.
- Try a different grade (octane) fuel, or a different fuel manufacturer.
- With electronic ignition, check ground connections, ignition components, and a possible sheared flywheel key (unlikely).

Reason:

To determine if spark plug cap is defective.

Equipment:

- Ohmmeter

Procedure:

1. Park machine safely in neutral with park brake locked.
2. Raise hood.
3. Disconnect spark plug cap.

M48364

4. Measure resistance across spark plug cap terminals. Resistance should be about the same as marked on the spark plug cap.

Specifications:

Spark plug cap resistance. . . . . marked on cap

Results:

- If resistance does not meet specification, replace spark plug cap.

Spark Plug Gap Adjustment

Reason:

To maintain the correct gap between the center electrode and the tab needed to produce a good spark.

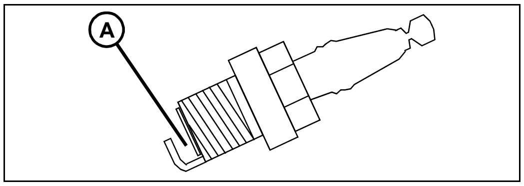
Equipment:

- Feeler gauge

Procedure:

> **IMPORTANT: Avoid damage! Do not clean spark plug with sand paper or abrasives. Engine scoring can result.**

1. Scrape or wire brush deposits from spark plug.
2. Inspect spark plug for:
    - Cracked porcelain.
    - Pitted or damaged electrodes.



MIF

3. Check spark plug gap (A) using a feeler gauge. Set gap to specifications.

4. Install and tighten spark plug to specifications.

Specifications:

Spark Plug Gap X465 . . . . 1.0 mm (0.040 in.)
Spark plug torque . . . . 22 N•m (195 lb-in.)
Spark plug gap X475, X575, X485, and X585. . . . . . . 0.75 mm (0.030 in.)
Spark plug torque . . . . 25 N•m (221 lb-in.)

Ignition Coil Test

Reason:

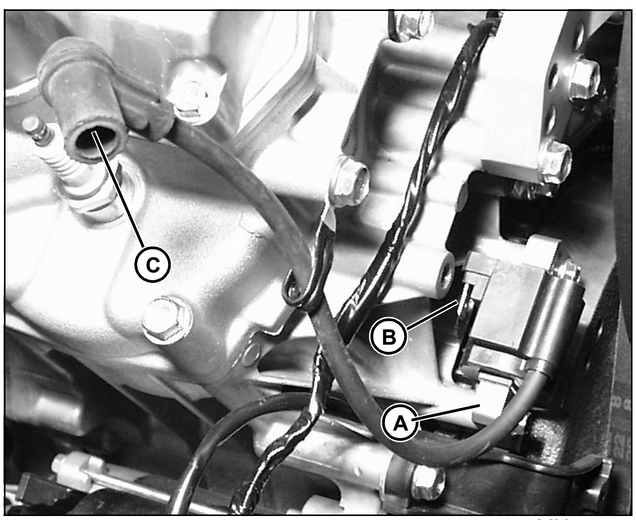
To determine condition of ignition coil windings.

Equipment:

- Ohmmeter

Procedure:

1. Put transmission in neutral. Put key switch in off position.
2. Remove spark plug cap from spark plug wire.
3. Disconnect wires from ignition coil terminal.



MX16073

4. Measure primary coil resistance between laminations (A) and terminal (B).
5. Measure secondary coil resistance between laminations (A) and spark plug wire (C).

Specifications:

Primary coil resistance . . . . 2.2 - 3.6 k-ohms
Secondary coil resistance . . . . 13 - 21 k-ohms

Results:

- If resistance does not meet specifications, replace the ignition coil.

Spark Test

Reason:

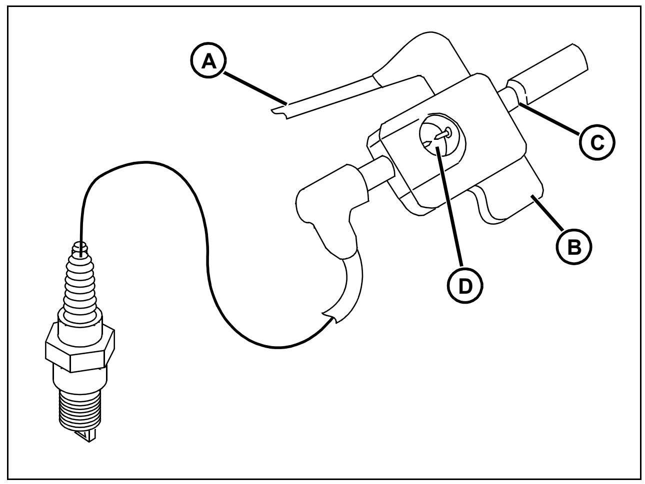
Check overall condition of ignition system.

Equipment:

- D-05351ST Spark Tester

Connections:

1. Brake locked. PTO disengaged.



MIF

2. Remove high tension lead (A) from spark plug and connect to spark tester (B).
3. Connect spark tester to spark plug.
4. Adjust spark tester gap to **4.2 mm (0.166 in.)** with screw (C).

NOTE: Do not adjust spark tester gap beyond 5.0 mm (0.200 in.) as damage to ignition system components could occur.

Procedure:

1. Turn key switch to start position and watch spark (D) at spark tester. If engine will start, watch spark with engine running.

Specifications:

- Steady, strong, blue spark.

Results:

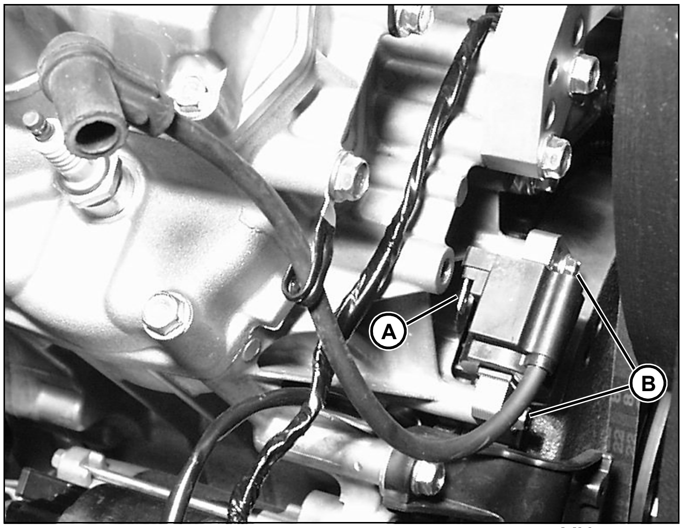
- If spark is weak, or if no spark, install a new spark plug and test again.
- If spark is still weak, or still no spark, run tests on individual components to find cause of malfunction.

Repair

Ignition Coil Replacement

Remove:

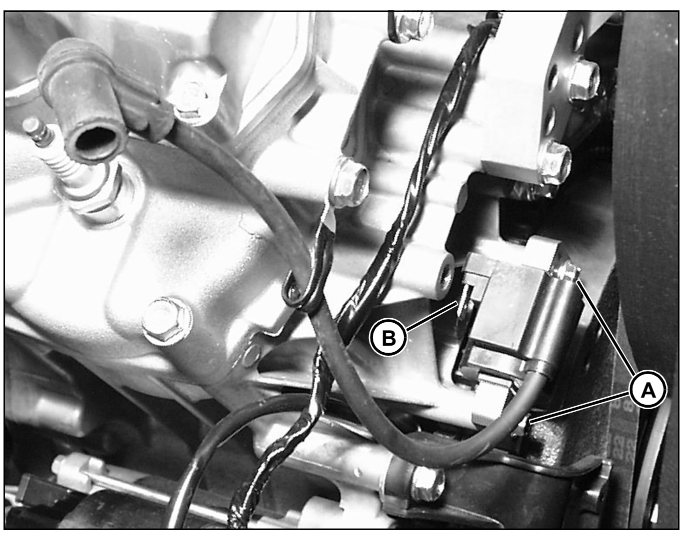
1. Park machine safely. See Parking Machine Safely in SAFETY SECTION.



MX16073

2. Disconnect plug wire from spark plug.
3. Disconnect ignition lead from primary coil terminal (A).
4. Remove ignition coil mounting capscrews (B).

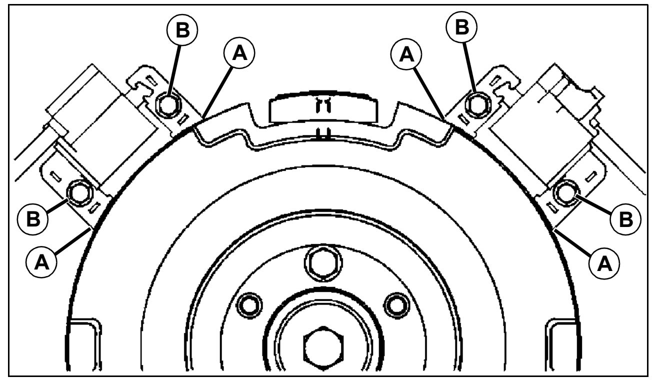
Install:



MX16073

1. Align ignition coil with mounting holes on crankcase.
2. Loosely install mounting capscrews (A).
3. Connect ignition lead from primary coil terminal (B).
4. Connect plug wire to spark plug.

Adjustment:



MX14236

1. Place feeler gauge between ignition coil legs (A) and flywheel.
2. Hold ignition coil in place and tighten the mounting capscrews (B) to **3.4 N•m (30 lb-in.)**.
3. Check air gap again after tightening. Readjust if needed.

Specification:

Air Gap . . . . 0.25 - 0.40 mm (0.010 - 0.016 in.)

Capscrew Torque . . . . 3.4 N•m (30 lb-in.)

Stator Replacement

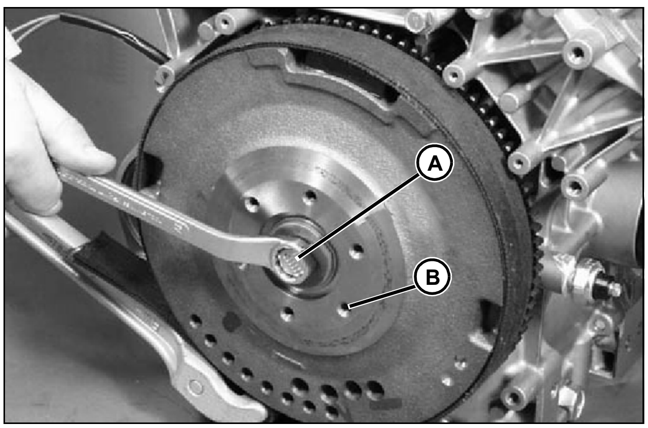
NOTE: The charging system is a permanent magnet and stator design. As the flywheel rotates, a permanent magnet in the flywheel induces AC current in the stator windings. This current flows to the regulator-rectifier where it is converted to DC current needed to charge the battery. Component may differ from illustration.

Removal:

1. Park machine safely. See "Park Machine Safely" on page 3 in Safety section.

NOTE: For X465 machines - See "Flywheel Removal and Installation" on page 79 in Engine section.

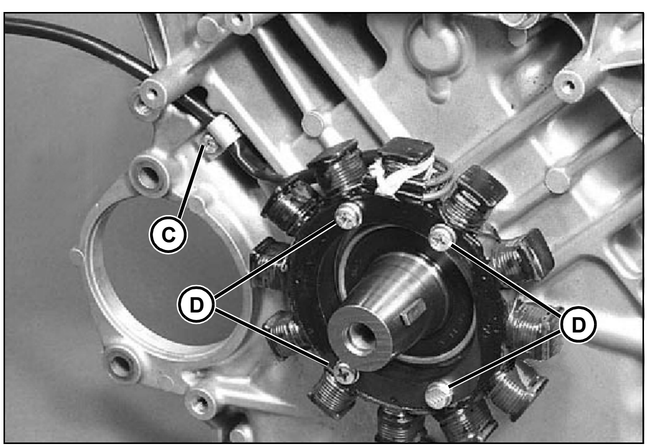
2. Remove engine from machine. See engine removal in appropriate engine section.



MX16077

3. Hold flywheel with strap and remove flywheel bolt (A).

4. Thread a flywheel puller into the pulling holes (B) and remove the flywheel.



MX16078

5. Remove the wire clamp (C).

6. Remove the four mounting capscrews (D) and remove stator from crankcase.

Installation:

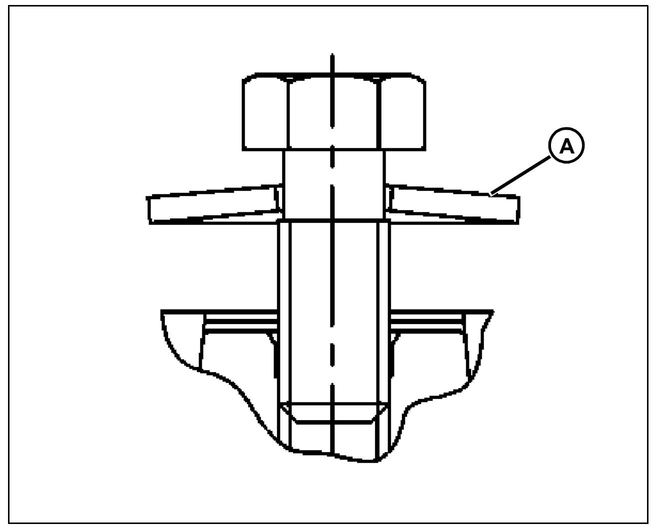
1. Place stator over crankshaft and onto crankcase.

Align the four mounting holes with the threaded holes in the crankcase and the stator wires routed towards the clamp.

2. Install and tighten the four capscrews.

3. Install and tighten the wire clamp.

4. Align the flywheel keyway with the key on the crankshaft and install flywheel.



MX16079

5. Install spring washer (A) as shown.

6. Install and tighten the flywheel bolt.

7. Install engine into machine.

Specification:

Stator Capscrew . . . . 3.4 N•m (30 lb-in.)

Wire Clamp Capscrew . . . . 3.4 N•m (30 lb-in.)

Flywheel Bolt . . . . 56 N•m (41 lb-ft)